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15 PLANNING AND SCHEDULING OPERATIONS

Operations planning and scheduling at an airline like Air New Zealand goes through several stages to match supply with demand, from aggregate plans to short-term schedules. Even after finalizing flights and crew roster schedules, severe weather conditions or mechanical failures can cause last-minute changes. Long-term competitive strength depends on how well it performs this process.

Air New Zealand

How important is scheduling to an airline company? Certainly, customer satisfaction regarding on-time schedule performance is critical in a highly competitive industry such as air transportation. In addition, airlines lose a lot of money when expensive equipment, such as an aircraft, is idle. Flight and crew scheduling, however, is a complex process. For example, Air New Zealand is a group of five airlines with a combined fleet of 96 aircraft, with another 22 more on order. The average utilization is 8:44 hours per day. It has undergone an $800 million upgrade to its long-haul service, refitting its Boeing 747 fleet and adding eight new Boeing 777-200 aircraft for flights to North America. It directly serves 50 ports—26 domestic and 24 international within 15 countries. It carries 11.7 million passengers annually, and its network incorporates flight times ranging from 15 minutes to 13 hours. Operations planning and scheduling at the aggregate level begins with a market plan that identifies the new and existing flight segments that are needed to remain competitive. This general plan is further refined to a three-year plan, and then is put into an annual budget in which flight segments have specific departure and arrival times.

Next, crew availability must be matched to the flight schedules. The two types of crews—pilots and attendants—each comes with its own set of constraints. Pilots, for example, cannot be scheduled for more than 35 hours in a 7-day week and no more than 100 hours in a 28-day cycle. They also must have a 36-hour break every 7 days and 30 days off in an 84-day cycle. Each pilot's tour

of duty begins and ends at a crew base and consists of an alternating sequence of duty periods and rest periods, with duty periods including one or more flights. The schedule must ensure that each flight has a qualified crew and that each crew member has a feasible tour of duty over the roster period. From the crew's point of view, it is also important to satisfy as many crew requests and preferences as possible.

Source: "Service Scheduling at Air New Zealand," *Operations Management 10e Video Library* (Upper Saddle River, NJ: Prentice Hall, 2010); **www.airnewzealand.com** (August, 2011). See video in MyOMLab.

LEARNING GOALS *After reading this chapter, you should be able to:*

1. Describe the operations planning and scheduling process.
2. Explain why the process of matching supply with demand begins with aggregation.
3. Identify the different demand and supply options.
4. Explain how operations plans and schedules relate to other plans.
5. Use spreadsheets to create sales and operations plans.
6. Develop employee schedules.
7. Develop schedules for single workstations.



Managing supply chains effectively requires more than just good demand forecasts. Demand is the first half of the equation, and the other half is supply. The firm must develop plans to supply the resources needed to meet the forecasted demand. These resources include the workforce, materials, inventories, dollars, and equipment capacity.

Operations planning and scheduling is the process of making sure that demand and supply plans are in balance, from the aggregate level down to the short-term scheduling level. As we saw with Air New Zealand, the process begins at the aggregate level and gets progressively more specific until all crew members know their tour of duty. Operations planning and scheduling lies at the core of supply chain integration, around which plans are made up and down the supply chain, from supplier deliveries to customer due dates and services. Table 15.1 defines several types of plans related to operations planning and scheduling.

In this chapter, we focus on two major parts of the overall process: (1) sales and operations planning and (2) scheduling. We begin with the purpose of aggregation in sales and operations planning. We examine how S&OP relates with other plans and functional areas within the firm. We describe a typical planning process, and various strategies to cope with uneven demand. We show how spreadsheets can help find good solutions. Then, we conclude with scheduling, including performance measures and some basic techniques for creating schedules. MyOMLab Supplement J, "Operations Scheduling", provides additional help with scheduling problems.

Operations Planning and Scheduling across the Organization

Operations planning and scheduling is meaningful for each organization along the supply chain. First, it requires managerial inputs from all of the firm's functions. Marketing provides inputs on demand and accounting provides important cost data and a firm's financial condition. Second, each function is affected by the plan. A plan that calls for expanding the workforce has a direct impact on the hiring and training requirements for the human resources function. As the plan is implemented, it creates revenue and cost streams that finance must deal with as it manages the firm's cash flows. Third, each department and group in a firm has its own workforce. Managers of these departments must make choices on hiring, overtime, and vacations.

Scheduling is important for both service and manufacturing processes. Whether the business is an airline, hotel, computer manufacturer, or university, schedules are a part of everyday life. Schedules involve an enormous amount of detail and affect every process in a firm. For example, service, product, and employee schedules determine specific cash flow requirements, trigger the firm's billing process, and initiate requirements for the employee training process. Firms use the scheduling process to lower their costs and improve their responsiveness, affecting operations up and down the supply chain worldwide.

TABLE 15.1 | TYPES OF PLANS WITH OPERATIONS PLANNING AND SCHEDULING

Key Term	Definition
Sales and operations plan (S&OP)	A plan of future aggregate resource levels so that supply is in balance with demand. It states a company's or department's production rates, workforce levels, and inventory holdings that are consistent with demand forecasts and capacity constraints. The S&OP is time-phased, meaning that it is projected for several time periods (such as months or quarters) into the future.
Aggregate plan	Another term for the sales and operations plan.
Production plan	A sales and operations plan for a *manufacturing firm* that centers on production rates and inventory holdings.
Staffing plan	A sales and operations plan for a *service firm*, which centers on staffing and on other human resource-related factors.
Resource plan	An intermediate step in the planning process that lies between S&OP and scheduling. It determines requirements for materials and other resources on a more detailed level than the S&OP. It is covered in the next chapter.
Schedule	A detailed plan that allocates resources over shorter time horizons to accomplish specific tasks.

Stages in Operations Planning and Scheduling

In this section, we explain why companies begin with plans that take a macro, or big picture, view of their business. We also describe how these plans relate to their other plans, and how the long-term plans ultimately are translated into detailed schedules ready for immediate action.

operations planning and scheduling

The process of balancing supply with demand, from the aggregate level down to the short-term scheduling level.

Aggregation

The sales and operations plan is useful because it focuses on a general course of action, consistent with the company's strategic goals and objectives, without getting bogged down in details. We must first aggregate, and then use the targets and resources from the plan to create effective, coordinated schedules. A company's managers must determine whether they can satisfy budgetary goals without having to schedule each of the company's thousands of products and employees individually. While schedules with such detail are the goal, the operations planning and scheduling process begins at the aggregate level.

In general, companies perform aggregation along three dimensions: (1) services or products, (2) workforce, and (3) time.

Product Families A group of customers, services, or products that have similar demand requirements and common process, workforce, and materials requirements is called a **product family**. Sometimes, product families relate to market groupings or to specific processes. A firm can aggregate its services or products into a set of relatively broad families, avoiding too much detail at this stage of the planning process. For instance, a manufacturer of bicycles that produces 12 different models of bikes might divide them into two groups, mountain bikes and road bikes, for the purpose of preparing the sales and operations plan. Common and relevant measurements should be used.

product family

A group of services or products that have similar demand requirements and common process, labor, and materials requirements.

Workforce A company can aggregate its workforce in various ways as well, depending on its flexibility. For example, if workers at the bicycle manufacturer are trained to work on either mountain bikes or road bikes, for planning purposes management can consider its workforce to be a single aggregate group, even though the skills of individual workers may differ.

Time The planning horizon covered by a sales and operations plan typically is one year, although it can differ in various situations. To avoid the expense and disruptive effect of frequent changes in output rates and the workforce, adjustments usually are made monthly or quarterly. In other words, the company looks at time in the aggregate—months, quarters, or seasons—rather than in weeks, days, or hours.

The Relationship of Operations Plans and Schedules to Other Plans

A financial assessment of the organization's near future—that is, for 1 or 2 years ahead—is called either a business plan (in for-profit firms) or an annual plan (in nonprofit service organizations). A **business plan** is a projected statement of income, costs, and profits. It usually is accompanied by budgets, a projected (pro forma) balance sheet, and a projected cash flow statement, showing sources and allocations of funds. The business plan unifies the plans and expectations of

business plan

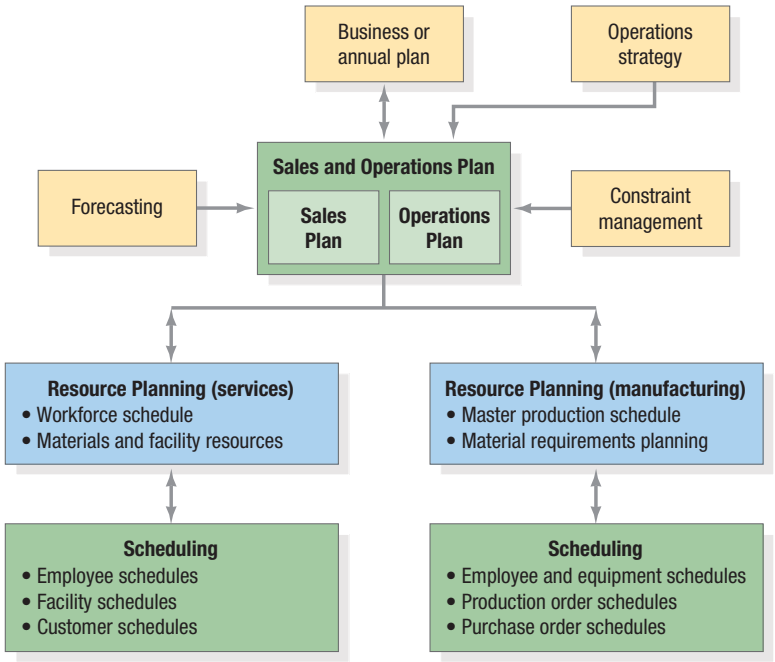
A projected statement of income, costs, and profits.

a firm's operations, finance, sales, and marketing managers. In particular, it reflects plans for market penetration, new product introduction, and capital investment. Manufacturing firms and for-profit service organizations, such as a retail store, a firm of attorneys, or a hospital, prepare such plans. A nonprofit service organization, such as the United Way or a municipal government, prepares a different type of plan for financial assessment, called an **annual plan or financial plan**.

annual plan or financial plan
A plan for financial assessment used by a nonprofit service organization.

Figure 15.1 illustrates the relationships among the business or annual plan, sales and operations plan, and detailed plans and schedules derived from it. For **service providers** in the supply chain, top management sets the organization's direction and objectives in the business plan (in a for-profit organization) or annual plan (in a not-for-profit organization). This plan then provides the framework for developing the sales and operations plan, which typically focuses on staffing and other human resource-related factors at a more aggregate level. It presents the number and types of employees needed to meet the objectives of the business or annual plan.

FIGURE 15.1 ▶
The Relationship of Sales and Operations Plans and Schedules to Other Plans



Based on the sales and operations plan for a service provider, the next planning level is *resource planning* to determine the firm's workforce schedules and other resource requirements, such as materials and facilities, on a more detailed level. The *workforce schedule* details the specific work schedule for each category of employee. For example, a sales and operations plan might allocate 10 police officers for the day shift in a particular district; the workforce schedule might assign 5 of them to work Monday through Friday and the other 5 to work Wednesday through Sunday to meet the varying daily needs for police protection in that district. The lowest planning level is *scheduling*, which puts together day-to-day schedules for individual employees and customers.

For manufacturing firms in the supply chain, top management sets the company's strategic objectives for at least the next year in the business plan. It provides the overall framework, along with inputs coming from operations strategy, forecasting, and capacity constraint management. The sales and operations plan specifies product family production rates, inventory levels, and workforce levels. The next planning level beneath the sales and operations plan is resource planning, which we cover in the next chapter. Resource planning gets specific as to individual products within each product family, purchased materials, and resources on a detailed level. The *master production schedule* specifies the timing and size of production quantities for each product in the product families. The *material requirements planning* process then derives plans for components, purchased materials, and workstations. As with service providers, the lowest and most detailed planning level is scheduling. It puts together day-to-day schedules or priorities for employees, equipment, and production or purchase orders. Thus, the sales and operations plan plays a key role in translating the strategies of the business plan into an operational plan for the manufacturing process.

As the arrows in Figure 15.1 indicate, information flows in two directions: from the top down (broad to detailed) and from the bottom up (detailed to broad). If a sales and operations plan

cannot be developed to satisfy the objectives of the business or annual plan with the existing resources, the business or annual plan might need some adjustment. Similarly, if a feasible master production schedule or workforce schedule cannot be developed, the sales and operations plan might need some adjustment. The planning process is dynamic, with periodic plan revisions or adjustments based on two-way information flows, typically on a monthly basis.

Managing Demand

Matching supply with demand becomes a challenge when forecasts call for uneven demand patterns—and uneven demand is more the rule than the exception. Demand swings can be from one month to the next, one week to the next, or even one hour to the next. Peaks and valleys in demand are costly or can cause poor customer service. Air New Zealand can lose sales because capacity is exceeded for one of its flights, while another of its flights to the same destination at about the same time has many empty seats. If nothing is done to even out demand, sales are lost or greater capacity cushions might be needed. For other companies, the supply options for handing uneven demand could be overtime, hiring and curtailing the workforce, and anticipation inventories. All come at an extra cost. Here we deal with **demand management**, the process of changing demand patterns using one or more demand options.

demand management
The process of changing demand patterns using one or more demand options.

Demand Options

Various options are available in managing demand, including complementary products, promotional pricing, prescheduled appointments, reservations, revenue management, backlogs, backorders, and stockouts. The manager may select one or more of them, as we illustrate below.

Complementary Products One demand option for a company to even out the load on resources is to produce **complementary products**, or services that have similar resource requirements but different demand cycles. For example, manufacturers of matzoh balls for the Jewish Passover holiday are in a seasonal business. The B. Manischewitz Company, a kosher foods manufacturer in Jersey City, New Jersey, previously experienced 40 percent of its annual sales for the 8-day Passover holiday alone. It expanded toward markets with year-round appeal such as low-carb, low-fat foods, including canned soups and crackers, borscht, cake mixes, dressing and spreads, juices, and condiments.

complementary products
Services or products that have similar resource requirements but different demand cycles.

For service providers, a city parks and recreation department can counterbalance seasonal staffing requirements for summer activities by offering ice skating, tobogganing, or indoor activities during the winter months. The key is to find services and products that can be produced with the existing resources and can level off the need for resources over the year.

Promotional Pricing Promotional campaigns are designed to increase sales with creative pricing. Examples include automobile rebate programs, price reductions for winter clothing in the late summer months, reduced prices for hotel rooms during off-peak periods, and "two-for-the-price-of-one" automobile tire sales. Lower prices can increase demand for the product or service from new and existing customers, take sales from competitors, or encourage customers to move up future buying. The first two outcomes increase overall demand, while the third shifts demand to the current period.

Prescheduled Appointments Service providers often can schedule customers for definite periods of order fulfillment. With this approach, demand is leveled to not exceed supply capacity. An appointment system assigns specific times for service to customers. The advantages of this method are timely customer service and the high utilization of service personnel.

Doctors, dentists, lawyers, and automobile repair shops are examples of service providers that use appointment systems. Doctors can use the system to schedule parts of their day to visit hospital patients, and lawyers can set aside time to prepare cases. Care must be taken to tailor the length of appointments to individual customer needs rather than merely scheduling customers at equal time intervals.



Thinkstock Images/Getty Images

This doctor's office uses the appointment system to schedule patients for definite periods of order fulfillment. Times are selected that are agreeable with the patient and fit with the doctor's schedule. Demand is leveled and does not exceed capacity.

Reservations Reservation systems, although quite similar to appointment systems, are used when the customer actually occupies or uses facilities associated with the service. For example, customers reserve hotel rooms, automobiles, airline seats, and concert seats. The major advantage of reservation systems is the lead time they give service managers and the ability to level demand. Managers can deal with no-shows with a blend of overbooking, deposits, and cancellation penalties. Sometimes overbooking means that a customer with reservations cannot be served as promised. In such cases, bonuses can be offered for compensation. For example, an airline passenger might not only get on the next available flight, but also may be given a free ticket for a second flight sometime in the future.

Revenue Management A specialized combination of the pricing and reservation options for service providers is revenue management. **Revenue management** (sometimes called *yield management*) is the process of varying price at the right time for different customer segments to maximize revenues generated from existing supply capacity. It works best if customers can be segmented, prices can be varied by segment, fixed costs are high, variable costs are low, service duration is predictable, and capacity is lost if not used (sometimes called *perishable capacity*). Airlines, hotels, cruise lines, restaurants (early-bird specials), and rental cars are good examples. Computerized reservation systems can make hour-by-hour updates, using decision rules for opening or closing price classes depending on the difference between supply and continually updated demand forecasts. In the airlines industry, prices are lowered if a particular airline flight is not selling as fast as expected, until more seats are booked. Alternately, if larger than expected demand is developing, prices for the remaining seats may be increased. Last-minute business travelers pay the higher prices, whereas leisure travelers making reservations well in advance and staying over the weekend get the bargain prices. Southwest Airlines now segments its customers by creating a "Business Select" ticket class that rewards more perks to frequent fliers willing to pay higher prices. Managerial Practice 15.1 describes an integrated approach toward segmenting customers and diversifying revenue streams at an airline.

revenue management

Varying price at the right time for different customer segments to maximize revenues yielded by existing supply capacity.

MANAGERIAL PRACTICE 15.1 Singapore Airlines Group

Singapore Airlines (SIA) is a global brand, known for its innovation, safety, and service excellence. Against the backdrop of strong competition from regional low-cost carriers, highly volatile fuel prices, security and safety concerns, SIA has consistently maintained above-industry average profit margins in the 10-year period between 2001 and 2009. One of the keys to its success is in its revenue management approach which included the key elements below.

Customer Segmentation. Singapore Airlines has a 49 percent stake in Tiger Airways, a budget airline, and in May 2011 announced the formation of a 100 percent owned low-cost carrier which will start operations in 2012. It has also introduced one of the lowest seating densities in the industry for its new planes, with plans to increase the number of business class seats from 60 to 86, while reducing the number of economy seats from 399 to 311. This allows SIA to cover different margin customer segments more effectively.

Business Ecosystems. Singapore Airlines recognizes it cannot compete in a vacuum and involves networks of players that support and sustain its revenue model. By tying up with leading hotels, restaurants, and retailers to offer discounts to its customers, it not only accentuates SIA's differentiation, but also adds revenue through commissions and cross selling.

Risk Management. Singapore Airlines flies to a vast spread of geography so that it does not rely on any one geographic market for a majority of its revenues. This strategy helps spread business risks arising from a change of flying habits and patterns, and human-induced and natural disasters.



Gordon Tipene | Dreamstime.com

Singapore Airlines consistently maintains an above-industry average profit, due, in part, to the revenue management approach undertaken by the company.

Source: Datamonitor (2011) "SWOT Analysis of Singapore Airlines." Flight International, May 31, 2011, "SIA to reduce A380 seating density." Heracleous, L. & J. Wirtz (2010) "Singapore Airlines' Balancing Act–Asia's Premier Carrier Successfully Executes a Dual Strategy: It Offers World-class Service and is a Cost Leader." *Harvard Business Review 88,* (7/8): 145–149.

Backlogs Much like the appointments or reservations of service providers, a **backlog** is an accumulation of customer orders that a manufacturer has promised for delivery at some future date. Manufacturers in the supply chain that maintain a backlog of orders as a normal business practice can allow the backlog to grow during periods of high demand and then reduce it during periods of low demand. Airplane manufacturers do not promise instantaneous delivery, as do wholesalers or retailers farther forward in the supply chain. Instead, they impose a lead time between when the order is placed and when it is delivered. For example, an automotive parts manufacturer may agree to deliver to the repair department of a car dealership a batch of 100 door latches for a particular car model next Tuesday. The parts manufacturer uses that due date to plan its production of door latches within its capacity limits. Firms that are most likely to use backlogs—and increase the size of them during periods of heavy demand—make customized products and tend to have a make-to-order strategy. Backlogs reduce the uncertainty of future production requirements and also can be used to level demand. However, they become a competitive disadvantage if they get too big.

backlog
An accumulation of customer orders that a manufacturer has promised for delivery at some future date.

Backorders and Stockouts A last resort in demand management is to set lower standards for customer service, either in the form of backorders or stockouts (see Chapter 10, "Designing Effective Supply Chains"). Not to be confused with a backlog, a *backorder* is a customer order that cannot be filled immediately but is filled as soon as possible. Demand may be too unpredictable or the item may be too costly to hold it in inventory. Although the customer is not pleased with the delay, the customer order is not lost and it is filled at a later date. In contrast, a *stockout* is much the same, except that the order is lost and the customer goes elsewhere. A backorder adds to the next period's demand requirement, whereas a stockout does not. Backorders and stockouts can lead dissatisfied customers to do their future business with another firm. Generally, backorders and stockouts are to be avoided.

Combinations of demand options can also be used. For example, a manufacturer of lighting equipment had several products characterized as "slow movers with spikes," where only 2 or 3 units were sold for several weeks, and then suddenly there was a huge order for 10,000 units the next week. The reason is that their product was purchased by commercial property managers who might be upgrading the lighting in a large office building. The result was a forecasting nightmare and having to resort to high cost supply options to meet the demand spikes. The breakthrough in solving this problem was to combine the pricing and backlog options. Contractors are now offered a 3 percent discount (the pricing option) on any order in excess of 10,000 units that are placed five or more weeks before they are needed (the backlog option). The advanced warning allows the manufacturer to smooth out its production processes, saving millions of dollars annually.

The left side of Table 15.2 summarizes the demand options for operations planning and scheduling, and the right side lists the supply options for balancing supply with demand. The following two sections on sales and operations planning and scheduling cover the supply options.

TABLE 15.2 DEMAND AND SUPPLY OPTIONS FOR OPERATIONS PLANNING AND SCHEDULING

Demand Options	Supply Options
Complementary products	Anticipation inventory
Promotional pricing	Workforce adjustment (hiring and layoffs)
Prescheduled appointments	Workforce utilization (overtime and undertime)
Reservations	Part-time workers and subcontractors
Revenue management	Vacation schedules
Backlogs	Workforce schedules
Backorders	Job and customer sequences
Stockouts	Expediting

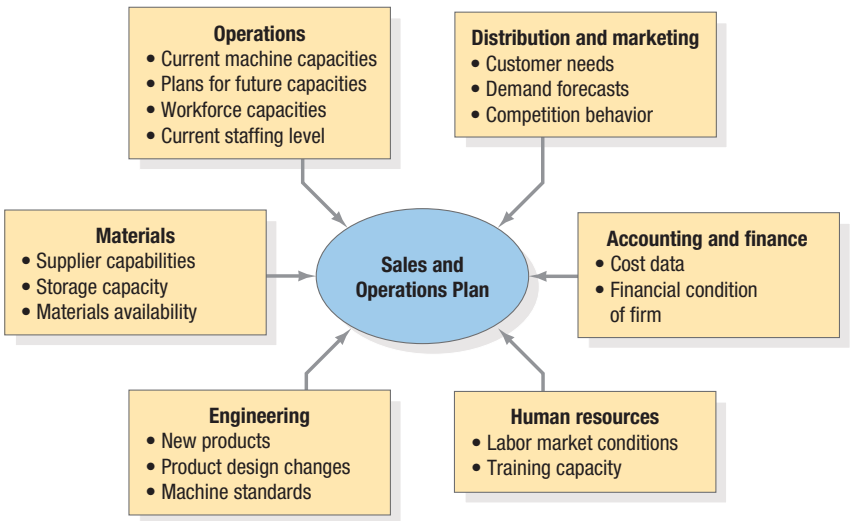
Sales and Operations Plans

Developing sales and operations plans means making decisions. In this section, we concentrate on the information inputs, the supply options themselves, and strategies that go into the sales and operations planning (S&OP) decisions.

Information Inputs

Just as it is needed to manage the demand side, consensus is needed among the firm's departments when decisions for the supply side are made. Information inputs are sought to create a plan that works for all. Figure 15.2 lists inputs from each functional area. They must be accounted for to make sure that the plan is a good one and also doable. Such coordination helps synchronize the flow of services, materials, and information through the supply chain to best balance supply with customer demand.

FIGURE 15.2 ▶
Managerial Inputs from Functional Areas to Sales and Operations Plans



Supply Options

Given demand forecasts, as modified by demand management choices, operations managers must develop a plan to meet the demand, drawing from the supply options listed in Table 15.2.



Photo by Michael Springer/Bloomberg via Getty Images

An employee stocks a Whirlpool air conditioner at a Lowe's store in Westborough, Massachusetts. The demand for window units is highly seasonal and also depends on variations in the weather. Typically, Whirlpool begins production of room air conditioners in the fall and holds them as inventory until they are shipped in the spring. Building anticipation inventory in the slack season allows the company to even out production rates over much of the year and still satisfy demand in the peak periods (spring and summer) when retailers are placing most of their orders.

Anticipation Inventory *Anticipation inventory* can be used to absorb uneven rates of demand or supply. For example, a plant facing seasonal demand can stock anticipation inventory during light demand periods and use it during heavy demand periods. Manufacturers of air conditioners, such as Whirlpool, can experience 90 percent of their annual demand during just three months of a year. Extra, or anticipation inventory, also can help when supply, rather than demand, is uneven. For example, a company can stock up on a certain purchased item if the company's suppliers expect severe capacity limitations. Despite its advantages, anticipation inventory can be costly to hold, particularly if stocked in its finished state. Moreover, when services or products are customized, anticipation inventory is not usually an option. Service providers in the supply chain generally cannot use anticipation inventory because services cannot be stocked.

Workforce Adjustment Management can adjust workforce levels by hiring or laying off employees. The use of this alternative can be attractive if the workforce is largely unskilled or semiskilled and the labor pool is large. These conditions are more likely found in some countries than in others. However, for a particular company, the size of the qualified labor pool may limit the number of new employees that can be hired at any one time. Also, new employees must be trained, and the capacity of the training facilities themselves might limit the number of new hires at any one time. In some industries, laying off employees is difficult or unusual for contractual reasons (unions); in other industries, such as tourism and agriculture, seasonal layoffs and hirings are the norm.

overtime
The time that employees work that is longer than the regular workday or workweek for which they receive additional pay.

Workforce Utilization An alternative to a workforce adjustment is a change in workforce utilization involving overtime and undertime. **Overtime** means that employees work longer than the regular workday or workweek and receive additional pay for the extra hours. It can be used to

satisfy output requirements that cannot be completed on regular time. Overtime is expensive (typically 150 percent of the regular-time pay rate) and workers often do not want to work a lot of overtime for an extended period of time. Excessive overtime also can result in declining quality and productivity. On the other hand, it helps avoid the costly fringe benefits (such as health insurance, dental care, Social Security, retirement funds, paid vacations, and holidays) that come with hiring a new full-time employee,

Undertime means that employees do not have enough work for the regular-time workday or workweek. For example, they cannot be fully utilized for eight hours per day or for five days per week. Undertime occurs when labor capacity exceeds demand requirements (net of anticipation inventory), and this excess capacity cannot or should not be used productively to build up inventory or to satisfy customer orders earlier than the delivery dates already promised.

undertime

The situation that occurs when employees do not have enough work for the regular-time workday or workweek.

Undertime can either be paid or unpaid. An example of *paid undertime* is when employees are kept on the payroll rather than being laid off. In this scenario, employees work a full day and receive their full salary but are not as busy because of the light workload. Some companies use paid undertime (though they do not call it that) during slack periods, particularly with highly skilled, hard-to-replace employees or when there are obstacles to laying off workers. The disadvantages of paid undertime include the cost of paying for work not performed and lowered productivity.

Part-Time Workers Another option apart from undertime is to hire part-time workers, who are paid only for the hours and days worked. Perhaps they only work during the peak times of the day or peak days of the week. Sometimes, part-time arrangements provide predictable work schedules, but in other cases workers are not called in if the workload is light. Such arrangements are more common in low-skill positions or when the supply of workers seeking such an arrangement is sufficient. Part-time workers typically do not receive fringe benefits.

Subcontractors Subcontractors can be used to overcome short-term capacity shortages, such as during peaks of the season or business cycle. Subcontractors can supply services, make components and subassemblies, or even assemble an entire product.

Vacation Schedules A manufacturer can shut down during an annual lull in sales, leaving a skeleton crew to cover operations and perform maintenance. Hospital employees might be encouraged to take all or part of their allowed vacation time during slack periods. The use of this alternative depends on whether the employer can mandate the vacation schedules of its employees. In any case, employees may be strongly discouraged from taking vacations during peak periods or encouraged to take vacations during slack periods.

chase strategy

A strategy that involves hiring and laying off employees to match the demand forecast.

level strategy

A strategy that keeps the workforce constant, but varies its utilization via overtime, undertime, and vacation planning to match the demand forecast.

Planning Strategies

Here we focus on supply options that define output rates and workforce levels. Two basic strategies are useful starting points in searching for the best plan.

1. *Chase Strategy.* The **chase strategy** involves hiring and laying off employees to match the demand forecast over the planning horizon. Varying the workforce's regular-time capacity to equate supply to demand requires no inventory investment, overtime, or undertime. The drawbacks are the expense of continually adjusting workforce levels, the potential alienation of the workforce, and the loss of productivity and quality because of constant changes in the workforce.
2. *Level Strategy.* The **level strategy** involves keeping the workforce constant (except possibly at the beginning of the planning horizon). It can vary its utilization to match the demand forecast via overtime, undertime (paid or unpaid), and vacation planning (i.e., paid vacations when demand is low). A constant workforce can be sized at many levels: Managers can choose to maintain a large workforce so as to minimize the planned use of overtime during peak periods (which, unfortunately, also maximizes the need for undertime during slack periods). Alternately, they can choose to maintain a smaller workforce and rely heavily on overtime during the peak periods (which places a strain on the workforce and endangers quality).



Justin Kase zsixz/Alamy

Even though Hallmark's business is seasonal, the company has never laid off employees. Instead the company's employees are trained to do different jobs at different times, and at different plants, if need be. Because they know they have greater job security, they work hard to keep setup times short and Hallmark's costs low.

mixed strategy
A strategy that considers the full range of supply options.

These two "pure" strategies used alone usually do not produce the best sales and operations plan. It might not be best to keep the workforce exactly level, or to vary it to exactly match forecasted demand on a period-by-period basis. The best strategy, therefore, usually is a **mixed strategy** that considers the full range of supply options. The chase strategy is limited to just hiring and laying off employees. The level strategy is limited to overtime, undertime, and vacation schedules. The mixed strategy opens things up to all options, including anticipation inventory, part-time workers, subcontractors, backorders, and stockouts.

Constraints and Costs

An acceptable sales and operations plan must recognize relevant constraints or costs. Constraints can be either physical limitations or related to managerial policies. Examples of physical constraints might be machine capacities that limit maximum output or inadequate inventory storage space. Policy constraints might include limitations on the number of backorders or the use of subcontractors or overtime, as well as the minimum inventory levels needed to achieve desired safety stocks. Ethical issues may also be involved, such as excessive layoffs or required overtime.

Typically, many plans can contain a number of constraints. Table 15.3 lists the costs that the planner considers when preparing sales and operations plans.

Sales and Operations Planning as a Process

Sales and operations planning is a decision-making process, involving both planners and management. It is dynamic and continuing, as aspects of the plan are updated periodically when new information becomes available and new opportunities emerge. It is a cross-functional process that seeks a set of plans that all of a firm's functions can support. For each product family, decisions are made based on cost trade-offs, recent history, recommendations by planners and middle management, and the executive team's judgment.

Figure 15.3 shows a typical plan for a manufacturer. The plan is for one of the manufacturer's make-to-stock product families expressed in aggregate units. This simple spreadsheet shows the interplay between demand and supply. The history on the left for January through March shows how forecasts are tracking actual sales, and how well actual production conforms to the plan. The inventory projections are of particular interest to finance because they significantly affect the manufacturer's cash requirements. The last two columns on the top right show how current fiscal year sales projections match up with the current business plan.

This particular plan is projected out for 18 months, beginning with April. The forecast, operations, and inventory sections for the first 6 months are shown on a month-by-month basis. They then are shown on a quarterly basis for the second 6 months. Finally, the totals for the last 6 months in the time horizon are given in just one column. This display gives more precision to the short term and yet gives coverage well into the future—all with a limited number of columns.

This particular make-to-stock family experiences highly seasonal demand. The operations plan is to build up seasonal inventory in the slack season, schedule vacations as much as

TABLE 15.3 | TYPES OF COSTS WITH SALES AND OPERATIONS PLANNING

Cost	Definition
Regular time	Regular-time wages paid to employees plus contributions to benefits, such as health insurance, dental care, Social Security, retirement funds, and pay for vacations, holidays, and certain other types of absences.
Overtime	Wages paid for work beyond the normal workweek, typically 150 percent of regular-time wages (sometimes up to 200 percent for Sundays and holidays), exclusive of fringe benefits. Overtime can help avoid the extra cost of fringe benefits that come with hiring another full-time employee.
Hiring and layoff	Costs of advertising jobs, interviews, training programs for new employees, scrap caused by the inexperience of new employees, loss of productivity, and initial paperwork. Layoff costs include the costs of exit interviews, severance pay, retaining and retraining remaining workers and managers, and lost productivity.
Inventory holding	Costs that vary with the level of inventory investment: the costs of capital tied up in inventory, variable storage and warehousing costs, pilferage and obsolescence costs, insurance costs, and taxes.
Backorder and stockout	Additional costs to expedite past-due orders, the costs of lost sales, and the potential cost of losing a customer to a competitor (sometimes called loss of goodwill).

Artic Air Company—April Sales and Operations Plan

Family: Medium window units (make-to-stock) *Unit of measure:* 100 units

	HISTORY									3rd	4th	Mos	Fiscal Year	Business
SALES	J	F	M	A*	M	J	J	A	S	3 Mos**	3 Mos	13–18	Projection ($000)	Plan ($000)
New forecast	45	55	60	70	85	95	130	110	70	150	176	275	$8,700	$8,560
Actual sales	52	40	63											
Diff for month	7	–15	3											
Cum		–8	–5											
OPERATIONS														
New Plan	75	75	75	75	75	85	85	85	75	177	225			
Actual	75	78	76											
Diff for month	0	3	1											
Cum		3	4											
INVENTORY														
Plan	85	105	120	125	115	105	60	35	40	198	321			
Actual	92	130	143											

DEMAND ISSUES AND ASSUMPTIONS

1. New product design to be launched in January of next year.

SUPPLY ISSUES

1. Vacations primarily in November and December.
2. Overtime in July–August.

* April is the first month of the planning horizon for this current plan. When next month's plan is developed, its first month in the planning horizon will be May, and the most recent month of the history will be April (with January no longer shown in the history).

** This column provides the sales, operations, and inventory totals for October through December. For example, the forecast of 150 units translates into an average of 50 units per month (or 150/3 = 50).

◀ FIGURE 15.3
Sales and Operations Plan for Make-to-Stock Product Family

possible in November and December, and use overtime in the peak season of June, July, and August. For example, the Operations plan increases monthly production from 75 to 85 for June through August, returns to 75 for September, and then drops to an average of only 59 (or 177/3) for October through December. Plan spreadsheets use different formats depending on production and inventory strategy. For an assemble-to-order strategy, the inventory does not consist of finished goods. Instead, it is inventory of standardized components and subassemblies built for the finishing and assembly operations. For the make-to-order strategy, the inventory section in the plan of Figure 15.3 is replaced by a section showing the planned and actual order backlog quantities.

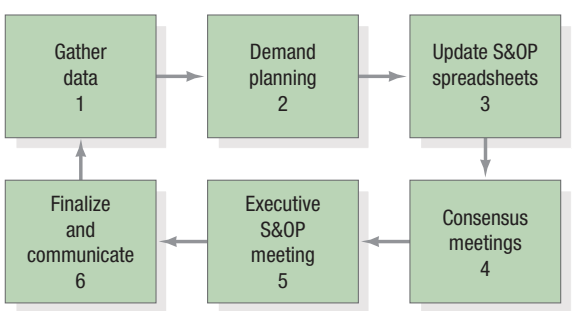
Plans for service providers are quite different. For one thing, their plan does not contain an inventory section, but focuses instead on the demand and supply of human resources. Forecasts are typically expressed in terms of employees required, with separate rows for regular time, overtime, vacations, part-time workers, and so on. Different departments or worker classifications replace product families.

The process itself, typically done on a monthly basis, consists of six basic steps. They are much like the steps we discussed in Chapter 14, "Forecasting Demand."

Step 1. Begin to "roll forward" the plan for the new planning horizon. Start preliminary work right after the month's end. Update files with actual sales, production, inventory, costs, and constraints.

Step 2. Participate in the forecasting and demand planning to create the authorized demand forecasts. For service providers, the forecasts are staff requirements for each workforce group. For example, a director of nursing in a hospital can develop a workload index for a nursing staff and translate a projection of the month-to-month patient load into an equivalent total amount of nursing care time—and thus the number of nurses—required for each month of the year.

Step 3. Update the sales and operations planning spreadsheet for each family, recognizing relevant constraints and costs including availability of materials from suppliers, training facilities capable of handling only so many new hires at a time, machine capacities, or limited storage space. Policy constraints might include limitations on the number of backorders, or the use of subcontractors or overtime, as well as the minimum inventory levels needed to achieve desired safety stocks. Typically, many plans can satisfy a specific set of constraints. The planner searches for a plan that best balances costs, customer service, workforce stability, and the like. This process may necessitate revising the plan several times.





Cultura Creative/Alamy

FUJIFILM Imaging Colorants makes inks and dyes, primarily for inkjet printer cartridges, and operates an effective S&OP process. It must coordinate between a U.S. finishing plant and a UK bulk manufacturing plant. Managers from all functions teleconference at the U.S. site with seven other UK managers. At this Partnership meeting (step 4 in the S&OP process), they review the demand, production, and inventory plans, as well as the projected working capital plan.

Step 4. Have one or more consensus meetings with the stakeholders on how best to balance supply with demand. Participants could include the supply chain manager, plant manager, controller, purchasing manager, production control manager, or logistics manager. The goal is one set of recommendations to present at the firm's executive sales and operations planning (S&OP) meeting. Where agreement cannot be reached, prepare scenarios of alternative plans. Also prepare an updated financial view of business by rolling up the plans for all product families into a spreadsheet expressed in total dollars.

Step 5. Present recommendations by product family at the executive S&OP meeting, which typically includes the firm's president and the vice presidents of functional areas. The plan is reviewed relative to the business plan, new product issues, special projects, and other relevant factors. The executives may ask for final changes to the plan, such as to balance conflicting objectives better. Acceptance of this authorized plan does not necessarily mean that everyone is in total agreement, but it does imply that everyone will work to achieve the plan.

Step 6. Update the spreadsheets to reflect the authorized plan, and communicate the plans to the important stakeholders for implementation. Important recipients include those who do resource planning, covered in the next chapter.

Using Spreadsheets

The sales and operations plan in Figure 15.3 does not show much on the supply options used in the operations plan or their cost implications. Here we discuss using spreadsheets that do just that. Supplement D, "Linear Programming Models," describes using the transportation method for production planning. Both techniques could be used on the side as a planner develops prospective plans in step 3 of the planning process.

Various spreadsheets can be used, including ones that you develop on your own. Here we work with the *Sales and Operations Planning with Spreadsheets* Solver in OM Explorer. Figure 15.4 shows a plan for a manufacturer, which uses all supply options except overtime.

FIGURE 15.4 ▶
Manufacturer's Plan Using a Spreadsheet and Mixed Strategy

	1	2	3	4	5	6	Total
Inputs							
Forecasted demand	24	142	220	180	136	168	870
Workforce level	120	158	158	158	158	158	910
Undertime	6	0	0	0	0	0	6
Overtime	0	0	0	0	0	0	0
Vacation time	20	6	0	0	4	10	40
Subcontracting time	0	0	0	0	0	6	6
Backorders	0	0	0	4	0	0	4
Derived							
Utilized time	94	152	158	158	154	148	864
Inventory	70	80	18	0	14	0	182
Hires	0	38	0	0	0	0	38
Layoffs	0	0	0	0	0	0	0
Calculated							
Utilized time cost	$376,000	$608,000	$632,000	$632,000	$616,000	$592,000	$3,456,000
Undertime cost	$24,000	$0	$0	$0	$0	$0	$24,000
Overtime cost	$0	$0	$0	$0	$0	$0	$0
Vacation time cost	$80,000	$24,000	$0	$0	$16,000	$40,000	$160,000
Inventory cost	$2,800	$3,200	$720	$0	$560	$0	$7,280
Backorders cost	$0	$0	$0	$4,000	$0	$0	$4,000
Hiring cost	$0	$91,200	$0	$0	$0	$0	$91,200
Layoff cost	$0	$0	$0	$0	$0	$0	$0
Subcontracting cost	$0	$0	$0	$0	$0	$43,200	$43,200
Total cost	$482,800	726,400	632,720	636,000	632,560	675,200	$3,785,680

Spreadsheets for a Manufacturer The top part of the spreadsheet shows the *input values* that give the forecasted demand requirements and the supply option choices period by period. Vary these "levers" as you search for better plans.

The next part of the spreadsheet (in green) shows the *derived values* that must follow from the input values. The first row of derived values is called *utilized time*, which is that portion of the workforce's regular time that is paid for and productively used. In any period, the utilized time equals the workforce level minus undertime and vacation time. For example, in period 1 the utilized time is 94 (or 120 – 6 – 20). The hires and layoffs rows can be derived from the workforce levels. In this example, the workforce is increased for period 2 from its initial size of 120 employees to 158, which means that 38 employees are hired. Because the workforce size remains constant at 158 throughout the rest of the planning horizon, no other hirings or layoffs happen. When additional alternatives, such as vacations, inventory, and backorders are all possible, the overtime and undertime cannot be derived just from information on forecasted demand and workforce levels. Thus, undertime and overtime are shown as input values (rather than derived values) in the spreadsheet, and the user must be careful to specify consistent input values.

The final part of the spreadsheet, the *calculated values* of the plan, shows the plan's cost consequences. Along with qualitative considerations, the cost of each plan determines whether the plan is satisfactory or whether a revised plan should be considered. When seeking clues about how to improve a plan already evaluated, we identify its highest cost elements. Revisions that would reduce these specific costs might produce a new plan with lower overall costs. Spreadsheet programs make analyzing these plans easy, and they present a whole new set of possibilities for developing sound sales and operations plans.

The plan in Figure 15.4 definitely is for a manufacturer because it uses inventory to advantage, particularly in the first two periods. It is a mixed strategy, and not just because it uses anticipation inventory, backorders, and subcontracting. The workforce level changes in period 2, but it does not exactly match the forecasted demand as with a chase strategy. It has some elements of the level strategy, because undertime and vacation time are part of the plan, but it does not rely exclusively on these supply options.

Care must be taken to recognize differences in how inputs are measured. The workforce level might be expressed as the number of employees, but the forecasted demand and inventory are expressed as units of the product. The OM Explorer spreadsheets require a common unit of measure, so we must translate some of the data prior to entering the input values. Perhaps the easiest approach is to express the forecasted demand and supply options as *employee-period equivalents*. If demand forecasts are given as units of product, we can convert them to employee-period equivalents by dividing them by the productivity of a worker. For example, if the demand is for 1,500 units of product and the average employee produces 100 units in one period, the demand requirement is 15 employee-period equivalents.

Spreadsheets for a Service Provider The same spreadsheets can be used by service providers, except anticipation inventory is not an option. You can unprotect the sheet and then hide the rows that are not relevant. It is useful not to hide the inventory row until the end, however, because positive or negative values signal an inconsistency in your plan. Whereas Figure 15.4 shows a good plan found after several revisions, here we illustrate with Example 15.1 how to find a good plan for a service provider beginning with the chase and level (ignoring vacations) strategies. These plans can provide insights that lead to even better mixed strategy plans.

EXAMPLE 15.1 Using the Chase and Level Strategies as Starting Points

The manager of a large distribution center must determine how many part-time stockpickers to maintain on the payroll. She wants to develop a staffing plan that minimizes total costs, and wants to begin with the chase strategy and level strategy. For the level strategy, she wants to first try the workforce level that meets demand with the minimum use of undertime and not consider vacation scheduling.

First, the manager divides the next year into six time periods, each one 2 months long. Each part-time employee can work a maximum of 20 hours per week on regular time, but the actual number can be less. Instead of paying undertime, each worker's day is shortened during slack periods. Once on the payroll, each worker is used each day, but they may work only a few hours. Overtime can be used during peak periods.

The distribution center's forecasted demand is shown as the number of part-time employees required for each time period at the maximum regular time of 20 hours per week. For example, in period 3, an estimated 18 part-time employees working 20 hours per week on regular time will be needed.

	1	2	3	4	5	6	Total
Forecasted demand*	6	12	18	15	13	14	78

*Number of part-time employees

Currently, 10 part-time clerks are employed. They have not been subtracted from the forecasted demand shown. Constraints and cost information are as follows:

a. The size of training facilities limits the number of new hires in any period to no more than 10.

b. No backorders are permitted; demand must be met each period.

c. Overtime cannot exceed 20 percent of the regular-time capacity (that is, 4 hours) in any period. Therefore, the most that any part-time employee can work is 1.20(20) = 24 hours per week.

d. The following costs can be assigned:

Regular-time wage rate	$2,000 per time period at 20 hours per week
Overtime wages	150 percent of the regular-time rate
Hires	$1,000 per person
Layoffs	$500 per person



AP Photos

Framed by thousands of ski poles, a part-time worker sorts and inventories new products in the receiving department of REI's distribution center in Sumner, Washington. REI employs a high percentage of part-time workers, many of whom are college students. They tend to be young people who participate in outdoor sports and are familiar with the equipment that REI sells.

MyOMLab

Tutor 15.1 in MyOMLab provides a new example for planning using the chase strategy with hiring and layoffs.

SOLUTION

a. Chase Strategy

This strategy simply involves adjusting the workforce as needed to meet demand, as shown in Figure 15.5. Rows in the spreadsheet that do not apply (such as inventory and vacations) are hidden. The workforce level row is identical to the forecasted demand row. A large number of hirings and layoffs begin with laying off four part-time employees immediately because the current staff is 10 and the staff level required in period 1 is only six. However, many employees, such as college students, prefer part-time work. The total cost is $173,500, and most of the cost increase comes from frequent hiring and layoffs, which add $17,500 to the cost of utilized regular-time costs.

FIGURE 15.5 ▶
Spreadsheet for Chase Strategy

	1	2	3	4	5	6	Total
Inputs							
Forecasted demand	6	12	18	15	13	14	78
Workforce level	6	12	18	15	13	14	78
Undertime	0	0	0	0	0	0	0
Overtime	0	0	0	0	0	0	0
Derived							
Utilized time	6	12	18	15	13	14	78
Hires	0	6	6	0	0	1	13
Layoffs	4	0	0	3	2	0	9
Calculated							
Utilized time cost	$12,000	$24,000	$36,000	$30,000	$26,000	$28,000	$156,000
Undertime cost	$0	$0	$0	$0	$0	$0	$0
Hiring cost	$0	$6,000	$6,000	$0	$0	$1,000	$13,000
Layoff cost	$2,000	$0	$0	$1,500	$1,000	$0	$4,500
Total cost	$14,000	30,000	42,000	31,500	27,000	29,000	$173,500

b. Level Strategy

In order to minimize undertime, the maximum use of overtime possible must occur in the peak period. For this particular level strategy (other workforce options are possible), the most overtime that the manager can use is 20 percent of the regular-time capacity, w, so

$$1.20w = 18 \text{ employees required in peak period (period 3)}$$

$$w = \frac{18}{1.20} = 15 \text{ employees}$$

A 15-employee staff size minimizes the amount of undertime for this level strategy. Because the staff already includes 10 part-time employees, the manager should immediately hire five more. The complete plan is shown in Figure 15.6. The total cost is \$164,000, which seems reasonable because the minimum conceivable cost is only \$156,000 (78 periods × \$2,000/period). This cost could be achieved only if the manager found a way to cover the forecasted demand for all 78 periods with regular time. The plan seems reasonable primarily because it involves the use of large amounts of undertime (15 periods), which in this example are unpaid.

MyOMLab

Tutor 15.2 in MyOMLab provides a new example for planning using the level strategy with overtime and undertime.

	1	2	3	4	5	6	Total
Inputs							
Forecasted demand	6	12	18	15	13	14	78
Workforce level	15	15	15	15	15	15	90
Undertime	9	3	0	0	2	1	15
Overtime	0	0	3	0	0	0	3
Derived							
Utilized time	6	12	15	15	13	14	75
Hires	5	0	0	0	0	0	5
Layoffs	0	0	0	0	0	0	0
Calculated							
Utilized time cost	\$12,000	\$24,000	\$30,000	\$30,000	\$26,000	\$28,000	\$150,000
Undertime cost	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Overtime cost	\$0	\$0	\$9,000	\$0	\$0	\$0	\$9,000
Hiring cost	\$5,000	\$0	\$0	\$0	\$0	\$0	\$5,000
Layoff cost	\$0	\$0	\$0	\$0	\$0	\$0	\$0
Total cost	\$17,000	24,000	39,000	30,000	26,000	28,000	\$164,000

◀ **FIGURE 15.6**
Spreadsheet for Level Strategy

DECISION POINT

The manager, now having a point of reference with which to compare other plans, decided to evaluate some other plans before making a final choice, beginning with the chase strategy. The only way to reduce costs is somehow to reduce the premium for three overtime employee periods (3 periods × \$3,000/period) or to reduce the hiring cost of five employees (5 hires × \$1,000/person). Nonetheless, better solutions may be possible. For example, undertime can be reduced by delaying the hiring until period 2 because the current workforce is sufficient until then. This delay would decrease the amount of unpaid undertime, which is a qualitative improvement. See Active Model 15.1 for additional insights.

MyOMLab

Active Model 15.1 in MyOMLab shows the impact of changing the workforce level, the cost structure, and overtime capacity.

Scheduling

Scheduling is the last step in Figure 15.1. It takes the operations and scheduling process from planning to execution, and is where the "rubber meets the road." This important aspect of supply chain management is itself a process. It requires gathering data from sources such as demand forecasts or specific customer orders, resource availability from the sales and operations plan, and specific constraints to be reckoned with from employees and customers. It then involves generating a work schedule for employees or sequences of jobs or customers at workstations. The schedule has to be coordinated with the employees and suppliers to make sure that all constraints are satisfied. Here we cover Gantt charts, employee schedules, job sequencing at workstations, and software support.

Gantt Charts

Schedules can be displayed in various ways. For different jobs or activities they can simply list their due dates, show in a table their start and finish times, or show in a graph their start and finish times. The *Gantt chart* uses the third approach. Figure 2.5 back in Chapter 2 demonstrates how a "picture can be worth a thousand words" in managing projects. Associates not familiar with scheduling techniques can still grasp the essence of the plan. This tool can be used to monitor the progress of work and to view the load on workstations. The chart takes two basic forms: (1) the job

or activity progress chart and (2) the workstation chart. The *Gantt progress chart* graphically displays the current status of each job or activity relative to its scheduled completion date. For example, suppose that an automobile parts manufacturer has three jobs under way, one each for Ford, Nissan, and Buick. The actual status of these orders is shown by the colored bars in Figure 15.7; the red lines indicate the desired schedule for the start and finish of each job. For the current date, April 21, this Gantt chart shows that the Ford order is behind schedule because operations has completed only the work scheduled through April 18. The Nissan order is exactly on schedule, and the Buick order is ahead of schedule.

FIGURE 15.7 ▶
Gantt Progress Chart for an Auto Parts Company

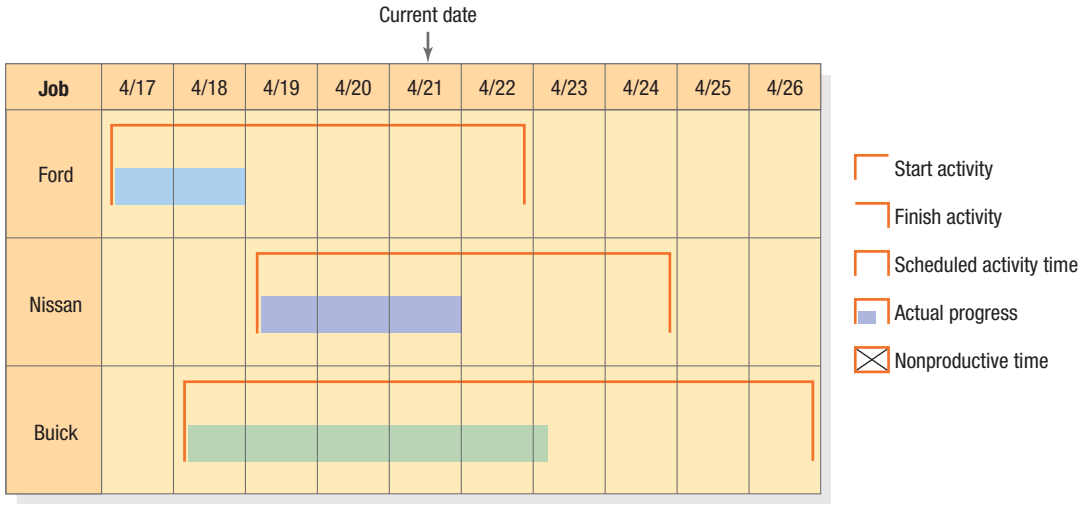
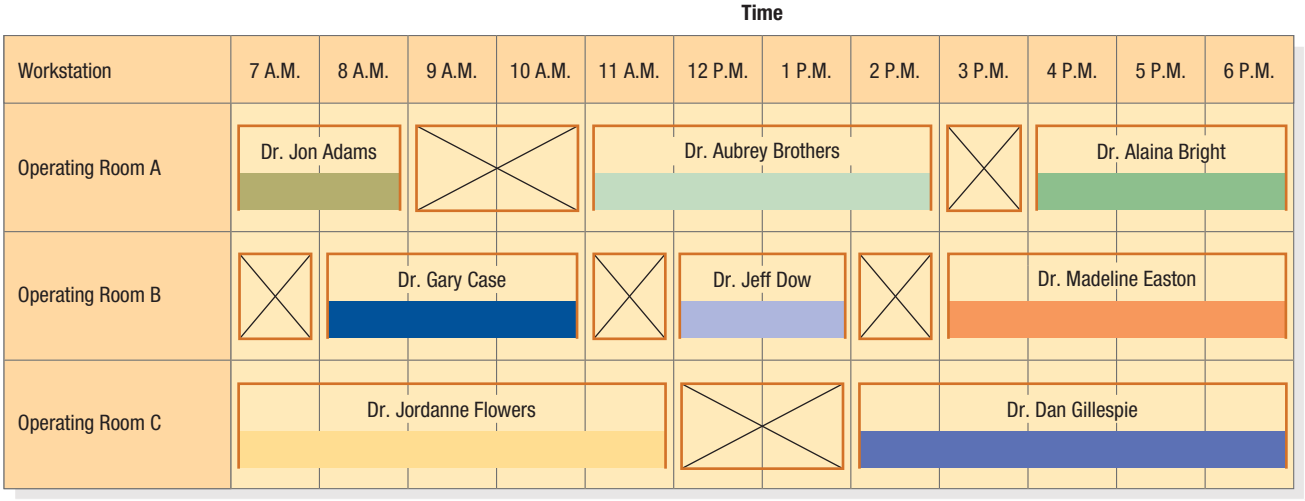


Figure 15.8 shows a *Gantt workstation chart* of the operating rooms at a hospital for a particular day. Using the same notation as in Figure 15.7, the chart shows the load on the operating rooms and the nonproductive time. The time slots assigned to each doctor include the time needed to clean the room prior to the next surgery. The chart can be used to identify time slots for unscheduled emergency surgeries. It can also be used to accommodate requests to change the time of surgeries. For example, Dr. Flowers may be able to change the start of her surgery to 2 P.M. by swapping time slots with Dr. Gillespie in operating room C or by asking Dr. Brothers to start her surgery one hour earlier in operating room A and asking Dr. Bright to schedule her surgery for the morning in operating room C. In any event, the hospital administrator would have to get involved in rescheduling the surgeries.

FIGURE 15.8 ▼
Gantt Workstation Chart for Operating Rooms at a Hospital



Scheduling Employees

workforce scheduling
A type of scheduling that determines when employees work.

Another way to manage capacity is **workforce scheduling**, which is a type of scheduling that determines when employees work. Of particular interest are situations when not all employees work the same five days a week, and same eight hours per day. The schedule specifies the on-duty and off-duty periods for each employee over a certain time period, as in assigning postal clerks, nurses, pilots, attendants, or police officers to specific workdays and shifts. This approach is used when

customers demand quick response and total demand can be forecasted with reasonable accuracy. In these instances, capacity is adjusted to meet the expected loads on the service system.

Workforce schedules translate the staffing plan into specific schedules of work for each employee. Determining the workdays for each employee in itself does not make the staffing plan operational. Daily workforce requirements, stated in aggregate terms in the staffing plan, must be satisfied. The workforce capacity available each day must meet or exceed daily workforce requirements. If it does not, the scheduler must try to rearrange days off until the requirements are met. If no such schedule can be found, management might have to change the staffing plan and hire more employees, authorize overtime hours, or allow for larger backlogs.



Digital Vision/Thinkstock

An air hostess serves coffee to the pilots in the cockpit. Airlines provide service for more than eight hours per day and five days per week, and there are many technical, behavioral, and legal constraints to be met. Such conditions make scheduling a challenging task.

Constraints The technical constraints imposed on the workforce schedule are the resources provided by the staffing plan and the requirements placed on the operating system. However, other constraints, including legal and behavioral considerations, also can be imposed. For example, Air New Zealand is required to have at least a minimum number of flight attendants on duty at all times. Similarly, a minimum number of fire and safety personnel must be on duty at a fire station at all times. Such constraints limit management's flexibility in developing workforce schedules.

The constraints imposed by the psychological needs of workers complicate scheduling even more. Some of these constraints are written into labor agreements. For example, an employer might agree to give employees a certain number of consecutive days off per week or to limit employees' consecutive workdays to a certain maximum. Other provisions might govern the allocation of vacations, days off for holidays, or rotating shift assignments. In addition, the preferences of the employees themselves need to be considered.

One way that managers deal with certain undesirable aspects of scheduling is to use a **rotating schedule**, which rotates employees through a series of workdays or hours. Thus, over a period of time, each person has the same opportunity to have weekends and holidays off and to work days, as well as evenings and nights. A rotating schedule gives each employee the next employee's schedule the following week. In contrast, a **fixed schedule** calls for each employee to work the same days and hours each week.

rotating schedule
A schedule that rotates employees through a series of workdays or hours.

fixed schedule
A schedule that calls for each employee to work the same days and hours each week.

Developing a Workforce Schedule Suppose that we are interested in developing a workforce schedule for a company that operates seven days a week and provides each employee with two consecutive days off. In this section, we demonstrate a method that recognizes this constraint. The objective is to identify the two consecutive days off for each employee that will minimize the amount of total slack capacity, thereby maximizing the utilization of the workforce. The work schedule for each employee, then, is the five days that remain after the two days off have been determined. The procedure involves the following steps.

Step 1. From the schedule of net requirements for the week, find all the pairs of consecutive days, excluding the day (or days) with the maximum daily requirement. Select the unique pair that has the lowest total requirements for the two days. In some unusual situations, all pairs may contain a day with the maximum requirements. If so, select the pair with the lowest total requirements. Suppose that the numbers of employees required are

Monday: 8	Thursday: 12	Saturday: 4
Tuesday: 9	Friday: 7	Sunday: 2
Wednesday: 2		

The maximum daily requirement is 12 employees, on Thursday. The consecutive pair with the lowest total requirements is Saturday and Sunday, with $4 + 2 = 6$.

Step 2. If a tie occurs, choose one of the tied pairs, consistent with the provisions written into the labor agreement, if any. Alternatively, the tie could be broken by asking the employee being scheduled to make the choice. As a last resort, the tie could be broken arbitrarily. For example, preference could be given to Saturday–Sunday pairs.

Step 3. Assign the employee the selected pair of days off. Subtract the requirements satisfied by the employee from the net requirements for each day the employee is to work. In this example, the employee is assigned Saturday and Sunday off. After requirements are subtracted, Monday's requirement is 7, Tuesday's is 8, Wednesday's is 1, Thursday's is 11, and Friday's is 6. Saturday's and Sunday's requirements do not change because no employee is yet scheduled to work those days.

Step 4. Repeat steps 1 through 3 until all the requirements have been satisfied or a certain number of employees have been scheduled.

This method reduces the amount of slack capacity assigned to days with low requirements and forces the days with high requirements to be scheduled first. It also recognizes some of the behavioral and contractual aspects of workforce scheduling in the tie-breaking rules.

EXAMPLE 15.2 Developing a Workforce Schedule

The Amalgamated Parcel Service is open seven days a week. The schedule of requirements is

Day	M	T	W	Th	F	S	Su
Required number of employees	6	4	8	9	10	3	2

The manager needs a workforce schedule that provides two consecutive days off and minimizes the amount of total slack capacity. To break ties in the selection of off days, the scheduler gives preference to Saturday and Sunday if it is one of the tied pairs. If not, she selects one of the tied pairs arbitrarily.

MyOMLab

Tutor 15.3 in MyOMLab provides a new example to practice workforce scheduling.

SOLUTION

Friday contains the maximum requirements, and the pair S–Su has the lowest total requirements. Therefore, Employee 1 is scheduled to work Monday through Friday.

Note that Friday still has the maximum requirements and that the requirements for the S–Su pair are carried forward because these are Employee 1's days off. These updated requirements are the ones the scheduler uses for the next employee.

The day-off assignments for the employees are shown in the following table.

SCHEDULING DAYS OFF

M	T	W	Th	F	S	Su	Employee	Comments
6	4	8	9	10	3	2	**1**	The S–Su pair has the lowest total requirements. Assign Employee **1** to a Monday through Friday schedule and update the requirements.
5	3	7	8	9	3	2	**2**	The S–Su pair has the lowest total requirements. Assign Employee **2** to a Monday through Friday schedule and update the requirements.
4	2	6	7	8	3	2	**3**	The S–Su pair has the lowest total requirements. Assign Employee **3** to a Monday through Friday schedule and update the requirements.
3	1	5	6	7	3	2	**4**	The M–T pair has the lowest total requirements. Assign Employee **4** to a Wednesday through Sunday schedule and update the requirements.
3	1	4	5	6	2	1	**5**	The S–Su pair has the lowest total requirements. Assign Employee **5** to a Monday through Friday schedule and update the requirements.
2	0	3	4	5	2	1	**6**	The M–T pair has the lowest total requirements. Assign Employee **6** to a Wednesday through Sunday schedule and update the requirements.
2	0	2	3	4	1	0	**7**	The S–Su pair has the lowest total requirements. Assign Employee **7** to a Monday through Friday schedule and update the requirements.
1	0	1	2	3	1	0	**8**	Four pairs have the minimum requirement and the lowest total: S–Su, Su–M, M–T, and T–W. Choose the S–Su pair according to the tie-breaking rule. Assign Employee **8** to a Monday through Friday schedule and update the requirements.
0	0	0	1	2	1	0	**9**	Arbitrarily choose the Su–M pair to break ties because the S–Su pair does not have the lowest total requirements. Assign Employee **9** to a Tuesday through Saturday schedule and update the requirements.
0	0	0	0	1	0	0	**10**	Choose the S–Su pair according to the tie-breaking rule. Assign Employee **10** to a Monday through Friday schedule.

In this example, Friday always has the maximum requirements and should be avoided as a day off. The final schedule for the employees is shown in the following table.

FINAL SCHEDULE

Employee	M	T	W	Th	F	S	Su	Total
1	X	X	X	X	X	off	off	
2	X	X	X	X	X	off	off	
3	X	X	X	X	X	off	off	
4	off	off	X	X	X	X	X	
5	X	X	X	X	X	off	off	
6	off	off	X	X	X	X	X	
7	X	X	X	X	X	off	off	
8	X	X	X	X	X	off	off	
9	off	X	X	X	X	X	off	
10	X	X	X	X	X	off	off	
Capacity, C	7	8	10	10	10	3	2	50
Requirements, R	6	4	8	9	10	3	2	42
Slack, C – R	1	4	2	1	0	0	0	8

DECISION POINT

With its substantial amount of slack capacity, the schedule is not unique. Employee 9, for example, could have Sunday and Monday, Monday and Tuesday, or Tuesday and Wednesday off without causing a capacity shortage. Indeed, the company might be able to get by with one fewer employee because of the total of eight slack days of capacity. However, all 10 employees are needed on Fridays. If the manager were willing to get by with only nine employees on Fridays or if someone could work one day of overtime on a rotating basis, he would not need Employee 10. As indicated in the table, the net requirement left for Employee 10 to satisfy amounts to only one day, Friday. Thus, Employee 10 can be used to fill in for vacationing or sick employees.



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Managers try to staff their call centers to meet certain performance measures. One such measure is the percentage of calls answered (PCA) within a specified time interval, termed the service objective (SO). Typically in a call center, the PCA is in the range of 80 to 90 percent, and the SO is 15 to 30 seconds. The problem is that the requirements for agents change over time, depending on the time of day and day of the year. Also, the callers are likely to speak different languages. Determining how many of each group of agents to have on hand at all times is not easy. Fortunately, employee scheduling software is available to estimate call volumes, project skill requirements, and identify employee start and end times or preferred days off.

Sequencing Jobs at a Workstation

Another aspect of scheduling is sequencing work at workstations. **Sequencing** determines the order in which jobs or customers are processed in the waiting line at a workstation. When combined with the expected processing time, the sequence allows you to estimate the start and finish times of each job.

sequencing

Determining the order in which jobs or customers are processed in the waiting line at a workstation.

priority sequencing rule
A rule that specifies the job or customer processing sequence when several jobs are waiting in line at a workstation.

first-come, first served (FCFS)
A priority sequencing rule that specifies that the job or customer arriving at the workstation first has the highest priority.

earliest due date (EDD)
A priority sequencing rule that specifies that the job or customer with the earliest due date is the next job to be processed.

expediting
The process of completing a job or finishing with a customer sooner than would otherwise be done.

flow time
The amount of time a job spends in the service or manufacturing system.

past due
The amount of time by which a job missed its due date.

tardiness
See past due.

Priority Sequencing Rules One way to determine what job or customer to process next is with the help of a **priority sequencing rule**. The following two priority sequencing rules are commonly used in practice.

- *First-Come, First-Served.* The job or customer arriving at the workstation first has the highest priority under a **first-come, first-served (FCFS)** rule. This rule is the most "democratic" in that each job is treated equally, with no one stepping ahead of others already in line. It is commonly used at service facilities, and is the rule that was assumed in Supplement B, "Waiting Line Models."
- *Earliest Due Date.* The job or customer with the **earliest due date (EDD)** is the next one to be processed. The *due date* specifies when work on a job or customer should be finished. Due dates are commonly used by manufacturers and suppliers in the supply chain. For example, a product cannot be assembled until all of its purchased and produced components are available. If these components were not already in inventory, they must be ordered prior to when the product assembly can begin. Their due date is the start date for assembling the product to be assembled. This simple relationship is fundamental to coordinating with suppliers and with the manufacturer's own shops in the supply chain. It is also the key to **expediting**, which is the process of completing a job sooner than would otherwise be done. Expediting can be done by revising the due date, moving the job to the front of the waiting line, making a special appeal by phone or e-mail to the supplier, adding extra capacity, or even putting a red tag on the job that says the job is urgent.

Neither rule guarantees finding an optimal solution. Different sequences found by trial and error can produce better schedules. In fact, there are multiple performance measures for judging a schedule. A schedule that does well on one measure may do poorly on another.

Performance Measures The quality of a schedule can be judged in various ways. Two commonly used performance measures are flow time and past due.

- *Flow Time.* The amount of time a job spends in the service or manufacturing system is called **flow time**. It is the sum of the waiting time for servers or machines; the process time, including setups; the time spent moving between operations; and delays resulting from machine breakdowns, unavailability of facilitating goods or components, and the like. Flow time is sometimes referred to as *throughput time* or *time spent in the system, including service.* For a set of jobs to be processed at a single workstation. a job's flow time is

$$\text{Flow time} = \text{Finish time} + \text{Time since job arrived at workstation}$$

  When using this equation, we assume for convenience that the first job scheduled starts at time zero (0). At time 0, all the jobs were available for processing at the workstation.
- *Past Due.* The measure **past due** can be expressed as the amount of time by which a job missed its due date (also referred to as **tardiness**) or as the percentage of total jobs processed over some period of time that missed their due dates. Minimizing these past due measures supports the competitive priorities of cost (penalties for missing due dates), quality (perceptions of poor service), and time (on-time delivery).

EXAMPLE 15.3 Using the Two Priority Sequencing Rules

Currently a consulting company has five jobs in its backlog. The time since the order was placed, processing time, and promised due dates are given in the following table. Determine the schedule by using the FCFS rule, and calculate the average days past due and flow time. How can the schedule be improved, if average flow time is the most critical?

Customer	Time Since Order Arrived (days ago)	Processing Time (days)	Due Date (days from now)
A	15	25	29
B	12	16	27
C	5	14	68
D	10	10	48
E	0	12	80

SOLUTION

a. The FCFS rule states that Customer A should be the first one in the sequence, because that order arrived earliest—15 days ago. Customer E's order arrived today, so it is processed last. The sequence is shown in the following table, along with the days past due and flow times.

Customer Sequence	Start Time (days)		Processing Time (days)		Finish Time (days)	Due Date	Days Past Due	Days Ago Since Order Arrived	Flow Time (days)
A	0	+	25	=	25	29	**0**	15	**40**
B	25	+	16	=	41	27	**14**	12	**53**
D	41	+	10	=	51	48	**3**	10	**61**
C	51	+	14	=	65	68	**0**	5	**70**
E	65	+	12	=	77	80	**0**	0	**77**

The *finish time* for a job is its start time plus the processing time. Its finish time becomes the start time for the next job in the sequence, assuming that the next job is available for immediate processing. The days past due for a job is zero (0) if its due date is equal to or exceeds the finish time. Otherwise it equals the shortfall. The flow time for each job equals its finish time plus the number of days ago since the order first arrived at the workstation. For example, Customer C's flow time is its scheduled finish time of 65 days plus the 5 days since the order arrived, or 70 days. The days past due and average flow time performance measures for the FCFS schedule are

$$\text{Average days past due} = \frac{0 + 14 + 3 + 0 + 0}{5} = 3.4 \text{ days}$$

$$\text{Average flow time} = \frac{40 + 53 + 61 + 70 + 77}{5} = 60.2 \text{ days}$$

b. The average flow time can be reduced. One possibility is the sequence shown in the following table, which uses the Shortest Processing Time (SPT) rule, which is one of several rules developed more fully in MyOMLab Supplement J, "Operations Scheduling." (For still another possibility, see Solved Problem 1, which applies the EDD rule.)

MyOMLab

Customer Sequence	Start Time (days)		Processing Time (days)		Finish Time (days)	Due Date	Days Past Due	Days Ago Since Order Arrived	Flow Time (days)
D	0	+	10	=	10	48	**0**	10	**20**
E	10	+	12	=	22	80	**0**	0	**22**
C	22	+	14	=	36	68	**0**	5	**41**
B	36	+	16	=	52	27	**25**	12	**64**
A	52	+	25	=	77	29	**48**	15	**92**

$$\text{Average days past due} = \frac{0 + 0 + 0 + 25 + 48}{5} = 14.6 \text{ days}$$

$$\text{Average flow time} = \frac{20 + 22 + 41 + 64 + 92}{5} = 47.8 \text{ days}$$

This schedule reduces the average flow time from 60.2 to 47.8 days—a 21 percent improvement. However, the past due times for jobs A and B have increased.

DECISION POINT

Management decided to use a modified version of the second schedule, adding overtime when Customer B is processed. Further, Customer A agreed to extend its due date to 77 days, because in this case the advanced warning allowed it to reschedule its own operations with little problem.



Northpoint/Nissan North America, Inc.

When Nissan introduced the Almera to the European market, it decided to produce it at the most efficient Sunderland plant in the United Kingdom. It already manufactured the Micra and Primera models there. Multi-vehicle scheduling is quite complex because of the many constraints involved. The painting portion of the job is particularly time consuming, but is only one of thousands of tasks the plant must efficiently schedule. A sophisticated software package, called ILOG Solver, assists the scheduling process for a coordinated flow.

Software Support

Computerized scheduling systems are available to cope with the complexity of workforce scheduling, such as the myriad constraints and concerns at Air New Zealand. In some types of firms, such as telephone companies, mail-order catalog houses, or emergency hotline agencies, employees must be on duty 24 hours a day, 7 days a week.

Sometimes a portion of the staff is part time, which allows management a great deal of flexibility but adds considerable complexity to the scheduling requirements. The flexibility comes from the opportunity to match anticipated loads closely through the use of overlapping shifts or odd shift lengths; the complexity comes from the need to evaluate the numerous possible alternatives. Management also must consider the timing of lunch breaks and rest periods, the number and starting times of shift schedules, and the days off for each employee. The programs select the schedule that minimizes the sum of expected costs of over- and understaffing.

Software is also available for sequencing jobs at workstations. They help firms design and manage the linkages between customers and suppliers in the supply chain. True integration requires the manipulation of large amounts of complex data in real time because the customer order work flow must be synchronized with the required material, manufacturing, and distribution activity. Coupled with the Internet and improved data storage and manipulation methods, such computer software has given rise to **advanced planning and scheduling (APS) systems**, which seek to optimize resources across the supply chain and align daily operations with strategic goals. A firm's ability to change its schedules quickly and still keep the goods and services flowing smoothly through the supply chain provides a competitive edge.

advanced planning and scheduling (APS) systems

Computer software systems that seek to optimize resources across the supply chain and align daily operations with strategic goals.

LEARNING GOALS IN REVIEW

1. **Describe the operations planning and scheduling process.** The section "Stages in Operations Planning and Scheduling," pp. 529–531, shows its various stages and how it relates with other plans in the organization. Pay particular attention to Figure 15.1.
2. **Explain why the process of matching supply with demand begins with aggregation.** Aggregation allows the general course of action to be established, without being bogged down in details. The section on "Aggregation," p. 529, shows that aggregation is performed across three dimensions: product families, workforce, and time.
3. **Identify the different demand and supply options.** Sections "Managing Demand" and "Sales and Operations Plans," pp. 531–541, describe the various options, along with their positive and negative aspects. Focus on Table 15.2 for the complete list.
4. **Explain how operations plans and schedules relate to other plans.** Figure 15.2 shows how Sales and Operations Plans are related with other functional areas. Also, check out the section "The Relationship of Operations Plans and Schedules to Other Plans," pp. 529–531.
5. **Use spreadsheets to create sales and operations plans.** Turn to "Using Spreadsheets" on pp. 538–541 to better understand how to use this tool. Solved Problem 3 is also helpful.
6. **Develop employee schedules.** The section on "Scheduling Employees," pp. 542–545, describes how to create a workforce schedule. Also see Example 15.2, pp. 544–545, and Solved Problem 2, pp. 550–551.
7. **Develop schedules for single workstations.** The FCFS and EDD rules, and the flow time and past due performance measures, are described in the "Sequencing Jobs at a Workstation," pp. 545–547. Example 15.3 illustrates their use, along with Solved Problem 1.

MyOMLab helps you develop analytical skills and assesses your progress with multiple problems on level, chase, and mixed strategy plans, total costs, anticipation inventory, break-even analysis, FCFS and EDD scheduling rules, average flow time, and average days past due.

MyOMLab Resources	Titles	Link to the Book
Video	Air New Zealand: Service Scheduling Sales and Operations Planning at Starwood	Operations Planning and Scheduling Across the Organization; Stages in Operations Planning and Scheduling; Stages in Operations Planning and Scheduling; Demand Options
Active Model Exercise	15.1 Level Strategy	Planning Strategies; Example 15.1 (pp. 539–541)
OM Explorer Solvers	Sales and Operations Planning with Spreadsheets Workforce Scheduler Single Work-Station Scheduler	Using Spreadsheets; Figure 15.3 (p. 530); Figure 15.4 (p. 538); Solved Problem 3 (pp. 551–553) Scheduling Employees; Example 15.2 (pp. 544–545); Solved Problem 2 Sequencing Jobs at a Workstation; Example 15.3 (pp. 546–547); Solved Problem 1 (p. 550)
OM Explorer Tutors	15.1 Chase Strategy 15.2 Level Strategy 15.3 Developing a Workforce Schedule 15.4 Staffing Strategies with Spreadsheets (4 pds)	Planning Strategies; Example 15.1 (pp. 539–541); Solved Problem 3 Planning Strategies; Example 15.1 (pp. 539–541); Solved Problem 3 Scheduling Employees; Example 15.2 (pp. 544–545); Solved Problem 2 Scheduling Employees; Example 15.2 (pp. 544–545); Solved Problem 2
Tutor Exercises	15.1 Results of Different Scenarios with a Level Strategy 15.2 Staffing for the Newest MBA Class	Planning Strategies; Example 15.1 (pp. 539–541); Solved Problem 2 Scheduling Employees; Example 15.2 (pp. 544–545)
POM for Windows	Scheduling	Sequencing Jobs at a Workstation; Example 15.3 (pp. 546–547); Solved Problem 1 (p. 550)
Virtual Tours	Stratton Furniture Company Stihl Chain Saws Stihl Factory Tour	Stages in Operations Planning and Scheduling; Demand Options; Supply Options; Planning Strategies
MyOMLab Supplements	H. Measuring Output Rates I. Learning Curve Analysis	MyOMLab Supplement H MyOMLab Supplement I
Internet Exercise	United Parcel Service United Parcel Service Centers Internal Revenue Service H&R Block Hoovers	Managing Demand; Supply Options; Planning Strategies Managing Demand; Supply Options; Planning Strategies
Additional Case	Food King	Scheduling Employees; Example 15.2 (pp. 544–545)
Key Equations		
Image Library		

Key Terms

advanced planning and scheduling (APS) systems 548
aggregate plan 529
annual plan or financial plan 530
backlog 533
backorder and stockout 536
business plan 529
chase strategy 535
complementary products 531
demand management 531
earliest due date (EDD) 546
expediting 546
first-come, first-served (FCFS) 546
fixed schedule 543
flow time 546
hiring and layoff 536
inventory holding 536
level strategy 535
mixed strategy 536
operations planning and scheduling 528
overtime 534
past due 546
priority sequencing rule 546
product family 529
production plan 529
regular time 536
resource plan 529
revenue management 532
rotating schedule 543
sales and operations planning (S&OP) 529
schedule 529
sequencing 545
staffing plan 529
tardiness 546
undertime 535
workforce scheduling 542

Solved Problem 1

Revisit Example 15.3, where the consulting company has five jobs in its backlog. Create a schedule using the EDD rule, calculating the average days past due and flow time. In this case, does EDD outperform the FCFS rule?

SOLUTION

Customer Sequence	Start Time (days)		Processing Time (days)		Finish Time (days)	Due Date	Days Past Due	Days Ago Since Order Arrived	Flow Time (days)
B	0	+	16	=	16	27	**0**	12	**28**
A	16	+	25	=	41	29	**12**	15	**56**
D	41	+	10	=	51	48	**3**	10	**61**
C	51	+	14	=	65	68	**0**	5	**70**
E	65	+	12	=	77	80	**0**	0	**77**

The days past due and average flow time performance measures for the EDD schedule are

$$\text{Average days past due} = \frac{0 + 12 + 3 + 0 + 0}{5} = 3.0 \text{ days}$$

$$\text{Average flow time} = \frac{28 + 56 + 61 + 70 + 77}{5} = 58.4 \text{ days}$$

By both measures, EDD outperforms the FCFS (3.0 versus 3.4 past due and 58.4 versus 60.2 flow time). However, the solution found in part (b) of Example 15.3 still has the best average flow time of only 47.8 days.

Solved Problem 2

The Food Bin grocery store operates 24 hours per day, 7 days per week. Fred Bulger, the store manager, has been analyzing the efficiency and productivity of store operations recently. Bulger decided to observe the need for checkout clerks on the first shift for a one-month period. At the end of the month, he calculated the average number of checkout registers that should be open during the first shift each day. His results showed peak needs on Saturdays and Sundays.

Day	M	T	W	Th	F	S	Su
Number of Clerks Required	3	4	5	5	4	7	8

Bulger now has to come up with a workforce schedule that guarantees each checkout clerk two consecutive days off, but still covers all requirements.

a. Develop a workforce schedule that covers all requirements while giving two consecutive days off to each clerk. How many clerks are needed? Assume that the clerks have no preference regarding which days they have off.

b. Plans can be made to use the clerks for other duties if slack or idle time resulting from this schedule can be determined. How much idle time will result from this schedule, and on what days?

SOLUTION

a. We use the method demonstrated in Example 15.2 to determine the number of clerks needed. The minimum number of clerks is eight.

	DAY						
	M	T	W	Th	F	S	Su
Requirements	3	4	5	5	4	7	8*
Clerk 1	off	off	X	X	X	X	X
Requirements	3	4	4	4	3	6	7*
Clerk 2	off	off	X	X	X	X	X
Requirements	3	4	3	3	2	5	6*
Clerk 3	X	X	X	off	off	X	X
Requirements	2	3	2	3	2	4	5*
Clerk 4	X	X	X	off	off	X	X
Requirements	1	2	1	3	2	3	4*
Clerk 5	X	off	off	X	X	X	X
Requirements	0	2	1	2	1	2	3*
Clerk 6	off	off	X	X	X	X	X
Requirements	0	2*	0	1	0	1	2*
Clerk 7	X	X	off	off	X	X	X
Requirements	0	1*	0	1*	0	0	1*
Clerk 8	X	X	X	X	off	off	X
Requirements	0	0	0	0	0	0	0

*Maximum requirements

b. Based on the results in part (a), the number of clerks on duty minus the requirements is the number of idle clerks available for other duties:

	M	T	W	Th	F	S	Su
Number on duty	5	4	6	5	5	7	8
Requirements	3	4	5	5	4	7	8
Idle clerks	2	0	1	0	1	0	0

The slack in this schedule would indicate to Bulger the number of employees he might ask to work part time (fewer than 5 days per week). For example, Clerk 7 might work Tuesday, Saturday, and Sunday and Clerk 8 might work Tuesday, Thursday, and Sunday. That would eliminate slack from the schedule.

Solved Problem 3

The Cranston Telephone Company employs workers who lay telephone cables and perform various other construction tasks. The company prides itself on good service and strives to complete all service orders within the planning period in which they are received.

Each worker puts in 600 hours of regular time per planning period and can work as many as an additional 100 hours of overtime. The operations department has estimated the following workforce requirements for such services over the next four planning periods:

MyOMLab

Tutor 15.4 in MyOMLab provides another example for practicing sales and operations planning using a variety of strategies.

Planning Period	1	2	3	4
Demand (hours)	21,000	18,000	30,000	12,000

Cranston pays regular-time wages of $6,000 per employee per period for any time worked up to 600 hours (including undertime). The overtime pay rate is $15 per hour over 600 hours. Hiring, training, and outfitting a new employee costs $8,000. Layoff costs are $2,000 per employee. Currently, 40 employees work for Cranston in this capacity. No delays in service, or backorders, are allowed. Use the spreadsheet approach to answer the following questions:

a. Prepare a chase strategy using only hiring and layoffs. What are the total numbers of employees hired and laid off?

b. Develop a workforce plan that uses the level strategy, relaying only on overtime and undertime. Maximize the use of overtime during the peak period so as to minimize the workforce level and amount of undertime.

c. Propose an effective mixed-strategy plan.

d. Compare the total costs of the three plans.

SOLUTION

a. The chase strategy workforce is calculated by dividing the demand for each period by 600 hours, or the amount or regular-time work for one employee during one period. This strategy calls for a total of 20 workers to be hired and 40 to be laid off during the four-period plan. Figure 15.9 shows the "chase strategy" solution that OM Explorer's *Sales and Operations Planning with Spreadsheets* Solver produces. We simply hide any unneeded columns and rows in this general-purpose solver.

FIGURE 15.9
Spreadsheet for Chase Strategy

	1	2	3	4	Total
Inputs					
Forecasted demand	35	30	50	20	135
Workforce level	35	30	50	20	135
Undertime	0	0	0	0	0
Overtime	0	0	0	0	0
Derived					
Utilized time	35	30	50	20	135
Hires	0	0	20	0	20
Layoffs	5	5	0	30	40
Calculated					
Utilized time cost	$210,000	$180,000	$300,000	$120,000	$810,000
Undertime cost	$0	$0	$0	$0	$0
Overtime cost	$0	$0	$0	$0	$0
Hiring cost	$0	$0	$160,000	$0	$160,000
Layoff cost	$10,000	$10,000	$0	$60,000	$80,000
Total cost	$220,000	190,000	460,000	180,000	$1,050,000

b. The peak demand is 30,000 hours in period 3. As each employee can work 700 hours per period (600 on regular time and 100 on overtime), the workforce level of the level strategy that minimizes undertime is 30,000/700 = 42.86, or 43 employees. This strategy calls for three employees to be hired in the first quarter and for none to be laid off. To convert the demand requirements into employee-period equivalents, divide the demand in hours by 600. For example, the demand of 21,000 hours in period 1 translates into 35 employee-period equivalents (21,000/600) and demand in period 3 translates into 50 employee-period equivalents (30,000/600). Figure 15.10 shows OM Explorer's spreadsheet for this level strategy that minimizes undertime.

FIGURE 15.10 ▶
Spreadsheet for Level Strategy

	1	2	3	4	Total
Inputs					
Forecasted demand	35	30	50	20	135
Workforce level	43	43	43	43	172
Undertime	8	13	0	23	44
Overtime	0	0	7	0	7
Derived					
Utilized time	35	30	43	20	128
Hires	3	0	0	0	3
Layoffs	0	0	0	0	0
Calculated					
Utilized time cost	$210,000	$180,000	$258,000	$120,000	$768,000
Undertime cost	$48,000	$78,000	$0	$138,000	$264,000
Overtime cost	$0	$0	$63,000	$0	$63,000
Hiring cost	$24,000	$0	$0	$0	$24,000
Layoff cost	$0	$0	$0	$0	$0
Total cost	$282,000	258,000	321,000	258,000	$1,119,000

c. The mixed-strategy plan that we propose uses a combination of hires, layoffs, and overtime to reduce total costs. The workforce is reduced by 5 at the beginning of the first period, increased by 8 in the third period, and reduced by 13 in the fourth period. Figure 15.11 shows the results.

	1	2	3	4	Total
Inputs					
Forecasted demand	35	30	50	20	135
Workforce level	35	35	43	30	143
Undertime	0	5	0	10	15
Overtime	0	0	7	0	7
Derived					
Utilized time	35	30	43	20	128
Hires	0	0	8	0	8
Layoffs	5	0	0	13	18
Calculated					
Utilized time cost	$210,000	$180,000	$258,000	$120,000	$768,000
Undertime cost	$0	$30,000	$0	$60,000	$90,000
Overtime cost	$0	$0	$63,000	$0	$63,000
Hiring cost	$0	$0	$64,000	$0	$64,000
Layoff cost	$10,000	$0	$0	$26,000	$36,000
Total cost	$220.000	210,000	385,000	206,000	$1,021,000

◀ **FIGURE 15.11** Spreadsheet for Mixed Strategy

d. The total cost of the chase strategy is $1,050,000. The level strategy results in a total cost of $1,119,000. The mixed-strategy plan was developed by trial and error and results in a total cost of $1,021,000. Further improvements are possible.

Discussion Questions

1. In your community, some employers maintain stable workforces at all costs, and others furlough and recall workers seemingly at the drop of a hat. What are the differences in markets, management, products, financial position, skills, costs, and competition that could explain these two extremes in personnel policy?
2. Quantitative methods can help managers evaluate alternative sales and operations plans on the basis of cost. These methods require cost estimates for each of the controllable variables, such as overtime, subcontracting, hiring, firing, and inventory investment. Say that the existing workforce is made up of 10,000 workers, each having skills valued at $40,000 per year. The plan calls for "creating alternative career opportunities"—in other words, laying off 500 employees. List the types of costs incurred when employees are laid off, and make a rough estimate of the length of time required for payroll savings to recover restructuring costs. If business is expected to improve in one year, are layoffs financially justified? What costs are incurred in a layoff that are difficult to estimate in monetary terms?
3. Explain why management should be concerned about priority systems in service and manufacturing organizations.

Problems

The OM Explorer and POM for Windows software is available to all students using the 10th edition of this textbook. Go to **www.pearsonglobaleditions.com/krajewski** to download these computer packages. If you purchased MyOMLab, you also have access to Active Models software and significant help in doing the following problems. Check with your instructor on how best to use these resources. In many cases, the instructor wants you to understand how to do the calculations by hand. At the least, the software provides a check on your calculations. When calculations are particularly complex and the goal is interpreting the results in making decision, the software entirely replaces the manual calculations.

1. The Barberton Municipal Division of Road Maintenance is charged with road repair in the city of Barberton and the surrounding area. Cindy Kramer, road maintenance director, must submit a staffing plan for the next year based on a set schedule for repairs and on the city budget. Kramer estimates that the labor hours required for the next four quarters are 6,000, 12,000, 19,000, and 9,000, respectively. Each of the 11 workers on the workforce can contribute 500 hours per quarter. Payroll costs are $6,000 in wages per worker for regular time worked up to 500 hours, with an overtime pay rate of $18 for each overtime hour. Overtime is limited to 20 percent of the

regular-time capacity in any quarter. Although unused overtime capacity has no cost, unused regular time is paid at $12 per hour. The cost of hiring a worker is $3,000, and the cost of laying off a worker is $2,000. Subcontracting is not permitted.

a. Find a level workforce plan that relies just on overtime and the minimum amount of undertime possible. Overtime can be used to its limits in any quarter. What is the total cost of the plan and how many undertime hours does it call for?

b. Use a chase strategy that varies the workforce level without using overtime or undertime. What is the total cost of this plan?

c. Propose a plan of your own. Compare your plan with those in part (a) and part (b) and discuss its comparative merits.

2. A manager faces peak (weekly) demand for one of her operations, but is not sure how long the peak will last. She can either use overtime from the current workforce, or hire/lay off and just pay regular-time wages. Regular-time pay is $500 per week, overtime is $825 per week, the hiring cost is $3,000, and the layoff cost is $2,000. Assuming that people are available seeking such a short-term arrangement, how many weeks must the surge in demand last to justify a temporary hire? *Hint:* Use break-even analysis (see Supplement A, "Decision Making Models"). Let w be the number of weeks of the high demand (rather than using Q for the break-even quantity). What is the fixed cost for the regular-time option? Overtime option?

3. Gerald Glynn manages the Michaels Distribution Center. After careful examination of his database information, he has determined the daily requirements for part-time loading dock personnel. The distribution center operates 7 days a week, and the daily part-time staffing requirements are

Day	M	T	W	Th	F	S	Su
Requirements	6	3	5	3	7	2	3

Find the minimum number of workers Glynn must hire. Prepare a workforce schedule for these individuals so that each will have two consecutive days off per week and all staffing requirements will be satisfied. Give preference to the S–Su pair in case of a tie.

4. The Donald Fertilizer Company produces industrial chemical fertilizers. The projected manufacturing requirements (in thousands of gallons) for the next four quarters are 80, 50, 80, and 130, respectively. A level workforce is desired, relying only on anticipation inventory as a supply option. Stockouts and backorders are to be avoided, as are overtime and undertime.

a. Determine the quarterly production rate required to meet total demand for the year, and minimize the anticipation inventory that would be left over at the end of the year. Beginning inventory is zero.

b. Specify the anticipation inventory that will be produced.

c. Suppose that the requirements for the next four quarters are revised to 80, 130, 50, and 80, respectively. If total demand is the same, what level of production rate is needed now, using the same strategy as part (a)?

5. The Hickory Company manufactures wooden desks. Management schedules overtime every weekend to reduce the backlog on the most popular models. The automatic routing machine is used to cut certain types of edges on the desktops. The following orders need to be scheduled for the routing machine:

Order	Time Since Order Arrived (hours ago)	Estimated Machine Time (hours)	Due Date (hours from now)
1	1	9	20
2	0	7	21
3	6	8	12
4	5	3	8
5	3	12	18

The due dates reflect the need for the order to be at its next operation.

a. Develop separate schedules by using the FCFS and EDD rules. Compare the schedules on the basis of average flow time and average past due hours.

b. Comment on the performance of the two rules relative to these measures.

6. Tax Prep Advisers, Inc., has forecasted the following staffing requirements for tax preparation associates over the next 12 months. Management would like three alternative staffing plans to be developed.

Month	1	2	3	4
Demand	5	8	10	13
Month	**5**	**6**	**7**	**8**
Demand	18	20	20	14
Month	**9**	**10**	**11**	**12**
Demand	12	8	2	1

The company currently has 10 associates. No more than 10 new hires can be accommodated in any month because of limited training facilities. No backorders are allowed, and overtime cannot exceed 25 percent of regular time capacity on any month. There is no cost for unused overtime capacity. Regular-time wages are $1,500 per month, and overtime wages are 150 percent of regular time wages. Undertime is paid at the same rate as regular time. The hiring cost is $2,500 per person, and the layoff cost is $2.000 per person.

a. Prepare a staffing plan utilizing a level workforce strategy, minimizing undertime. The plan may call for a one-time adjustment of the workforce before month 1.

b. Using a chase strategy, prepare a plan that is consistent with the constraint on hiring and minimizes use of overtime.

c. Prepare a mixed strategy in which the workforce level is slowly increased by two employees per month through month 5 and is then decreased by two employees per month starting in month 6 and continuing through month 12. Does this plan violate the hiring or overtime constraints set the company?

d. Contrast these three plans on the basis of annual costs.

7. Climate Control, Inc., makes expedition-quality rain gear for outdoor enthusiasts. Management prepared a forecast of sales (in suits) for next year and now must prepare a production plan. The company has traditionally maintained a level workforce strategy. All nine workers are treated like family and have been employed by the company for a number of years. Each employee can produce 2,000 suits per month. At present, finished goods inventory holds 24,000 suits. The demand forecast follows:

Month	**1**	**2**	**3**	**4**
Demand	25,000	16,000	15,000	19,000
Month	**5**	**6**	**7**	**8**
Demand	32,000	29,000	27,000	22,000
Month	**9**	**10**	**11**	**12**
Demand	14,000	15,000	20,000	6,000

a. Management is willing to authorize overtime in periods for which regular production and current levels of anticipation inventory do not satisfy demand. However, overtime must be strictly limited to no more than 20 percent of regular time capacity. Management wants to avoid stockouts and backorders and is not willing to accept a plan that calls for shortages. Is it feasible to hold the workforce constant, assuming that overtime is only used in periods for which shortages would occur?

b. Assume that management is not willing to authorize any overtime. Instead, management is willing to negotiate with customers so that backorders may be used as a supply option. However, management is not willing to carry more than 5,000 suits from one month to the next in backorder. Is it feasible to hold the workforce constant, assuming that a maximum backorder of 5,000 suits may be maintained from month to month?

c. Assume management is willing to authorize the use of overtime over the next four months to build additional anticipation inventory. However, overtime must be strictly limited to no more than 20 percent of regular time capacity. Management wants to avoid stockouts and backorders and is not willing to accept a plan that calls for shortages. Is it feasible to hold the workforce constant, assuming that overtime is only used in months 1–4? If not, in which months would additional overtime be required?

8. Gretchen's Kitchen is a fast-food restaurant located in an ideal spot near the local high school. Gretchen Lowe must prepare an annual staffing plan. The only menu items are hamburgers, chili, soft drinks, shakes, and French fries. A sample of 1,000 customers taken at random revealed that they purchased 2,100 hamburgers, 200 pints of chili, 1,000 soft drinks and shakes, and 1,000 bags of French fries. Thus, for purposes of estimating staffing requirements, Lowe assumes that each customer purchases 2.1 hamburgers, 0.2 pint of chili, 1 soft drink or shake, and 1 bag of French fries. Each hamburger requires 4 minutes of labor, a pint of chili requires 3 minutes, and a soft drink or shake and a bag of fries each take 2 minutes of labor.

The restaurant currently has 10 part-time employees who work 80 hours a month on staggered shifts. Wages are $400 per month for regular time and $7.50 per hour for overtime. Hiring and training costs are $250 per new employee, and layoff costs are $50 per employee.

Lowe realizes that building up seasonal inventories of hamburgers (or any of the products) would not be wise because of shelf-life considerations. Also, any demand not satisfied is a lost sale and must be avoided. Three strategies come to mind.

- Use a level strategy relying on overtime and undertime, with up to 20 percent of regular-time capacity on overtime.
- Maintain a base of 10 employees, hiring and laying off as needed to avoid any overtime.
- Utilize a chase strategy, hiring and laying off employees as demand changes to avoid overtime.

When performing her calculations, Lowe always rounds to the next highest integer for the number of employees. She also follows a policy of not using an employee more than 80 hours per month, except when overtime is needed. The projected demand by month (number of customers) for next year is as follows:

Jan.	3,200	July	4,800
Feb.	2,600	Aug.	4,200
Mar.	3,300	Sept.	3,800
Apr.	3,900	Oct.	3,600
May	3,600	Nov.	3,500
June	4,200	Dec.	3,000

a. Develop the schedule of service requirements (hours per month) for the next year.

b. Which strategy is most effective?

c. Suppose that an arrangement with the high school enables the manager to identify good prospective employees without having to advertise in the local newspaper. This source reduces the hiring cost to $50, which is mainly the cost of charred hamburgers during training. If cost is her only concern, will this method of hiring change Gretchen Lowe's strategy? Considering other objectives that may be appropriate, do you think she should change strategies?

9. The Mowry Machine Shop still has five jobs to be processed as of 8 A.M. today (day 23) at its bottleneck operation. The day when the order arrived, processing time, and promised due dates are given in the following table. The jobs are listed in the order of arrival.

Job	Time Since Order Arrived (days ago)	Processing Time (days)	Due Date (days from now)
A	13	8	36
B	15	4	42
C	17	4	39
D	22	3	53
E	24	10	45

a. Develop separate schedules by using the FCFS and EDD rules. Compare the schedules on the basis of average flow time and average days past due.

b. Which rule gives the best schedule, in your judgment? Why?

10. Bob Carlton's golf camp estimates the following workforce requirements for its services over the next 2 years.

Quarter	1	2	3	4
Demand (hours)	4,200	6,400	3,000	4,800
Quarter	5	6	7	8
Demand (hours)	4,400	6,240	3,600	4,800

Each certified instructor puts in 480 hours per quarter regular time and can work an additional 120 hours overtime. Regular-time wages and benefits cost Carlton $7,200 per employee per quarter for regular time worked up to 480 hours, with an overtime cost of $20 per hour. Unused regular time for certified instructors is paid at $15 per hour. There is no cost for unused overtime capacity. The cost of hiring, training, and certifying a new employee is $10,000. Layoff costs are $4,000 per employee. Currently, eight employees work in this capacity.

a. Find a workforce plan using the level strategy that allows for no delay in service. It should rely only on overtime and the minimum amount of undertime necessary. What is the total cost of this plan?

b. Use a chase strategy that varies the workforce level without using overtime or undertime. What is the total cost of this plan?

c. Propose a better plan and calculate its total cost.

11. Continuing Problem 10, now assume that Carlton is permitted to employ some uncertified, part-time instructors, provided they represent no more than 15 percent of the total workforce hours in any quarter. Each part-time instructor can work up to 240 hours per quarter, with no overtime or undertime cost. Labor costs for part-time instructors are $12 per hour. Hiring and training costs are $2,000 per uncertified instructor, and there are no layoff costs.

a. Propose a low-cost, mixed-strategy plan and calculate its total cost.

b. What are the primary advantages and disadvantages of having a workforce consisting of both regular and temporary employees?

12. Cara Ryder manages a ski school in a large resort and is trying to develop a schedule for instructors. The instructors receive little salary and work just enough to earn room and board. They receive free skiing and spend most of their free time tackling the resort's notorious double black-diamond slopes. Hence, the instructors work only 4 days a week. One of the lesson packages offered at the resort is a 4-day beginner package. Ryder likes to keep the same instructor with a group over the 4-day period, so she schedules the instructors for 4 consecutive days and then 3 days off. Ryder uses years of experience with demand forecasts provided by management to formulate her instructor requirements for the upcoming month.

Day	M	T	W	Th	F	S	Su
Requirements	7	5	4	5	6	9	8

a. Determine how many instructors Ryder needs to employ. Give preference to Saturday and Sunday off. (*Hint:* Look for the group of *3* days with the lowest requirements.)

b. Specify the work schedule for each employee. How much slack does your schedule generate for each day?

13. The mayor of Cambridge, Colorado, wanting to be environmentally progressive, decides to implement a recycling plan. All residents of the city will receive a special three-part bin to separate their glass, plastic, and aluminum, and the city will be responsible for picking up the materials. A young city and regional planning graduate, Michael Duffy, has been hired to manage the recycling program. After carefully studying the city's population density, Duffy decides that the following numbers of recycling collectors will be needed:

Day	M	T	W	Th	F	S	Su
Requirements	12	7	9	9	5	3	6

The requirements are based on the populations of the various housing developments and subdivisions in the city and surrounding communities. To motivate residents of some areas to have their pickups scheduled on weekends, a special tax break will be given.

a. Find the minimum number of recycling collectors required if each employee works 5 days a week and has two consecutive days off. Give preference to the S–Su pair when that pair is involved in a tie.

b. Specify the work schedule for each employee. How much slack does your schedule generate for each day?

c. Suppose that Duffy can smooth the requirements further through greater tax incentives. The requirements then will be eight collectors on Monday and seven on the other days of the week. How many collectors will be needed now? Does smoothing of requirements have capital investment implications? If so, what are they?

14. Little 6, Inc., an accounting firm, forecasts the following weekly workload during the tax season:

	DAY						
	M	T	W	Th	F	S	Su
Personal Tax Returns	24	14	18	18	10	28	16
Corporate Tax Returns	16	10	12	15	24	12	4

Corporate tax returns each require 4 hours of an accountant's time, and personal returns each require 90 minutes. During tax season, each accountant can work up to 10 hours per day. However, error rates increase to unacceptable levels when accountants work more than five consecutive days per week.

Hint: Read Supplement D before doing this problem. Let x_i = number for each working schedule, e.g., x_1 = number for Tuesday through Saturday.

a. Create an effective and efficient work schedule by formulating the problem as a Linear Program and solve using POM for Windows.

b. Assume that management has decided to offer a pay differential to those accountants who are scheduled to work on a weekend day. Normally, accountants earn $1,200 per week, but management will pay a bonus of $100 for Saturday work and $150 for Sunday work. What schedule will cover all demand as well as minimize payroll cost?

c. Assume that Little 6 has three part-time employees available to work Friday, Saturday, and Sunday at a rate of $800. Could these employees be cost effectively utilized?

15. Return to Problem 3 and the workforce schedule for part-time loading dock workers. Suppose that each part-time worker can work only 3 days, but the days must be consecutive. Formulate and solve this workforce scheduling problem as a Linear Program and solve it using POM for Windows. Your objective is to minimize total slack capacity. What is the minimum number of loaders needed now and what are their schedules?

Hint: Read Supplement D before doing this problem. Let x_i = number for each 3-day schedule, e.g., x_1 = number for Tuesday through Thursday.

16. Management at the Kerby Corporation has determined the following aggregated demand schedule (in units):

Month	1	2	3	4
Demand	500	800	1,000	1,400
Month	5	6	7	8
Demand	2,000	3,000	2,700	1,500
Month	9	10	11	12
Demand	1,400	1,500	2,000	1,200

An employee can produce an average of 10 units per month. Each worker on the payroll costs $2,000 in regular-time wages per month. Undertime is paid at the same rate as regular time. In accordance with the labor contract in force, Kerby Corporation does not work overtime or use subcontracting. Kerby can hire and train a new employee for $2,000 and lay off one for $500. Inventory costs $32 per unit on hand at the end of each month. At present, 140 employees are on the payroll and anticipation inventory is zero.

a. Prepare a production plan that only uses a level workforce and anticipation inventory as its supply options. Minimize the inventory left over at the end of the year. Layoffs, undertime, vacations, subcontracting, backorders, and stockouts are not options. The plan may call for a one-time adjustment of the workforce before month 1 begins.

b. Prepare a production plan using a chase strategy, relying only on hiring and layoffs.

c. Prepare a mixed-strategy production plan that uses only a level workforce and anticipation inventory through month 7 (an adjustment of the workforce may be made before month 1 begins) then switches to a chase strategy for months 8–12.

d. Contrast these three plans on the basis of annual costs.

17. Currently a company that designs Web sites has five customers in its backlog. The day when the order arrived, processing time, and promised due dates are given in the following table. The customers are listed in the order of when they arrived. They are ready to be scheduled today, which is the start of day 190.

Customer	Time Since Order Arrived (days ago)	Processing Time (days)	Due Date (days from now)
A	10	20	26
B	8	12	50
C	6	28	66
D	3	24	58
E	2	32	100

a. Develop separate schedules by using the FCFS and EDD rules. Compare the schedules on the basis of average flow time and average days past due.

b. Comment on the performance of the two rules relative to these measures. Which one gives the best schedule? Why?

18. The Kool King Company has followed a policy of no layoffs for most of the manufacturer's life, even though the demand for its air conditioners is highly seasonal. Management wants to evaluate the cost-effectiveness of this policy. Competitive pressures are increasing, and ways need to be found to reduce costs. The following demand (expressed in employee–month equivalents) has been forecast for next year:

Jan.	70	May	130	Sept.	110
Feb.	90	June	170	Oct.	60
Mar.	100	July	170	Nov.	20
Apr.	100	Aug.	150	Dec.	40

Additional planning data follow, with costs, inventory, and backorders expressed in employee–month equivalents:

Regular-time production cost	$1,500	Hire cost	$500/ person
Overtime production cost	150% of regular-time production cost	Layoff cost	$2,000/ person
Subcontracting cost	$2,500	Current backorders	10
Inventory holding cost	$100	Current inventory	0
Backorder cost	$1,000	Desired ending inventory	0
Maximum overtime	20% of regular-time capacity	Current employment	130 employees

Hiring costs are lower than layoff costs because the facility is located near a Technical Training School. Undertime is paid at the rate equivalent to regular-time production. Each employee who has been with the company at least one year also received 0.5 months of paid vacation. All 130 employees currently employed qualify for vacations next year, assuming that they remain on the workforce. Answer the following questions using *Sales and Operations Planning with Spreadsheets* Solver in OM Explorer, or an Excel spreadsheet that you developed on your own.

a. Develop an S&OP with the level strategy using overtime, undertime, and vacations as the only supply options. Use the maximum amount of overtime so as to minimize undertime. What is the total cost of this plan, and what are its advantages and disadvantages?

b. Develop an S&OP with the chase strategy. Part of your decision will be when and how many vacation periods to grant. What is the total cost of this plan, and what are its advantages and disadvantages?

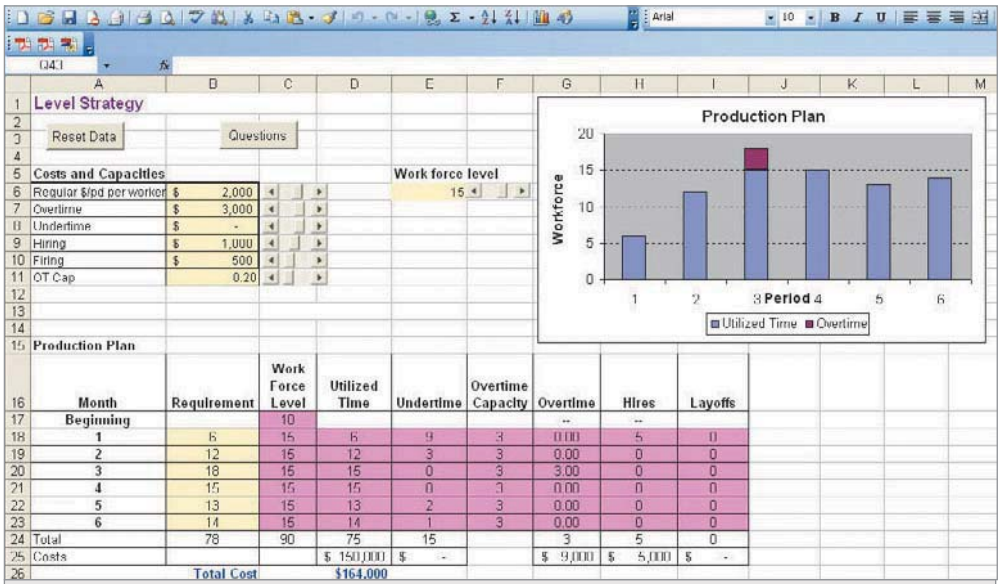
c. Develop an S&OP with a lower cost than found with either the level or chase strategy, being open to the full range of supply options (including anticipation inventory). Subcontractors can supply up to 50 employee–month equivalents. What is the total cost of this plan, and what are its advantages and disadvantages?

Active Model Exercise

This Active Model appears in MyOMLab. It allows you to evaluate the effects of modifying the size of a constant workforce.

QUESTIONS

1. If we use the same number of workers in each period, what happens as the number of workers increases from 15?
2. If the overtime capacity increases to 30 percent, what is the minimum number of workers that meets the demand in every month?
3. Suppose the overtime cost is $3,300. What happens as the number of workers increases?
4. Suppose the hiring cost is $1,100. What happens as the number of workers increases?
5. Suppose the undertime cost is the same as the regular-time cost (i.e., paid undertime). What is the best number of workers to have in each month and still meet the demand?
6. If we use the same number of workers in each period, what happens as the number of workers decreases from 15?



Sales and Operations Planning Using Data from Example 15.1 Spreadsheet for Level Strategy

VIDEO CASE Sales and Operations Planning at Starwood

Business travel often means staying overnight in a hotel. Upon arrival, you may be greeted by a doorman or valet to assist you with your luggage. Front desk staff awaits your check-in. Behind the scenes, housekeeping, maintenance, and culinary staff prepare for your stay. Making a reservation gives the hotel notice of your plan to stay, but even before your trip is ever conceived, the hotel is staffed and ready. How? Through a process called *sales and operations planning*.

Sales and operations planning is a process every organization performs to some degree. Called a staffing plan (or service resource plan if more detailed) in service organizations, the plan must strike the right level of customer service while maintaining workforce stability and cost control so as to achieve the organization's profit expectations. So where do companies begin? Let us take a look at Starwood Hotels and Resorts to see how it is done.

Starwood operates in more than 750 locations around the globe. At the highest levels, Starwood engages in sales and operations planning on an annual basis, with adjustments made as needed each month by region and by property. Budgeted revenues and other projections come from headquarters; the regions and individual properties then break down the forecasts to meet their expected occupancies. Typically, the director of human resources determines the staffing mix needed across divisions such as food and beverage service, rooms (including housekeeping, spa, and guest services), engineering, Six Sigma (see Chapter 5, "Managing Quality"), revenue management, and accounting.

At the property level, general managers and their staff must provide input into next year's plan while implementing and monitoring activity in the current year. For most properties, payroll is close to 40 percent of budgeted revenues and represents the largest single expense the hotel incurs. It is also the most controllable expense. Many of Starwood's hotels and most resorts experience patterns of seasonality that affect demand for rooms and services. This seasonality, in turn, significantly affects the organization's staffing plan.

To determine the staffing levels, the company uses a proprietary software program that models occupancy demand based on historical data. The key drivers of staffing are occupied rooms and restaurant meals, called "covers." Starwood knows on a *per room* and *per cover* basis how many staff are required to function properly. When occupancy and covers are entered into the software program, the output models a recommended staffing level for each division. This recommendation is then reviewed by division managers and adjusted, as needed, to be sure staffing is in line with budgeted financial plans. Job fairs to recruit nonmanagement staff are held several times a year so a qualified candidate pool of both part-time and full-time staff is ready when needed. Most hotels maintain a pool of part-time workers who can contract or expand the hours worked if required by property guest levels. Vacations for management are scheduled for the low season. Overtime will be worked as needed, but is less desirable than scheduling the appropriate level of staff in each division.

The program also takes into account both the complexity and positioning of the property within Starwood. For example, a 400-room city hotel that is essentially a high-rise building is not as complex as a 400-room sprawling resort with golf, spa, convention, and other services not offered by the city hotel. Positioning also is important. A five-star resort hotel's customer service expectations are much greater than a three-star airport hotel location and requires much higher ratios of staff to guests. Finally, if the hotel is a brand new property, historical data from similar properties is used to model staffing for the first year or two of operation.



Pearson

A software program that forecasts occupancy based on historical data helps Starwood maintain proper staffing levels at its hotels. Managers know on a per-room, and "per-cover," basis how many hotel employees should be scheduled so that customers get good service.

Starwood attempts to modify demand and smooth out the peaks and valleys of its demand patterns. Many of the company's hotels experience three seasons: high, mid (called "shoulder"), and low season. Starwood, like its competitors, offers special rates, family packages, and weekend specials to attract different segments of the market during slower business periods. Staff is cross-trained to work in multiple areas, such as front reception and the concierge desk, so additional staff does not have to be added across seasons. Employees may also be temporarily redeployed among Starwood's properties to help out during peak periods. For example, when occupancy is forecast to be high in one region of the country, staff from areas entering their low season will be assigned to cover the demand.

QUESTIONS

1. At what points in the planning process would you expect accounting/finance, marketing, information systems, and operations to play a role? What inputs should these areas provide, and why?
2. Does Starwood employ a chase, level, or mixed strategy? Why is this approach the best choice for the company?
3. How would staffing for the opening of a new hotel or resort differ from that of an existing property? What data might Starwood rely upon to make sure the new property is not over- or understaffed in its first year of operation?

CASE Memorial Hospital

Memorial Hospital is a 265-bed regional health care facility located in the mountains of western North Carolina. The mission of the hospital is to provide quality health care to the people of Ashe County and the six surrounding counties. To accomplish this mission, Memorial Hospital's CEO has outlined three objectives: (1) maximize customer service to increase customer satisfaction, (2) minimize costs to remain competitive, and (3) minimize fluctuations in workforce levels to help stabilize area employment.

The hospital's operations are segmented into eight major wards for the purposes of planning and scheduling the nursing staff. These wards are listed in Table 15.4, along with the number of beds, targeted patient-to-nurse ratios, and average patient census for each ward. The overall demand for hospital services remained relatively constant over the past few years even though the population of the seven counties served increased. This stable demand can be attributed to increased competition from other hospitals in the area and the rise in alternative health care delivery systems, such as health maintenance organizations (HMOs). However, demand for Memorial Hospital's services does vary considerably by type of ward and time of year. Table 15.5 provides a historical monthly breakdown of the average daily patient census per ward.

The director of nursing for Memorial Hospital is Darlene Fry. Each fall she confronts one of the most challenging aspects of her job: planning the nurse-staffing levels for the next calendar year. Although the average demand for nurses has remained relatively stable over the past couple of years, the staffing plan usually changes because of changing work policies, changing pay structures, and temporary nurse availability and cost. With fall quickly approaching, Fry is collecting information to plan next year's staffing levels.

The nurses at Memorial Hospital work a regular schedule of four 10-hour days per week. The average regular-time pay across all nursing grades is $12.00 per hour. Overtime may be scheduled when necessary. However, because of the intensity of the demands placed on nurses, only a limited amount of overtime is permitted per week. Nurses may be scheduled for as many as 12 hours per day, for a maximum of 5 days per week. Overtime is compensated at a rate of $18.00 per hour. In periods of extremely high demand, temporary part-time nurses may be hired for a limited period of time. Temporary nurses are paid $15.00 per hour. Memorial Hospital's policy limits the proportion of temporary nurses to 15 percent of the total nursing staff.

Finding, hiring, and retaining qualified nurses is an ongoing problem for hospitals. One reason is that various forms of private practice lure many nurses away from hospitals with higher pay and greater flexibility. This situation has caused Memorial to guarantee its full-time staff nurses pay for a minimum of 30 hours per week, regardless of the demand placed on nursing services. In addition, each nurse receives 4 weeks of paid vacation each year. However, vacation scheduling may be somewhat restricted by the projected demand for nurses during particular times of the year.

TABLE 15.4 | WARD CAPACITY DATA

Ward	Number of Beds	Patients per Nurse	Patient Census*
Intensive Care	20	2	10
Cardiac	25	4	15
Maternity	30	4	10
Pediatric	40	4	22
Surgery	5	†	†
Post-Op	15	5	8 (T–F daily equivalent)‡
Emergency	10	3	5 (daily equivalent)‡
General	120	8	98

*Yearly average per day

†The hospital employs 20 surgical nurses. Routine surgery is scheduled on Tuesdays and Fridays; five surgeries can be scheduled per day per operating room (bed) on these days. Emergency surgery is scheduled as needed.

‡Daily equivalents are used to schedule nurses because patients flow through these wards in relatively short periods of time. A daily equivalent of 5 indicates that throughout a typical day, an average of five patients are treated in the ward.

At present, the hospital employs 130 nurses, including 20 surgical nurses. The other 110 nurses are assigned to the remaining seven major areas of the hospital. The personnel department informed Fry that the average cost to the hospital for hiring a new full-time nurse is $400 and for laying off or firing a nurse is $150. Although layoffs are an option, Fry is aware of the hospital's objective of maintaining a level workforce.

After looking over the information that she collected, Darlene Fry wants to consider staffing changes in all areas except the surgery ward, which is already correctly staffed.

QUESTIONS

1. Explain the alternatives available to Darlene Fry as she develops a nurse staffing plan for Memorial Hospital. How does each alternative plan meet the objective stated by the CEO?
2. Based on the data presented, develop a nurse staffing plan for Memorial Hospital. Explain your rationale for this plan.

TABLE 15.5 | AVERAGE DAILY PATIENT CENSUS PER MONTH

	MONTH											
Ward	**J**	**F**	**M**	**A**	**M**	**J**	**J**	**A**	**S**	**O**	**N**	**D**
Intensive Care	13	10	8	7	7	6	11	13	9	10	12	14
Cardiac	18	16	15	13	14	12	13	12	13	15	18	20
Maternity	8	8	12	13	10	8	13	13	14	10	8	7
Pediatric	22	23	24	24	25	21	22	20	18	20	21	19
Surgery*	20	18	18	17	16	16	22	21	17	18	20	22
Post-Op†	10	8	7	7	6	6	10	10	7	8	9	10
Emergency†	6	4	4	7	8	5	5	4	4	3	4	6
General	110	108	100	98	95	90	88	92	98	102	107	94

Source: This case was prepared by Dr. Brooke Saladin, Wake Forest University, North Carolina, as a basis for classroom discussion. Copyright © Brooke Saladin. Used with permission.

*Average surgeries per day on Tuesday and Thursday.

†Daily equivalents

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16 PLANNING SUFFICIENT RESOURCES

Dow Corning scientists research next generation solar technologies at the company's Solar Application Center

Dow Corning

Dow Corning, created as a joint venture between Corning Incorporated and the Dow Chemical Company in 1943, is a global leader that offers over 7,000 innovative products and services using silicon-based technology in diverse industries such as electronics, aviation and aerospace, textile, automotive, and health care, among others. As an example, Dow Corning's Electrical Insulating Compounds are used for creating moisture proof seals for aircraft, automotive, and marine ignition systems. With 11,500 employees, over 25,000 customers, and 45 manufacturing and warehouse locations worldwide, its annual revenues of $6 billion generated a net income of $866 million in 2010. More than half its sales are outside the United States. In order to integrate different business functions and enhance resource planning across the entire firm and its supply chain, Dow Corning turned to SAP, a leading provider of enterprise resource planning (ERP) software solutions. SAP delivers role based access to crucial data, applications, and analytical tools, and has a suite of applications for Business Processes (like Financial Management, Customer Relationship Management, Human Capital Management, and Supply Chain Management), Business Analytics, and Technology. Dow installed the SAP R/3 and mySAP Supply Chain Management solution, with SAP Advanced Planner and Optimizer (APO) at its core.

Prior to the ERP implementation, limited transparency and redundancy within the existing legacy systems made it difficult to access and analyze data

needed for effective resource planning, and also hampered decision making and responsiveness. A sequenced implementation of the SAP modules facilitated the linking of key processes from order generation to production planning to warehousing to delivery and final billing. The SAP APO solution enabled the SCOR model of plan, source, make, and deliver (see Chapter 12, "Integrating the Supply Chain"), which allowed the linkage of shop floor processes and manufacturing operations with the rest of the business. With a transparent view of orders, materials, equipment, product quality, and cost information, Dow Corning can now coordinate plants and processes with greater ease and better match production to market requirements on a global scale. Employee productivity and satisfaction are also up, due mostly to faster response times and accurate on-time deliveries.

Source: End to End Supply Chain Management at Dow Corning, **http://www.sap.com/usa/solutions/business-suite/erp/operations/pdf/CS_Dow_Corning.pdf**; Dow Corning, Optimizing Operational Performance to Sharpen Competitive Advantage, **www.sap.com**; **http://www.dowcorning.com/content/about/aboutmedia/fastfacts.asp**; May 29, 2011; **http://www12.sap.com/index.epx#/solutions/index.epx**; May 29, 2011.

LEARNING GOALS *After reading this chapter, you should be able to:*

1. Explain how enterprise resource planning (ERP) systems can foster better resource planning.
2. Explain how the concept of dependent demand is fundamental to resource planning.
3. Describe a master production schedule (MPS) and compute available-to-promise quantities.
4. Apply the logic of a material requirements planning (MRP) system to identify production and purchase orders needed for dependent demand items.
5. Apply MRP principles to the provision of services and distribution inventories.

Dow Corning demonstrates that companies can gain a competitive edge by using an effective information system to help with their resource planning. Companies must ensure that all of the resources they need to produce finished services or products are available at the right time. If they are not, a firm risks losing business. For a manufacturer, this task can mean keeping track of thousands of subassemblies, components, and raw materials as well as key equipment capacities. For a service provider, this task can mean keeping track of numerous supplies and carefully scheduling the time and capacity requirements of different employees and types of equipment.

We begin this chapter by describing enterprise resource planning (ERP) systems, which have become a valuable tool for, among other things, resource planning. We then examine a specific approach to resource planning, called material requirements planning (MRP). The concluding section of the chapter illustrates how service providers manage their supplies, human resources, equipment, and financial resources.

resource planning
A process that takes sales and operations plans; processes information in the way of time standards, routings, and other information on how services or products are produced; and then plans the input requirements.

enterprise process
A companywide process that cuts across functional areas, business units, geographical regions, and product lines.

enterprise resource planning (ERP) systems
Large, integrated information systems that support many enterprise processes and data storage needs.

Resource Planning across the Organization

Resource planning lies at the heart of any organization, cutting across all of its different functional areas. It takes sales and operations plans; processes information in the way of time standards, routings, and other information on how services or products are produced; and then plans the input requirements. It also can create reports for managers of the firm's major functional areas, such as human resources, purchasing, sales and marketing, and finance and accounting. In essence, resource planning is a process in and of itself that can be analyzed relative to the firm's competitive priorities.

Enterprise Resource Planning

An **enterprise process** is a companywide process that cuts across functional areas, business units, geographic regions, product lines, suppliers, and customers. **Enterprise resource planning (ERP) systems** are large, integrated information systems that support many enterprise processes and

data storage needs. By integrating the firm's functional areas, ERP systems allow an organization to view its operations as a whole rather than having to try to put together the different information pieces produced by its various functions and divisions. Today, ERP systems are being used by traditional brick-and-mortar organizations such as manufacturers, restaurants, hospitals, and hotels, as well as by Internet companies that rely extensively on Web connectivity to link their customers and suppliers.

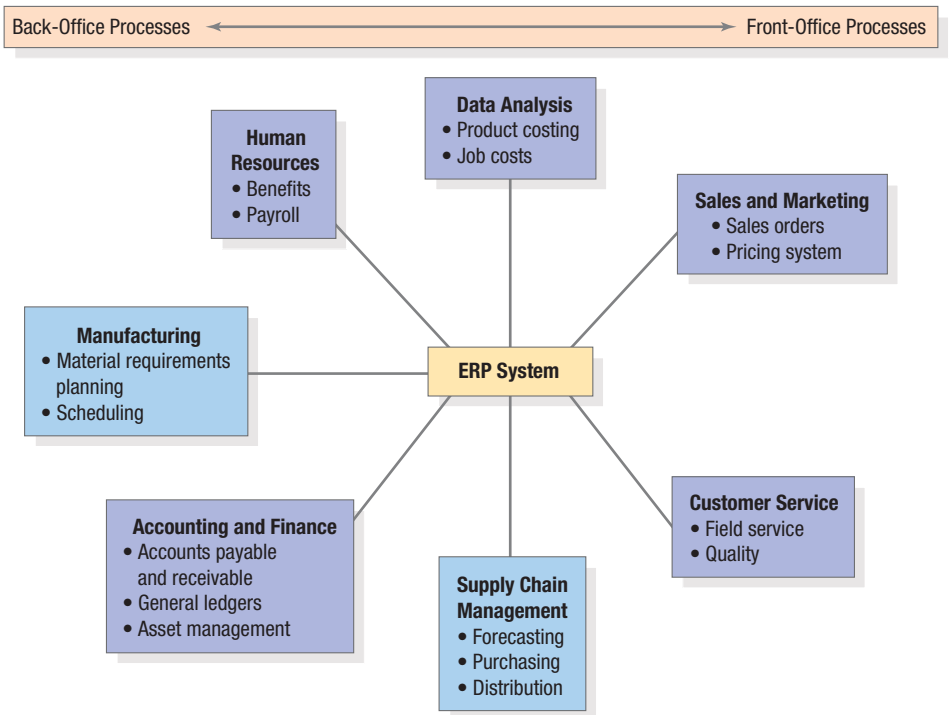
How ERP Systems Are Designed

ERP revolves around a single comprehensive database that can be made available across the entire organization (or enterprise). Passwords are generally issued to allow certain personnel to access certain areas of the system. Having a single database for all of the firm's information makes it much easier for managers to monitor all of the company's products at all locations and at all times. The database collects data and feeds them into the various modular applications (or suites) of the software system. As new information is entered as a *transaction* in one application, related information is automatically updated in the other applications, including the firm's financial and accounting databases, its human resource and payroll databases, sales, supplier and customer databases, and so forth. In this way, the ERP system streamlines the data flows throughout the organization and supply chain and provides employees with direct access to a wealth of real-time operating information scattered across different functions in the organization. Figure 16.1 shows some of the typical applications with a few subprocesses nested within each one. Some of the applications are for back-office operations such as manufacturing and payroll, while others are for front-office operations such as customer service.

Amazon.com is one company that uses an ERP system. The supply chain application of Amazon's system is particularly important because it allows Amazon.com to link customer orders to warehouse shipments and, ultimately, to supplier replenishment orders. Other applications are more important in other businesses. For example, universities put particular emphasis on the human resources and accounting and finance applications, and manufacturers have an interest in almost every application suite. Not all applications in Figure 16.1 need to be integrated into an ERP system, but those left out will not share their information with the ERP system. Sometimes, however, ERP systems are designed to interface with a firm's existing, older information systems (called "legacy systems").

Designing an ERP system requires that a company carefully analyze its major processes so that appropriate decisions about the coordination of legacy systems and new software can be made. Sometimes, a company's processes that involve redundancies and convoluted information flows must be completely reengineered before the firm can enjoy the benefits of an integrated

Competing with Operations
Using Operations to Compete
Managing Effective Projects
Designing and Managing Processes
Developing a Process Strategy
Analyzing Processes
Managing Quality
Planning Capacity
Managing Process Constraints
Designing Lean Systems
Designing and Managing Supply Chains
Managing Inventories
Designing Effective Supply Chains
Locating Facilities
Integrating the Supply Chain
Managing Sustainable Supply Chains
Forecasting Demand
Planning and Scheduling Operations
Planning Sufficient Resources



◀ **FIGURE 16.1**
ERP Application Modules
Source: Based on *Enterprise Resource Planning (ERP)* by Scalle and Cotteleer, Harvard Business School Press. Boston, MA, 1999, No. 9-699-020.



Stefan Kiefer/Imagebroker/Alamy

Headquarters of the software company SAP AG in Walldorf, Baden-Wuerttemberg, Germany, Europe

information system. However, a recent study showed that companies reap the greatest rewards when they keep their ERP implementations simple, work with a small number of software vendors, and use standardized systems rather than customizing them extensively. Firms can otherwise end up spending excessive amounts of money on ERP systems that are complex to use and costly to manage. UK confectionary giant Cadbury had to take a 12-million pound hit to their profits due to a build-up of excess inventory of chocolate bars caused by information technology problems arising from the roll out of a new SAP-based ERP system.

Most ERP systems today use a graphical user interface, although the older, keyboard-driven, text-based systems are still popular because of their dependability and technical simplicity. Users navigate through various screens and menus. Training, such as during ERP implementation, focuses on these screens and how users can utilize them to get their jobs done. The biggest suppliers of these off-the-shelf commercial ERP packages are SAP AG, a German company that was also used by Dow Corning in the opening vignette, followed by Oracle Corporation.

Material Requirements Planning

material requirements planning (MRP)

A computerized information system developed specifically to help manufacturers manage dependent demand inventory and schedule replenishment orders.

MRP explosion

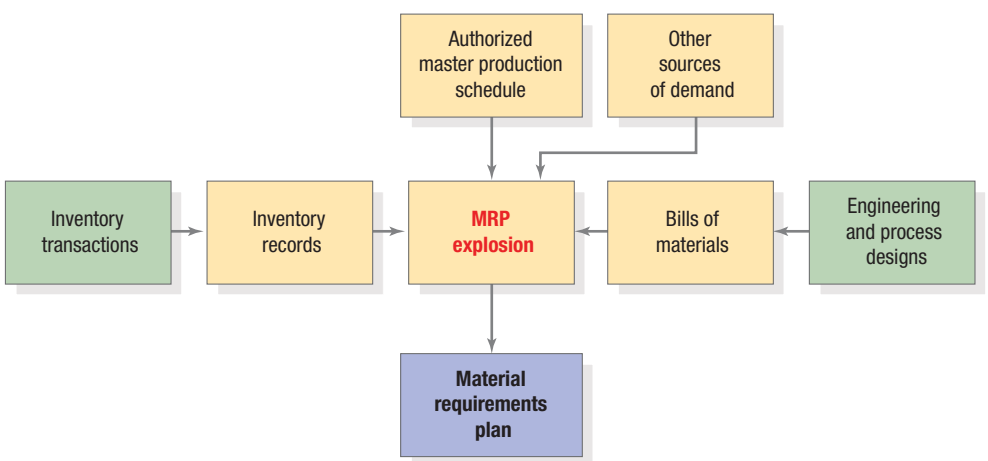
A process that converts the requirements of various final products into a material requirements plan that specifies the replenishment schedules of all the subassemblies, components, and raw materials needed to produce final products.

The Manufacturing and Supply Chain Management modules in Figure 16.1 deal with resource planning. Understanding resource planning begins with the concept of *dependent demand*, which sets it apart from the techniques covered in Chapter 9, "Managing Inventories." **Material requirements planning (MRP)** is a computerized information system developed specifically to help manufacturers manage dependent demand inventory and schedule replenishment orders. The key inputs of an MRP system are a bill of materials database, a master production schedule, and an inventory record database, as shown in Figure 16.2. Using this information, the MRP system identifies the actions planners must take to stay on schedule, such as releasing new production orders, adjusting order quantities, and expediting late orders.

An MRP system translates the master production schedule and other sources of demand, such as independent demand for replacement parts and maintenance items, into the requirements for all subassemblies, components, and raw materials needed to produce the required parent items. This process is called an **MRP explosion** because it converts the requirements of various final products into a material requirements plan that specifies the replenishment schedules of all the subassemblies, components, and raw materials needed to produce final products.

We first explore the nature of dependent demand and how it differs from independent demand, followed by a discussion of each of the key inputs to the MRP system shown in Figure 16.2.

FIGURE 16.2 ▶
Material Requirements Plan Inputs



Dependent Demand

For years, many companies tried to manage production and their dependent demand inventories using independent demand systems similar to those discussed in Chapter 9, "Managing Inventories," but the outcome was seldom satisfactory because dependent demand is fundamentally different from independent demand. To illustrate the concept of dependent demand, let us consider a Huffy bicycle produced for retail outlets. Demand for a final product, such as a bicycle, is called *independent demand* because it is influenced only by market conditions. In contrast, the demand for spokes going into the bicycle "depends" on the production planned for its wheels. Huffy must forecast this demand using techniques such as those discussed in Chapter 14, "Forecasting Demand." However, Huffy also keeps many other items in inventory—handlebars, pedals, frames, and wheel rims—used to make completed bicycles. Each of these items has a **dependent demand** because the quantity required varies with the production plans for other items held in the firm's inventory—finished bikes, in this case. For example, the demand for frames, pedals, and wheel rims is *dependent* on the production of completed bicycles. Operations can calculate the demand for dependent demand items once the bicycle production levels are laid out in the sales and operations plan. For example, every bicycle needs two wheel rims, so 1,000 completed bicycles need 1,000(2) = 2,000 rims. Forecasting techniques are not needed for the rims.

dependent demand

The demand for an item that occurs because the quantity required varies with the production plans for other items held in the firm's inventory.

The bicycle, or any other product that is manufactured from one or more components, is called a **parent**. The wheel rim is an example of a **component**—an item that goes through one or more operations to be transformed into or become part of one or more parents. A wheel rim, for example, will have several different parents if the rim is used to make more than one style of bicycle. This parent–component relationship can cause erratic dependent demand patterns for components. Suppose that every time inventory falls to 500 units (a reorder point), an order for 1,000 more bicycles is placed, as shown in Figure 16.3(a). The assembly supervisor then authorizes the withdrawal of 2,000 rims from inventory, along with other components for the finished product. The demand for the rim is shown in Figure 16.3(b). So, even though customer demand for the finished bicycle is continuous and reasonably uniform, the production demand for wheel rims is "lumpy"; that is, it occurs sporadically, usually in relatively large quantities. Thus, the production decisions for the assembly of bicycles, which account for the costs of assembling the bicycles and the projected assembly capacities at the time the decisions are made, determine the demand for wheel rims.

parent

Any product that is manufactured from one or more components.

component

An item that goes through one or more operations to be transformed into or become part of one or more parents.

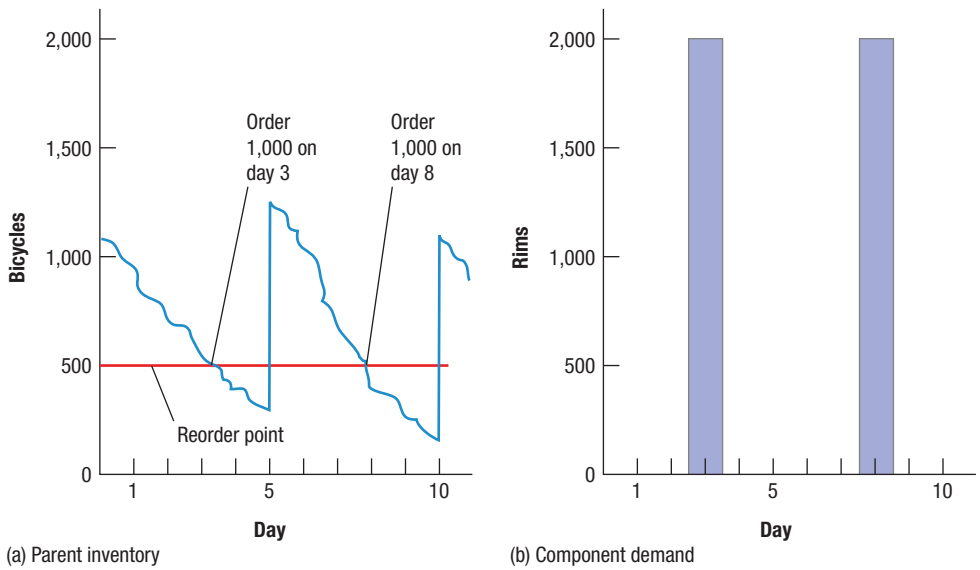
Managing dependent demand inventories is complicated because some components may be subject to both dependent and independent demand. For example, the shop floor needs 2,000 wheel rims for the new bicycles, but the company also sells replacement rims for old bicycles directly to retail outlets. This practice places an independent demand on the inventory of wheel rims. Materials requirements planning can be used in complex situations involving components that may have independent demand as well as dependent demand inventories.

Bill of Materials

The replenishment schedule for a component is determined from the production schedules of its parents. Hence, the system needs accurate information on parent–component relationships. A **bill of materials (BOM)** is a record of all the components of an item, the parent–component

bill of materials (BOM)

A record of all the components of an item, the parent–component relationships, and the usage quantities derived from engineering and process designs.



◀ **FIGURE 16.3**
Lumpy Dependent Demand Resulting from Continuous Independent Demand

relationships, and the usage quantities derived from engineering and process designs. In Figure 16.4, the BOM of a simple ladder-back chair shows that the chair is made from a ladder-back subassembly, a seat subassembly, front legs, and leg supports. In turn, the ladder-back subassembly is made from back legs and back slats, and the seat subassembly is made from a seat frame and a seat cushion. Finally, the seat frame is made from seat-frame boards. For convenience, we refer to these items by the letters shown in Figure 16.4.

usage quantity
The number of units of a component that are needed to make one unit of its immediate parent.

end item
The final product sold to a customer.

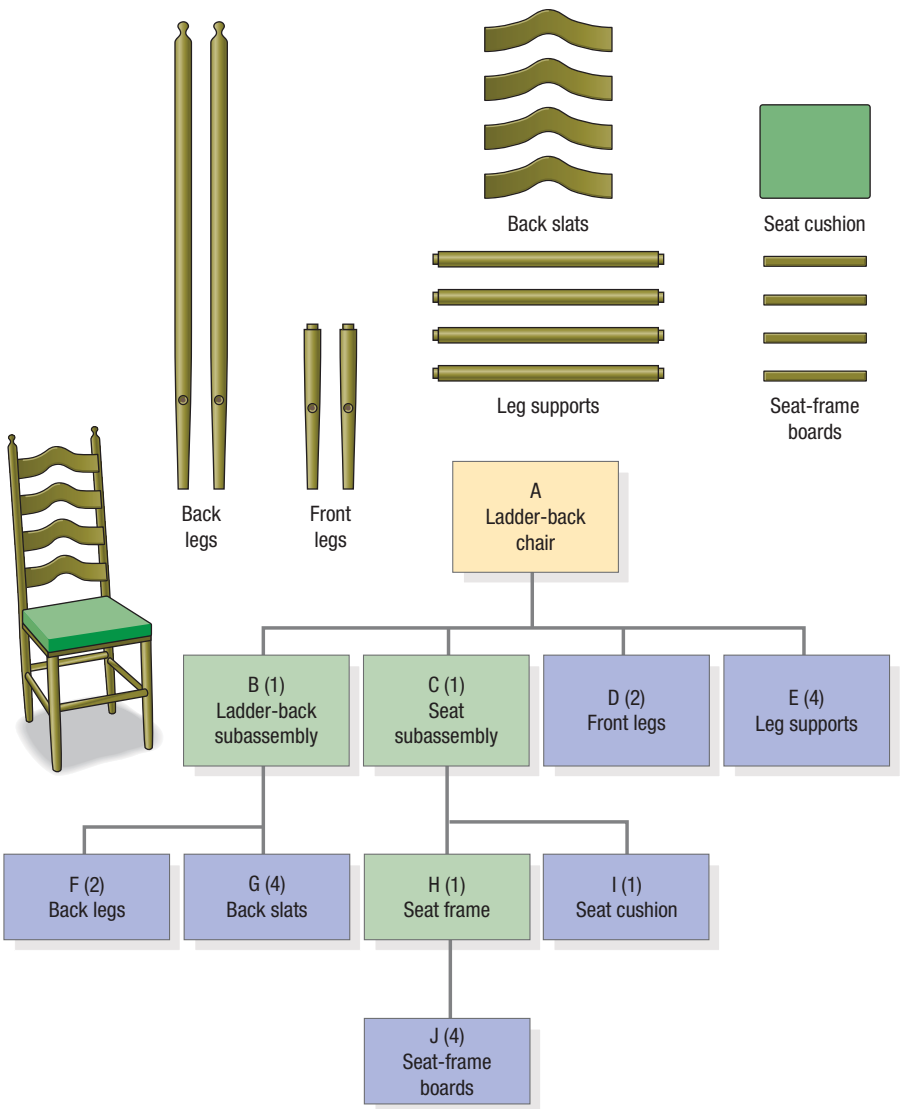
intermediate item
An item that has at least one parent and at least one component.

subassembly
An intermediate item that is *assembled* (as opposed to being transformed by other means) from more than one component.

All items except item A are components because they are needed to make a parent. Items A, B, C, and H are parents because they all have at least one component. The BOM also specifies the **usage quantity**, or the number of units of a component that are needed to make one unit of its immediate parent. Figure 16.4 shows usage quantities for each parent–component relationship in parentheses. Note that one chair (item A) is made from one ladder-back subassembly (item B), one seat subassembly (item C), two front legs (item D), and four leg supports (item E). In addition, item B is made from two back legs (item F) and four back slats (item G). Item C needs one seat frame (item H) and one seat cushion (item I). Finally, item H needs four seat-frame boards (item J).

Four terms frequently used to describe inventory items are *end items*, *intermediate items*, *subassemblies*, and *purchased items*. An **end item** typically is the final product sold to the customer; it is a parent but not a component. Item A in Figure 16.4, the completed ladder-back chair, is an end item. Accounting statements classify inventory of end items as either work-in-process (WIP), if work remains to be done, or finished goods. An **intermediate item**, such as item B, C, or H, has at least one parent and at least one component. Some products have several levels of intermediate items; the parent of one intermediate item can also be an intermediate item. Inventory of intermediate items—whether completed or still on the shop floor—is classified as WIP. A **subassembly** is an intermediate item that is assembled (as opposed to being transformed by other means) from more than

FIGURE 16.4 ▶
BOM for a Ladder-Back Chair



one component. Items B and C are subassemblies. A **purchased item** has no components because it comes from a supplier, but it has one or more parents. Examples are items D, E, F, G, I, and J in Figure 16.4. Inventory of purchased items is treated as raw materials in accounting statements.

purchased item

An item that has one or more parents but no components because it comes from a supplier.

A component may have more than one parent. **Part commonality**, sometimes called *standardization of parts* or *modularity*, is the degree to which a component has more than one immediate parent. As a result of commonality, the same item may appear in several places in the bill of materials for a product, or it may appear in the bills of materials for several different products. For example, the seat subassembly in Figure 16.4 is a component of the ladder-back chair and of a kitchen chair that is part of the same family of products. The usage quantity specified in the bill of materials relates to a specific parent–component relationship. The usage quantity for any component can therefore change, depending on the parent item. Part commonality, or using the same part in many parents, increases its volume and repeatability, which provides several process advantages and helps minimize inventory costs.

part commonality

The degree to which a component has more than one immediate parent.

A careful dissection of the popular iPAD and iPAD 2 products from Apple shows that their bill of materials differ in the enclosure and battery, but are otherwise remarkably similar in their components and design. Same suppliers are used for many components, and costs are very comparable for new revisions of chips found in the previous iPad and iPhones. Standardization of designs and components across different products and generations allows Apple to be very competitive and profitable.

Master Production Scheduling

The second input into a material requirements plan is the **master production schedule (MPS)**, which details how many end items will be produced within specified periods of time. It breaks the sales and operations plan into specific product schedules. Figure 16.5 shows how a sales and operations plan for a family of chairs breaks down into the weekly MPS for each specific chair type (the time period can be hours, days, weeks, or months). The chair example demonstrates the following aspects of master scheduling:

master production schedule (MPS)

A part of the material requirements plan that details how many end items will be produced within specified periods of time.

1. The sums of the quantities in the MPS must equal those in the sales and operations plan. This consistency between the plans is desirable because of the economic analysis done to arrive at the sales and operations plan.
2. The production quantities must be allocated efficiently over time. The specific mix of chair types—the number of each type as a percent of the total family's quantity—is based on historic demand and on marketing and promotional considerations. The planner must select lot sizes for each chair type, taking into consideration economic factors such as production setup costs and inventory carrying costs.

iPad total product costs $270		iPad 2 total product costs $267
	LCD display – 9.7-inch multi-touch screen **$59**	LCD display – 9.7-inch multi-touch screen **$50**
	Camera – None **$0**	Camera – Front & Rear **$4.50**
	Memory – Samsung **$47**	Memory – Samsung/Toshiba **$30**
	Apple Processor – Apple A4 **$17**	Apple Processor – Apple A5 **$25**
	Radio Components Infineon/Broadcom **$26**	Radio Components Infineon/Broadcom or Qualcomm/Broadcom **$25**
	Sensors – STMicroelectronics **$1.00**	Sensors – STMicroelectronics **$2.50**
	Battery **$23**	Battery **$20 – $25**
		

Bill of Materials for Apple's iPad and iPad2.

D. Hurst/Alamy and PSL Images/Alamy

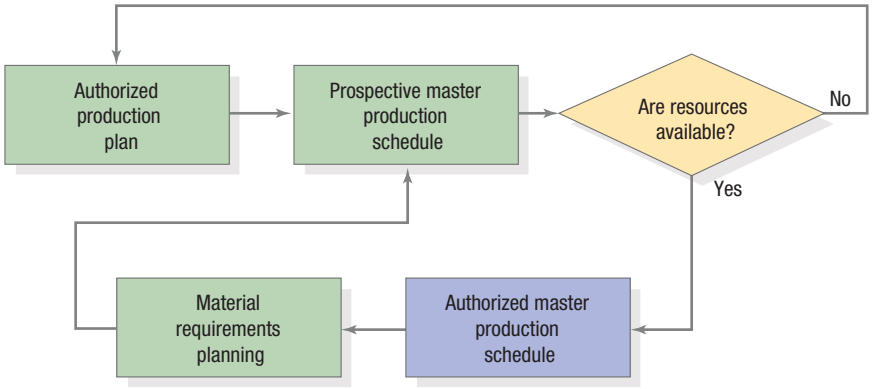
FIGURE 16.5 ▶
MPS for a Family of Chairs

	April				May			
	1	2	3	4	5	6	7	8
Ladder-back chair	150					150		
Kitchen chair				120			120	
Desk chair		200	200		200			200
Sales and operations plan for chair family	670				670			

3. Capacity limitations and bottlenecks, such as machine or labor capacity, storage space, or working capital, may determine the timing and size of MPS quantities. The planner must acknowledge these limitations by recognizing that some chair styles require more resources than others and setting the timing and size of the production quantities accordingly.

Figure 16.6 shows the master production scheduling process. Operations must first create a prospective MPS to test whether it meets the schedule with the resources (e.g., machine capacities, workforce, overtime, and subcontractors) provided for in the sales and operations plan. Operations then revises the MPS until a schedule that satisfies all of the resource limitations is developed or until it is determined that no feasible schedule can be developed. In the latter event, the production plan must be revised to adjust production requirements or increase authorized resources. Once a feasible prospective MPS has been accepted by the firm's managers, operations uses the authorized MPS as input to material requirements planning. Operations can then determine specific schedules for component production and assembly. Actual performance data such as inventory levels and shortages are inputs to preparing the prospective MPS for the next period, and so the master production scheduling process is repeated from one period to the next.

FIGURE 16.6 ▶
Master Production Scheduling Process



Developing a Master Production Schedule The process of developing a master production schedule includes (1) calculating the projected on-hand inventory and (2) determining the timing and size of the production quantities of specific products. We use the manufacturer of the ladder-back chair to illustrate the process. For simplicity, we assume that the firm does not utilize safety stocks for end items, even though many firms do. In addition, we use weeks as our planning periods, even though hours, days, or months could be used.

Step 1. *Calculate Projected On-Hand Inventories.* The first step is to calculate the projected on-hand inventory, which is an estimate of the amount of inventory available each week after demand has been satisfied:

$$\begin{pmatrix}\text{Projected on-hand}\\ \text{inventory at end}\\ \text{of this week}\end{pmatrix} = \begin{pmatrix}\text{On-hand}\\ \text{inventory at}\\ \text{end of last week}\end{pmatrix} + \begin{pmatrix}\text{MPS quantity}\\ \text{due at start}\\ \text{of this week}\end{pmatrix} - \begin{pmatrix}\text{Projected}\\ \text{requirements}\\ \text{this week}\end{pmatrix}$$

In some weeks, no MPS quantity for a product may be needed because sufficient inventory already exists. For the projected requirements for this week, the scheduler uses whichever is

larger—the forecast or the customer orders booked—recognizing that the forecast is subject to error. If actual booked orders exceed the forecast, the projection will be more accurate if the scheduler uses the booked orders because booked orders are a known quantity. Conversely, if the forecast exceeds booked orders for a week, the forecast will provide a better estimate of the requirements needed for that week because some orders are yet to come in.

The manufacturer of the ladder-back chair produces the chair to stock and needs to develop an MPS for it. Marketing has forecasted a demand of 30 chairs for the first week of April, but actual customer orders booked are for 38 chairs. The current on-hand inventory is 55 chairs. No MPS quantity is due in week 1. Figure 16.7 shows an MPS record with these quantities listed. Because actual orders for week 1 are greater than the forecast, the scheduler uses that figure for actual orders to calculate the projected inventory balance at the end of week 1:

Item: Ladder-back chair

Quantity on Hand: 55	April 1	April 2
Forecast	30	30
Customer orders (booked)	38	27
Projected on-hand inventory	17	–13
MPS quantity	0	0
MPS start		

Explanation: Forecast is less than booked orders in week 1; projected on-hand inventory balance = 55 + 0 – 38 = 17.

Explanation: Forecast exceeds booked orders in week 2; projected on-hand inventory balance = 17 + 0 – 30 = –13. The shortage signals a need to schedule an MPS quantity for completion in week 2.

◀ **FIGURE 16.7** Master Production Schedule for Weeks 1 and 2

$$\text{Inventory} = \begin{pmatrix}55 \text{ chairs}\\ \text{currently}\\ \text{in stock}\end{pmatrix} + \begin{pmatrix}\text{MPS quantity}\\ (0 \text{ for week 1}\end{pmatrix} - \begin{pmatrix}38 \text{ chairs already}\\ \text{promised for}\\ \text{delivery in week 1}\end{pmatrix} = 17 \text{ chairs}$$

In week 2, the forecasted quantity exceeds actual orders booked, so the projected on-hand inventory for the end of week 2 is 17 + 0 − 30 = −13. The shortage signals the need for more chairs to be produced and available for week 2.

Step 2. *Determine the Timing and Size of MPS Quantities.* The goal of determining the timing and size of MPS quantities is to maintain a nonnegative projected on-hand inventory balance. As shortages in inventory are detected, MPS quantities should be scheduled to cover them. The first MPS quantity should be scheduled for the week when the projected on-hand inventory reflects a shortage, such as week 2 in Figure 16.7.[1] The scheduler adds the MPS quantity to the projected on-hand inventory and searches for the next period when a shortage occurs. This shortage signals a need for a second MPS quantity, and so on.

Figure 16.8 shows a master production schedule for the ladder-back chair for the next 8 weeks. The order policy requires production lot sizes of 150 units. A shortage of 13 chairs in week 2 will occur unless the scheduler provides for an MPS quantity for that period. Our convention is to show blanks instead of zeroes in all rows, which improves

▼ **FIGURE 16.8** Master Production Schedule for Weeks 1–8

Item: Ladder-back chair **Order Policy:** 150 units **Lead Time:** 1 week

Quantity on Hand: 55	April 1	April 2	April 3	April 4	May 5	May 6	May 7	May 8
Forecast	30	30	30	30	35	35	35	35
Customer orders (booked)	38	27	24	8	0	0	0	0
Projected on-hand inventory	17	137	107	77	42	7	122	87
MPS quantity		150					150	
MPS start	150					150		

Explanation: The time needed to assemble 150 chairs is 1 week. The assembly department must start assembling chairs in week 1 to have them ready by week 2.

Explanation: On-hand inventory balance = 17 + 150 – 30 = 137. The MPS quantity is needed to avoid a shortage of 30 – 17 = 13 chairs in week 2.

[1] In some cases, new orders will be planned before a shortage is encountered. Two such instances occur when safety stocks and anticipation inventories are built up.

readability and is often used in practice. The only exception is in the projected on-hand inventory row, where a number is always shown, even if it is a 0 or negative number.

Once the MPS quantity is scheduled, the updated projected inventory balance for week 2 is

$$\text{Inventory} = \begin{pmatrix}\text{17 chairs in}\\ \text{inventory at the}\\ \text{end of week 1}\end{pmatrix} + \begin{pmatrix}\text{MPS quantity}\\ \text{of 150 chairs}\end{pmatrix} - \begin{pmatrix}\text{Forecast of}\\ \text{30 chairs}\end{pmatrix} = 137 \text{ chairs}$$

The scheduler proceeds column by column through the MPS record until it reaches the end, filling in the MPS quantities as needed to avoid shortages. The 137 units will satisfy forecasted demands until week 7, when the inventory shortage in the absence of an MPS quantity is $7 + 0 - 35 = -28$. This shortage signals the need for another MPS quantity of 150 units. The updated inventory balance is $7 + 150 - 35 = 122$ chairs for week 7.

The last row in Figure 16.8 indicates the periods in which production of the MPS quantities must *begin* so that they will be available when indicated in the MPS quantity row. In the upper-right portion of the MPS record, a lead time of 1 week is indicated for the ladder-back chair; that is, 1 week is needed to assemble 150 ladder-back chairs, assuming that items B, C, D, and E are available. For each MPS quantity, the scheduler works backward through the lead time to determine when the assembly department must start producing chairs. Consequently, a lot of 150 units must be started in week 1 and another in week 6.

available-to-promise (ATP) inventory

The quantity of end items that marketing can promise to deliver on specified dates.

Available-to-Promise Quantities In addition to providing manufacturing with the timing and size of production quantities, the MPS provides marketing with information useful for negotiating delivery dates with customers. The quantity of end items that marketing can promise to deliver on specified dates is called **available-to-promise (ATP) inventory**. It is the difference between the customer orders already booked and the quantity that operations is planning to produce. As new customer orders are accepted, the ATP inventory is reduced to reflect the commitment of the firm to ship those quantities, but the actual inventory stays unchanged until the order is removed from inventory and shipped to the customer. An available-to-promise inventory is associated with each MPS quantity because the MPS quantity specifies the timing and size of new stock that can be earmarked to meet future bookings.

▼ **FIGURE 16.9**
MPS Record with an ATP Row

Item: Ladder-back chair — **Order Policy:** 150 units — **Lead Time:** 1 week

	April				May			
Quantity on Hand: 55	1	2	3	4	5	6	7	8
Forecast	30	30	30	30	35	35	35	35
Customer orders (booked)	38	27	24	8	0	0	0	0
Projected on-hand inventory	17	137	107	77	42	7	122	87
MPS quantity		150					150	
MPS start	150					150		
Available-to-promise (ATP) inventory	17	91					150	

Explanation: The total of customer orders booked until the next MPS receipt is 38 units. The ATP = 55 (on-hand) + 0 (MPS quantity) – 38 = 17.

Explanation: The total of customer orders booked until the next MPS receipt is 27 + 24 + 8 = 59 units. The ATP = 150 (MPS quantity) – 59 = 91 units.

Figure 16.9 shows an MPS record with an additional row for the available-to-promise quantities. The ATP in week 2 is the MPS quantity minus booked customer orders until the next MPS quantity, or $150 - (27 + 24 + 8 + 0 + 0) = 91$ units. The ATP indicates to marketing that, of the 150 units scheduled for completion in week 2, 91 units are uncommitted, and total new orders up to that quantity can be promised for delivery as early as week 2. In week 7, the ATP is 150 units because there are no booked orders in week 7 and beyond.

The procedure for calculating available-to-promise information is slightly different for the first (current) week of the schedule than for other weeks because it accounts for the inventory currently in stock. The ATP inventory for the first week equals *current on-hand inventory* plus the MPS quantity for the first week, minus the cumulative total of booked orders up to (but not including) the week in which the next MPS quantity arrives. So, in Figure 16.9, the ATP for the first week is $55 + 0 - 38 = 17$. This information indicates to the sales department that it can promise as many as 17 units this week, 91 more units sometime in weeks 2

through 6, and 150 more units in week 7 or 8. If customer order requests exceed ATP quantities in those time periods, the MPS must be changed before the customer orders can be booked or the customers must be given a later delivery date—when the next MPS quantity arrives. See Solved Problem 2 at the end of this chapter for an example of decision making using the ATP quantities.

Inventory planners do not create master production plans manually, although they thoroughly understand the logic built into them. Figure 16.10 is typical of the computer support available. It was created with the *Master Production Scheduling* Solver in OM Explorer, and confirms the output shown in Figure 16.9.

Lot Size	150								
Lead Time	1								
Quantity on Hand	55	1	2	3	4	5	6	7	8
Forecast		30	30	30	30	35	35	35	35
Customer Orders (Booked)		38	27	24	8				
Projected On-Hand Inventory		17	137	107	77	42	7	122	87
MPS Quantity			150					150	
MPS Start		150					150		
Available-to-Promise Inv (ATP)		17	91					150	

FIGURE 16.10 ▲
Master Production Scheduling Solver Output Using *OM Explorer*

Freezing the MPS The master production schedule is the basis of all end item, subassembly, component, and materials schedules. For this reason, changes to the MPS can be costly, particularly if they are made to MPS quantities soon to be completed. Increases in an MPS quantity can result in material shortages, delayed shipments to customers, and excessive expediting costs. Decreases in MPS quantities can result in unused materials or components (at least until another need for them arises) and valuable capacity being used to create products not needed. Similar costs occur when forecasted need dates for MPS quantities are changed. For these reasons, many firms, particularly those with a make-to-stock strategy and a focus on low-cost operations, *freeze*, or disallow changes to, the near-term portion of the MPS.

Reconciling the MPS with Sales and Operations Plans Because the master production schedule is based on both forecasts as well as actual orders received, it can differ from the sales and operations plan when summed across different periods in a month. For instance, in Figure 16.5, if the sum total MPS quantities of the three models of chairs in the month of April was 725 instead of 670, then either the management must revise the sales and operations plan upwards by authorizing additional resources to match supply with demand, or reduce the quantities of MPS in the month of April to match the sales and operations plan. Master production schedules drive plant and supplier activity, so they must be synchronized with actual customer demands and the sales and operations plans to ensure that the firm's planning decisions are actually being implemented on an ongoing basis.

MyOMLab

Tutor 16.1 in MyOMLab provides a new example to practice completing a master production schedule.

inventory record

A record that shows an item's lot-size policy, lead time, and various time-phased data.

gross requirements

The total demand derived from *all* parent production plans.

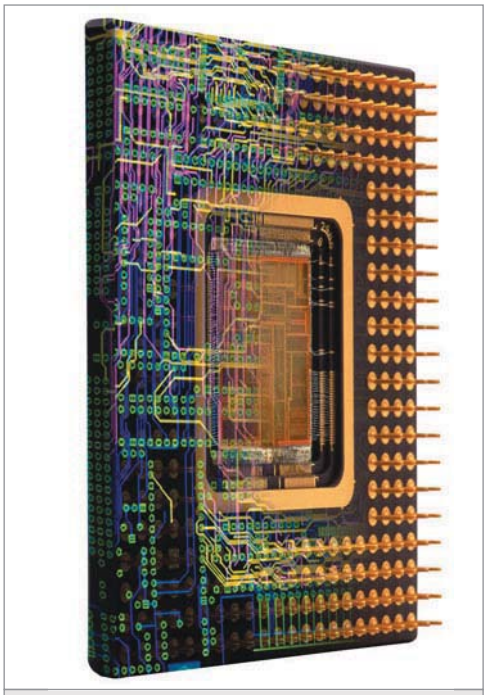
Inventory Record

Inventory records are a third major input to MRP, and inventory transactions are the basic building blocks of up-to-date records (see Figure 16.2). These transactions include releasing new orders, receiving scheduled receipts, adjusting due dates for scheduled receipts, withdrawing inventory, canceling orders, correcting inventory errors, rejecting shipments, and verifying scrap losses and stock returns. Recording the transactions accurately is essential if the firm's on-hand inventory balances are to be correct and its MRP system is to operate effectively.

The **inventory record** divides the future into time periods called *time buckets*. In our discussion, we use weekly time buckets for consistency with our MPS example, although other time periods could as easily be used. The inventory record shows an item's lot-size policy, lead time, and various time-phased data. The purpose of the inventory record is to keep track of inventory levels and component replenishment needs. The time-phased information contained in the inventory record consists of (1) *gross requirements*, (2) *scheduled receipts*, (3) *projected on-hand inventory*, (4) *planned receipts*, and (5) *planned order releases*.

We illustrate the discussion of inventory records with the seat subassembly, item C that was shown in Figure 16.4. Suppose that it is used in two products: a ladder-back chair and a kitchen chair.

Gross Requirements The **gross requirements** are the total demand derived from *all* parent production plans. They also include demand not otherwise accounted for, such as demand for replacement parts for units already sold. Figure 16.11 shows an inventory record for item C, the seat subassembly. Item C is produced in lots of 230 units and has a lead time of 2 weeks. The inventory



WidStock/Alamy

Cypress Semiconductor, a California-based company that manufactures logic devices, USB controllers, general-purpose programmable clocks, memories, and wireless connectivity solutions for consumer and automotive markets, uses commercial software solutions to manage the complexity of its master production scheduling processes.

Item: C
Description: Seat subassembly
Lot Size: 230 units
Lead Time: 2 weeks

		Week							
		1	2	3	4	5	6	7	8
Gross requirements		150	0	0	120	0	150	120	0
Scheduled receipts		230	0	0	0	0	0	0	0
Projected on-hand inventory	37	117	117	117	–3	–3	–153	–273	–273
Planned receipts									
Planned order releases									

Explanation:
Gross requirements are the total demand for the two chairs. Projected on-hand inventory in week 1 is 37 + 230 – 150 = 117 units.

FIGURE 16.11 ▲
MRP Record for the Seat Subassembly

record also shows item C's gross requirements for the next 8 weeks, which come from the master production schedule for the ladder-back and kitchen chairs (see Figure 16.5). The MPS start quantities for each parent are added to arrive at each week's gross requirements. The seat subassembly's gross requirements exhibit lumpy demand: Operations will withdraw seat subassemblies from inventory in only 4 of the 8 weeks.

The MRP system works with release dates to schedule production and delivery for components and subassemblies. Its program logic anticipates the removal of all materials required by a parent's production order from inventory at the beginning of the parent item's lead time—when the scheduler first releases the order to the shop.

Scheduled Receipts Recall that *scheduled receipts* (sometimes called open orders) are orders that have been placed but not yet completed. For a purchased item, the scheduled receipt could be in one of several stages: being processed by a supplier, being transported to the purchaser, or being inspected by the purchaser's receiving department. If the firm is making the item in-house, the order could be on the shop floor being processed, waiting for components, waiting for a machine to become available, or waiting to be moved to its next operation. According to Figure 16.11, one 230-unit order of item C is due in week 1. Given the 2-week lead time, the inventory planner probably released the order 2 weeks ago. Scheduled receipts due in beyond the item's lead time are unusual, caused by events such as a last-minute change in the MPS.

projected on-hand inventory
An estimate of the amount of inventory available each week after gross requirements have been satisfied.

Projected On-Hand Inventory The **projected on-hand inventory** is an estimate of the amount of inventory available each week after gross requirements have been satisfied. The beginning inventory, shown as the first entry (37) in Figure 16.11, indicates the on-hand inventory available at the time the record was computed. As with scheduled receipts, entries are made for each actual withdrawal and receipt to update the MRP database. Then, when the MRP system produces the revised record, the correct inventory will appear.

Other entries in the row show inventory expected in future weeks. Projected on-hand inventory is calculated as

$$\begin{pmatrix}\text{Projected on-hand}\\ \text{inventory balance}\\ \text{at end of week } t\end{pmatrix} = \begin{pmatrix}\text{Inventory on}\\ \text{hand at end of}\\ \text{week } t-1\end{pmatrix} + \begin{pmatrix}\text{Scheduled}\\ \text{or planned}\\ \text{reeipts in}\\ \text{week } t\end{pmatrix} - \begin{pmatrix}\text{Gross}\\ \text{requirements}\\ \text{in week } t\end{pmatrix}$$

planned receipts
Orders that are not yet released to the shop or the supplier.

The projected on-hand calculation includes the consideration of **planned receipts**, which are orders not yet released to the shop or the supplier. Planned receipts should not be confused with scheduled receipts. Planned receipts are still at the planning stage and can still change from one week to the next, whereas scheduled receipts are actual orders that are being acted upon by the shop or supplier. In Figure 16.11, the planned receipts are all zero. The on-hand inventory calculations for each week are as follows:

Week 1: $37 + 230 - 150 = 117$
Weeks 2 and 3: $117 + 0 - 0 = 117$
Week 4: $117 + 0 - 120 = -3$
Week 5: $-3 + 0 - 0 = -3$
Week 6: $-3 + 0 - 150 = -153$
Week 7: $-153 + 0 - 120 = -273$
Week 8: $-273 + 0 - 0 = -273$

In week 4, the balance drops to −3 units, which indicates that a shortage of 3 units will occur unless more seat subassemblies are built. This condition signals the need for a planned receipt to arrive in week 4. In addition, unless more stock is received, the shortage will grow to 273 units in weeks 7 and 8.[2]

Item: C
Description: Seat subassembly
Lot Size: 230 units
Lead Time: 2 weeks

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	150	0	0	120	0	150	120	0
Scheduled receipts	230	0	0	0	0	0	0	0
Projected on-hand inventory (37)	117	117	117	227	227	77	187	187
Planned receipts				230			230	
Planned order releases		230			230			

Explanation:
Without a planned receipt in week 4, a shortage of 3 units will occur: 117 + 0 + 0 – 120 = –3 units. Adding the planned receipt brings the balance to 117 + 0 + 230 – 120 = 227 units. Offsetting for a 2-week lead time puts the corresponding planned order release back to week 2.

Explanation:
The first planned receipt lasts until week 7, when projected inventory would drop to 77 + 0 + 0 – 120 = –43 units. Adding the second planned receipt brings the balance to 77 + 0 + 230 – 120 = 187 units.
The corresponding planned order release is for week 5 (or week 7 minus 2 weeks).

▲ **FIGURE 16.12**
Completed Inventory Record for the Seat Subassembly

Planned Receipts Planning for the receipt of new orders will keep the projected on-hand balance from dropping below zero. The planned receipt row is developed as follows:

1. Weekly on-hand inventory is projected until a shortage appears. Completion of the initial planned receipt is scheduled for the week in which the shortage is projected. The addition of the newly planned receipt should increase the projected on-hand balance so that it equals or exceeds zero. It will exceed zero when the lot size exceeds requirements in the week it is planned to arrive.
2. The projection of on-hand inventory continues until the next shortage occurs. This shortage signals the need for the second planned receipt.

This process is repeated until the end of the planning horizon by proceeding column by column through the MRP record—filling in planned receipts as needed and completing the projected on-hand inventory row. Figure 16.12 shows the planned receipts for the seat subassembly. In week 4, the projected on-hand inventory will drop below zero, so a planned receipt of 230 units is scheduled for week 4. The updated inventory on-hand balance is 117 (inventory at end of week 3) + 230 (planned receipts) − 120 (gross requirements) = 227 units. The projected on-hand inventory remains at 227 for week 5 because no scheduled receipts or gross requirements are anticipated. In week 6, the projected on-hand inventory is 227 (inventory at end of week 5) − 150 (gross requirements) = 77 units. This quantity is greater than zero, so no new planned receipt is needed. In week 7, however, a shortage will occur unless more seat subassemblies are received. With a planned receipt in week 7, the updated inventory balance is 77 (inventory at end of week 6) + 230 (planned receipts) − 120 (gross requirements) = 187 units.

Planned Order Releases A **planned order release** indicates when an order for a specified quantity of an item is to be issued. We must place the planned order release quantity in the proper time bucket. To do so, we must assume that all inventory flows—scheduled receipts, planned receipts, and gross requirements—occur at the same point of time in a time period. Some firms assume that all flows occur at the beginning of a time period; other firms assume that they occur at the end of a time period or at the middle of the time period. Regardless of when the flows are assumed to occur, we find the release date by subtracting the lead time from the receipt date. For example, the release date for the first planned order release in Figure 16.12 is 4 (planned receipt date) −2 (lead time) = 2 (planned order release date). Figure 16.12 shows the planned order releases for the seat subassembly. If all goes according to the plan, we will release an order for 230 seat assemblies next week (in week 2). This order release sets off a series of updates to the inventory record. First, the planned order release for the order is removed. Next, the planned receipt for 230 units in week 4 is also removed. Finally, a new scheduled receipt for 230 units will appear in the scheduled receipt row for week 4.

planned order release
An indication of when an order for a specified quantity of an item is to be issued.

[2]There is an exception to the rule of scheduling a planned receipt whenever the projected inventory otherwise becomes negative. When a scheduled receipt is coming in *after* the inventory becomes negative, the first recourse is to expedite the scheduled receipt (giving it an earlier due date), rather than scheduling a new planned receipt.

Planning Factors

The planning factors in a MRP inventory record play an important role in the overall performance of the MRP system. By manipulating these factors, managers can fine-tune inventory operations. In this section, we discuss planning lead time, lot-sizing rules, and safety stock.

Planning Lead Time Planning lead time is an estimate of the time between placing an order for an item and receiving the item in inventory. Accuracy is important in planning lead time. If an item arrives in inventory sooner than needed, inventory holding costs increase. If an item arrives too late, stockouts, excessive expediting costs, or both may occur.

For purchased items, the planning lead time is the time allowed for receiving a shipment from the supplier after the order has been sent, including the normal time to place the order. Often, the purchasing contract stipulates the delivery date. For items manufactured in-house, a rough-cut estimate of the planning lead time can be obtained by keeping track of the actual lead times for recent orders and computing an average. A more extensive estimating process consists of breaking down each of the following factors:

- Setup time
- Processing time
- Materials handling time between operations
- Waiting time

Each of these times must be estimated for every operation along the item's route. Estimating setup, processing, and materials handling times can be relatively easy, but estimating the waiting time for materials handling equipment or for a workstation to perform a particular operation can be more difficult. In a facility that uses a make-to-order strategy, such as a machine shop, the load on the shop varies considerably over time, causing actual waiting times for a particular order to fluctuate widely. Therefore, being able to accurately estimate the waiting time is especially important when it comes to estimating the planning lead time. However, in a facility that uses a make-to-stock strategy, such as an assembly plant, product routings are more standard and waiting time is more predictable; hence, waiting time generally is a less-troublesome part of planning lead times.

Lot-Sizing Rules A lot-sizing rule determines the timing and size of order quantities. A lot-sizing rule must be assigned to each item before planned receipts and planned order releases can be computed. The choice of lot-sizing rules is important because they determine the number of setups required and the inventory holding costs for each item. We present three lot-sizing rules: (1) fixed order quantity, (2) periodic order quantity, and (3) lot-for-lot.

fixed order quantity (FOQ)

A rule that maintains the same order quantity each time an order is issued.

Fixed Order Quantity The **fixed order quantity (FOQ)** rule maintains the same order quantity each time an order is issued.[3] For example, the lot size might be the size dictated by equipment capacity limits, such as when a full lot must be loaded into a furnace at one time. For purchased items, the FOQ could be determined by the quantity discount level, truckload capacity, or minimum purchase quantity. Alternatively, the lot size could be determined by the economic order quantity (EOQ) formula (see Chapter 9, "Managing Inventories"). Figure 16.12 illustrated the FOQ rule. However, if an item's gross requirement within a week is particularly large, the FOQ might be insufficient to avoid a shortage. In such unusual cases, the inventory planner must increase the lot size beyond the FOQ, typically to a size large enough to avoid a shortage. Another option is to make the order quantity an integer multiple of the FOQ. This option is appropriate when capacity constraints limit production to FOQ sizes (at most).

periodic order quantity (POQ)

A rule that allows a different order quantity for each order issued but issues the order for predetermined time intervals.

Periodic Order Quantity The **periodic order quantity (POQ)** rule allows a different order quantity for each order issued but issues the order for predetermined time intervals, such as every two weeks. The order quantity equals the amount of the item needed during the predetermined time between orders and must be large enough to prevent shortages. Specifically, the POQ is

$$\begin{pmatrix}\text{POQ lot size}\\ \text{to arrive in}\\ \text{week } t\end{pmatrix} = \begin{pmatrix}\text{Total gross requirements}\\ \text{for } P \text{ weeks, including}\\ \text{week } t\end{pmatrix} - \begin{pmatrix}\text{Projected on-hand}\\ \text{inventory balance at}\\ \text{end of week } t-1\end{pmatrix}$$

This amount exactly covers P weeks' worth of gross requirements. That is, the projected on-hand inventory should equal zero at the end of the Pth week.

[3] The *kanban* system essentially uses a FOQ rule, except that the order quantity is very small.

Suppose that we want to switch from the FOQ rule used in Figure 16.12 to the POQ rule. Figure 16.13 was created with the *Single-Item MRP* Solver in OM Explorer. It shows the application of the POQ rule, with $P = 3$ weeks, to the seat subassembly inventory. The first order is required in week 4 because it is the first week that projected inventory balance will fall below zero. The first order using $P = 3$ weeks is

$$\text{(POQ lot size)} = \begin{pmatrix}\text{Gross requirements}\\ \text{for weeks}\\ 4, 5, \text{and } 6\end{pmatrix} - \begin{pmatrix}\text{Inventory at}\\ \text{end of week 3}\end{pmatrix}$$

$$= (120 + 0 + 150) - 117 = 153 \text{ units}$$

The second order must arrive in week 7 with a lot size of $(120 + 0) - 0 = 120$ units. This second order reflects only two weeks' worth of gross requirements—to the end of the planning horizon.

MyOMLab

Tutor 16.2 in MyOMLab provides a new example to practice lot-sizing decisions using FOQ, POQ, and L4L rules.

Periods 8

Item Description: Seat Assembly | Period (P) for POQ: 3 | Lot Size (FOQ): | Lead Time: 2

POQ Rule

		1	2	3	4	5	6	7	8
Gross requirements		150			120		150	120	
Scheduled receipts		230							
Projected on-hand inventory	37	117	117	117	150	150			
Planned receipts					153			120	
Planned order releases			153			120			

◀ **FIGURE 16.13**
The POQ ($P = 3$) Rule for the Seat Subassembly

The POQ rule does not mean that the planner must issue a new order every P weeks. Rather, when an order is planned, its lot size must be enough to cover P successive weeks. One way to select a P value is to divide the average lot size desired, such as the EOQ or some other applicable lot size, by the average weekly demand. That is, express the target lot size as the desired weeks of supply (P) and round to the nearest integer.

Lot for Lot A special case of the POQ rule is the **lot-for-lot (L4L) rule**, under which the lot size ordered covers the gross requirements of a single week. Thus, $P = 1$, and the goal is to minimize inventory levels. This rule ensures that the planned order is just large enough to prevent a shortage in the single week it covers. The L4L lot size is

lot-for-lot (L4L) rule
A rule under which the lot size ordered covers the gross requirements of a single week.

$$\begin{pmatrix}\text{L4L lot size}\\ \text{to arrive in}\\ \text{week } t\end{pmatrix} = \begin{pmatrix}\text{Gross requirements}\\ \text{for week } t\end{pmatrix} - \begin{pmatrix}\text{Projected on-hand}\\ \text{inventory balance at}\\ \text{end of week } t-1\end{pmatrix}$$

The projected on-hand inventory combined with the new order will equal zero at the end of week t. Following the first planned order, an additional planned order will be used to match each subsequent gross requirement.

This time we want to switch from the FOQ rule to the L4L rule. Figure 16.14 shows the application of the L4L rule to the seat subassembly inventory. As before, the first order is needed in week 4:

$$\text{(L4L lot size)} = \begin{pmatrix}\text{Gross requirements}\\ \text{in week 4}\end{pmatrix} - \begin{pmatrix}\text{Inventory balance}\\ \text{at end of week 3}\end{pmatrix}$$

$$= 120 - 117 = 3$$

Periods 8

Item Description: Seat Assembly | Period (P) for POQ: | Lot Size (FOQ): | Lead Time: 2

L4L Rule

		1	2	3	4	5	6	7	8
Gross requirements		150			120		150	120	
Scheduled receipts		230							
Projected on-hand inventory	37	117	117	117					
Planned receipts					3		150	120	
Planned order releases			3		150	120			

◀ **FIGURE 16.14**
The L4L Rule for the Seat Subassembly

The stockroom must receive additional orders in weeks 6 and 7 to satisfy each of the subsequent gross requirements. The planned receipt for week 6 is 150 and for week 7 is 120.

Comparing Lot-Sizing Rules Choosing a lot-sizing rule can have important implications for inventory management. Lot-sizing rules affect inventory costs and setup and ordering costs. The FOQ, POQ, and L4L rules differ from one another in one or both respects. In our example, each rule took effect in week 4, when the first order was placed. Let us compare the projected on-hand inventory averaged over weeks 4 through 8 of the planning horizon. The data are shown in Figures 16.12, 16.13, and 16.14, respectively.

$$\text{FOQ: } \frac{227 + 227 + 77 + 187 + 187}{5} = 181 \text{ units}$$

$$\text{POQ: } \frac{150 + 150 + 0 + 0 + 0}{5} = 60 \text{ units}$$

$$\text{L4L: } \frac{0 + 0 + 0 + 0 + 0}{5} = 0 \text{ units}$$

The performance of the L4L rule with respect to average inventory levels comes at the expense of an additional planned order and its accompanying setup time and cost. We can draw three conclusions from this comparison:

1. The FOQ rule generates a high level of average inventory because it creates inventory *remnants*. A remnant is inventory carried into a week, but it is too small to prevent a shortage. Remnants occur because the FOQ does not match requirements exactly. For example, according to Figure 16.12, the stockroom must receive a planned order in week 7, even though 77 units are on hand at the beginning of that week. The remnant is the 77 units that the stockroom will carry for 3 weeks, beginning with receipt of the first planned order in week 4. Although they increase average inventory levels, inventory remnants introduce stability into the production process by buffering unexpected scrap losses, capacity bottlenecks, inaccurate inventory records, or unstable gross requirements.
2. The POQ rule reduces the amount of average on-hand inventory because it does a better job of matching order quantity to requirements. It adjusts lot sizes as requirements increase or decrease. Figure 16.13 shows that in week 7, when the POQ rule has fully taken effect, the projected on-hand inventory is zero—no remnants.
3. The L4L rule minimizes inventory investment, but it also maximizes the number of orders placed. This rule is most applicable to expensive items or items with small ordering or setup costs. It is the only rule that can be used for a low-volume item made to order. It can also approximate the small-lot inventory levels of a lean system.

By avoiding remnants, both the POQ and the L4L rule may introduce instability by tying the lot-sizing decision so closely to requirements. If any requirement changes, so must the lot size, which can disrupt component schedules. Last-minute increases in parent orders may be hindered by missing components.

Safety Stock An important managerial decision is the quantity of safety stock to carry. It is more complex for dependent demand items than for independent demand items. Safety stock for dependent demand items with lumpy demand (gross requirements) is helpful only when future gross requirements, the timing or size of scheduled receipts, and the amount of scrap that will be produced are uncertain. As these uncertainties are resolved, safety stock should be reduced and ultimately eliminated. The usual policy is to use safety stock for end items and purchased items to protect against fluctuating customer orders and unreliable suppliers of components but to avoid using it as much as possible for intermediate items. Safety stocks can be incorporated in the MRP logic by using the following rule: Schedule a planned receipt whenever the projected on-hand inventory balance drops below the desired safety stock level (rather than zero, as before). The objective is to keep a minimum level of planned inventories equal to the safety stock quantity. Figure 16.15 shows what happens when the safety stock requirement has just been increased from 0 units to 80 units of safety stock for the seat assembly using an FOQ of 230 units. The beginning projected on-hand quantity is still 37 units when the safety stock policy is introduced, and cannot fall below 80 units in any future period thereafter. Compare the

▼ **FIGURE 16.15**
Inventory Record for the Seat Subassembly Showing the Application of a Safety Stock

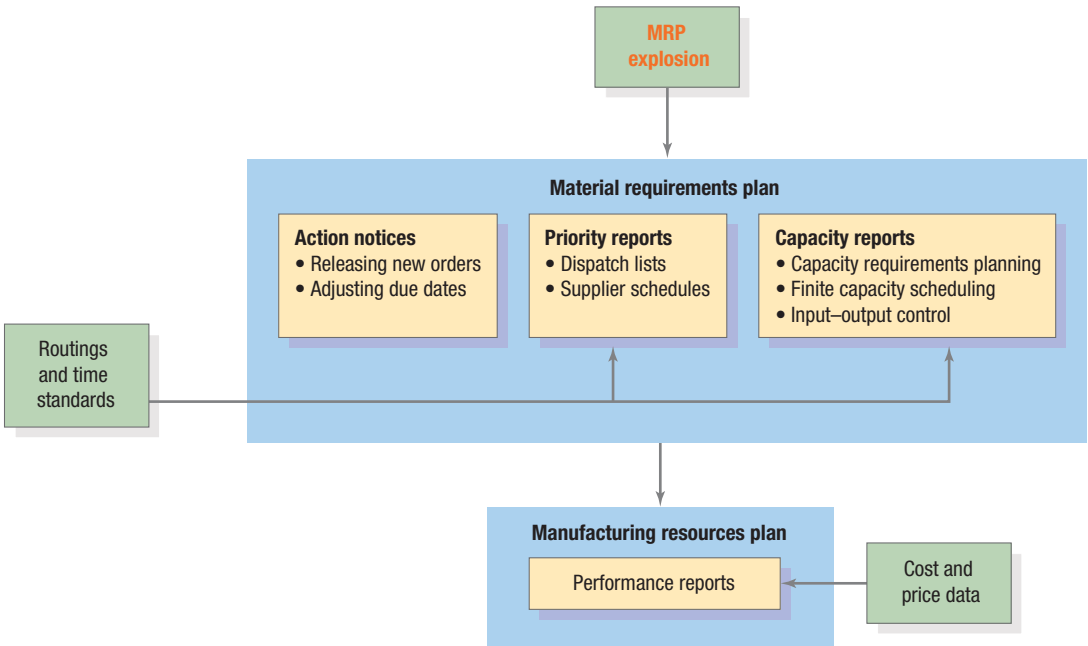
FOQ Rule — **Lot Size:** 230 units; **Lead Time:** 2 weeks; **Safety Stock:** 80 units

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	150	0	0	120	0	150	120	0
Scheduled receipts	230	0	0	0	0	0	0	0
Projected on-hand inventory 37	117	117	117	227	227	307	187	187
Planned receipts	0	0	0	230	0	230	0	0
Planned order releases	0	230	0	230	0	0	0	0

results in Figure 16.15 to Figure 16.12. The net effect is to move the second planned order release from week 5 to week 4 to avoid dropping below 80 units in week 6.

Outputs from MRP

MRP systems provide many reports, schedules, and notices to help planners control dependent demand inventories, as indicated in Figure 16.16. In this section, we discuss the MRP explosion process, notices that alert planners to items needing attention, resource requirement reports, and performance reports.



◀ **FIGURE 16.16**
MRP Outputs

MRP Explosion MRP translates, or *explodes*, the MPS and other sources of demand into the requirements needed for all of the subassemblies, components, and raw materials the firm needs to produce parent items. This process generates the material requirements plan for each component item.

An item's gross requirements are derived from three sources:

1. The MPS for immediate parents that are end items
2. The planned order releases (*not* the gross requirements, scheduled receipts, or planned receipts) for immediate parents below the MPS level
3. Any other requirements not originating in the MPS, such as the demand for replacement parts

Consider the seat subassembly and its inventory record shown in Figure 16.12. The seat subassembly requires a seat cushion and a seat frame, which in turn needs four seat-frame boards. Its BOM is shown in Figure 16.17 (see also Figure 16.4, which shows how the seat subassembly BOM relates to the product as a whole). How many seat cushions should we order from the supplier? How many seat frames should we produce to support the seat subassembly schedule? How many seat-frame boards do we need to make? The answers to these questions depend on the existing inventories of these items and the replenishment orders already in progress. MRP can help answer these questions through the explosion process.

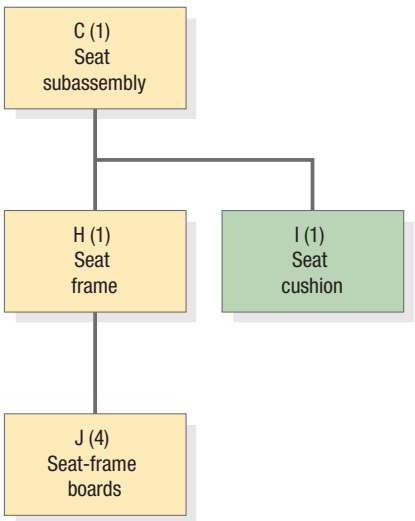
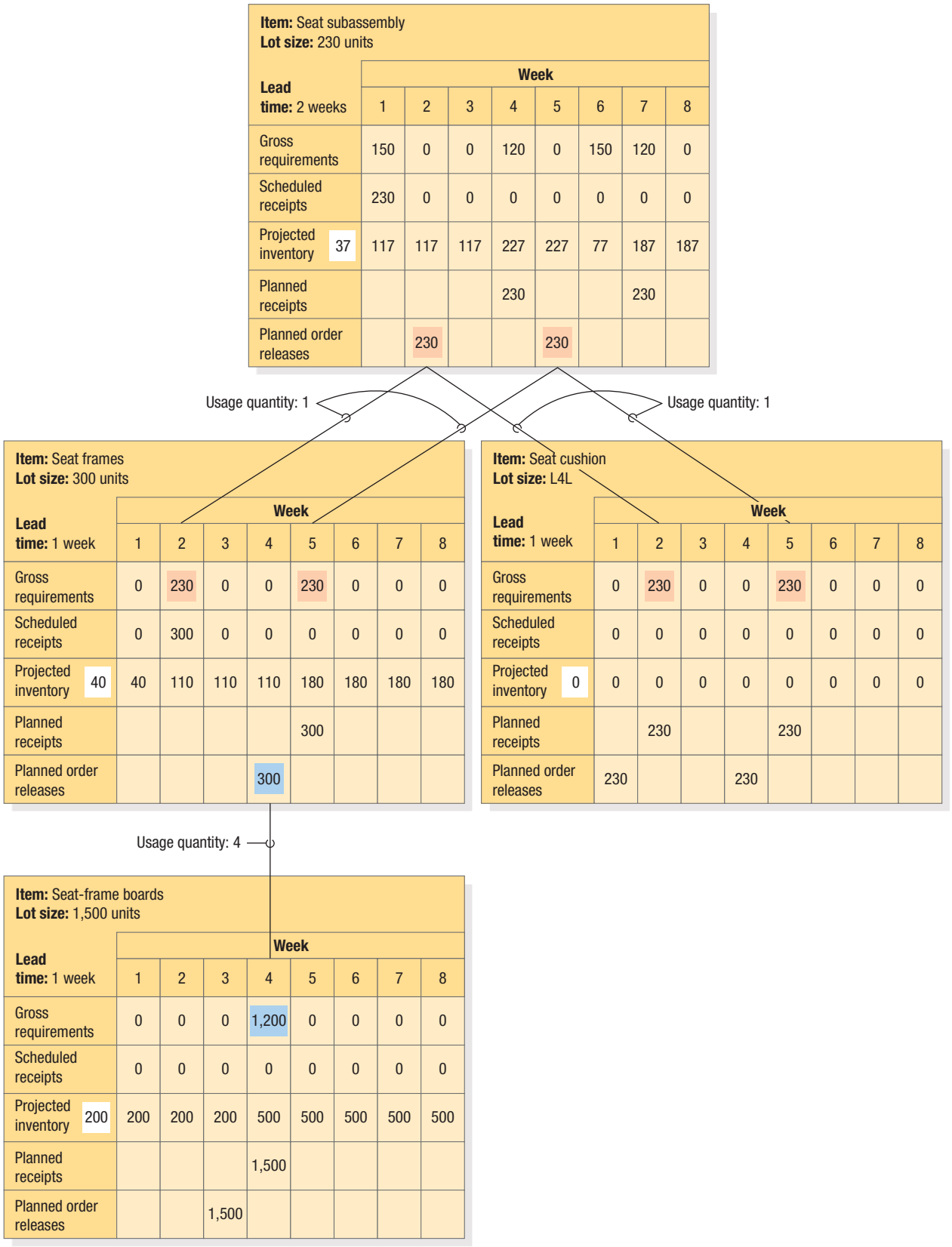


FIGURE 16.17 ▲
BOM for the Seat Subassembly

Figure 16.18 shows the MRP records for the seat subassembly and its components. We already showed how to develop the MRP record for the seat subassembly. We now concentrate on the MRP records of its components. The lot-size rules are an FOQ of 300 units for the seat frame, L4L for the seat cushion, and an FOQ of 1,500 for the seat-frame boards. All three components have a 1-week lead time. The key to the explosion process is to determine the proper timing and size of the gross requirements for each component. After we make those determinations, we can derive the planned order release schedule for each component by using the logic already demonstrated.



Item: Seat subassembly
Lot size: 230 units
Lead time: 2 weeks

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	150	0	0	120	0	150	120	0
Scheduled receipts	230	0	0	0	0	0	0	0
Projected inventory 37	117	117	117	227	227	77	187	187
Planned receipts				230			230	
Planned order releases		230			230			

Usage quantity: 1

Usage quantity: 1

Item: Seat frames
Lot size: 300 units
Lead time: 1 week

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	0	230	0	0	230	0	0	0
Scheduled receipts	0	300	0	0	0	0	0	0
Projected inventory 40	40	110	110	110	180	180	180	180
Planned receipts					300			
Planned order releases				300				

Item: Seat cushion
Lot size: L4L
Lead time: 1 week

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	0	230	0	0	230	0	0	0
Scheduled receipts	0	0	0	0	0	0	0	0
Projected inventory 0	0	0	0	0	0	0	0	0
Planned receipts		230			230			
Planned order releases	230			230				

Usage quantity: 4

Item: Seat-frame boards
Lot size: 1,500 units
Lead time: 1 week

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	0	0	0	1,200	0	0	0	0
Scheduled receipts	0	0	0	0	0	0	0	0
Projected inventory 200	200	200	200	500	500	500	500	500
Planned receipts				1,500				
Planned order releases			1,500					

FIGURE 16.18 ▲
MRP Explosion of Seat Assembly Components

In our example, the components have no independent demand for replacement parts. Consequently, in Figure 16.18 the gross requirements of a component come from the planned order releases of its parents. The seat frame and the seat cushion get their gross requirements from the planned order release schedule of the seat subassembly. Both components have gross requirements of 230 units in weeks 2 and 5, the same weeks in which we will be releasing orders to make more seat subassemblies. In week 2, for example, the materials handler for the assembly department will withdraw 230 seat frames and 230 seat cushions from inventory so that the assembly department can produce the seat subassemblies in time to avoid a stockout in week 4. The materials plans for the seat frame and the seat cushion must allow for that.

Using the gross requirements in weeks 2 and 5, we can develop the MRP records for the seat frame and the seat cushion, as shown in Figure 16.18. For a scheduled receipt of 300 seat frames in week 2, an on-hand quantity of 40 units, and a lead time of 1 week, we need to release an order of 300 seat frames in week 4 to cover the assembly schedule for the seat subassembly. The seat cushion has no scheduled receipts and no inventory on hand; consequently, we must place orders for 230 units in weeks 1 and 4, using the L4L logic with a lead time of 1 week.

After determining the replenishment schedule for the seat frame, we can calculate the gross requirements for the seat-frame boards. We plan to begin producing 300 seat frames in week 4. Each frame requires 4 boards, so we need to have $300(4) = 1{,}200$ boards available in week 4. Consequently, the gross requirement for seat-frame boards is 1,200 in week 4. Given no scheduled receipts, 200 boards in stock, a lead time of 1 week, and an FOQ of 1,500 units, we need a planned order release of 1,500 in week 3.

The questions posed earlier can now be answered. We should plan to release the following orders: 300 seat frames in week 4; 230 seat cushions in each of weeks 1 and 4; and 1,500 seat-frame boards in week 3. If MRP plans are updated weekly, only the planned order for week 1 should be released now. Releasing it creates a scheduled receipt of 230 seat cushions that will appear in the updated inventory record. The other orders remain in the planning stage, and even might be revised by the MRP explosion done next week.

Computer Support In practice a company can have thousands of dependent demand items with an average of six bills of materials levels. Time horizons often stretch out for 30 or more time periods into the future. Doing a MRP explosion by hand (as shown in Figure 16.18) would be impractical. What is needed is massive data processing, the very thing that computers do best, leaving the decision making to the inventory analyst. The *Material Requirements Planning* Solver of OM Explorer in MyOMLab represents a small example of what is done on a much larger scale by commercial packages, can compute requirements for up to two end items. It has the ability to develop inventory records up to 18 items deep with little effort, and can easily re-compute these requirements if there is any change in planning parameters.

MyOMLab

Figure 16.19 shows the output from the *Material Requirements Planning* Solver of OM Explorer. It confirms the same results as Figure 16.18. Based on this output, only one thing would be brought to the attention of the inventory planner—the planned order release of 230 units for seat cushions. Its planned order release is now "mature" for release this week. Unless the planner knows of a problem, the planner would place the order with the supplier. At the same time, the planner would input a transaction that automatically eliminates the planned order release in period 1, removes the planned receipt in period 2, and inserts a scheduled receipt for 230 units in period 2. The planner need not look at the records for seat frames or seat-frame boards, because no action is needed for them.

Other Important Reports Once computed, inventory records for any item appearing in the BOM can be printed in hard copy or displayed on a computer video screen. Inventory planners use a computer-generated memo called an **action notice** to make decisions about releasing new orders and adjusting the due dates of scheduled receipts. These notices are generated every time the system is updated, typically once per week. The action notice alerts planners to only the items that need their attention, such as those items that have a planned order release in the current period or a scheduled receipt that needs its due date revised. Planners can then view the full records for those items and take the necessary actions. An action notice can simply be a list of part numbers for items that need attention; or it can be the full record for such items, with a note at the bottom identifying the action needed.

action notice

A computer-generated memo alerting planners about releasing new orders and adjusting the due dates of scheduled receipts.

By itself, the MRP system does not recognize capacity limitations when computing planned orders; that is, it may call for a planned order release that exceeds the amount that can be physically produced. An essential role of planners is to monitor the capacity requirements of material requirements plans, adjusting a plan when it cannot be met. Particular attention is paid to bottlenecks. The planner can apply theory of constraints (TOC) principles (see Chapter 7, "Managing Process Constraints") to keep bottleneck operations fed by adjusting some lot sizing rules or occasionally overriding planned order releases. To facilitate this process, various types of capacity reports can be provided. For example, **capacity requirements planning (CRP)** reports

capacity requirements planning (CRP)

A technique used for projecting time-phased capacity requirements for work stations; its purpose is to match the materials requirements plan with the capacity of key processes.

FIGURE 16.19 ▶
Material Requirements Planning Solver Output of *OM Explorer* for Seat Assembly Components

Master Production Schedule

		1	2	3	4	5	6	7	8	9	10
Item A MPS Start	Descr: Seat		230			230					
Item B MPS Start	Descr:										

☑ Use second finished item

Material Requirements Planning

Item C — Descr: Seat Frames — Period (P) for POQ — Lot Size (FOQ): 300; Lead Time: 1; Safety Stock
FOQ Rule
Usage Quantity for Item: A 1 B

		1	2	3	4	5	6	7	8	9	10
Gross Requirements			230			230					
Scheduled Receipts			300								
Projected On-Hand Inventory	40	40	110	110	110	180	180	180	180	180	180
Planned Receipts						300					
Planned Order Releases					300						

Item D — Descr: Seat Cushion — Period (P) for POQ: 1 — Lot Size (FOQ); Lead Time: 1; Safety Stock
L4L Rule
Usage Quantity for Item: A 1 B C

		1	2	3	4	5	6	7	8	9	10
Gross Requirements			230			230					
Scheduled Receipts											
Projected On-Hand Inventory	0	0	0	0	0	0	0	0	0	0	0
Planned Receipts			230			230					
Planned Order Releases		230			230						

Item E — Descr: Seat-frame boards — Period (P) for POQ — Lot Size (FOQ): 1500; Lead Time: 1; Safety Stock
FOQ Rule
Usage Quantity for Item: A B C 4 D

		1	2	3	4	5	6	7	8	9	10
Gross Requirements					1200						
Scheduled Receipts											
Projected On-Hand Inventory	200	200	200	200	500	500	500	500	500	500	500
Planned Receipts					1500						
Planned Order Releases				1500							

project time-phased capacity requirements for workstations. They calculate workload according to the work required to complete the scheduled receipts already in the shop and to complete the planned order releases not yet released. Bottlenecks are those workstations at which the projected loads exceed station capacities.

Other types of outputs are also possible, such as priority reports on orders already placed to the shop or with suppliers. Priority reports begin with the due dates assigned to scheduled receipts, which planners keep up to date so that they continue to reflect when receipt is really needed. On a broader scale, the information in an MRP system is useful to functional areas other than operations. MRP evolved into **manufacturing resource planning (MRP II)**, a system that ties the basic MRP system to the company's financial system and to other core and supporting processes. For example, management can project the dollar value of shipments, product costs, overhead allocations, inventories, backlogs, and profits by using the MRP plan along with prices and product and activity costs from the accounting system. Also, information from the MPS, scheduled receipts, and planned orders can be converted into cash flow projections, which are broken down by product families. Similar computations are possible for other performance measures of interest to management. In fact, MRP II ultimately evolved into enterprise resource planning (ERP), which was introduced at the beginning of this chapter. Some firms may, however, forego the cost of vendor-delivered MRP and ERP systems because of the huge budgets and company resources involved in their deployment, and instead create their own MRP system implementations in-house. Managerial Practice 16.1 illustrates how Winnebago created and adapted its own homegrown software for the MRP system to achieve business self sufficiency.

manufacturing resource planning (MRP II)
A system that ties the basic MRP system to the company's financial system and to other core and supporting processes.

MANAGERIAL PRACTICE 16.1 Material Requirements Planning at Winnebago Industries

Winnebago Industries, based in Forest City, Iowa, is a leading manufacturer of motor homes and related products and services in the United States since 1958. In a challenging business environment, the firm improves the quality of its products at lowered production costs by emphasizing employee teamwork and involvement. So it is no surprise that Winnebago has for decades kept away from software packages, preferring instead to create and adapt its own applications. It can thereby achieve a closer fit to the company's business needs, an objective that is at times not met by the ERP vendors described at the beginning of this chapter. It can also be more responsive to users of its systems at lowered costs since it can reuse its code as needed.

An example of Winnebago's homegrown approach is its MRP system, which runs on an IBM zSeries mainframe and is used to plan material needs and schedule production orders. When the company added a new model to its fleet of motor homes, it took only a few hundred hours of development, including a new bill of materials, to make the required changes to the MRP system in order to support production of the new vehicle. Such agility in adapting the MRP system to manufacturing and supplier needs can be especially rewarding in an environment where the change in product variety is constant. In 2007 alone, Winnebago offered 20 different models in 86 different floor plans, each of which can be outfitted with a variety of options including colors, wood stains, and drawer pulls. For the 2011 model year, several new floorplans were added. When you are in the business of building customized homes-on-wheels ranging in prices from $60,000 to $285,000, combining cost consciousness with common sense and employee involvement in developing your own MRP, sales order management, and purchasing systems can prove valuable.



CRAIG BORCK KRT/Newscom

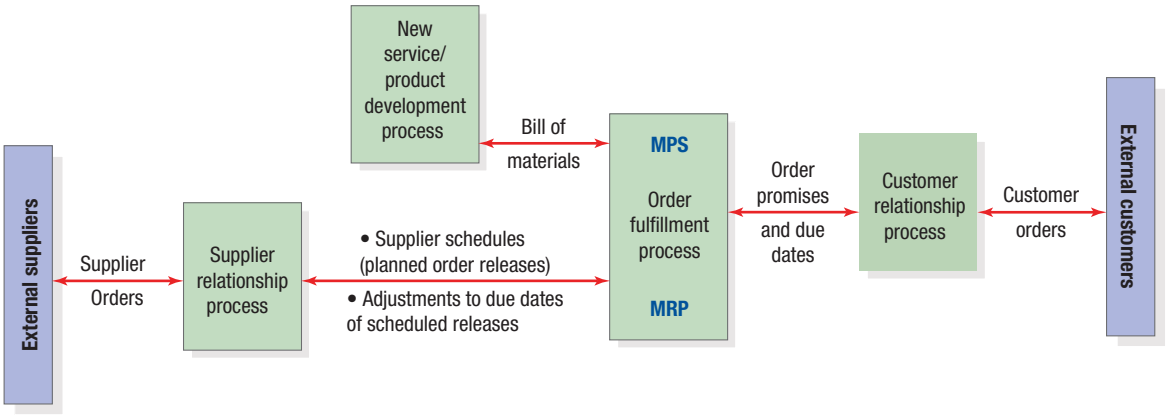
Workmen at the Winnebago factory in Forest City, Iowa, lift a completed side of a Winnebago, with interior and exterior surfaces, wiring, framework and foam insulation all in place, onto a cart so it can be hauled to the next step in the assembly process. Since 1995, Winnebago has spent approximately $5 million annually to automate some of its assembly-line processes, up from $2 million per year previously.

Source: "Road Rules: Creating and Adapting Homegrown Software is the Key to Winnebago's Drive for Business Self Sufficiency," October 15, 2006; **http://www.winnebagoind.com/** (May 28, 2011).

MRP, Core Processes, and Supply Chain Linkages

Among the four core processes of an organization that link activities within and across firms in a supply chain, the MRP system interacts with all of them either through its inputs or its outputs. It all begins with customer orders, which consist of orders for end items as well as replacement parts. MRP and resource planning typically reside inside the order fulfillment process. Master Production Scheduling is an integral part of MRP (see Figure 16.2). As shown schematically in Figure 16.20, the MPS drives the feedback between the order fulfillment process and the customer relationship process through confirmation of order receipts and promised due dates. MPS also provides guidance to the sales group within the customer relationship process with respect to when future orders can be promised, and whether the due dates for existing orders can be adjusted in the time frame requested. The new service and product development process provides an updated bill of materials to the MRP system, and makes sure that every component and assembly needed for manufacturing of end items is properly recognized.

FIGURE 16.20 ▶
MRP Related Information Flows in the Supply Chain



In a similar vein on the inbound side, orders to external suppliers are based on the planned order releases, which come directly from the output of MRP reports. The power of MRP, however, becomes evident when changes to an existing schedule are needed. These changes can be generated, for example, by changes to the MPS because a customer wants to change the timing or size of future orders, by some internal failure such as material shortages or unexpected machine downtime, or by supplier failure. In a supply chain, schedule changes have implications for customers as well as suppliers. Some firms, in partnership with their suppliers, have ERP/MRP systems that can actually "see" into their suppliers' inventory to determine if a particular item is in stock or, if not, when it can be expected. This is an advantage when contemplating a change to the original schedule of order releases. While systems such as this are powerful tools for making changes, care must be taken to avoid unnecessary fluctuations in the timing and size of PORs because of the choice of lot sizing policy. As we have seen, lot sizing rules such as POQ or L4L are susceptible to changes in requirements, and using them indiscriminately can cause instability in replenishment orders. In turn, this instability can be transmitted up the supply chain if the firm's MRP system is electronically linked to the production planning and control systems of its immediate suppliers.

Execution of MRP-based plans using the information flows between core processes as shown in Figure 16.20 properly link a firm with its upstream and downstream supply chain partners. Valid customer and supplier priorities would not be effectively recognized without such an MRP-based framework, which in many firms is actually implemented through an ERP system discussed earlier in the chapter.

MRP and the Environment

Consumer and governmental concern about the deterioration of the natural environment has driven manufacturers to reengineer their processes to become more environmentally friendly. The recycling of base materials is becoming more commonplace, and products are being designed in such a way that they can be remanufactured after their useful lives. Nonetheless, manufacturing processes often produce a number of waste materials that need to be properly disposed of. Wastes come in many forms:

- Effluents, such as carbon monoxide, sulfur dioxide, and hazardous chemicals associated with the processes used to manufacture the product
- Materials, such as metal shavings, oils, and chemicals associated with specific operations
- Packaging materials, such as unusable cardboard and plastics associated with certain products or purchased items
- Scrap associated with unusable products or component defects generated by the manufacturing process

Companies can modify their MRP systems to help them track these wastes and plan for their disposal. The type and amount of waste associated with each item can be entered into its BOM by treating the waste much like you would a component of the item. When the MPS is developed for a product, reports can be generated that project the amount of waste expected during the production process and when it will occur. Although this approach can require that a firm's BOM be modified substantially, the benefits are also substantial. Firms can identify their waste problems in advance to eliminate them in some cases (through process improvement efforts) or plan for their proper disposal in others. It also gives the firm a way to generate any formal documentation required by the government to verify that it has complied with environmental laws and policies.

Resource Planning for Service Providers

We have seen how manufacturing companies can disaggregate an MPS of finished products, which in turn must be translated into the needs for resources, such as staff, equipment, components, and financial assets. The driver for these resource requirements is a material requirements plan. Service providers, of course, must plan their resources just as manufacturers do. However, unlike finished goods, services cannot be inventoried. They must be provided on demand. In terms of resource planning then, service organizations must focus on maintaining the *capacity* to serve their customers. In this section, we will discuss how service providers use the concept of dependent demand and a bill of resources in managing capacity.

Dependent Demand for Services

When we discussed planning and control systems for manufacturers earlier in this chapter, we introduced the concept of *dependent demand*, which is demand for an item that is a function of the production plans for some other item the company produces. For service resource planning, it is useful to define the concept of dependent demand to include demands for resources that are driven by forecasts of customer requests for services or by plans for various activities in support of the services the company provides. Here are some other examples of dependent demands for service providers.

Restaurants Every time you order from the menu at a restaurant, you initiate the restaurant's need for certain types of goods (uncooked food items, plates, and napkins), staff (chef, servers, and dishwashers), and equipment (stoves, ovens, and cooking utensils). Using a forecast of the demand for each type of meal, the manager of the restaurant can estimate the need for these resources. Many restaurants, for example, feature "specials" on certain days, say, fish on Fridays or prime rib on Saturdays. Specials improve the accuracy of the forecasts managers need to make for different types of meals (and the food products that are required to make them) and typically signal the need for above-average staffing levels. How much of these resources will be needed, however, depends on the number of meals the restaurant ultimately expects to serve. As such, these items—food products and staff members—are dependent demands.

Airlines Whenever an airline schedules a flight, certain supporting goods are needed (beverages, snacks, and fuel), labor (pilots, flight attendants, and airport services), and equipment (a plane and airport gate). The number of flights and passengers the airline forecasts it will serve determines the amount of these resources needed. Just like a manufacturer, the airline can explode its master schedule of flights to make this determination.

Hospitals With the exception of the emergency room services, hospitals can use their admission appointments to create a master schedule. The master schedule can be exploded to determine the resources the hospital will need during a certain period. For example, when you schedule a surgical procedure, you generate a need for facilitating goods such as medicines, surgical gowns, linens, staff (a surgeon, nurses, and anesthesiologist), and equipment (an operating room, surgical tools, and recovery bed). As they build their master schedules, hospitals must ensure that certain equipment and personnel do not become overcommitted—that capacity is maintained, in other words. For example, an appointment for a key operation might have to be scheduled in advance at a time a surgeon is available to do it, even though the hospital's other resources—operating room, nurses, and so forth—might be currently be available.



Doctors and nurses in operation room during surgery

ochen Tack/PhotoLibrary

Hotels A traveler who makes a reservation at a hotel generates demand for facilitating goods (soap and towels), staff (front desk, housekeeping, and concierge), and equipment (fax, television, and exercise bicycle). To determine its dependent resource needs, a hotel adds the number of reservations already booked to the number of "walk-in" customers it forecasts it will have. This figure is used to create the hotel's master schedule. One resource a hotel cannot easily adjust, however, is the number of rooms it has. If the hotel is overbooked, for instance, it cannot simply add more rooms. If it has too few guests, it cannot "downsize" its number of rooms. Given the high capital costs needed

for this resource, hotels try to maintain as high a utilization rate as possible by offering group rates or special promotions at certain times of the year. In other words, they try to drive up dependent demand for this particular resource.

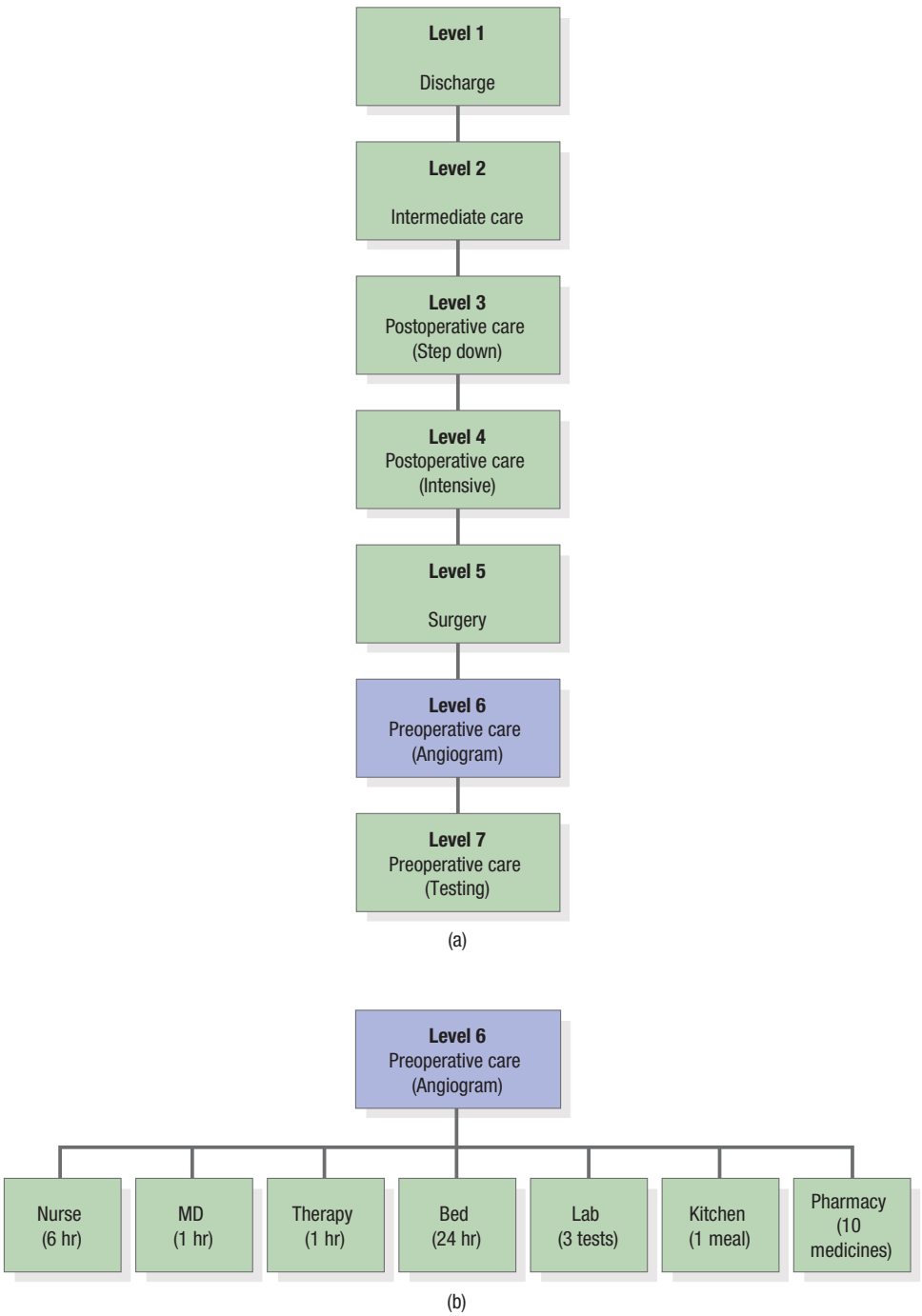
Bill of Resources

bill of resources (BOR)

A record of a service firm's parent-component relationships and all of the materials, equipment time, staff, and other resources associated with them, including usage quantities.

The service analogy to the bill of materials in a manufacturing company is the **bill of resources (BOR)**, which is a record of a service firm's parent–component relationships and all of the materials, equipment time, staff, and other resources associated with them, including usage quantities. Once the service firm has completed its master schedule, the BOR can be used to determine what resources the firm will need, how much of them it will need, and when. A BOR for a service provider can be as complex as a BOM for a manufacturer. Consider a hospital that just scheduled treatment of a patient with an aneurysm. As shown in Figure 16.21 (a), the BOR for treatment of an aneurysm has seven levels, starting at the top (end item): (1) discharge, (2) intermediate care, (3) postoperative care (step down), (4) postoperative care (intensive), (5) surgery, (6) preoperative

FIGURE 16.21 ▶
BOR for Treating an Aneurysm



care (angiogram), and (7) preoperative care (testing). Each level of the BOR has a set of material and resource requirements and an associated lead time. For example, at level 6, shown in Figure 16.21(b), the patient needs 6 hours of nurses' time, 1 hour of the primary MD's time, 1 hour of the respiratory therapist's time, 24 hours of bed time, 3 different lab tests, 1 meal, and 10 different medicines from the pharmacy. The lead time for this level is 1 day. The lead time for the entire stay for treatment of the aneurysm is 12.2 days. A master schedule of patient admissions and the BORs for each illness enable the hospital to manage its critical resources. Reports analogous to the MRP II reports we discussed earlier in the chapter can be generated for the people who manage the various functional areas of the hospital.

One resource every service provider needs, however, is cash. Service organizations have to forecast the number of customers they expect to serve so that they have enough cash on hand to purchase materials that support the services—labor and other products. Purchasing these items increases the firm's accounts payable. As services are actually completed for customers, the firm's accounts receivable increases. The firm's master schedule and its accounts receivable and payable help a company predict the amount and timing of its cash flows.

LEARNING GOALS IN REVIEW

1. **Explain how enterprise resource planning (ERP) systems can foster better resource planning**. Review the opening vignette on Dow Corning, and the section on "Enterprise Resource Planning," pp. 564–566. Pay attention to Figure 16.1 to understand how different application modules come together to create functionality and value in the ERP systems.
2. **Explain how the concept of dependent demand is fundamental to resource planning**. See the section on "Dependent Demand," p. 567, which shows how continuous independent demand can lead to lumpy requirements for dependent demand. Then, a separate system, called material requirements planning, is needed to manage dependent demand situations.
3. **Describe a master production schedule (MPS) and compute available-to-promise quantities.** The section "Master Production Scheduling," pp. 569–573, shows you how firms break down a production plan into more detailed schedules. Understand the key relationships between Figures 16.5, 16.6, and 16.8.
4. **Apply the logic of a material requirements planning (MRP) system to identify production and purchase orders needed for dependent demand items**. Using Figure 16.12, page 575, understand how an inventory record is created for a given lot size rule. The section on "Planning Factors," pp. 576–579, shows you how choice of different managerial policies affect material plans. Finally, focus on understanding the MRP explosion process as illustrated in Figure 16.18 on page 580 and Solved Problem 3 on pages 590–591.
5. **Apply MRP principles to the provision of services and distribution inventories.** The section, "Resource Planning for Service Providers," pp. 585–587, illustrates how the Bill of Resources can be used to plan dependent demand for services in settings such as the restaurants, airlines, hospitals, and hotels.

MyOMLab helps you develop analytical skills and assesses your progress with multiple problems on bills of material, parents, intermediate items, components, purchased items, lead times, master production schedules, MRP record, FOQ, L4L, POQ, scheduled receipts, and multi-level material requirements plans.

MyOMLab Resources	Titles	Link to the Book
Video	*Nantucket Nectars: ERP*	Enterprise Resource Planning
Active Model Exercise	16.1 Material Requirements Plan	Bill of Resources
OM Explorer Solvers	Master Production Scheduling Material Requirements Planning Single-Item MRP	Master Production Scheduling; Solved Problem 2 (pp. 589–590) Material Requirements Planning; Solved Problem 3 (pp. 590–591) Material Requirements Planning; Solved Problem 3 (pp. 590–591)
OM Explorer Tutors	16.1 Master Production Scheduling 16.2 FOQ, POQ, and L4L Rules	Master Production Scheduling; Solved Problem 2 (pp. 589–590) Planning Factors; Solved Problem 3 (pp. 590–591)
Tutor Exercises	16.1 Applying Different MRP Lot Sizing Rules	Planning Factors; Solved Problem 3 (pp. 590–591)
Virtual Tours	Vaughn Hockey Equipment Winslow Life Rafts	Bill of Materials Enterprise Resource Planning

MyOMLab Resources	Titles	Link to the Book
MyOMLab Supplements	H. Measuring Output Rates I. Learning Curve Analysis	MyOMLab Supplement H MyOMLab Supplement I
Internet Exercise	Schwinn Hewlett-Packard	Material Requirements Planning Material Requirements Planning
Key Equations		
Image Library		

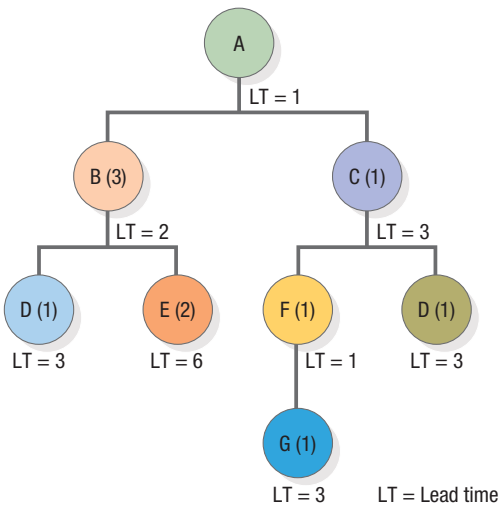
Key Terms

action notice 581
available-to-promise (ATP) inventory 572
bill of materials (BOM) 567
bill of resources (BOR) 586
capacity requirements planning (CRP) 581
component 567
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Solved Problem 1

Refer to the bill of materials for product A shown in Figure 16.22.

FIGURE 16.22 ▶
BOM for Product A



If there is no existing inventory and no scheduled receipts, how many units of items G, E, and D must be purchased to produce 5 units of end item A?

SOLUTION

Five units of item G, 30 units of item E, and 20 units of item D must be purchased to make 5 units of A. The usage quantities shown in Figure 16.22 indicate that 2 units of E are needed to make 1 unit of B and that 3 units of B are needed to make 1 unit of A; therefore, 5 units of A require 30 units of E ($2 \times 3 \times 5 = 30$). One unit of D is consumed to make 1 unit of B, and 3 units of B per unit of A result in 15 units of D ($1 \times 3 \times 5 = 15$); 1 unit of D in each unit of C and 1 unit of C per unit of A result in another 5 units of D ($1 \times 1 \times 5 = 5$). The total requirements to make 5 units of A are 20 units of D ($15 + 5$). The calculation of requirements for G is simply $1 \times 1 \times 1 \times 5 = 5$ units.

Solved Problem 2

The order policy is to produce end item A in lots of 50 units. Using the data shown in Figure 16.23 and the FOQ lot-sizing rule, complete the projected on-hand inventory and MPS quantity rows. Then, complete the MPS start row by offsetting the MPS quantities for the final assembly lead time. Compute the available-to-promise inventory for item A. Finally, assess the following customer requests for new orders. Assume that these orders arrive consecutively and their affect on ATP is cumulative. Which of these orders can be satisfied without altering the MPS Start quantities?

a. Customer C requests 10 units in week 3.

b. Customer B requests 30 units in week 4.

c. Customer A requests 30 units in week 1.

d. Customer D requests 50 units in week 5.

Item: A — **Order Policy:** 50 units — **Lead Time:** 1 week

Quantity on Hand: 5	Week 1	2	3	4	5	6	7	8	9	10
Forecast	20	10	40	10	0	0	30	20	40	20
Customer orders (booked)	30	20	5	8	0	2	0	0	0	0
Projected on-hand inventory	25									
MPS quantity	50									
MPS start										
Available-to-promise (ATP) inventory										

▲ **FIGURE 16.23** MPS Record for End Item A

SOLUTION

The projected on-hand inventory for the second week is

$$\begin{pmatrix}\text{Projected on-hand inventory at end of week 2}\end{pmatrix} = \begin{pmatrix}\text{On-hand inventory in week 1}\end{pmatrix} + \begin{pmatrix}\text{MPS quantity due in week 2}\end{pmatrix} - \begin{pmatrix}\text{Requirements in week 2}\end{pmatrix}$$

$$= 25 + 0 - 20 = 5 \text{ units}$$

where requirements are the larger of the forecast or actual customer orders booked for shipment during this period. No MPS quantity is required.

Without an MPS quantity in the third period, a shortage of item A will occur: $5 + 0 - 40 = -35$. Therefore, an MPS quantity equal to the lot size of 50 must be scheduled for completion in the third period. Then, the projected on-hand inventory for the third week will be $5 + 50 - 40 = 15$.

Figure 16.24 shows the projected on-hand inventories and MPS quantities that would result from completing the MPS calculations. The MPS start row is completed by simply shifting a copy of the MPS quantity row to the left by one column to account for the 1-week final assembly lead time. Also shown are the available-to-promise quantities. In week 1, the ATP is

$$\begin{pmatrix}\text{Available-to-Promise in week 1}\end{pmatrix} = \begin{pmatrix}\text{On-hand quantity in week 1}\end{pmatrix} + \begin{pmatrix}\text{MPS quantity in week 1}\end{pmatrix} - \begin{pmatrix}\text{Orderes booked up to week 3 when the next MPS arrives}\end{pmatrix}$$

$$= 5 + 50 - (30 + 20) = 5 \text{ units}$$

The ATP for the MPS quantity in week 3 is

$$\begin{pmatrix}\text{Available-to-Promise in week 3}\end{pmatrix} = \begin{pmatrix}\text{MPS quantity in week 3}\end{pmatrix} - \begin{pmatrix}\text{Orderes booked up to week 7 when the next MPS arrives}\end{pmatrix}$$

$$= 50 - (5 + 8 + 0 + 2) = 35 \text{ units}$$

The other ATPs equal their respective MPS quantities because no orders are booked for those weeks. As for the new orders, Customer A's request for 30 units in week 1 cannot be accommodated; the earliest it can be shipped is week 3 because the ATP for week 1 is insufficient. Assuming that Customer A's order is rejected, Customer B's request may be satisfied. The ATP for week 1 will

Lot Size: 50 units
Lead Time: 1 week

Quantity on Hand: 5	Week 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Forecast	20	10	40	10			30	20	40	20					
Customer orders (booked)	30	20	5	8		2									
Projected on-hand inventory	25	5	15	5	5	3	23	3	13	43					
MPS quantity	50		50				50		50	50					
MPS start		50				50		50	50						
Available-to-promise (ATP) inventory	5		35				50		50	50					

FIGURE 16.24 ▲
Completed MPS Record for End Item A

stay at 5 units and the ATP for week 3 will be reduced to 5 units. This acceptance allows the firm the flexibility to immediately satisfy an order for 5 units or less, if one comes in. When the MPS is updated next, the customer orders booked for week 4 will be increased to 38 to reflect the new order's shipping date. Customer C's order for 10 units in week 3 is likewise accepted. The ATP for weeks 1 and 3 will be reduced to 0, and when the MPS is updated, the customer orders booked for week 3 will be increased to 15. Finally, Customer D's order for 50 units in week 5 cannot be satisfied without changing the MPS.

Solved Problem 3

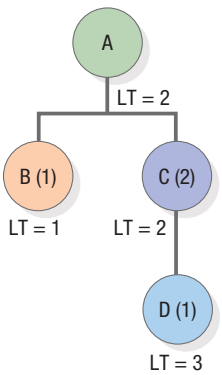


FIGURE 16.25 ▲
BOM for Product A

MyOMLab
Active Model 16.1 in MyOMLab provides additional insight on lotsizing decisions for this problem.

The MPS start quantities for product A calls for the assembly department to begin final assembly according to the following schedule: 100 units in week 2; 200 units in week 4; 120 units in week 6; 180 units in week 7; and 60 units in week 8. Develop a material requirements plan for the next 8 weeks for items B, C, and D. The BOM for A is shown in Figure 16.25, and data from the inventory records are shown in Table 16.1.

SOLUTION

We begin with items B and C and develop their inventory records, as shown in Figure 16.26. The MPS for product A must be multiplied by 2 to derive the gross requirements for item C because of the usage quantity. Once the planned order releases for item C are found, the gross requirements for item D can be calculated.

TABLE 16.1 | INVENTORY RECORD DATA

	ITEM		
Data Category	**B**	**C**	**D**
Lot-sizing rule	POQ ($P = 3$)	L4L	FOQ = 500 units
Lead time (LT)	1 week	2 weeks	3 weeks
Scheduled receipts	None	200 (week 1)	None
Beginning (on-hand) inventory	20	0	425

Item: B
Description:
Lot Size: POQ (P = 3)
Lead Time: 1 week

		Week									
		1	2	3	4	5	6	7	8	9	10
Gross requirements			100		200		120	180	60		
Scheduled receipts											
Projected on-hand inventory	20	20	200	200	0	0	240	60	0	0	0
Planned receipts			280				360				
Planned order releases		280				360					

Item: C
Description:
Lot Size: L4L
Lead Time: 2 weeks

		Week									
		1	2	3	4	5	6	7	8	9	10
Gross requirements			200		400		240	360	120		
Scheduled receipts		200									
Projected on-hand inventory	0	200	0	0	0	0	0	0	0	0	0
Planned receipts					400		240	360	120		
Planned order releases			400		240	360	120				

Item: D
Description:
Lot Size: FOQ = 500 units
Lead Time: 3 weeks

		Week									
		1	2	3	4	5	6	7	8	9	10
Gross requirements			400		240	360	120				
Scheduled receipts											
Projected on-hand inventory	425	425	25	25	285	425	305	305	305	305	305
Planned receipts					500	500					
Planned order releases		500	500								

FIGURE 16.26
Inventory Records for Items B, C, and D

Discussion Questions

1. Consider a service provider that is in the delivery business, such as UPS or FedEx. How can the principles of MRP be useful to such a company?
2. Form a group in which each member represents a different functional area of a firm. Provide a priority list of the information that could be generated from an MPS, from the most important to the least important, for each functional area. Rationalize the differences in the lists.
3. Consider the master flight schedule of a major airline, such as Air New Zealand. Discuss the ways in which it is analogous to a master production schedule for a manufacturer.
4. For an organization of your choice, such as where you previously worked, discuss how an ERP system could be used and whether it would increase effectiveness.

Problems

The OM Explorer and POM for Windows software is available to all students using the 10th edition of this textbook. Go to **www.pearsonglobaleditions.com/krajewski** website to download these computer packages. If you purchased MyOMLab, you also have access to Active Models software and significant help in doing the following problems. Check with your instructor on how best to use these resources. In many cases, the instructor wants you to understand how to do the calculations by hand. At the least, the software provides a check on your calculations. When calculations are particularly complex and the goal is interpreting the results in making decision, the software entirely replaces the manual calculations.

1. Consider the bill of materials (BOM) in Figure 16.27.
   - **a.** How many immediate parents (one level above) does item I have? How many immediate parents does item E have?
   - **b.** How many unique components does product A have at all levels?
   - **c.** Which of the components are purchased items?
   - **d.** How many intermediate items does product A have at all levels?
   - **e.** Given the lead times (LT) in weeks noted on Figure 16.27, how far in advance of shipment must a purchase commitment be for any of the purchased items identified in part (c)?
2. Product A is made from components B, C, and E. Item E is a subassembly that requires 2 units of D and 3 unit of C. Item B also is an intermediate item, made from F. All other usage quantities are 1. Draw the BOM for product A.
3. What is the lead time (in weeks) to respond to a customer order for product A, based on the BOM shown in Figure 16.28, assuming no existing inventories or scheduled receipts?
4. Product A is made from components B and C. Item B, in turn, is made from D and E. Item C also is an intermediate item, made from F and H. Finally, intermediate item E is made from H and G. Note that item H has two parents. The following are item lead times:

Item	A	B	C	D	E	F	G	H
Lead Time (weeks)	1	2	2	6	5	6	4	3

   - **a.** What lead time (in weeks) is needed to respond to a customer order for product A, assuming no existing inventories or scheduled receipts?
   - **b.** What is the customer response time if all purchased items (i.e., D, F, G, and H) are in inventory?
   - **c.** If you are allowed to keep just one purchased item in stock, which one would you choose?
5. Refer to Figure 16.22 and Solved Problem 1. If inventory consists of 2 units of B, 1 unit of F, and 3 units of G, how many units of G, E, and D must be purchased to produce 5 units of product A?

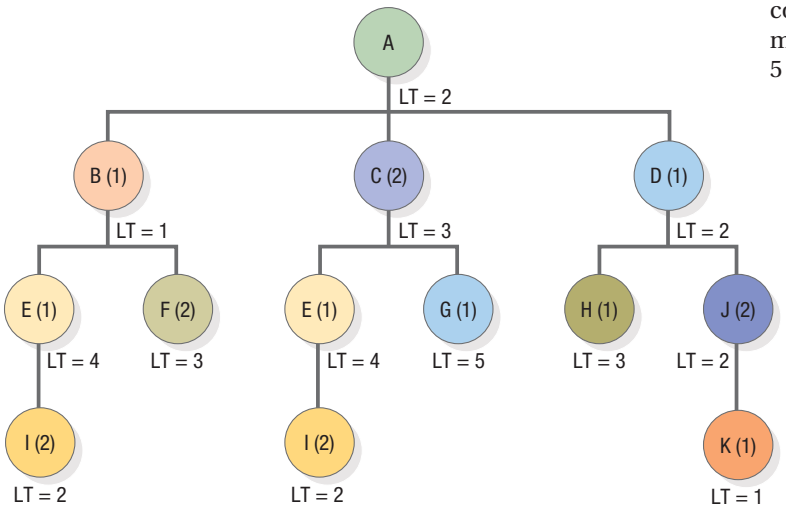


FIGURE 16.27 ▲
BOM for Product A

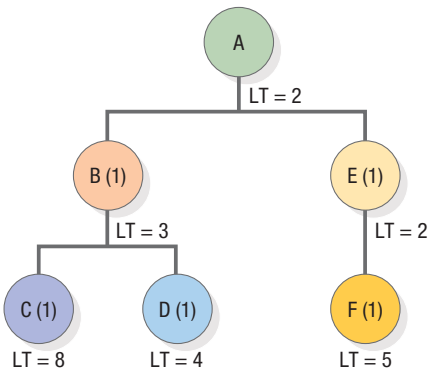


FIGURE 16.28 ▲
BOM for Product A

6. Complete the MPS record in Figure 16.29 for a single item.

FIGURE 16.29
MPS Record for Single Item

Item: A — **Order Policy:** 100 units — **Lead Time:** 1 week

Quantity on Hand: 30	**Week**							
	1	2	3	4	5	6	7	8
Forecast	20	18	31	28	26	30	31	35
Customer orders (booked)	15	17	9	17	11	0	11	0
Projected on-hand inventory								
MPS quantity								
MPS start								

7. Complete the MPS record shown in Figure 16.30 for a single item.

FIGURE 16.30
MPS Record for Single Item

Item: A — **Order Policy:** 100 units — **Lead Time:** 1 week

Quantity on Hand: 70	**January**				**February**			
	1	2	3	4	5	6	7	8
Forecast	70	70	70	50	55	50	50	55
Customer orders (booked)	40	15	90	0	35	65	0	0
Projected on-hand inventory								
MPS quantity								
MPS start								

8. An end item's demand forecasts for the next 10 weeks are 30, 20, 35, 50, 25, 25, 0, 40, 0, and 50 units. The current on-hand inventory is 80 units. The order policy is to produce in lots of 100. The booked customer orders for the item, starting with week 1, are 22, 30, 15, 9, 0, 0, 5, 3, 7, and 0 units. At present, no MPS quantities are on-hand for this item. The lead time is 2 weeks. Develop an MPS for this end item.

9. Figure 16.31 shows a partially completed MPS record for ball bearings.

a. Develop the MPS for ball bearings.

b. Four customer orders arrived in the following sequence:

Order	Quantity	Week Desired
1	500	4
2	400	5
3	300	1
4	300	7

Assume that you must commit to the orders in the sequence of arrival and cannot change the desired shipping dates or your MPS. Which orders should you accept?

10. The forecasted requirements for an electric hand drill for the next 6 weeks are 15, 40, 10, 20, 50, and 30 units. The marketing department has booked orders totaling 20, 25, 10, and 20 units for delivery in the first (current), second, third, and fourth weeks. Currently, 30 hand drills are in stock. The policy is to order in lots of 60 units. Lead time is one week.

a. Develop the MPS record for the hand drills.

b. A distributor of the hand drills places an order for 15 units. What is the appropriate shipping date for the entire order?

11. Figure 16.32 shows a partially completed MPS record for 2" pneumatic control valves. Suppose that you receive the following orders for the valves (shown in the order of their arrival). As they arrive, you must decide whether to accept or reject them. Which orders would you accept for shipment?

Order	Amount (Units)	Week Requested
1	15	2
2	30	5
3	25	3
4	75	7

12. Tabard Industries forecasted the following demand for one of its most profitable products for the next 8 weeks: 120, 120, 120, 100, 100, 100, 80, and 80 units. The booked customer orders for this product, starting in week 1 are: 100, 80, 60, 40, 10, 10, 0, and 0 units. The current on-hand inventory is 150 units, the order quantity is 200 units, and the lead time is one week.

a. Develop a MPS for this product.

b. The marketing department revised its forecast. Starting with week 1, the new forecasts are: 120, 120, 120, 150, 150, 150, 100, and 100 units. Assuming that the prospective MPS you developed in part (a) does not change, prepare a revised MPS record. Comment on the situation that Tabard now faces.

c. Returning to the original forecasted demand level and the MPS record you developed in part (a), assume that marketing accepted a new customer order for 200 units in week 2 and thereby booked orders in week 2 is now 280 units. Assuming that the prospective MPS you developed in part (a) does not change, prepare a revised MPS record. Comment on the situation that Tabard now faces.

13. A forecast of 240 units in January, 320 units in February, and 240 units in March has been approved for the

FIGURE 16.31 ▶
MPS Record for Ball Bearings

Item: Ball bearings — **Order Policy:** 500 units; **Lead Time:** 1 week

Quantity on Hand: 400	Week 1	2	3	4	5	6	7	8	9	10
Forecast	550	300	400	450	300	350	200	300	450	400
Customer orders (booked)	300	350	250	250	200	150	100	100	100	100
Projected on-hand inventory										
MPS quantity	500									
MPS start										
Available-to-promise (ATP) inventory										

Item: 2" Pneumatic control valve **Order Policy:** 75 units
Lead Time: 1 week

Quantity on Hand: 10	Week 1	2	3	4	5	6	7	8
Forecast	40	40	40	40	30	30	50	50
Customer orders (booked)	60	45	30	35	10	5	5	0
Projected on-hand inventory								
MPS quantity	75	75						
MPS start	75							
Available-to-promise (ATP) inventory								

◀ **FIGURE 16.32**
MPS Record for 2" Pneumatic Control Valve

seismic-sensory product family manufactured at the Rockport facility of Maryland Automated, Inc. Three products, A, B, and C, comprise this family. The product mix ratio for products A, B, and C for the past 2 years has been 35 percent, 40 percent, and 25 percent, respectively. Management believes that the monthly forecast requirements are evenly spread over the 4 weeks of each month. Currently, 5 units of product C are on hand. The company produces product C in lots of 40, and the lead time is 2 weeks. A production quantity of 40 units from the previous period is scheduled to arrive in week 1. The company has accepted orders of 20, 9, 8, 15, 2, and 6 units of product C in weeks 1 through 6, respectively. Prepare a prospective MPS for product C and calculate the available-to-promise inventory quantities.

14. The partially completed inventory record for the tabletop subassembly in Figure 16.33 shows gross requirements, scheduled receipts, lead time, and current on-hand inventory.

a. Complete the last three rows of the record for an FOQ of 120 units.

b. Complete the last three rows of the record by using the L4L lot-sizing rule.

c. Complete the last three rows of the record by using the POQ lot-sizing rule, with $P = 2$.

Item: M405—X **Lot Size:**
Description: Tabletop subassembly **Lead Time:** 2 weeks

Week	1	2	3	4	5	6	7	8	9	10
Gross requirements	95		80		80		40	95		
Scheduled receipts	120									
Projected on-hand inventory 30										
Planned receipts										
Planned order releases										

◀ **FIGURE 16.33**
Inventory Record for the Tabletop Subassembly

15. The partially completed inventory record for the driveshaft subassembly in Figure 16.34 shows gross requirements, scheduled receipts, lead time, and current on-hand inventory.

a. Complete the last three rows of the record for an FOQ of 60 units.

b. Complete the last three rows of the record by using the L4L lot-sizing rule.

c. Complete the last three rows of the record by using the POQ lot-sizing rule, with $P = 4$.

FIGURE 16.34 ▶
Inventory Record for the Driveshaft Subassembly

Item: Driveshaft subassembly — **Lot Size:** 60 units, **Lead Time:** 3 weeks

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	30	30	20	15	35	35	55	55
Scheduled receipts	80							
Projected on-hand inventory 10								
Planned receipts								
Planned order releases								

16. The partially completed inventory record for the rotor subassembly in Figure 16.35 shows gross requirements, scheduled receipts, lead time, and current on-hand inventory.

a. Complete the last three rows of the record for an FOQ of 140 units.

b. Complete the last three rows of the record by using the L4L lot-sizing rule.

c. Complete the last three rows of the record by using the POQ lot-sizing rule, with $P = 2$.

17. Figure 16.36 shows a partially completed inventory record for the real wheel subassembly. Gross requirements, scheduled receipts, lead time, and current on-hand inventory are shown.

a. Complete the last three rows of the record for an FOQ of 200 units.

b. Complete the last three rows of the record by using an FOQ of 100 units.

c. Complete the last three rows of the record by using the L4L rule.

FIGURE 16.35 ▶
Inventory Record for the Rotor Subassembly

Item: Rotor subassembly — **Lot Size:** 140 units, **Lead Time:** 2 weeks

	Week							
	1	2	3	4	5	6	7	8
Gross requirements	55	20	45	40	70	70	70	70
Scheduled receipts	140							
Projected on-hand inventory 20								
Planned receipts								
Planned order releases								

Item: MQ–09 **Lot Size:**
Description: Rear wheel subassembly **Lead Time:** 1 week

	Week									
	1	2	3	4	5	6	7	8	9	10
Gross requirements	25	105	110	90		45	110	60		
Scheduled receipts										
Projected on-hand inventory 50										
Planned receipts										
Planned order releases										

◀ **FIGURE 16.36**
Inventory Record for the Rear Wheel Subassembly

18. A partially completed inventory record for the motor subassembly is shown in Figure 16.37.

a. If it costs the company $1 to hold a unit in inventory from one week to the next, and the cost to release an order is $50, which of the lot sizing rules used above will provide the lowest inventory holding + order release cost?

b. Complete the last three rows of the record by using the POQ rule with $P = 2$.

c. Complete the last three rows of the record by using the POQ rule with $P = 4$.

d. Complete the last three rows of the record by using the L4L rule.

Item: GF–4 **Lot Size:**
Description: Motor subassembly **Lead Time:** 2 weeks

	Week									
	1	2	3	4	5	6	7	8	9	10
Gross requirements		80	50	35	20	55	15	30	25	10
Scheduled receipts	60									
Projected on-hand inventory 20										
Planned receipts										
Planned order releases										

▲ **FIGURE 16.37**
Inventory Record for the Motor Subassembly

Advanced Problems

Depending on the time available to cover this chapter, your instructor may not assign these problems. Perhaps, your assignment for the multi-level problems allows the use of the *Material Requirements Planning* Solver of OM Explorer, with main requirement of looking for situations that need the inventory planner's attention (i.e., an action notice).

19. The BOM for product A is shown in Figure 16.38, and data from the inventory records are shown in Table 16.2. In the master production schedule for product A, the MPS start row has 100 units in week 3 and 200 in week 6. Develop the material requirements plan for the next 6 weeks for items B, C, and D.

a. Develop the material requirements plan for the next 6 weeks for items B, C, and D.

b. What specific managerial actions are required in week 1?

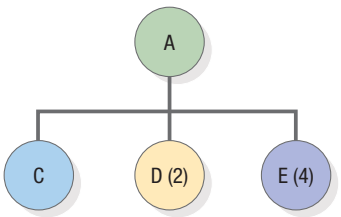


FIGURE 16.38 ▲
BOM for Product A

TABLE 16.2 | INVENTORY RECORD DATA

	ITEM		
Data Category	**C**	**D**	**E**
Lot-Sizing Rule	L4L	FOQ = 200	POQ (P = 3 weeks)
Lead time	2 weeks	1 week	1 week
Scheduled receipts	None	200 (in week 3)	200 (in week 3)
Beginning inventory	50	200	0

20. The BOMs for products A & B and data from the inventory records are shown in Figure 16.39. Data from the inventory records are shown in Table 16.3. In the master production schedule for product A, the MPS start row has 85 units in week 2 and 200 in week 4 and 50 in week 8. In the master production schedule for product B, the MPS start row has 65 units in week 3 and 50 in week 4 and 50 in week 5 and 75 in week 8.

a. Develop the material requirements plan for the next 8 weeks for items C, D, and E. Note any difficulties you observe in the inventory records.

b. Can the difficulties noted in part a be rectified by expediting any Scheduled Receipts?

FIGURE 16.39 ▶
BOMs for Product A and Product B

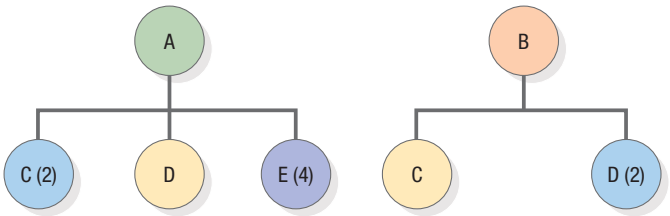
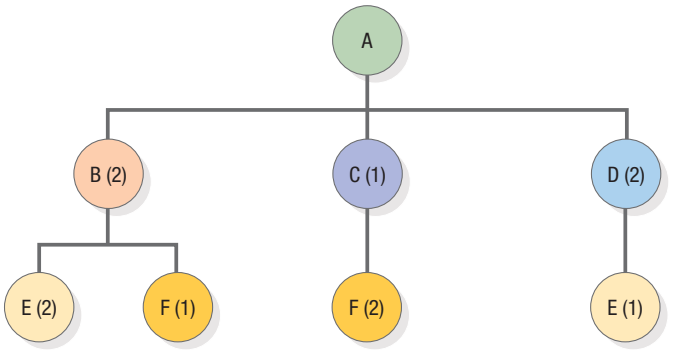


TABLE 16.3 | INVENTORY RECORD DATA

	ITEM		
Data Category	**C**	**D**	**E**
Lot-sizing rule	L4L	FOQ = 200	POQ (P = 2 weeks)
Lead time	2 weeks	1 week	1 week
Scheduled receipts	200 (in week 3)	0	0
Beginning inventory	0	0	200

21. Figure 16.40 illustrates the BOM for product A. The MPS start row in the master production schedule for product A calls for 50 units in week 2, 65 units in week 5, and 80 units in week 8. Item C is produced to make A and to meet the forecasted demand for replacement parts. Past replacement part demand has been 20 units per week (add 20 units to C's gross requirements). The lead times for items F and C are 1 week, and for the other items the lead time is 2 weeks. No safety stock is required for items B, C, D, E, and F. The L4L lot-sizing rule is used for items B and F; the POQ lot-sizing rule (P = 3) is used for C. Item E has an FOQ of 600 units, and D has an FOQ of 250 units. On-hand inventories are 50 units of B, 50 units of C, 120 units of D, 70 units of E, and 250 units of F. Item B has a scheduled receipt of 50 units in week 2. Develop a material requirements plan for the next 8 weeks for items B, C, D, E, and F.

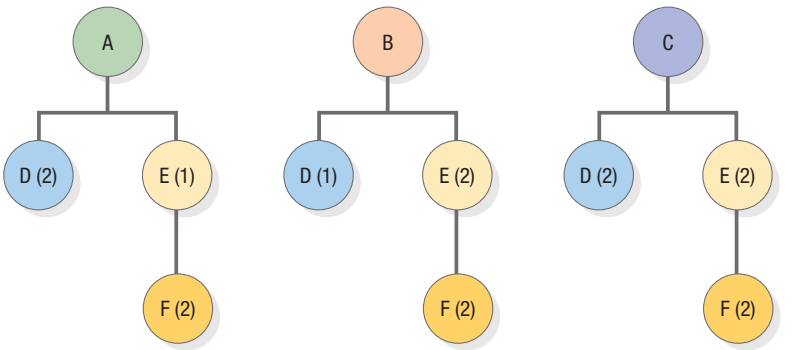


▲ **FIGURE 16.40**
BOM for Product A

22. The following information is available for three MPS items:

Product A	An 80-unit order is to be started in week 3. A 55-unit order is to be started in week 6.
Product B	A 125-unit order is to be started in week 5.
Product C	A 60-unit order is to be started in week 4.

Develop the material requirements plan for the next 6 weeks for items D, E, and F. The BOMs are shown in Figure 16.41, and data from the inventory records are shown in Table 16.4. (*Warning:* A safety stock requirement applies to item F. Be sure to plan a receipt for any week in which the projected on-hand inventory becomes less than the safety stock.)

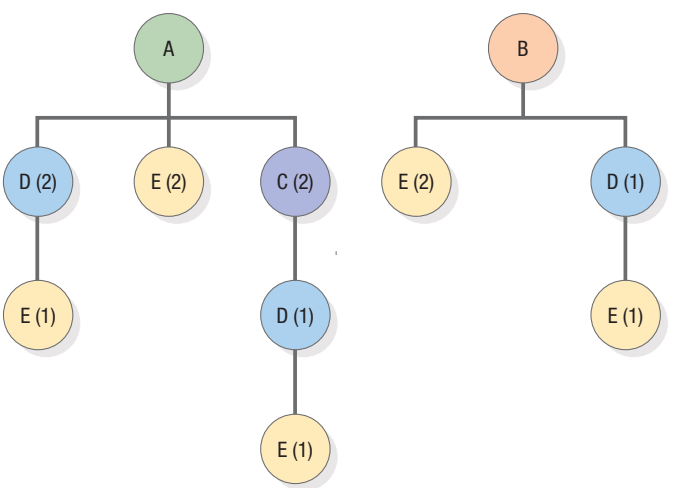


◀ **FIGURE 16.41**
BOMs for Products A, B, and C

TABLE 16.4 | **INVENTORY RECORD DATA**

	ITEM		
Data Category	**D**	**E**	**F**
Lot-sizing rule	FOQ = 150	L4L	POQ ($P = 2$)
Lead time	3 weeks	1 week	2 weeks
Safety stock	0	0	30
Scheduled receipts	150 (week 3)	120 (week 2)	None
Beginning inventory	150	0	100

▼ **FIGURE 16.42**
BOMs for Products A and B



23. Figure 16.42 shows the BOMs for two products, A and B. Table 16.5 shows the MPS quantity start date for each one. Table 16.6 contains data from inventory records for items C, D, and E. There are no safety stock requirements for any of the items.

a. Determine the material requirements plan for items C, D, and E for the next 8 weeks.

b. What specific managerial actions are required in week 1?

c. Suppose that a very important customer places an emergency order for a quantity of product A. In order to satisfy this order, a new MPS of 200 units of product A is now required in week 5. Determine the changes to the material requirements plan if this order is accepted and note any problems that you detect.

TABLE 16.5 | MPS QUANTITY START DATES

	DATE							
Product	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
A		125		95		150		130
B			80			70		

TABLE 16.6 | INVENTORY RECORD DATA

	ITEM		
Data Category	**C**	**D**	**E**
Lot-sizing rule	L4L	POQ ($P = 3$)	FOQ = 800
Lead time	3 weeks	2 weeks	1 week
Scheduled receipts	200 (week 2)	None	800 (week 1)
Beginning inventory	85	625	350

24. The BOM for product A is shown in Figure 16.43. The MPS for product A calls for 120 units to be started in weeks 2, 4, 5, and 8. Table 16.7 shows data from the inventory records.

a. Develop the material requirements plan for the next 8 weeks for each item.

b. What specific managerial actions are required in week 1? Make sure you address any specific difficulties you encounter in the inventory records.

FIGURE 16.43 ▶
BOM for Product A

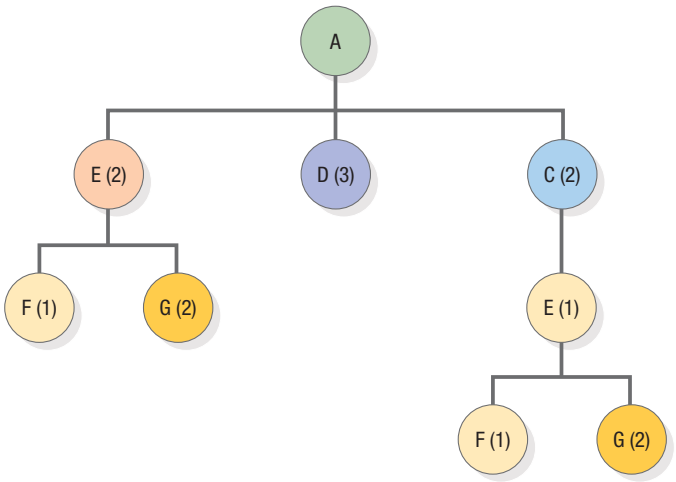


TABLE 16.7 | INVENTORY RECORD DATA

	ITEM				
Data Category	**C**	**D**	**E**	**F**	**G**
Lot-sizing rule	L4L	FOQ = 700	FOQ = 700	L4L	L4L
Lead time	3 weeks	3 weeks	4 weeks	2 weeks	1 week
Safety stock	0	0	0	50	0
Scheduled receipts	150 (week 2)	450 (week 2)	700 (week 1)	None	1,400 (week 1)
Beginning inventory	125	0	235	750	0

25. Refer to Solved Problem 1 (Figure 16.22) for the bill of materials and Table 16.8 for component inventory record information. Develop the material requirements plan for all components and intermediate items associated with product A for the next 10 weeks. The MPS for product A calls for 60 units to be started in weeks 2, 6, 8, and 9. (*Warning:* Safety stock requirements apply to items B and C.)

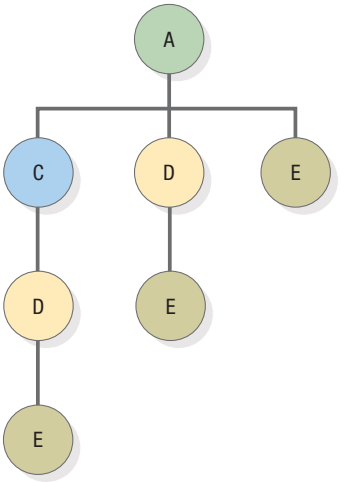
TABLE 16.8 | INVENTORY RECORD DATA

	ITEM					
Data Category	**B**	**C**	**D**	**E**	**F**	**G**
Lot-sizing rule	L4L	L4L	POQ ($P = 2$)	L4L	L4L	FOQ = 480
Lead time	2 weeks	3 weeks	3 weeks	6 weeks	1 week	3 weeks
Safety stock	20	15	0	0	0	0
Scheduled receipts	120 (week 2)	50 (week 2)	None	200 (week 6)	90 (week 3)	None
Beginning inventory	20	60	60	300	0	0

26. The bill of materials and the data from the inventory records for product A are shown in Figure 16.44. Assume that the MPS start quantities for A are 100 units in weeks 1, 2, 3, 4, 7, 8, 9, and 10.

Derive an MRP plan for the components going into product A using the data in Table 16.9.

What specific managerial actions are required in week 1? Make sure you address any specific difficulties you encounter in the inventory records.



▲ **FIGURE 16.44**
BOM for product A

TABLE 16.9 | INVENTORY RECORD DATA

	ITEM		
Data Category	**C**	**D**	**E**
Lot-sizing rule	L4L	L4L	FOQ = 300
Lead time	1 week	2 weeks	2 weeks
Scheduled receipts		100 (in week 2)	300 (in week 2)
Beginning Inventory	225	350	100

27. An end item's demand forecasts for the next 10 weeks are 35, 35, 35, 35, 20, 20, 30, 30, 30, and 30 units. The current on-hand inventory is 100 units. The order policy is to produce in lots of 75. The booked customer orders for the item, starting with week 1, are 15, 38, 7, 5, 0, 3, 10, 0, 0, and 0 units. The lead time is 2 weeks.

a. Develop an MPS for this end item.

b. The marketing department has received five orders for this item in the following sequence:

Order 1 is for 40 units to be delivered in period 1
Order 2 is for 75 units to be delivered in period 4
Order 3 is for 100 units to be delivered in period 6
Order 4 is for 125 units to be delivered in period 7
Order 5 is for 90 units to be delivered in period 10

Assuming that the prospective MPS you developed in part (a) does not change, which orders would you be able to accept based on the available to promise (ATP)?

28. An end item's demand forecasts for the next 6 weeks are 35 units, followed by forecasts of 30 units for weeks 7 though 10. The current on-hand inventory is 70 units. The order policy is to produce in lots of 120. The booked customer orders for the item, starting with week 1, are 17, 35, 15, 14, 0, 0, 12, 0, 0, and 0 units. The lead time is 2 weeks.

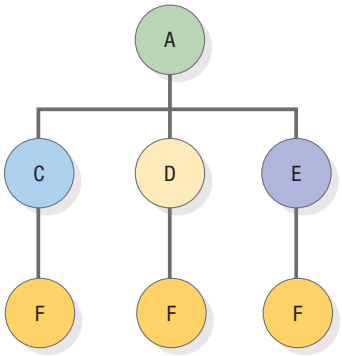
a. Develop an MPS for this end item.

b. The marketing department has received six orders for this item in the following sequence:

Order 1 is for 40 units to be delivered in period 3
Order 2 is for 60 units to be delivered in period 4
Order 3 is for 70 units to be delivered in period 6
Order 4 is for 40 units to be delivered in period 3
Order 5 is for 20 units to be delivered in period 5
Order 6 is for 115 units to be delivered in period 9

Assuming that the prospective MPS you developed in part (a) does not change, which orders would you be able to accept based on the available to promise (ATP)?

29. The bill of materials and the data from the inventory records for product A are shown in Figure 16.45. Assume that the MPS start quantities for A are 50 units in weeks 1, 2 and 3, and 150 units in weeks 6 and 8. Derive an MRP plan for the components going into product A using the data in Table 16.10.



▲ **FIGURE 16.45**
BOM for Product A

TABLE 16.10 | INVENTORY RECORD DATA

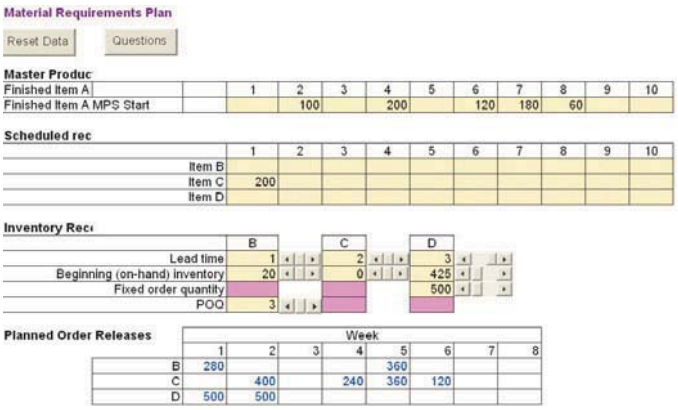
	ITEM			
Data Category	**C**	**D**	**E**	**F**
Lot-sizing rule	POQ ($P = 2$)	L4L	FOQ = 300	FOQ = 400
Lead time	1 week	1 week	2 weeks	4 weeks
Scheduled receipts		100 (week 2)		400 (week 1)
Beginning inventory	100	0	110	40

Active Model Exercise

This Active Model appears in MyOMLab. It allows you to evaluate the relationship between the inventory record data and the planned order releases.

QUESTIONS

1. As the lead time for item C changes, what happens to the order releases for items B, C, and D?
2. As the on-hand inventory for item C increases from 0 to 200, what happens to the order releases for items B, C, and D?
3. As the fixed order quantity (FOQ) for item D increases from 500 to 750, what happens to the order releases for items B, C, and D?
4. Suppose that the POQ for item B is changed from 3 weeks to 2 weeks. How does this change affect the order releases for items B, C, and D?



▲ **ACTIVE MODEL 16.1**
Material Requirements Planning Using Data from Solved Problem 3 and Table 16.1

CASE Flashy Flashers, Inc.

Flashy Flashers is a medium-sized firm employing 900 persons and 125 managerial and administrative personnel. The firm produces a line of automotive electrical components. It supplies about 95 auto parts stores and Moonbird Silverstreak car dealers in its region. Johnny Bennett, who serves as the president, founded the company by producing cable assemblies in his garage. By working hard, delivering consistent product quality, and by providing good customer service, he expanded his business to produce a variety of electrical components. Bennett's commitment to customer service is so strong that his company motto, "Love Thy Customers as Thyself," is etched on a big cast-iron plaque under his giant oil portrait in the building's front lobby.

The company's two most profitable products are the automotive front sidelamp and the headlamp. With the rising popularity of Eurosport sedans, such as the Moonbird Silverstreak, Flashy Flashers has enjoyed substantial demand for these two lamp items.

Last year, Kathryn Marley, the vice president of operations and supply chain management, approved the installation of a new MRP system. It is a first important step toward the eventual goal of a full-fledged ERP system. Marley worked closely with the task force that was created to bring MRP online. She frequently attended the training sessions for selected employees, emphasizing how MRP should help Flashy Flashers secure a better competitive edge.

A year later, the MRP system is working fairly well. However, Marley believes that there is always a better way and seeks to continually improve the company's processes. To get a better sense for potential improvements, she met with the production and inventory control manager, the shop supervisor, and the purchasing manager. Here are some of their observations.

Production and Inventory Control Manager

Inventory records and BOM files are accurate and well maintained. Inventory transactions are faithfully made when inventory is replenished or removed from the stockroom so that current on-hand balances are credible. There is a MRP explosion each week, which gives the company the new material

requirements plan. It provides information that helps identify when new orders need to be launched. Information can also be searched to help identify which scheduled receipts need to be expedited and which ones can be delayed by assigning them a later due date, thereby making room for more urgent jobs.

One planner suggested that the MRP outputs should be extended to provide priority and capacity reports, with pointers as to which items need their attention. The original plan was to get the order-launching capability implemented first. However, there is no formal system of priority planning, other than the initial due date assigned to each scheduled receipt when it is released, transforming it from a planned order release into a scheduled receipt. The due dates do not get updated later even when there are unexpected scrap losses, capacity shortages, short shipments, or last-minute changes in the MPS (responding to requests from favorite customers). Jobs are scheduled on the shop floor and by suppliers according to the EDD rule, based on their due dates. If due dates assigned to scheduled receipts were updated, it might help get open orders done when they are really needed. Furthermore, planned order releases in the action bucket are translated into scheduled receipts (using inventory transactions), after checking that its components are available. The current system does not consider possible capacity problems when releasing new orders.

Shop Supervisor

His primary complaint is that the shop workloads are anything but level. One week, they hardly have any work, and the supervisor overproduces (more than called for by the scheduled receipts) just to keep everyone busy. The next week can be just the opposite—so many new orders with short fuses that almost everyone needed to work overtime or else the scheduled receipt quantities are reduced to cover immediate needs. It is feast or famine, unless they make things work on the shop floor! They do make inventory transactions to report deviations from plan for the scheduled receipts, but these "overrides" make the scheduled receipt information in the MRP records more uncertain for the planners. A particular concern is to make sure that the bottleneck workstations are kept busy.

Purchasing

Buyers are putting out too many fires, leaving little time for creative buying. In such cases, their time is spent following up on orders that are required in the very near future or that are even late. Sometimes, the MRP plan shows planned order releases for purchased items that that are needed almost immediately, not allowing for the planned lead time. In checking the MRP records, the planned lead times are realistic and what the suppliers expect. Last week, things were fine for an item, and this week a rush order needs to be placed. What is the problem?

Marley tried to assimilate all this information. She decided to collect all the required information about the sidelamps and headlamps (shown in Table 16.11 through Table 16.14 and in Figure 16.46) to gain further insight into possible problems and identify areas for improvement.

Your Assignment

Put yourself in Marley's place and prepare a report on your findings. Specifically, you are required to do a manual MRP explosion for the sidelamps and headlamps for the next 6 weeks (beginning with the current week). Assume that it is now the start of week 1. Fill in the planned order releases form provided in Table 16.15. It should show the planned order releases for all items for the next 6 weeks. Include it in your report.

TABLE 16.11 | PART NUMBERS AND DESCRIPTIONS

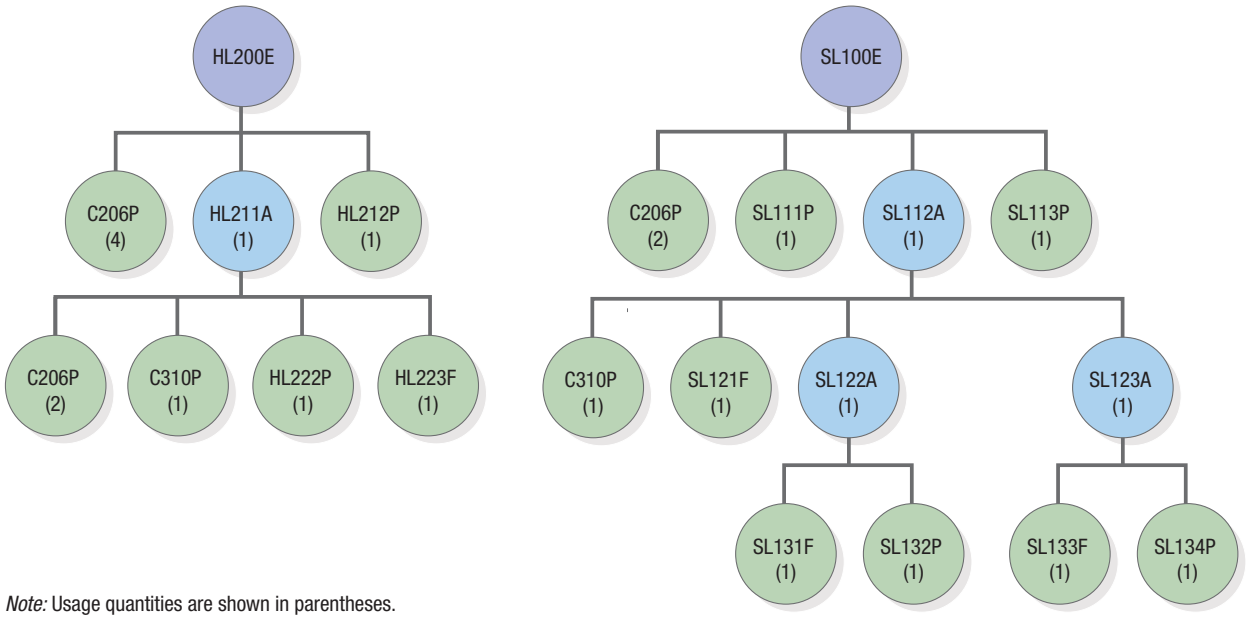
Part Number	Description
C206P	Screws
C310P	Back rubber gasket
HL200E	Headlamp
HL211A	Head frame subassembly
HL212P	Head lens
HL222P	Headlamp module
HL223F	Head frame
SL100E	Sidelamp
SL111P	Side lens
SL112A	Side frame subassembly
SL113P	Side lens rubber gasket
SL121F	Side frame
SL121F	Side frame
SL122A	Side bulb subassembly
SL123A	Flasher bulb subassembly
SL131F	Side cable grommet and receptacle
SL132P	Side bulb
SL133F	Flasher cable grommet and receptacle
SL134P	Flasher bulb

TABLE 16.12 | MASTER PRODUCTION SCHEDULE

Item Description and Part Number	Quantity	MPS Start Date
Headlamp (HL200E)	120	Week 4
	90	Week 5
	75	Week 6
Sidelamp (SL100E)	100	Week 3
	80	Week 5
	110	Week 6

TABLE 16.13 | REPLACEMENT PART DEMAND

Item Description and Part Number	Quantity	Date
Side lens (SL111P)	40	Week 3
	35	Week 6



Note: Usage quantities are shown in parentheses.

▲ **FIGURE 16.46**
BOMs for Headlamps and Sidelamps

Supplement your report with worksheets on the manual MRP explosion, and list the actions that planners should consider this week to (1) release new orders, (2) expedite scheduled receipts, and (3) delay a scheduled receipt's due date.

Finally, identify the good and bad points of MRP implementation at Flashy Flashers. Conclude by making suggestions on ways to improve its resource planning process.

TABLE 16.14 | SELECTED DATA FROM INVENTORY RECORDS

Part Number	Lead Time (Weeks)	Safety Stock (Units)	Lot-Sizing Rule	On-Hand (Units)	Scheduled Receipt (Units and Due Dates)
C206P	1	30	FOQ = 2,500	270	—
C310P	1	20	FOQ = 180	40	180 (week 1)
HL211A	3	0	L4L	0	
HL212P	2	15	FOQ = 350	15	—
HL222P	3	10	POQ (P = 4 weeks)	10	285 (week 1)
HL223F	1	0	POQ (P = 4 weeks)	0	120 (week 1)
SL111P	2	0	FOQ = 350	15	—
SL112A	3	0	L4L	20	80 (week 2)
SL113P	1	20	FOQ = 100	20	—
SL121F	2	0	L4L	0	80 (week 2)
SL122A	2	0	L4L	0	80 (week 2)
SL123A	2	0	FOQ = 200	0	—
SL131F	2	0	POQ (P = 2 weeks)	0	110 (week 1)
SL132P	1	25	FOQ = 100	35	100 (week 1)
SL133F	2	0	FOQ = 250		—
SL134P	1	0	FOQ = 400	100	—

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TABLE 16.15 | PLANNED ORDER RELEASE FORM

FILL IN THE PLANNED ORDER RELEASES FOR ALL COMPONENTS.

	Week					
Item Description and Part Number	**1**	**2**	**3**	**4**	**5**	**6**
Side lens (SL111P)						
Side lens rubber gasket (SL113P)						
Side frame subassembly (SL112A)						
Side frame (SL121F)						
Side bulb subassembly (SL122A)						
Flasher bulb subassembly (SL123A)						
Side cable grommet and receptacle (SL131F)						
Flasher cable grommet and receptacle (SL133F)						
Side bulb (SL132P)						
Flasher bulb (SL134P)						
Head frame subassembly (HL211A)						
Head lens (HL212P)						
Headlamp module (HL222P)						
Head frame (HL223F)						
Back rubber gasket (C310P)						
Screws (C206P)						

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SUPPLEMENT

D

LINEAR PROGRAMMING MODELS

In many business situations, resources are limited and demand for them is great. For example, a limited number of vehicles may have to be scheduled to make multiple trips to customers, or a staffing plan may have to be developed to cover expected variable demand with the fewest employees. In this supplement, we describe a technique called **linear programming**, which is useful for allocating scarce resources among competing demands. The resources may be time, money, or materials, and the limitations are known as constraints. Linear programming can help managers find the best allocation solution and provide information about the value of additional resources.

linear programming

A technique that is useful for allocating scarce resources among competing demands.

LEARNING GOALS *After reading this supplement, you should be able to:*

1. Describe formulating models for various problems.
2. Demonstrate graphic analysis and solutions for two-variable problems.
3. Define slack variables, surplus variables, and sensitivity analysis.
4. Interpret computer output of a linear programming solution.
5. Apply the transportation method to Sales and Operations Planning (S&OP) problems.

Basic Concepts

Before we can demonstrate how to solve problems in operations and supply chain management with linear programming, we must first explain several characteristics of all linear programming models and mathematical assumptions that apply to them: (1) objective function, (2) decision variables, (3) constraints, (4) feasible region, (5) parameters, (6) linearity, and (7) nonnegativity.

Linear programming is an *optimization* process. A single **objective function** states mathematically what is being maximized (e.g., profit or present value) or minimized (e.g., cost or scrap). The objective function provides the scorecard on which the attractiveness of different solutions is judged.

objective function

An expression in linear programming models that states mathematically what is being maximized or minimized.

Decision variables represent choices that the decision maker can control. Solving the problem yields their optimal values. For example, a decision variable could be the number of units of a product to make next month or the number of units of inventory to hold next month. Linear programming is based on the assumption that decision variables are *continuous*; they can be fractional quantities and need not be whole numbers. Often, this assumption is realistic, as when the decision variable is expressed in dollars, hours, or some other continuous measure. Even when the decision variables represent nondivisible units, such as workers, tables, or trucks, we

decision variables

Variables that represent the choices the decision maker can control.

sometimes can simply round the linear programming solution up or down to get a reasonable solution that does not violate any constraints, or we can use a more advanced technique, called *integer programming*.

constraints
In linear programming, the limitations that restrict the permissible choices for the decision variables.

Constraints are limitations that restrict the permissible choices for the decision variables. Each limitation can be expressed mathematically in one of three ways: a less-than-or-equal-to ($\leq$), an equal-to ($=$), or a greater-than-or-equal-to ($\geq$) constraint. A $\leq$ constraint puts an upper limit on some function of decision variables and most often is used with maximization problems. For example, a $\leq$ constraint may specify the maximum number of customers who can be served or the capacity limit of a machine. An $=$ constraint means that the function must equal some value. For example, 100 (not 99 or 101) units of one product must be made. An $=$ constraint often is used for certain mandatory relationships, such as the fact that ending inventory always equals beginning inventory plus production minus sales. A $\geq$ constraint puts a lower limit on some function of decision variables. For example, a $\geq$ constraint may specify that production of a product must exceed or equal demand.

feasible region
A region that represents all permissible combinations of the decision variables in a linear programming model.

Every linear programming problem must have one or more constraints. Taken together, the constraints define a **feasible region**, which represents all permissible combinations of the decision variables. In some unusual situations, the problem is so tightly constrained that there is only one possible solution—or perhaps none. However, in the usual case, the feasibility region contains infinitely many possible solutions, assuming that the feasible combinations of the decision variables can be fractional values. The goal of the decision maker is to find the best possible solution.

parameter
A value that the decision maker cannot control and that does not change when the solution is implemented.

The objective function and constraints are functions of decision variables and parameters. A **parameter**, also known as a *coefficient* or *given constant*, is a value that the decision maker cannot control and that does not change when the solution is implemented. Each parameter is assumed to be known with **certainty**. For example, a computer programmer may know that running a software program will take 30 minutes—no more, no less.

certainty
The word that is used to describe that a fact is known without doubt.

linearity
A characteristic of linear programming models that implies proportionality and additivity—there can be no products or powers of decision variables.

The objective function and constraint equations are assumed to be linear. **Linearity** implies proportionality and additivity—there can be no products (e.g., $10x_1x_2$) or powers (e.g., x_1^3) of decision variables. Suppose that the profit gained by producing two types of products (represented by decision variables x_1 and x_2) is $2x_1 + 3x_2$. Proportionality implies that one unit of x_1 contributes \$2 to profits and two units contribute \$4, regardless of how much of x_1 is produced. Similarly, each unit of x_2 contributes \$3, whether it is the first or the tenth unit produced. Additivity means that the total objective function value equals the profits from x_1 plus the profits from x_2.

nonnegativity
An assumption that the decision variables must be positive or zero.

Finally, we make an assumption of **nonnegativity**, which means that the decision variables must be positive or zero. A firm that makes spaghetti sauce, for example, cannot produce a negative number of jars. To be formally correct, a linear programming formulation should show a ≥ 0 constraint for each decision variable.

Although the assumptions of linearity, certainty, and continuous variables are restrictive, linear programming can help managers analyze many complex resource allocation problems. The process of building the model forces managers to identify the important decision variables and constraints, which is a useful step in its own right. Identifying the nature and scope of the problem represents a major step toward solving it. In a later section, we show how sensitivity analysis can help the manager deal with uncertainties in the parameters and answer "what-if" questions.

Formulating a Problem

Linear programming applications begin with the formulation of a *model* of the problem with the general characteristics just described. We illustrate the modeling process here with the **product-mix problem**, which is a one-period type of planning problem, the solution of which yields optimal output quantities (or product mix) of a group of services or products subject to resource capacity and market demand constraints. This problem was first introduced in Chapter 7, "Managing Process Constraints," and now we take it up more formally. Formulating a model to represent each unique problem, using the following three-step sequence, is the most creative and perhaps the most difficult part of linear programming.

product-mix problem
A one-period type of planning problem, the solution of which yields optimal output quantities (or product mix) of a group of services or products subject to resource capacity and market demand constraints.

Step 1. Define the Decision Variables. What must be decided? Define each decision variable specifically, remembering that the definitions used in the objective function must be equally useful in the constraints. The definitions should be as specific as possible. Consider the following two alternative definitions:

$x_1 =$ product 1
$x_1 =$ number of units of product 1 to be produced and sold next month

The second definition is much more specific than the first, making the remaining steps easier.

Step 2. Write Out the Objective Function. What is to be maximized or minimized? If it is next month's profits, write out an objective function that makes next month's profits a linear function of the decision variables. Identify parameters to go with each decision variable. For example, if each unit of x_1 sold yields a profit of \$7, the total profit from product $x_1 = 7x_1$. If a variable has no impact on the objective function, its objective function coefficient is 0. The objective function often is set equal to Z, and the goal is to maximize or minimize Z.

Step 3. Write Out the Constraints. What limits the values of the decision variables? Identify the constraints and the parameters for each decision variable in them. As with the objective function, the parameter for a variable that has no impact in a constraint is 0. To be formally correct, also write out the nonnegativity constraints.

As a consistency check, make sure that the same unit of measure is being used on both sides of each constraint and in the objective function. For example, suppose that the right-hand side of a constraint is hours of capacity per month. Then, if a decision variable on the left-hand side of the constraint measures the number of units produced per month, the dimensions of the parameter that is multiplied by the decision variable must be hours per unit because

$$\left(\frac{\text{Hours}}{\cancel{\text{Unit}}}\right)\left(\frac{\cancel{\text{Units}}}{\text{Month}}\right) = \left(\frac{\text{Hours}}{\text{Month}}\right)$$

Of course, you can also skip around from one step to another, depending on the part of the problem that has your attention. If you cannot get past step 1, try a new set of definitions for the decision variables. Often the problem can be modeled correctly in more than one way.

EXAMPLE D.1 Formulating a Linear Programming Model

The Stratton Company produces two basic types of plastic pipe. Three resources are crucial to the output of pipe: extrusion hours, packaging hours, and a special additive to the plastic raw material. The following data represent next week's situation. All data are expressed in units of 100 feet of pipe.

	PRODUCT		
Resource	**Type 1**	**Type 2**	**Resource Availability**
Extrusion	4 hr	6 hr	48 hr
Packaging	2 hr	2 hr	18 hr
Additive mix	2 lb	1 lb	16 lb

The contribution to profits and overhead per 100 feet of pipe is \$34 for type 1 and \$40 for type 2. Formulate a linear programming model to determine how much of each type of pipe should be produced to maximize contribution to profits and to overhead, assuming that everything produced can be sold.

SOLUTION

Step 1. To define the decision variables that determine product mix, we let

x_1 = amount of type 1 pipe to be produced and sold next week, measured in 100-foot increments (e.g., $x_1 = 2$ means 200 feet of type 1 pipe)

and

x_2 = amount of type 2 pipe to be produced and sold next week, measured in 100-foot increments

Step 2. Next, we define the objective function. The goal is to maximize the total contribution that the two products make to profits and overhead. Each unit of x_1 yields \$34, and each unit of x_2 yields \$40. For specific values of x_1 and x_2, we find the total profit by multiplying the number of units of each product produced by the profit per unit and adding them. Thus, our objective function becomes

$$\text{Maximize: } \$34x_1 + \$40x_2 = Z$$

Step 3. The final step is to formulate the constraints. Each unit of x_1 and x_2 produced consumes some of the critical resources. In the extrusion department, a unit of x_1 requires 4 hours and a unit of x_2 requires

6 hours. The total must not exceed the 48 hours of capacity available, so we use the $\leq$ sign. Thus, the first constraint is

$$4x_1 + 6x_2 \leq 48$$

Similarly, we can formulate constraints for packaging and raw materials:

$$2x_1 + 2x_2 \leq 18 \text{ (packaging)}$$
$$2x_1 + x_2 \leq 16 \text{ (additive mix)}$$

These three constraints restrict our choice of values for the decision variable because the values we choose for x_1 and x_2 must satisfy all of the constraints. Negative values for x_1 and x_2 do not make sense, so we add nonnegativity restrictions to the model:

$$x_1 \geq 0 \text{ and } x_2 \geq 0 \text{ (nonnegativity restrictions)}$$

We can now state the entire model, made complete with the definitions of variables.

$$\text{Maximize: } \$34x_1 + \$40x_2 = Z$$
$$\text{Subject to: } 4x_1 + 6x_2 \leq 48$$
$$2x_1 + 2x_2 \leq 18$$
$$2x_1 + x_2 \leq 16$$
$$x_1 \geq 0 \text{ and } x_2 \geq 0$$

where

x_1 = amount of type 1 pipe to be produced and sold next week, measured in 100-foot increments

x_2 = amount of type 2 pipe to be produced and sold next week, measured in 100-foot increments

Graphic Analysis

graphic method of linear programming

A type of graphic analysis that involves the following five steps: plotting the constraints, identifying the feasible region, plotting an objective function line, finding a visual solution, and finding the algebraic solution.

With the model formulated, we now seek the optimal solution. In practice, most linear programming problems are solved with a computer. However, insight into the meaning of the computer output—and linear programming concepts in general—can be gained by analyzing a simple two-variable problem with the **graphic method of linear programming**. Hence, we begin with the graphic method, even though it is not a practical technique for solving problems that have three or more decision variables. The five basic steps are (1) *plot the constraints,* (2) *identify the feasible region,* (3) *plot an objective function line,* (4) *find the visual solution,* and (5) *find the algebraic solution.*

Plot the Constraints

We begin by plotting the constraint equations, disregarding the inequality portion of the constraints ($<$ or $>$). Making each constraint an equality ($=$) transforms it into the equation for a straight line. The line can be drawn as soon as we identify two points on it. Any two points reasonably spread out may be chosen; the easiest ones to find are the *axis intercepts,* where the line intersects each axis. To find the x_1 axis intercept, set x_2 equal to 0 and solve the equation for x_1. For the Stratton Company in Example D.1, the equation of the line for the extrusion process is

$$4x_1 + 6x_2 = 48$$

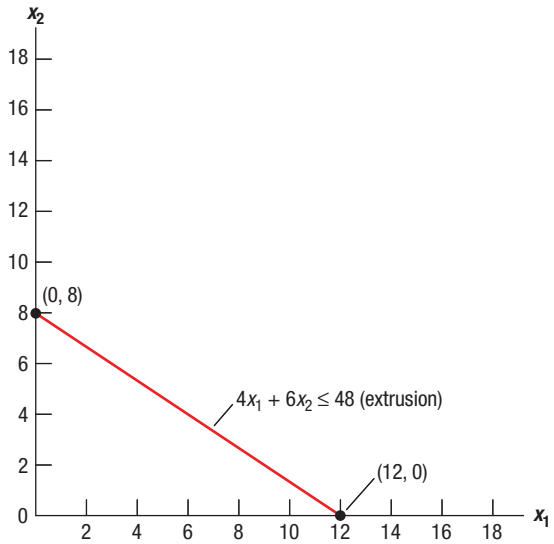
For the x_1 axis intercept, $x_2 = 0$, so

$$4x_1 + 6(0) = 48$$
$$x_1 = 12$$

To find the x_2 axis intercept, set $x_1 = 0$ and solve for x_2:

$$4(0) + 6x_2 = 48$$
$$x_2 = 8$$

We connect points (0, 8) and (12, 0) with a straight line, as shown in Figure D.1.



◀ **FIGURE D.1**
Graph of the Extrusion Constraint

EXAMPLE D.2 Plotting the Constraints

For the Stratton Company problem, plot the other constraints: one constraint for packaging and one constraint for the additive mix.

MyOMLab
Active Model D.1 in MyOMLab offers many insights on graphic analysis and sensitivity analysis. Use it when studying Examples D.2 through D.4.

MyOMLab
Tutor D.1 in MyOMLab provides a new practice example for plotting the constraints.

SOLUTION

The equation for the packaging process's line is $2x_1 + 2x_2 = 18$. To find the x_1 intercept, set $x_2 = 0$:

$$2x_1 + 2(0) = 18$$
$$x_1 = 9$$

To find the x_2 axis intercept, set $x_1 = 0$:

$$2(0) + 2x_2 = 18$$
$$x_2 = 9$$

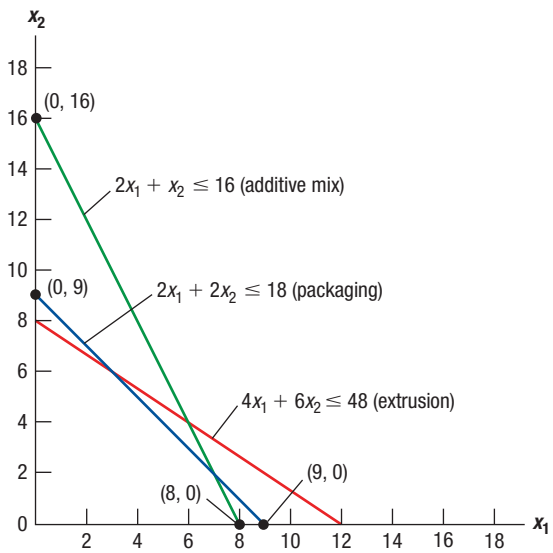
The equation for the additive mix's line is $2x_1 + x_2 = 16$. To find the x_1 intercept, set $x_2 = 0$:

$$2x_1 + 0 = 16$$
$$x_1 = 8$$

To find the x_2 axis intercept, set $x_1 = 0$:

$$2(0) + x_2 = 16$$
$$x_2 = 16$$

With a straight line, we connect points (0, 9) and (9, 0) for the packaging constraint and points (0, 16) and (8, 0) for the additive mix constraint. Figure D.2 shows the graph with all three constraints plotted.



◀ **FIGURE D.2**
Graph of the Three Constraints

Identify the Feasible Region

The feasible region is the area on the graph that contains the solutions that satisfy all the constraints simultaneously, including the nonnegativity restrictions. To find the feasible region, first locate the feasible points for each constraint and then the area that satisfies all constraints. Generally, the following three rules identify the feasible points for a given constraint:

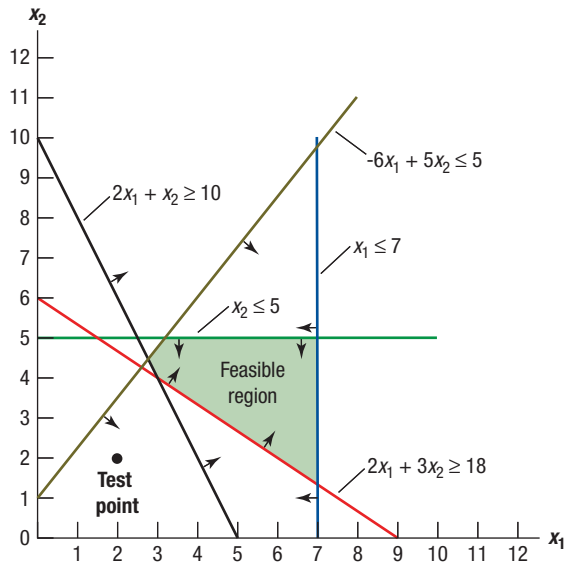
1. For the $=$ constraint, only the points on the line are feasible solutions.
2. For the $\leq$ constraint, the points on the line and the points below or to the left of the line are feasible solutions.
3. For the $\geq$ constraint, the points on the line and the points above or to the right of the line are feasible solutions.

Exceptions to these rules occur when one or more of the parameters on the left-hand side of a constraint are negative. In such cases, we draw the constraint line and test a point on one side of it. If the point does not satisfy the constraint, it is in the infeasible part of the graph. Suppose that a linear programming model has the following five constraints plus the two nonnegativity constraints:

$$2x_1 + x_2 \geq 10$$
$$2x_1 + 3x_2 \geq 18$$
$$x_1 \leq 7$$
$$x_2 \leq 5$$
$$-6x_1 + 5x_2 \leq 5$$
$$x_1, x_2 \geq 0$$

The feasible region is the shaded portion of Figure D.3. The arrows shown on each constraint identify which side of each line is feasible. The rules work for all but the fifth constraint, which has a negative parameter, -6, for x_1. We arbitrarily select (2, 2) as the test point, which Figure D.3 shows is below the line and to the right. At this point, we find $-6(2) + 5(2) = -2$. Because -2 does not exceed 5, the portion of the figure containing (2, 2) is feasible, at least for this fifth constraint.

FIGURE D.3 ▶
Identifying the Feasible Region

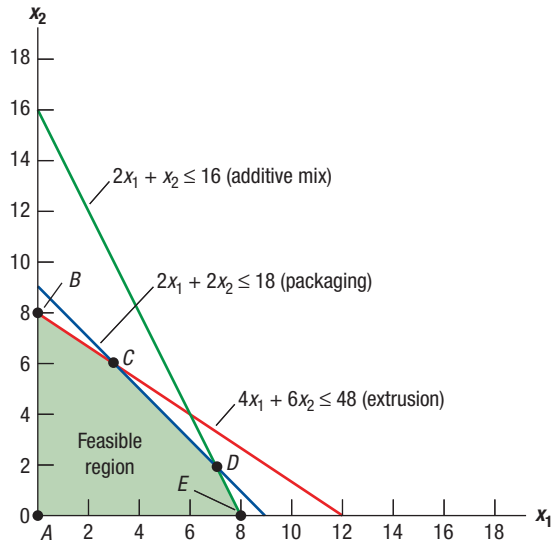


EXAMPLE D.3 **Identifying the Feasible Region**

Identify the feasible region for the Stratton Company problem.

SOLUTION

Because the problem contains only $\leq$ constraints, and the parameters on the left-hand side of each constraint are not negative, the feasible portions are to the left of and below each constraint. The feasible region, shaded in Figure D.4, satisfies all three constraints simultaneously.



◀ **FIGURE D.4**
Identifying the Feasible Region

Plot an Objective Function Line

Now we want to find the solution that optimizes the objective function. Even though all the points in the feasible region represent possible solutions, we can limit our search to the corner points. A **corner point** lies at the intersection of two (or possibly more) constraint lines on the boundary of the feasible region. No interior points in the feasible region need be considered because at least one corner point is better than any interior point. Similarly, other points on the boundary of the feasible region can be ignored because a corner point is at least as good as any of them.

corner point
A point that lies at the intersection of two (or possibly more) constraint lines on the boundary of the feasible region.

In Figure D.4, the five corner points are marked *A*, *B*, *C*, *D*, and *E*. Point *A* is the origin (0, 0) and can be ignored because any other feasible point is a better solution. We could try each of the other corner points in the objective function and select the one that maximizes *Z*. For example, corner point *B* lies at (0, 8). If we substitute these values into the objective function, the resulting *Z* value is 320:

$$34x_1 + 40x_2 = Z$$
$$34(0) + 40(8) = 320$$

However, we may not be able to read accurately the values of x_1 and x_2 for some of the points (e.g., *C* or *D*) on the graph. Algebraically solving two linear equations for each corner point also is inefficient when there are many constraints and, thus, many corner points.

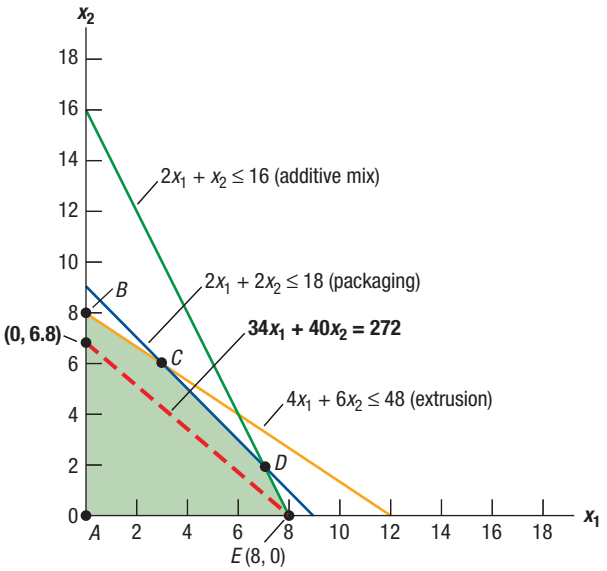
The best approach is to plot the objective function on the graph of the feasible region for some arbitrary *Z* values. From these objective function lines, we can spot the best solution visually. If the objective function is profits, each line is called an *iso-profit line* and every point on that line will yield the same profit. If *Z* measures cost, the line is called an *iso-cost line* and every point on it represents the same cost. We can simplify the search by plotting the first line in the feasible region—somewhere near the optimal solution, we hope. For the Stratton Company example, let us pass a line through point *E* (8, 0). This point is a corner point. It might even be the optimal solution because it is far from the origin. To draw the line, we first identify its *Z* value as $34(8) + 40(0) = 272$. Therefore, the equation for the objective function line passing through *E* is

$$34x_1 + 40x_2 = 272$$

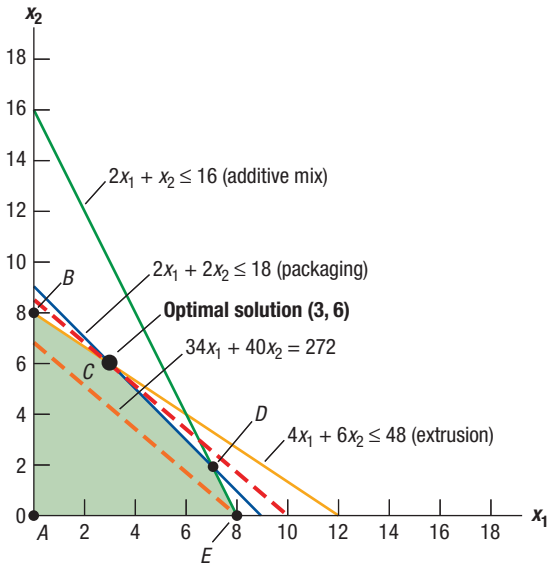
Every point on the line defined by this equation has an objective function *Z* value of 272. To draw the line, we need to identify a second point on it and then connect the two points. Let us use the x_2 intercept, where $x_1 = 0$:

$$34(0) + 40x_2 = 272$$
$$x_2 = 6.8$$

Figure D.5 shows the iso-profit line that connects points (8, 0) and (0, 6.8). A series of other dashed lines could be drawn parallel to this first line. Each would have its own *Z* value. Lines above the first line we drew would have higher *Z* values. Lines below it would have lower *Z* values.



▲ **FIGURE D.5**
Passing an Iso-Profit Line Through (8, 0)



▲ **FIGURE D.6**
Drawing the Second Iso-Profit Line

Find the Visual Solution

We now eliminate corner points A and E from consideration as the optimal solution because better points lie above and to the right of the $Z = 272$ iso-profit line. Our goal is to maximize profits, so the best solution is a point on the iso-profit line *farthest* from the origin, but still touching the feasible region. (For minimization problems, it is a point in the feasible region on the iso-cost line *closest* to the origin.)[1] To identify which of the remaining corner points is optimal (*B*, *C*, or *D*), we draw, parallel to the first line, one or more iso-profit lines that give better *Z* values (higher for maximization and lower for minimization). The line that just touches the feasible region identifies the optimal solution. For the Stratton Company problem, Figure D.6 shows the second iso-profit line. The optimal solution is the last point touching the feasible region: point *C*. It appears to be in the vicinity of (3, 6), but the visual solution is not exact.

A linear programming problem can have more than one optimal solution. This situation occurs when the objective function is parallel to one of the faces of the feasible region. Such would be the case if our objective function in the Stratton Company problem were $\$38x_1 + \$38x_2$. Points (3, 6) and (7, 2) would be optimal, as would any other point on the line connecting these two corner points. In such a case, management probably would base a final decision on nonquantifiable factors. It is important to understand, however, that we need to consider only the corner points of the feasible region when optimizing an objective function.

Find the Algebraic Solution

To find an exact solution, we must use algebra. We begin by identifying the pair of constraints that define the corner point at their intersection. We then list the constraints as equations and solve them simultaneously to find the coordinates x_1, x_2 of the corner point. Simultaneous equations can be solved several ways. For small problems, the easiest way is as follows:

Step 1. Develop an equation with just one unknown. Start by multiplying both sides of one equation by a constant so that the coefficient for one of the two decision variables is *identical* in both equations. Then, subtract one equation from the other and solve the resulting equation for its single unknown variable.

Step 2. Insert this decision variable's value into either one of the original constraints and solve for the other decision variable.

[1]The statements "farthest from the origin" or "closest to the origin" would no longer be true if there are negative coefficients in the objective function.

EXAMPLE D.4 Finding the Optimal Solution Algebraically

Find the optimal solution algebraically for the Stratton Company problem. What is the value of Z when the decision variables have optimal values?

MyOMLab
Tutor D.2 in MyOMLab provides a new practice example for finding the optimal solution.

SOLUTION

Step 1. Figure D.6 showed that the optimal corner point lies at the intersection of the extrusion and packaging constraints. Listing the constraints as equalities, we have

$$4x_1 + 6x_2 = 48 \text{ (extrusion)}$$
$$2x_1 + 2x_2 = 18 \text{ (packaging)}$$

We multiply each term in the packaging constraint by 2. The packaging constraint now is $4x_1 + 4x_2 = 36$. Next, we subtract the packaging constraint from the extrusion constraint. The result will be an equation from which x_1 has dropped out. (Alternatively, we could multiply the second equation by 3 so that x_2 drops out after the subtraction.) Thus,

$$\begin{aligned} 4x_1 + 6x_2 &= 48 \\ -(4x_1 + 4x_2 &= 36) \\ \hline 2x_2 &= 12 \\ x_2 &= 6 \end{aligned}$$

Step 2. Substituting the value of x_2 into the extrusion equation, we get

$$\begin{aligned} 4x_1 + 6(6) &= 48 \\ 4x_1 &= 12 \\ x_1 &= 3 \end{aligned}$$

Thus, the optimal point is (3, 6). This solution gives a total profit of $34(3) + 40(6) = \$342$.

DECISION POINT

Management at the Stratton Company decided to produce 300 feet of type 1 pipe and 600 feet of type 2 pipe for the next week for a total profit of $342.

Slack and Surplus Variables

Figure D.6 shows that the optimal product mix will exhaust all the extrusion and packaging resources because at the optimal corner point (3, 6) the two constraints are equalities. Substituting the values of x_1 and x_2 into these constraints shows that the left-hand sides equal the right-hand sides:

$$4(3) + 6(6) = 48 \text{ (extrusion)}$$
$$4(3) + 2(6) = 18 \text{ (packaging)}$$

A constraint (such as the one for extrusion) that helps form the optimal corner point is called a **binding constraint** because it limits the ability to improve the objective function. If a binding constraint is relaxed, or made less restrictive, a better solution is possible. Relaxing a constraint means increasing the right-hand-side parameter for a $\leq$ constraint or decreasing it for a $\geq$ constraint. No improvement is possible from relaxing a constraint that is not binding, such as the additive mix constraint in Figure D.6. If the right-hand side was increased from 16 to 17 and the problem solved again, the optimal solution would not change. In other words, there is already more additive mix than needed.

binding constraint
A constraint that helps form the optimal corner point; it limits the ability to improve the objective function.

For nonbinding inequality constraints, knowing how much the left and right sides differ is helpful. Such information tells us how close the constraint is to becoming binding. For a $\leq$ constraint, the amount by which the left-hand side falls short of the right-hand side is called **slack**. For a $\geq$ constraint, the amount by which the left-hand side exceeds the right-hand side is called **surplus**. To find the slack for a $\leq$ constraint algebraically, we *add* a slack variable to the constraint and convert it to an equality. Then, we substitute in the values of the decision variables and solve for the slack. For example, the additive mix constraint in Figure D.6, $2x_1 + x_2 \leq 16$, can be rewritten by adding slack variable s_1:

$$2x_1 + x_2 + s_1 = 16$$

We then find the slack at the optimal solution (3, 6):

$$\begin{aligned} 2(3) + 6 + s_1 &= 16 \\ s_1 &= 4 \end{aligned}$$

slack
The amount by which the left-hand side of a linear programming constraint falls short of the right-hand side.

surplus
The amount by which the left-hand side of a linear programming constraint exceeds the right-hand side.

The procedure is much the same to find the surplus for a $\geq$ constraint, except that we *subtract* a surplus variable from the left-hand side. Suppose that $x_1 + x_2 \geq 6$ was another constraint in the Stratton Company problem, representing a lower bound on the number of units produced. We would then rewrite the constraint by subtracting a surplus variable s_2:

$$x_1 + x_2 - s_2 = 6$$

MyOMLab

Tutor D.3 in MyOMLab provides another practice example for finding slack.

The surplus at the optimal solution (3, 6) would be

$$3 + 6 - s_2 = 6$$
$$s_2 = 3$$

Sensitivity Analysis

Rarely are the parameters in the objective function and constraints known with certainty. Often, they are just estimates of actual values. For example, the available packaging and extrusion hours for the Stratton Company are estimates that do not reflect the uncertainties associated with absenteeism or personnel transfers, and the required hours per unit to package and extrude may be time estimates that essentially are averages. Likewise, profit contributions used for the objective function coefficients do not reflect uncertainties in selling prices and such variable costs as wages, raw materials, and shipping.

MyOMLab

Despite such uncertainties, initial estimates are needed to solve the problem. Accounting, marketing, and time-standard information systems (see MyOMLab Supplement H) often provide these initial estimates. After solving the problem using these estimated values, the analyst can determine how much the optimal values of the decision variables and the objective function value Z would be affected if certain parameters had different values. This type of postsolution analysis for answering "what-if" questions is called *sensitivity analysis*.

One way of conducting sensitivity analysis for linear programming problems is the brute-force approach of changing one or more parameter values and resolving the entire problem. This approach may be acceptable for small problems, but it is inefficient if the problem involves many parameters. For example, brute-force sensitivity analysis using 3 separate values for each of 20 objective function coefficients requires 3^{20}, or 3,486,784,401, separate solutions! Fortunately, efficient methods are available for getting sensitivity information without resolving the entire problem, and they are routinely used in most linear programming computer software packages. Table D.1 describes the four basic types of sensitivity analysis information provided by linear programming.

TABLE D.1 | SENSITIVITY ANALYSIS INFORMATION PROVIDED BY LINEAR PROGRAMMING

Key Term	Definition
Reduced cost	How much the objective function coefficient of a decision variable must improve (increase for maximization or decrease for minimization) before the optimal solution changes and the decision variable "enters" the solution with some positive number
Shadow price	The marginal improvement in Z (increase for maximization and decrease for minimization) caused by relaxing the constraint by one unit
Range of optimality	The interval (lower and upper bounds) of an objective function coefficient over which the optimal values of the decision variables remain unchanged
Range of feasibility	The interval (lower and upper bounds) over which the right-hand-side parameter of a constraint can vary while its shadow price remains valid

Computer Solution

Most real-world linear programming problems are solved on a computer, so we concentrate here on understanding the use of linear programming and the logic on which it is based. The solution procedure in computer codes is some form of the **simplex method**, which is an iterative algebraic procedure for solving linear programming problems.

simplex method

An iterative algebraic procedure for solving linear programming problems.

Simplex Method

The graphic analysis gives insight into the logic of the simplex method, beginning with the focus on corner points. If there is any feasible solution to a problem, at least one corner point will always be the optimum, even when multiple optimal solutions are available. Thus, the simplex method starts with an initial corner point and then systematically evaluates other corner points in such a way that the objective function improves (or, at worst, stays the same) at each iteration. In the Stratton Company problem, an improvement would be an increase in profits. When no more

improvements are possible, the optimal solution has been found.[2] The simplex method also helps generate the sensitivity analysis information that we developed graphically.

Each corner point has no more than m variables that are greater than 0, where m is the number of constraints (not counting the nonnegativity constraints). The m variables include slack and surplus variables, not just the original decision variables. Because of this property, we can find a corner point by simultaneously solving m constraints, where all but m variables are set equal to 0. For example, point B in Figure D.6 has three nonzero variables: x_2, the slack variable for packaging, and the slack variable for the additive mix. Their values can be found by simultaneously solving the three constraints, with x_1 and the slack variable for extrusion equal to 0. After finding this corner point, the simplex method applies information similar to the reduced costs to decide which new corner point to find next that gives an even better Z value. It continues in this way until no better corner point is possible. The final corner point evaluated is the optimal one.

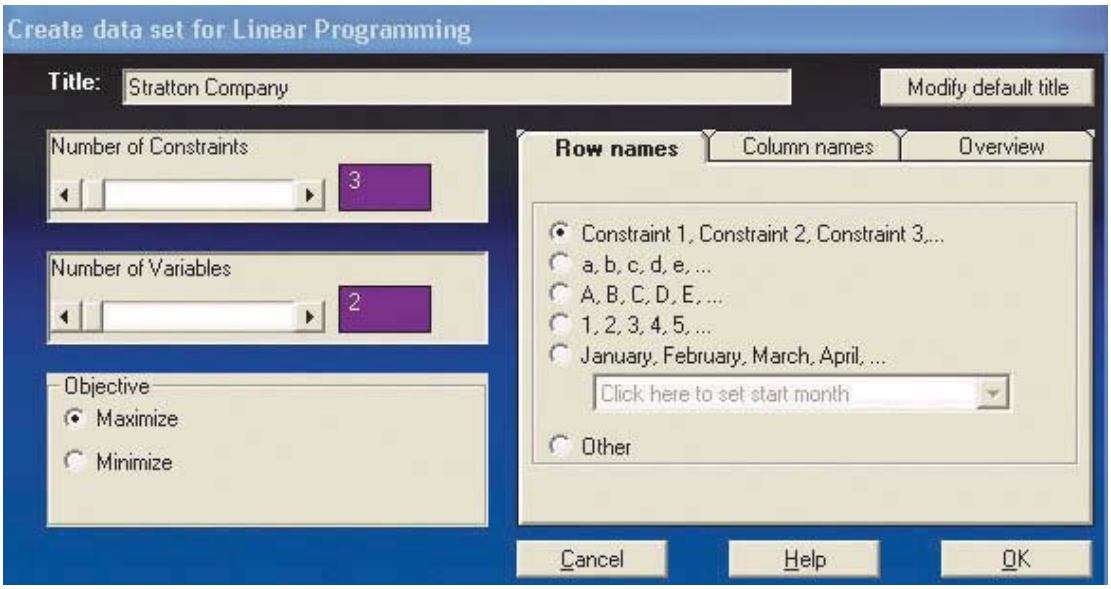
Computer Output

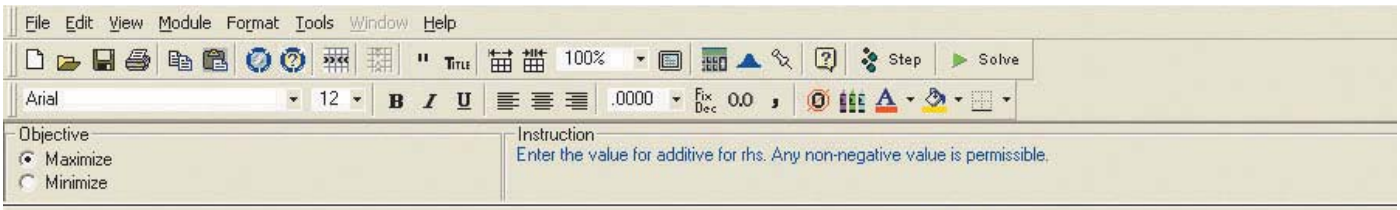
Computer programs dramatically reduce the amount of time required to solve linear programming problems. The capabilities and displays of software packages are not uniform. For example, POM for Windows in MyOMLab can handle small- to medium-sized linear programming problems. Inputs are made easily and nonnegativity constraints need not be entered. Microsoft's *Excel* Solver offers a second option for similar problem sizes. More advanced software for larger problems is available from multiple sources.

MyOMLab

Here we show output from POM for Windows when applied to the Stratton Company. Figure D.7 shows the two *data entry* screens. The first screen allows you to enter the problem's

▼ **FIGURE D.7**
Data Entry Screens





Stratton Company

	X1	X2		RHS	Equation form
Maximize	34	40			Max 34X1 + 40X2
Extrusion	4	6	<=	48	4X1 + 6X2 <= 48
Packaging	2	2	<=	18	2X1 + 2X2 <= 18
Additive	3	1	<=	16	3X1 + X2 <= 0

[2] For more information on how to perform the simplex method manually, see Render, Barry, Ralph M. Stair, and Michael E. Hanna. *Quantitative Analysis for Management,* 11th ed. or any other current textbook on management science.

name, specify the number of constraints and decision variables, and choose between maximization and minimization. After making these inputs and clicking the OK button, the data table screen is shown. Enter the parameters, give names to each constraint and decision variable (as desired), and specify the type of relationship ($\leq$, $=$, $\geq$) for each constraint. The second screen in Figure D.7 shows the completed data table. The user may customize labels for the decision variables, right-hand-side values, and constraints. Here, the first decision variable is labeled as "X1," the right-hand-side values as "RHS," and the extrusion constraint as "Extrusion." Slack and surplus variables will be added automatically as needed. When all of the inputs are made, click the green arrow labeled "Solve" in the upper-right corner.

Figure D.8 displays the solution with the *Results* screen. All output confirms our earlier calculations and the graphic analysis. Of particular interest is the bottom row that gives the optimal values of the decision variables ($x_1 = 3$ and $x_2 = 6$), and also the optimal value of the objective function ($342). The shadow prices for each constraint are given in the last column.

▼ **FIGURE D.8**
Results Screen

	Stratton Company Solution				
	X1	X2		RHS	Shadow Price
Maximize	34	40			
Constraint 1	4	6	<=	48	3
Constraint 2	2	2	<=	18	11
Constraint 3	2	1	<=	16	0
Solution->	3	6		342	

Click on the Window icon and select the second option in the drop down menu to switch to the *Ranging* screen. It is shown in Figure D.9. The top half deals with the decision variables. Of particular interest are the reduced costs and the lower and upper bounds. Two tips on interpreting the *reduced cost* information are as follows:

1. It is relevant only for a decision variable that is 0 in the optimal solution. If the decision variable is greater than 0, ignore the reduced cost number. Thus, for the Stratton Company problem the reduced cost numbers provide no new insight because they are always 0 when decision variables have positive values in the optimal solution. Look instead at the lower and upper bounds on the objective function coefficients.
2. It tells how much the objective function coefficient of a decision variable that is 0 in the optimal solution must *improve* (increase for maximization problems or decrease for minimization problems) before the optimal solution would change. At that point, the decision variable associated with the coefficient enters the optimal solution at some positive level. To learn the new solution, apply POM for Windows again with a coefficient improved by slightly more than the reduced cost number.

▼ **FIGURE D.9**
Ranging Screen

Variable	Value	Reduced Cost	Original Val	Lower Bound	Upper Bound
X1	3	0	34	26.6667	40
X2	6	0	40	34	51
Constraint	**Shadow Price**	**Slack/Surplus**	**Original Val**	**Lower Bound**	**Upper Bound**
Constraint 1	3	0	48	40	54
Constraint 2	11	0	18	16	20
Constraint 3	0	4	16	12	Infinity

The top half of this screen also gives the *range of optimality*, or the lower and upper bound over which the objective function coefficients can range without affecting the optimal values of the decision variables. Note that the objective function coefficient for x_1, which currently has a value of $34, has a range of optimality from $26.6667 to $40. While the objection function's Z value would change with coefficient changes over this range, the optimal values of the decision variables remain the same.

The bottom half of Figure D.9 deals with the constraints, including the slack or surplus variables and the original right-hand-side values. Of particular interest are the shadow prices. Two tips on interpreting a *shadow price* follow:

1. The number is relevant only for a binding constraint, where the slack or surplus variable is 0 in the optimal solution. For a nonbinding constraint, the shadow price is 0.
2. The sign of the shadow price can be a positive or negative. The sign depends on whether the objective function is being maximized or minimized, and whether it is a $\leq$ constraint or $\geq$ constraint. If you simply ignore the sign, interpret the shadow price as the benefit of increasing the right-hand side by one unit for a $\leq$ constraint and reducing it by one unit for a $\geq$ constraint. The benefit is a reduction in the objective function value for minimization problems, and an increase for maximization problems. The shadow price can also be interpreted as the marginal loss (or penalty) in Z caused by making the constraint more restrictive by one unit.

Thus, the Stratton Company problem has 4 pounds of the additive mix slack, so the shadow price is $0. Packaging, on the other hand, is a binding constraint because it has no slack. The shadow price of one more packaging hour is $11.

Finally, Figure D.9 reports the lower and upper bounds for the *range of feasibility*, over which the right-hand-side parameters can range without changing the shadow prices. For example, the $11 shadow price for packaging is valid over the range from 16 to 20 hours.

The number of variables in the optimal solution (counting the decision variables, slack variables, and surplus variables) that are greater than 0 never exceeds the number of constraints. Such is the case for the Stratton Company problem, with its three constraints (not counting the implicit nonnegativity constraints) and three nonzero variables in the optimal solution (x_1, x_2, and the additive mix slack variable). On some rare occasions, the number of nonzero variables in the optimal solution can be less than the number of constraints—a condition called **degeneracy**. When degeneracy occurs, the sensitivity analysis information is suspect. If you want more "what-if" information, simply run your software package again using the new parameter values that you want to investigate.

degeneracy

A condition that occurs when the number of nonzero variables in the optimal solution is less than the number of constraints.

EXAMPLE D.5 Using Shadow Prices for Decision Making

The Stratton Company needs answers to three important questions: (1) Would increasing capacities in the extrusion or packaging area pay if it cost an extra $8 per hour over and above the normal costs already reflected in the objective function coefficients? (2) Would increasing packaging capacity pay if it cost an additional $6 per hour? (3) Would buying more raw materials pay?

SOLUTION

Expanding extrusion capacity would cost a premium of $8 per hour, but the shadow price for that capacity is only $3 per hour. However, expanding packaging hours would cost only $6 per hour more than the price reflected in the objective function, and the shadow price is $11 per hour. Finally, buying more raw materials would not pay because a surplus of 4 pounds already exists; the shadow price is $0 for that resource.

DECISION POINT

Management decided to increase its packaging hour's capacity but did not decide to expand extrusion capacity or buy more raw materials.

The Transportation Method

A special case of linear programming is the **transportation problem**, which can be represented as a standard table, sometimes called a *tableau*. Figure D.10 is an example, where the rows are supply sources and the columns are demands. Both the supplies and demands can be broken down into several periods into the future. Typically, the rows of the table are linear constraints that impose capacity limitations, and the columns are linear constraints that require certain demand levels to be met. Each cell in the tableau is a decision variable, and a per-unit cost is shown in the upper-right corner of each cell. Figure D.10 implies 52 decision variables (13 rows $\times$ 4 columns) and five constraints (13 rows + 4 columns).

Transportation problems can be formulated as a conventional linear programming problem and solved as usual. The **transportation method** simplifies data input and is a more efficient solution technique, but does not provide sensitivity analysis as in Figure D.9. Here we show

transportation problem

A special case of linear programming that has linear constraints for capacity limitations and demand requirements.

transportation method

A more efficient solution technique than the simplex method for solving transportation problems.

FIGURE D.10 ▶
Example of Transportation Tableau

Source of Supply		Time Period 1	2	3	4	Capacity
Period	Initial Inventory					
1	Type 1					
	Type 2					
	Type 3					
2	Type 1					
	Type 2					
	Type 3					
3	Type 1					
	Type 2					
	Type 3					
4	Type 1					
	Type 2					
	Type 3					
Demand						

how production planning problems can be formulated as transportation problems. Chapter 11, "Locating Facilities" shows an entirely different application of the transportation method: how to solve location problems. We focus on the setup and interpretation of the problem, leaving the rest of the solution process to a computer software package.

Transportation Method for Production Planning

Making sure that demand and supply are in balance is central to sales and operations planning (SOP), so it is no surprise that the transportation method can be applied to it. The *transportation method for production planning* is particularly helpful in determining anticipation inventories. Thus, it relates more to manufacturers' production plans than to service providers' staffing plans. In fact, the workforce levels for each period are inputs to the transportation method rather than outputs from it. Different workforce adjustment plans should be evaluated. Thus, several transportation method solutions may be obtained before a final plan is selected.

Using the transportation method for production planning is based on the assumption that a demand forecast is available for each period, along with a possible workforce adjustment plan. Capacity limits on overtime and the use of subcontractors also are needed for each period. Another assumption is that all costs are linearly related to the amount of goods produced; that is, a change in the amount of goods produced creates a proportionate change in costs.

To develop a sales and operations plan for a manufacturer, we do the following:

1. Obtain the demand forecasts for each period to be covered by the sales and operations plan and identify the initial inventory level currently available that can be used to meet future demand.
2. Select a candidate workforce adjustment plan, using a chase strategy, level strategy, or a mixed strategy. Specify the capacity limits of each production alternative (regular time, overtime, and subcontracting) for each period covered by the plan.
3. Estimate the cost of holding inventory and the cost of possible production alternatives (regular-time production, overtime production, and subcontracting). Identify the cost of undertime, if idle regular-time capacity is paid.

4. Input the information gathered in steps 1–3 into a computer routine that solves the transportation problem. After getting the solution, calculate the anticipation inventory levels and identify high-cost elements of the plan.
5. Repeat the process with other plans for regular-time, overtime, and subcontracting capacities until you find the solution that best balances cost and qualitative considerations. Even though this process involves trial and error, the transportation method yields the best mix of regular time, overtime, and subcontracting for each supply plan.

Example D.6 demonstrates this approach using the *Transportation Method (Production Planning)* module in the POM for Windows package.

EXAMPLE D.6 Preparing a Production Plan with the Transportation Method

The Tru-Rainbow Company produces a variety of paint products for both commercial and private use. The demand for paint is highly seasonal, peaking in the third quarter. Initial inventory is 250,000 gallons, and ending inventory should be 300,000 gallons.

Tru-Rainbow's manufacturing manager wants to determine the best production plan using the following demand requirements and capacity plan. Demands and capacities here are expressed in thousands of gallons (rather than employee-period equivalents). The manager knows that the regular-time cost is $1.00 per unit, overtime cost is $1.50 per unit, subcontracting cost is $1.90 per unit, and inventory holding cost is $0.30 per unit per quarter. Undertime is paid and the cost is $0.50 per unit. It is less than the regular-time cost because only labor costs are involved, not materials and variable overhead going into paint production.

	Demand	Regular-time Capacity	Overtime Capacity	Subcontractor Capacity
Quarter 1	300	450	90	200
Quarter 2	850	450	90	200
Quarter 3	1,500	750	150	200
Quarter 4	350	450	90	200
Totals	3,000	2,100	420	800

The following constraints apply:

a. The maximum allowable overtime in any quarter is 20 percent of the regular-time capacity in that quarter.

b. The subcontractor can supply a maximum of 200,000 gallons in any quarter. Production can be subcontracted in one period and the excess held in inventory for a future period to avoid a stockout.

c. No backorders or stockouts are permitted.

SOLUTION

Figure D.11 shows the POM for Windows screen of data inputs. Figure D.12 shows the POM for Windows screen that displays the optimal solution for this particular workforce adjustment plan. It looks much like the table shown previously, but with one exception. The demand for quarter 4 is shown to be 650,000 gallons rather than the demand forecast of only 350,000. The larger number reflects the desire of the manager to have an ending inventory in quarter 4 of 300,000 gallons. Some points to note in Figures D.11 and D.12 include the following:

1. There is a row in Figure D.12 for each supply alternative (instead of the "source of supply" in Figure D.10) on a quarter-by-quarter basis. The last column in each row indicates the maximum amount that can be used

▼ **FIGURE D.11**
POM for Windows Screens for Tru-Rainbow Company

Period	Demand	Regular tm Capacity	Overtime Capacity	Subcontract Capacity		Unit costs	Value
Quarter 1	300	450	90	200		Regular time	1
Quarter 2	850	450	90	200		Overtime	1.5
Quarter 3	1500	750	150	200		Subcontracting	1.9
Quarter 4	650	450	90	200		Holding cost	.3
						Lost sales cost	Not allowed
						Idle RT Capacity	.5
						Initial Inventory	250

Optimal cost = $4,010	Quarter 1	Quarter 2	Quarter 3	Quarter 4	Excess Capacity	Capacity
Init Inventory	230		20			250
Quarter 1 RegTime	50	400				450
Quarter 1 Overtime			90			90
Quarter 1 Subcontract	20				180	200
Quarter 2 RegTime		450				450
Quarter 2 Overtime			90			90
Quarter 2 Subcontract			200			200
Quarter 3 RegTime			750			750
Quarter 3 Overtime			150			150
Quarter 3 Subcontract			200			200
Quarter 4 RegTime				450		450
Quarter 4 Overtime				90		90
Quarter 4 Subcontract				110	90	200
Demand	300	850	1500	650	270	

▲ **FIGURE D.12**
Solution Screen for Prospective Tru-Rainbow Company Production Plan

to meet demand. The first row is the initial inventory available, and the rows that follow are for regular-time, overtime, and subcontracting production in each of the four quarters. The initial inventory can be used to satisfy demand in any of the four quarters. The second row (regular-time production in period 1) can also be used to satisfy demand in any of the four periods the plan will cover, and so on. The numbers in the last column give the maximum capacity made available for the supply alternatives. For example, the regular-time capacity for quarter 3 increases from the usual 450,000 gallons to 750,000 gallons, to help with the peak demand forecasted to be 1,500,000 gallons.

2. A column indicates each future quarter of demand and the last row gives its demand forecast. The demand for the fourth quarter is shown to be 650 units, because it includes the desired amount of ending inventory. The Excess Capacity column shows the cost of unused capacity, and the number in the last row (270 units) is the amount by which total capacity exceeds total demand. The unit costs for unused capacity are 0, except for the $0.50 unit cost of unused regular-time capacity.
3. The numbers in the other cells (excluding the cells in the last row or last column) show the cost of producing a unit in one period and, in some cases, carrying the unit in inventory for sale in a future period. These numbers correspond to the costs in the upper-right corners of the cells in Figure D.10. For example, the cost per unit of regular-time production in quarter 1 is $1.00 per gallon if it is used to meet the demand in quarter 1. This cost is found in row 2 and column 1 of the tableau. However, if it is produced to meet demand in quarter 2, the cost increases to $1.30 (or $1.00 + $0.30) because we must hold the unit in inventory for one quarter. Satisfying a unit of demand in quarter 3 by producing in quarter 1 on regular time and carrying the unit for two quarters costs $1.60, or [$1.00 + (2 × $0.30)], and so on. A similar approach is used for the costs of overtime and subcontracting.
4. The cells in the bottom left portion of the tableau with a cost of $9,999 are associated with backorders (or producing in a period to satisfy demand in a period after it was needed). Here we disallow backorders by making the backorder cost an arbitrarily large number, in this case $9,999 per unit. If backorder costs are so large, the transportation method will try to avoid backorders because it seeks a solution that minimizes total cost. If that is not possible, we increase the staffing plan and the overtime and subcontracting capacities.
5. The least expensive alternatives are those in which the output is produced and sold in the same period. For example, the cost for quarter 2 overtime production is only $1.50 per gallon if it is designated to meet demand in quarter 2 (row 6, column 2). The cost increases to $2.10 if it is designated for quarter 4 demand. However, we may not always be able to avoid alternatives that create inventory because of capacity restrictions.
6. Finally, the per-unit holding cost for the beginning inventory in period 1 is 0 because it is a function of previous production-planning decisions.

The first row in Figure D.12 shows that 230 units of the initial inventory are used to help satisfy the demand in quarter 1. The remaining 20 units in the first row are earmarked for helping supply the demand in quarter 3. The sum of the allocations across row 1 for the four quarters (230 + 0 + 20 + 0) does not exceed the

maximum capacity of 250, given in the right column. With the transportation method, this result must occur with each row. Any shortfalls are unused capacity, given in the "Excess Capacity" column. In this case, the undertime cost of $0.50 per unit was sufficiently large that no regular time capacity went unused.

Similarly, the sum of the allocations down each column must equal the total demand for the quarter. For example, the demand for quarter 1 is supplied from 230 units of the initial inventory, 50 units of quarter 1 regular-time production, and 20 units of quarter 1 subcontracting production. Summed together, they equal the forecasted demand of 300 units.

To further interpret the solution, we can convert Figure D.12 into the following table. For example, the total regular-time production in quarter 1 is 450,000 gallons (50,000 gallons to help meet demand in quarter 1 and 400,000 gallons to help satisfy demand in quarter 2).

The anticipation inventory held at the end of each quarter is obtained in the last column. For any quarter, it is the quarter's beginning inventory plus total supply (regular-time and overtime production, plus subcontracting) minus demand. For example, for quarter 1 the beginning inventory (250,000) plus the total from production and subcontracting (560,000) minus quarter 1 demand (300,000) results in an ending inventory of 510,000, which also is the beginning inventory for quarter 2.

Quarter	Regular-time Production	Overtime Production	Subcontracting	Total Supply	Anticipation Inventory
1	450	90	20	560	250 + 560 − 300 = 510
2	450	90	200	740	510 + 740 − 850 = 400
3	750	150	200	1,100	400 + 1,100 − 1,500 = 0
4	450	90	110	650	0 + 650 − 350 = 300
Totals	2,100	420	530	3,050	

Note: Anticipation inventory is the amount at the end of each quarter, where Beginning inventory + Total production − Actual Demand = Ending inventory.

The breakdown of costs can be found by multiplying the allocation in each cell of Figure D.12 by the cost per unit in that cell in Figure D.11 (b). Computing the cost column by column (it can also be done on a row-by-row basis) yields a total cost of $4,010,000, or $4,010 × 1,000.

Cost Calculations by Column		
Quarter 1	230($0) + 50($1.00) + 20($1.90)	= $ 88
Quarter 2	400($1.30) + 450($1.00)	= 970
Quarter 3	20($0.60) + 90($2.10) + 90($1.80) + 200($2.20) + 750($1.00) + 150($1.50) + 200($1.90)	= 2,158
Quarter 4	450($1.00) + 90($1.50) + 110($1.90)	= 794
		Total = $4,010

DECISION POINT

This plan requires too much overtime and subcontracting, and the anticipation inventory cost is substantial. The manager decided to search for a better capacity plan—with increases in the workforce to boost regular-time production capacity—that could lower production costs, perhaps even low enough to offset the added capacity costs.

Applications

Many problems in operations and supply chain management, and in other functional areas, lend themselves to linear programming and the transportation method. In addition to the examples already introduced, applications also exist in process management, constraint management, shipping assignments, inventory control, and shift scheduling. The review problems at the end of this supplement and at the end of previous chapters illustrate many of these types of problems. Once the decision maker knows how to formulate a problem generally, he or she can then adapt it to the situation at hand.

LEARNING GOALS IN REVIEW

1. **Describe formulating models for various problems**. Review the section "Formulating a Problem," pp. 608–610, and pay particular attention to Example D.1.
2. **Demonstrate graphic analysis and solutions for two-variable problems.** See the section "Graphic Analysis," pp. 610–616, on plotting the constraints, identifying the feasible region, plotting the objective function, and finding the solution.
3. **Define slack variables, surplus variables, and sensitivity analysis**. The sections on "Slack and Surplus Variables" and "Sensitivity Analysis" are found on pp. 615–616.
4. **Interpret computer output of a linear programming solution**. Computer output from POM for Windows is displayed and interpreted on pp. 617–619. Key information relates to the optimal values of the decision variables, objective function value, slack variables, and surplus variables. The shadow prices, reduced costs, lower bounds, and upper bounds can be valuable information for sensitivity analysis.
5. **Apply the transportation method to Sales and Operations Planning (S&OP) problems.** The section "Transportation Method for Production Planning," pp. 619–623, gives a step-by-step description on how to set up the problem as a transportation problem and then how to interpret the POM for Windows output. Pay particular attention to Figures D.11 and D.12. Example D.6 and Solved Problem 2 are also helpful.

MyOMLab helps you develop analytical skills and assesses your progress with multiple problems on graphic analysis, slack and surplus variables, optimal solution, computer solution, shadow prices, the transportation method, production plans, and shift schedules.

MyOMLab Resources	Titles	Link to the Book
Active Model Exercise	D.1 LP Graph	Graphic Analysis; Solved Problem 1 (p. 625)
OM Explorer Tutors	D.1 Plotting the Constraints	Plot the Constraints; Figure D.1 (p. 611; Example D.2 (p. 611); Figure D.2 (p. 611) Example D.3 (pp. 612–613)
	D.2 Finding the Optimal Solution	Plot the Objective Function Line; Figures D.5 and D.6 (p. 614); Find the Visual Solution; Find the Algebraic Solution; Example D.4 (p. 615)
	D.3 Finding Slack at Optimal Solution	Slack and Surplus Variables; Solved Problem 1 (p. 625)
	D.4 Graphic and Algebraic Solution	Graphic Analysis; Solved Problem 1 (p. 625)
POM for Windows	Linear Programming Results	Computer Solution
	Ranging	Computer Output; Figures D.7, D.8, and D.9 (pp. 617– 618)
	Graph	Graphic Analysis
	Transportation Method (Production Planning)	Transportation Method for Production; Example D.6; Figures D.11, D.12, and D.14 (pp. 621, 622, and 626); Solved Problem 2 (pp. 625–627)
Virtual Tours	Virgin Atlantic	Formulating a Problem
	Nike	Formulating a Problem
	Purina	Formulating a Problem
MyOMLab Supplements	F. Financial Analysis	MyOMLab Supplement F
Internet Exercise	Whirlpool	Formulating a Problem
	Marriott	Formulating a Problem
	Starbucks	Formulating a Problem
Key Equations		
Image Library		

Key Terms

binding constraint 615
certainty 608
constraints 608
corner point 613
decision variables 607
degeneracy 619
feasible region 608
graphic method of linear programming 610
linearity 608
linear programming 607
nonnegativity 608
objective function 607
parameter 608
product-mix problem 608
simplex method 616
range of feasibility 616
range of optimality 616
reduced cost 616
shadow price 616
slack 615
surplus 615
transportation method 619
transportation problem 619

Solved Problem 1

MyOMLab
Tutor D.4 in MyOMLab provides a practice example for finding the graphic and algebraic solution.

O'Connel Airlines is considering air service from its hub of operations in Cicely, Alaska, to Rome, Wisconsin, and Seattle, Washington. O'Connel has one gate at the Cicely Airport, which operates 12 hours per day. Each flight requires 1 hour of gate time. Each flight to Rome consumes 15 hours of pilot crew time and is expected to produce a profit of $2,500. Serving Seattle uses 10 hours of pilot crew time per flight and will result in a profit of $2,000 per flight. Pilot crew labor is limited to 150 hours per day. The market for service to Rome is limited to nine flights per day.

a. Identify positive slack and surplus variables, if any.

b. Use the graphic method of linear programming to maximize profits for O'Connel Airlines.

SOLUTION

a. The objective function is to maximize profits, Z:

$$\text{Maximize: } \$2{,}500x_1 + \$2{,}000x_2 = Z$$

where

x_1 = number of flights per day to Rome, Wisconsin
x_2 = number of flights per day to Seattle, Washington

The constraints are

$$x_1 + x_2 \leq 12 \text{ (gate capacity)}$$
$$15x_1 + 10x_2 \leq 150 \text{ (labor)}$$
$$x_1 \leq 9 \text{ (market)}$$
$$x_1 \geq 0 \text{ and } x_1 \geq 0$$

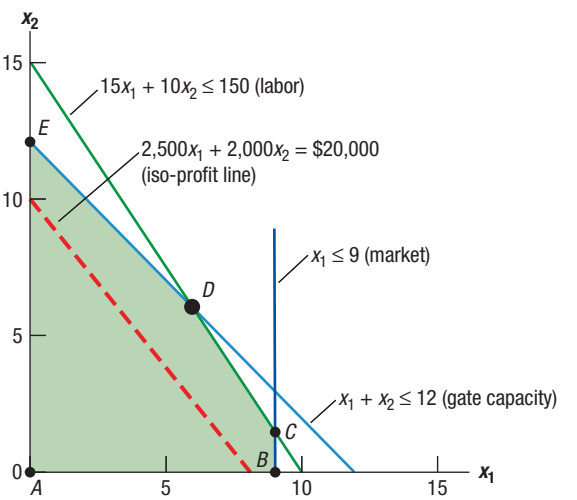
A careful drawing of iso-profit lines parallel to the one shown in Figure D.13 will indicate that point D is the optimal solution. It is at the intersection of the labor and gate capacity constraints. Solving algebraically, we get

$$15x_1 + 10x_2 = 150 \text{ (labor)}$$
$$-10x_1 - 10x_2 = -120 \text{ (gate} \times -10\text{)}$$
$$5x_1 + 0x_2 = 30$$
$$x_1 = 6$$
$$6 + x_2 = 12 \text{ (gate)}$$
$$x_2 = 6$$

The maximum profit results from making six flights to Rome and six flights to Seattle:

$$\$2{,}500(6) + \$2{,}000(6) = \$27{,}000$$

▼ **FIGURE D.13**
Graphic Solution for O'Connel Airlines



b. The market constraint has three units of slack, so the demand for flights to Rome is not fully met:

$$x_1 \leq 9$$
$$x_1 + s_3 = 9$$
$$6 + s_3 = 9$$
$$s_3 = 3$$

Solved Problem 2

The Arctic Air Company produces residential air conditioners. The manufacturing manager wants to develop a sales and operations plan for the next year based on the following demand and capacity data (in hundreds of product units):

	Demand	Regular-time Capacity	Overtime Capacity	Subcontractor Capacity
Jan–Feb (1)	50	65	13	10
Mar–Apr (2)	60	65	13	10
May–Jun (3)	90	65	13	10
Jul–Aug (4)	120	80	16	10
Sep–Oct (5)	70	80	16	10
Nov–Dec (6)	40	65	13	10
Totals	430	420	84	60

Undertime is unpaid, and no cost is associated with unused overtime or subcontractor capacity. Producing one air conditioning unit on regular time costs $1,000, including $300 for labor. Producing a unit on overtime costs $1,150. A subcontractor can produce a unit to Arctic Air specifications for $1,250. Holding an air conditioner in stock costs $60 for each 2-month period, and 200 air conditioners are currently in stock. The plan calls for 400 units to be in stock at the end of period 6. No backorders are allowed. Use the transportation method to develop a plan that minimizes costs.

SOLUTION

The following tables identify the optimal production and inventory plans. Figure D.14 shows the tableau that corresponds to this solution. An arbitrarily large cost ($99,999 per period) was used for backorders, which effectively ruled them out. Again, all production quantities are in hundreds of units. Note that demand in period 6 is 4,400. That amount is the period 6 demand plus the desired ending inventory of 400. The anticipation inventory is measured as the amount at the end of each period. Cost calculations are based on the assumption that workers are not paid for undertime or are productively put to work elsewhere in the organization whenever they are not needed for this work.

FIGURE D.14 ▶
Tableau for Optimal Production and Inventory Plans

Alternatives		Time Period 1	2	3	4	5	6	Unused Capacity	Total Capacity
Period	Initial Inventory	0	60	120	180 2	240	300	0	2
1	Regular time	1,000 50	1,060 15	1,120	1,180	1,240	1,300	0	65
	Overtime	1,150	1,210	1,270	1,330	1,390	1,450	13	13
	Subcontract	1,250	1,310	1,370	1,430	1,490	1,550	10	10
2	Regular time	99,999	1,000 41	1,060 12	1,120 12	1,180	1,240	0	65
	Overtime	99,999	1,150 4	1,210	1,270	1,330	1,390	9	13
	Subcontract	99,999	1,250	1,310	1,370	1,430	1,490	10	10
3	Regular time	99,999	99,999	1,000 65	1,060	1,120	1,180	0	65
	Overtime	99,999	99,999	1,150 13	1,210	1,270	1,330	0	13
	Subcontract	99,999	99,999	1,250	1,310	1,370	1,430	10	10
4	Regular time	99,999	99,999	99,999	1,000 80	1,060	1,120	0	80
	Overtime	99,999	99,999	99,999	1,150 16	1,210	1,270	0	16
	Subcontract	99,999	99,999	99,999	1,250 10	1,310	1,370	0	10
5	Regular time	99,999	99,999	99,999	99,999	1,000 70	1,060	10	80
	Overtime	99,999	99,999	99,999	99,999	1,150	1,210	16	16
	Subcontract	99,999	99,999	99,999	99,999	1,250	1,310	10	10
6	Regular time	99,999	99,999	99,999	99,999	99,999	1,000 44	21	65
	Overtime	99,999	99,999	99,999	99,999	99,999	1,150	13	13
	Subcontract	99,999	99,999	99,999	99,999	99,999	1,250	10	10
Demand		50	60	90	120	70	44	132	566

One initially puzzling aspect of this solution is that it allocates the initial inventory of 200 units to meet demand in period 4 rather than in period 1. The explanation is that multiple optimal solutions exist and this solution is only one of them. However, all solutions result in the same production and anticipation inventory plans derived as follows:

PRODUCTION PLAN

Period	Regular-time Production	Overtime Production	Subcontracting	Total
1	6,500	—	—	6,500
2	6,500	400	—	6,900
3	6,500	1,300	—	7,800
4	8,000	1,600	1,000	10,600
5	7,000	—	—	7,000
6	4,400	—	—	4,400

ANTICIPATION INVENTORY

Period	Beginning Inventory Plus Total Production Minus Demand	Anticipation (Ending) Inventory
1	200 + 6,500 − 5,000	1,700
2	1,700 + 6,900 − 6,000	2,600
3	2,600 + 7,800 − 9,000	1,400
4	1,400 + 10,600 − 12,000	0
5	0 + 7,000 − 7,000	0
6	0 + 4,400 − 4,000	400

Discussion Questions

1. Linear programming and the transportation method promise optimal solutions. However, wise managers sometimes, after seeing the optimal results such as in Figures D.8 or D.12, might decide to implement different plans. How do you explain such decision making?
2. A particular linear programming maximization problem has the following less-than-or-equal-to constraints: (1) raw materials, (2) labor hours, and (3) storage space. The optimal solution occurs at the intersection of the raw materials and labor hours constraints, so those constraints are binding. Management is considering whether to authorize overtime. What useful information could the linear programming solution provide to management in making this decision? Suppose a warehouse becomes available for rent at bargain rates. What would management need to know in order to decide whether to rent the warehouse? How could the linear programming model be helpful?

Problems

The OM Explorer and POM for Windows software is available to all students using the 10th edition of this textbook. Go to **www.pearsonhighered.com/krajewski** to download these computer packages. Check with your instructor on how best to use it. For linear programming problems with more than two variables, and for transportation problems, use POM for Windows to solve your model formulations. The emphasis in these cases is on modeling problems and interpreting computer output.

1. A business student at Nowledge College must complete a total of 65 courses to graduate. The number of business courses must be greater than or equal to 23. The number of nonbusiness courses must be greater than or equal to 20. The average business course requires a textbook costing $60 and 120 hours of study. Nonbusiness courses require a textbook costing $24 and 200 hours of study. The student has $3,000 to spend on books.
   a. Create a set of linear equations to describe the objective function and the constraints.
   b. Use graphic analysis to find the visual solution.
   c. What combination of business and nonbusiness courses minimizes total hours of study?
   d. Identify the slack or surplus variables.

2. In Problem 1, suppose that the objective is to minimize the cost of books and that the student's total study time is limited to 12,600 hours.

a. Use graphic analysis to determine the combination of courses that minimizes the total cost of books.

b. Identify the slack or surplus variables.

3. The Sports Shoe Company is a manufacturer of basketball and football shoes. The manager of marketing must decide the best way to spend advertising resources.
Each football team sponsored requires 120 pairs of shoes.
Each basketball team requires 32 pairs of shoes.
Football coaches receive $300,000 for shoe sponsorship, and basketball coaches receive $1,000,000. The manager's promotional budget is $30,000,000. The company has a limited supply (4 liters, or 4,000 cubic centimeters) of flubber, a rare and costly compound used in promotional athletic shoes. Each pair of basketball shoes requires 3 cc of flubber, and each pair of football shoes requires 1 cc. The manager wants to sponsor as many basketball and football teams as resources will allow.

a. Create a set of linear equations to describe the objective function and the constraints.

b. Use graphic analysis to find the visual solution.

c. What is the maximum number of each type of team that the company can sponsor?

4. The plant manager of a plastic pipe manufacturer has the opportunity to use two different routings for a particular type of plastic pipe. Routing 1 uses extruder A, and routing 2 uses extruder B. Both routings require the same melting process. The following table shows the time requirements and capacities of these processes:

	TIME REQUIREMENTS (HR/100 FT)		
Process	**Routing 1**	**Routing 2**	**Capacity (hr)**
Melting	1	1	45
Extruder A	3	0	90
Extruder B	0	1	160

Each 100 feet of pipe processed on routing 1 uses 5 pounds of raw material, whereas each 100 feet of pipe processed on routing 2 used only 4 pounds. This difference results from differing scrap rates of the extruding machines. Consequently, the profit per 100 feet of pipe processed on routing 1 is $60 and on routing 2 is $80. A total of 200 pounds of raw material is available.

a. Create a set of linear equations to describe the objective function and the constraints.

b. Use graphic analysis to find the visual solution.

c. What is the maximum profit?

5. Mile-High Microbrewery makes a light beer and a dark beer. Mile-High has a limited supply of barley, limited bottling capacity, and a limited market for light beer. Profits are $0.20 per bottle of light beer and $0.50 per bottle of dark beer.

a. The following table shows resource availability of products at the Mile-High Microbrewery. Use the graphic method of linear programming to maximize profits. How many bottles of each product should be produced per month?

	PRODUCT		
Resource	**Light Beer (x_1)**	**Dark Beer (x_2)**	**Resource Availability (per month)**
Barley	0.1 gram	0.6 gram	2,000 grams
Bottling	1 bottle	1 bottle	6,000 bottles
Market	1 bottle	—	4,000 bottles

b. Identify any constraints with slack or surplus.

6. The Trim-Look Company makes several lines of skirts, dresses, and sport coats. Recently, a consultant suggested that the company reevaluate its South Islander line and allocate its resources to products that would maximize contribution to profits and to overhead. Each product requires the same polyester fabric and must pass through the cutting and sewing departments. The following data were collected for the study:

	PROCESSING TIME (HR)		
Product	**Cutting**	**Sewing**	**Material (yd)**
Skirt	1	1	1
Dress	3	4	1
Sport coat	4	6	4

The cutting department has 100 hours of capacity, sewing has 180 hours of capacity, and 60 yards of material are available. Each skirt contributes $5 to profits and overhead; each dress, $17; and each sport coat, $30.

a. Specify the objective function and constraints for this problem.

b. Use a computer package such as POM for Windows to solve the problem.

7. A manufacturer of textile dyes can use two different processing routings for a particular type of dye. Routing 1 uses drying press A, and routing 2 uses drying press B. Both routings require the same mixing vat to blend chemicals for the dye before drying. The following table shows the time requirements and capacities of these processes:

	TIME REQUIREMENTS (HR/KG)		
Process	**Routing 1**	**Routing 2**	**Capacity (hr)**
Mixing	2	2	54
Dryer A	6	0	120
Dryer B	0	8	180

Each kilogram of dye processed on routing 1 uses 20 liters of chemicals, whereas each kilogram of dye processed on routing 2 uses only 15 liters. The difference results from differing yield rates of the drying presses. Consequently, the profit per kilogram processed on routing 1 is $50 and on routing 2 is $65. A total of 450 liters of input chemicals is available.

a. Write the constraints and objective function to maximize profits.

b. Use the graphic method of linear programming to find the optimal solution.

c. Identify any constraints with slack or surplus.

8. Polly Astaire makes fine clothing for big and tall men. A few years ago Astaire entered the sportswear market with the Sunset line of shorts, pants, and shirts. Management wants to make the amount of each product that will maximize profits. Each type of clothing is routed through two departments, A and B. The relevant data for each product are as follows:

	PROCESSING TIME (HR)		
Product	**Department A**	**Department B**	**Material (yd)**
Shirts	2	1	2
Shorts	2	3	1
Pants	3	4	4

Department A has 120 hours of capacity, department B has 160 hours of capacity, and 90 yards of material are available. Each shirt contributes $10 to profits and overhead; each pair of shorts, $10; and each pair of pants, $23.

a. Specify the objective function and constraints for this problem.

b. Use a computer package such as POM for Windows to solve the problem.

c. How much should Astaire be willing to pay for an extra hour of department A capacity? How much for an extra hour of department B capacity? For what range of right-hand values are these shadow prices valid?

9. Consider Problem 6 further.

a. How much would you be willing to pay for an extra hour of cutting time? For an extra hour of sewing time? For an extra yard of material? Explain your response to each question.

b. Determine the range of right-hand-side values over which the shadow price would be valid for the cutting constraint and for the material constraint.

10. The Nutmeg Corporation produces three different products, each in a 1-pound can: Almond-Lovers Mix, Walnut-Lovers Mix, and the Thrifty Mix. Three types of nuts are used in Nutmeg's products: almonds, walnuts, and peanuts. Nutmeg currently has 350 pounds of almonds, 150 pounds of walnuts, and 1000 pounds of peanuts. Each of Nutmeg's products must contain a certain percentage of each type of nut, as shown in the following table. The table also shows the revenue per can as well as the cost per pound to purchase nuts.

	PERCENTAGE REQUIREMENTS PER CAN			
	Almonds	**Walnuts**	**Peanuts**	**Revenue per can**
Almond-Lovers Mix	80%	20%	0%	$8.00
Walnut-Lovers Mix	20%	80%	0%	$10.00
Thrifty Mix	10%	10%	80%	$4.50
Cost per pound	$4.50	$6.00	$3.00	

a. Given Nutmeg's current stock of nuts, how many cans of each product should be produced to maximize revenue?

b. Does the solution you developed in part a change if Nutmeg is interested in maximizing contribution margin (defined as revenue per unit – raw material cost)?

c. If 50 additional pounds of walnuts became available, how would your contribution-margin maximizing solution from part (b) change?

11. The Butterfield Company makes a variety of knives. Each knife is processed on four machines. The processing times required are as follows. Machine capacities (in hours) are 1,500 for machine 1; 1,400 for machine 2; 1,600 for machine 3; and 1,500 for machine 4.

	PROCESSING TIME (HR)			
Knife	**Machine 1**	**Machine 2**	**Machine 3**	**Machine 4**
A	0.05	0.10	0.15	0.05
B	0.15	0.10	0.05	0.05
C	0.20	0.05	0.10	0.20
D	0.15	0.10	0.10	0.10
E	0.05	0.10	0.10	0.05

Each product contains a different amount of two basic raw materials. Raw material 1 costs $0.50 per ounce, and raw material 2 costs $1.50 per ounce. There are 75,000 ounces of raw material 1 and 100,000 ounces of raw material 2 available.

	REQUIREMENTS (OZ/UNIT)		
Knife	**Raw Material 1**	**Raw Material 2**	**Selling Price ($/unit)**
A	4	2	15.00
B	6	8	25.50
C	1	3	14.00
D	2	5	19.50
E	6	10	27.00

a. If the objective is to maximize profit, specify the objective function and constraints for the problem. Assume that labor costs are negligible.

b. Solve the problem with a computer package such as POM for Windows.

12. A small fabrication firm makes three basic types of components for use by other companies. Each component is processed on three machines. The processing times follow. Total capacities (in hours) are 1,600 for machine 1; 1,400 for machine 2; and 1,500 for machine 3.

	PROCESSING TIME (HR)		
Component	**Machine 1**	**Machine 2**	**Machine 3**
A	0.25	0.10	0.05
B	0.20	0.15	0.10
C	0.10	0.05	0.15

Each component contains a different amount of two basic raw materials. Raw material 1 costs $0.20 per ounce, and raw material 2 costs $0.35 per ounce. At present, 200,000 ounces of raw material 1 and 85,000 ounces of raw material 2 are available.

	REQUIREMENT (OZ/UNIT)		
Component	Raw Material 1	Raw Material 2	Selling Price ($/unit)
A	32	12	40
B	26	16	28
C	19	9	24

a. Assume that the company must make at least 1,200 units of component B, that labor costs are negligible, and that the objective is to maximize profits. Specify the objective function and constraints for the problem.

b. Use a computer package such as POM for Windows to solve the problem.

13. A problem often of concern to managers in processing industries is blending. The Nutmeg Corporation, from Problem 10, is considering a new product it intends to sell to active and health-concerned adults. This new product will be a 4-ounce package of nuts that conforms to specific heath requirements. First, the 4 ounce package can contain no more than 720 calories. It must deliver at least 20 grams of protein. Finally, the package must provide at least 15 percent of the adult daily requirement (ADR) of calcium and 20 percent of the ADR of iron. Nutmeg would like to use only almonds, walnuts, and peanuts in this new product. The following table provides nutritional data on each of these ingredients as well as their cost to Nutmeg.

Ingredients	Calories per ounce	Grams of protein per ounce	Percent ADR of calcium per ounce	Percent ADR of iron per ounce	Cost per ounce
Almonds	180	6	8%	6%	$0.28
Walnuts	190	4	2%	6%	$0.38
Peanuts	170	7	0%	4%	$0.12

a. Use linear programming to find the cost minimizing number of ounces of each ingredient that Nutmeg should use in each 4 ounce package. What is the per package cost of raw materials?

b. The marketing department at Nutmeg insists that each package should contain at least $\frac{1}{2}$ ounce of almonds, at least $\frac{1}{2}$ ounce of walnuts, and no more than 1 ounce of peanuts. Does the solution developed for part (a) satisfy these new constraints? If not, use linear programming to find a solution that includes these marketing requirements. What is the new cost of raw materials?

14. The following is a linear programming model for analyzing the product mix of Maxine's Hat Company, which produces three hat styles:

$$\text{Maximize: } \$7x_1 + \$8x_2 + \$6x_3 = Z$$

$$\text{Subject to: } 2x_1 + 4x_2 + 2x_3 \leq 120 \text{ (machine A time)}$$
$$5x_1 + 3x_2 + 2x_3 \leq 400 \text{ (machine B time)}$$
$$2x_1 + 2x_2 + 4x_3 \leq 110 \text{ (machine C time)}$$
$$x_1 \geq 0 x_2 \geq 0, \text{ and } x_3 \geq 0$$

The POM for Windows printout in Figure D.15 shows the optimal solution to the problem. Consider each of the following statements independently, and state whether it is true or false. Explain each answer.

a. If the price of hat 3 were increased to $11.50, it would be part of the optimal product mix. *Hint:* Hat 3 is represented by x_3 and its optimal value currently is 0, which means that it is not to be produced (and not part of the optimal product mix).

b. The capacity of machine B can be reduced to 280 hours without affecting profits.

c. If machine C had a capacity of 115 hours, the production output would remain unchanged.

15. The Warwick Manufacturing Company produces shovels for industrial and home use. Sales of the shovels are seasonal, and Warwick's customers refuse to stockpile them during slack periods. In other words, the customers want to minimize inventory, insist on shipments according to their schedules, and will not accept backorders.

Warwick employs manual, unskilled laborers who require only basic training. Producing 1,000 shovels costs $3,500 on regular time and $3,700 on overtime. These amounts include materials, which account for more than 85 percent of the cost. Overtime is limited to production of 15,000 shovels per quarter. In addition, subcontractors can be hired at $4,200 per thousand shovels, but Warwick's labor contract restricts this type of production to 5,000 shovels per quarter.

The current level of inventory is 30,000 shovels, and management wants to end the year at that level. Holding 1,000 shovels in inventory costs $280 per quarter. The latest annual demand forecast is as follows:

Quarter	Demand
1	70,000
2	150,000
3	320,000
4	100,000
Totals	640,000

Build a linear programming model to determine the *best* regular-time capacity plan. Assume the following:

- The firm has 30 workers now, and management wants to have the same number in quarter 4.
- Each worker can produce 4,000 shovels per quarter.
- Hiring a worker costs $1,000, and laying off a worker costs $600.

16. The Washington Chemical Company produces chemicals and solvents for the glue industry. The production process is divided into several "focused factories," each producing a specific set of products. The time has come to prepare the production plan for one of the focused factories. This particular factory produces five products, which

(a) Results Screen

Maximize	7	8	6			
machine A time	2	4	2	<=	120	.5
machine B time	5	3	2	<=	400	0
machine C time	2	2	4	<=	110	3
Solution->	50	5	0		390	

(b) Ranging Screen

Variable	Value	Reduced Cost	Original Val	Lower Bound	Upper Bound
X1	50	0	7	4.6667	8
X2	5	0	8	7	14
X3	0	7	6	-Infinity	13
Constraint	Shadow Price	Slack/Surplus	Original Val	Lower Bound	Upper Bound
machine A time	.5	0	120	110	220
machine B time	0	135	400	265	Infinity
machine C time	3	0	110	60	120

▲ **FIGURE D.15**
Solution Screens for Maxine's Hat Company

must pass through both the reactor and the separator. Each product also requires a certain combination of raw materials. Production data are shown in Table D.2.

The Washington Chemical Company has a long-term contract with a major glue manufacturer that requires annual production of 3,000 pounds of both products 3 and 4. More of these products could be produced because demand currently exceeds production capacity.

a. Determine the annual production quantity of each product that maximizes contribution to profits. Assume the company can sell all it can produce.

b. Specify the lot size for each product.

17. The Bull Grin Company produces a feed supplement for animal foods produced by a number of other companies. Producing 1,000 pounds of supplement costs $810 on regular time and $900 on overtime. These amounts include materials, which account for more than 80 percent of the cost. The plant can produce 400,000 pounds of supplement per quarter using regular time but overtime is limited to the production of 40,000 pounds per quarter. The current level of inventory is 40,000 pounds and management wants to end the year at that level. Holding 1,000 pounds of feed supplement in inventory costs $110 per quarter. The latest annual demand forecast follows:

Quarter	Demand (in Pounds)
1	100,000
2	410,000
3	770,000
4	440,000

a. Formulate this production planning problem as a linear program after defining all decision variables.

b. Solve your formulation using a computer package such as POM for Windows.

TABLE D.2 | PRODUCTION DATA FOR WASHINGTON CHEMICAL

	PRODUCT					
Resource	**1**	**2**	**3**	**4**	**5**	**Total Resources Available**
Reactor (hr/lb)	0.05	0.10	0.80	0.57	0.15	7,500 hr*
Separator (hr/lb)	0.20	0.02	0.20	0.09	0.30	7,500 hr*
Raw material 1 (lb)	0.20	0.50	0.10	0.40	0.18	10,000 lb
Raw material 2 (lb)	—	0.70	—	0.50	—	6,000 lb
Raw material 3 (lb)	0.10	0.20	0.40	—	—	7,000 lb
Profit contribution ($/lb)	4.00	7.00	3.50	4.00	5.70	

*The total time available has been adjusted to account for setups. The five products have a prescribed sequence owing to the cost of changeovers between products. The company has a 35-day cycle (or 10 changeovers per year per product). Consequently, the time for these changeovers has been deducted from the total time available for these machines.

c. Assume that subcontractors can be hired at $1,100 per thousand pounds to produce as much supplement as Bull Grin requires. Does this change the cost minimizing solution found in part (b)?

d. If Bull Grin realizes that the current level of inventory is actually 0 pounds, are the resources assumed in part (c) adequate to satisfy all demand and still end the year with 40,000 pounds in ending inventory? If so, how much will the cost of Bull Grin's production plan increase?

18. The management of Warwick Manufacturing Company is willing to give price breaks to its customers as an incentive to purchase shovels in advance of the traditional seasons. Warwick's sales and marketing staff estimates that the demand for shovels resulting from the price breaks would be as follows:

Quarter	Demand	Original Demand
1	120,000	70,000
2	180,000	150,000
3	180,000	320,000
4	160,000	100,000
Totals	640,000	640,000

Calculate the optimal production plan (including the workforce staffing plan) under the new demand schedule. Compare it to the optimal production plan under the original demand schedule. Evaluate the potential effects of demand management.

19. JPMorgan Chase has a scheduling problem. Operators work 8-hour shifts and can begin work at midnight, 4 A.M., 8 A.M., noon, 4 P.M., or 8 P.M. Operators are needed to satisfy the following demand pattern. Formulate a linear programming model to cover the demand requirements with the minimum number of operators.

Time Period	Operators Needed
Midnight to 4 A.M.	4
4 A.M. to 8 A.M.	6
8 A.M. to noon	90
Noon to 4 P.M.	85
4 P.M. to 8 P.M.	55
8 P.M. to 12 midnight	20

20. Supertronics, Inc., would like to know how the firm's profitability is altered by product mix. Currently, product mix is determined by giving priority to the product with the highest per unit contribution margin (defined as the difference between price and material cost). Details on the Supertronics product line, including processing time at each workstation, follow:

	PRODUCT			
	Alpha	Beta	Delta	Gamma
Price	$350.00	$320.00	$400.00	$500.00
Material Cost	$50.00	$40.00	$125.00	$150.00
Weekly Demand in Units	100	60	50	80
Processing Time at Machine 1 in Minutes	20	0	40	10
Processing Time at Machine 2 in Minutes	25	20	0	50
Processing Time at Machine 3 in Minutes	0	20	60	30

a. Assume that Supertronics has 5,500 minutes of capacity available at each workstation each week. Develop a linear program to define the production mix that maximizes contribution margin.

b. Solve your formulation using a computer package such as POM for Windows.

c. Given your solution for part (b), which machine is the bottleneck?

d. How would your formulation and solution in part (b) change if 50 units of each product were already committed to customers and thereby had to be produced?

21. The Holloway Calendar Company produces a variety of printed calendars for both commercial and private use. The demand for calendars is highly seasonal, peaking in the third quarter. Current inventory is 165,000 calendars, and ending inventory should be 200,000 calendars.

Ann Ritter, Holloway's manufacturing manager, wants to determine the best production plan for the demand requirements and capacity plan shown in the following table. (Here, demand and capacities are expressed as thousands of calendars rather than as employee-period equivalents.) Ritter knows that the regular-time cost is $0.50 per unit, overtime cost is $0.75 per unit, subcontracting cost is $0.90 per unit, and inventory holding cost is $0.10 per calendar per quarter. Unused regular-time capacity is not paid.

	QUARTER				
	1	2	3	4	Total
Demand	250	515	1,200	325	2,290
Capacities					
Regular time	300	300	600	300	1,500
Overtime	75	75	150	75	375
Subcontracting	150	150	150	150	600

a. Recommend a production plan to Ritter, using the Transportation Method (Production Planning) module of POM for Windows. (Do not allow any stockouts or backorders to occur.)

b. Interpret and explain your recommendation.

c. Calculate the total cost of your recommended production plan.

22. The Cut Rite Company is a major producer of industrial lawn mowers. The cost to Cut Rite for hiring a semiskilled worker for its assembly plant is $3,000, and the cost for laying off one is $2,000. The plant averages an output of 36,000 mowers per quarter with its current workforce of 720 employees. Regular-time capacity is directly proportional to the number of employees. Overtime is limited to a maximum of 3,000 mowers per quarter, and subcontracting is limited to 1,000 mowers per quarter. The costs to produce one mower are $2,430 on regular time (including materials), $2,700 on overtime, and $3,300 via subcontracting. Unused regular-time capacity costs $270 per mower. No additional cost is incurred for unused overtime or subcontractor capacity. The current level of inventory is 4,000 mowers, and management wants to end the year at that level. Customers do not tolerate backorders, and holding a mower in inventory per quarter costs $300. The demand for mowers this coming year is as follows:

Quarter	1	2	3	4
Demand	10,000	41,000	77,000	44,000

Two workforce plans have been proposed, and management is uncertain as to which one to use. The following table shows the number of employees per quarter under each plan:

Quarter	1	2	3	4
Plan 1	720	780	920	720
Plan 2	860	860	860	860

a. Which plan would you recommend to management? Explain, supporting your recommendation with an analysis using the transportation method of production planning.

b. If management used creative pricing to get customers to buy mowers in nontraditional time periods, the following demand schedule would result:

Quarter	1	2	3	4
Demand	20,000	54,000	54,000	44,000

Which workforce plan would you recommend now?

23. Revisit Problem 17 on the Bull Grin Company. Some cost and demand parameters have changed. Producing 1,000 pounds of supplement now costs $830 on regular time and $910 on overtime. No additional cost is incurred for unused regular time, overtime, or subcontractor capacity. Overtime is limited to production of a total of 20,000 pounds per quarter. In addition, subcontractors can be hired at $1,000 per thousand pounds, but only 30,000 pounds per quarter can be produced this way.

The current level of inventory is 40,000 pounds, and management wants to end the year at that level. Holding 1,000 pounds of feed supplement in inventory per quarter costs $100. The latest annual forecast is shown in Table D.3.

Use the transportation method of production planning in POM for Windows to find the optimal production plan and calculate its cost, or use the spreadsheet approach to find a good production plan and calculate its cost.

TABLE D.3 | FORECASTS AND CAPACITIES

	PERIOD				
	Quarter 1	Quarter 2	Quarter 3	Quarter 4	Total
Demand (pounds)	130,000	400,000	800,000	470,000	1,800,000
Capacities (pounds)					
Regular time	390,000	400,000	460,000	380,000	1,630,000
Overtime	20,000	20,000	20,000	20,000	80,000
Subcontract	30,000	30,000	30,000	30,000	30,000

CASE R.U. Reddie for Location

The R.U. Reddie Corporation, located in Chicago, manufactures clothing specially designed for stuffed cartoon animals such as Snoopy and Wile E. Coyote. Among the popular products are a wedding tuxedo for Snoopy and a flak jacket for Wile E. Coyote. The latter is capable of stopping an Acme rocket at close range . . . sometimes.

For many sales, the company relies upon the help of spoiled children who refuse to leave the toy store until their parents purchase a wardrobe for their stuffed toys. Rhonda Ulysses Reddie, owner of the company, is concerned over the market projections that indicate demand for the product is substantially greater than current plant capacity. The "most likely" projections indicate that the company will be short by 400,000 units next year, and thereafter 700,000 units annually. As such, Rhonda is considering opening a new plant to produce additional units.

Background

The R.U. Reddie Corporation currently has three plants located in Boston, Cleveland, and Chicago. The company's first plant was the Chicago location, but as sales grew in the Midwest and Northeast, the Cleveland and Boston plants were built in short order. As the demand for wardrobes for stuffed animals moved west, warehouse centers were opened in St. Louis and Denver. The capacities of the three plants were increased to accommodate the demand. Each plant has its own warehouse to satisfy demands in its own area. Extra capacity was used to ship the product to St. Louis or Denver.

The new long-term forecasts provided by the sales department contain both good news and bad news. The added revenues will certainly help Rhonda's profitability, but the company would have to buy another plant to realize the added profits. Space is not available at the existing plants, and the benefits of the new technology for manufacturing stuffed animal wardrobes are tantalizing. These factors motivate the search for the best location for a new plant. Rhonda identifies Denver and St. Louis as possible locations for the new plant.

Rhonda's Concerns

A plant addition is a big decision. Rhonda has started to think about the accuracy of the data she was able to obtain. She has market, financial, and operations concerns.

Market

The projected demands for years 2 through 10 show an annual increase of 700,000 units to a total of 2,000,000 units for each year. Rhonda has two concerns here. First, what if the projections for each city are off plus or minus 10 percent equally across the board? That is, total annual demands could be as low as 1,800,000 or as high as 2,200,000, with each city being affected the same as the others. Second, the marketing manager has expressed a concern over a possible market shift from the Midwest and Northeast to the West. Under this scenario, additional demand would reach 50,000 units in St. Louis and 150,000 units in Denver, with the other cities staying at the "most likely" demand projections.

Financial

Rhonda has realized that the net present value (NPV) of each alternative is an important input to the final decision. However, the accuracy of the estimates for the various costs is critical to determining good estimates of cash flows. She wonders whether her decision will change if the COGS (variable production plus transportation costs) for each option are off by ± 10 percent. That is, what if the variable production costs and transportation costs of St. Louis are 10 percent higher than estimated while the variable production costs and transportation costs for Denver are 10 percent lower than estimated? Or vice versa? Furthermore, what if the estimate for fixed costs are off by ± 10 percent? For example, suppose St. Louis is 10 percent higher while Denver is 10 percent lower, or vice versa. Would the recommendation change under any of these situations?

Operations

The ultimate location of the new plant will determine the distribution assignments and the level of utilization of each plant in the network. Cutting back production in any of the plants will change the distribution assignments of all plants. Because the "most likely" demand projections with a new plant will mean excess capacity in the system, the capacity of the Cleveland plant could be cut in year 2 and beyond. Suppose Cleveland cuts back production by 50(000) units a year from year 2 and beyond. Will this action affect the choice between Denver and St. Louis? What is the impact on the distribution assignments of the plant? Some nonquantifiable concerns also need consideration. First, the availability of a good workforce is much better in Denver than St. Louis because of the recent shutdown of a Beanie Baby factory. The labor market is much tighter in St. Louis and the prognosis is for continued short supply in the foreseeable future. Second, the Denver metropolitan area just instituted strict environmental regulations. Rhonda's new plant would adhere to existing laws, but the area is highly environmentally conscious and more regulations may be coming in the future. It is costly to modify a plant once operations begin. Finally, Denver has a number of good suppliers with the capability to assist in production design (new wardrobe fashions). St. Louis also has suppliers, but they cannot help with product development. Proximity to suppliers with product development capability is a "plus" for this industry.

Data

The following data have been gathered for Rhonda:

1. The per-unit shipping cost based on the average tonmile rates for the most efficient carriers is $0.0005 per mile. The average revenue per outfit is $8.00.
2. The company currently has the following capacity (in thousands) constraints:

	Capacity
Boston	400
Cleveland	400
Chicago	500

3. Data concerning the various locations are found in Table D.4.
4. New plant information:

Alternative	Building and Equipment[2]	Annual Fixed Costs (SGA)[1,3]	Variable Production Costs/Unit	Land[1]
Denver	$12,100	$550	$3.15	$1,200
St. Louis	10,800	750	3.05	800

[1] Figures are given in thousands.
[2] Net book value of plant and equipment with remaining depreciable life of 10 years.
[3] Annual fixed costs do not include depreciation on plant and equipment.

5. The road mileage between the cities is as follows:

	Boston	Cleveland	Chicago	St. Louis	Denver
Boston	—	650	1,000	1,200	2,000
Cleveland		—	350	600	1,400
Chicago			—	300	1,000
St. Louis				—	850
Denver					—

6. Basic assumptions you should follow:
   - Terminal value (in 10 years) of the new investment is 50 percent of plant, equipment, and land cost.
   - The tax rate is 40 percent.
   - Straight-line depreciation is used for all assets over a 10-year life.
   - R.U. Reddie is a 100 percent equity company with all equity financing and a weighted average cost of capital (WACC) of 11 percent.
   - Capacity of the new plant production for the first year will be 500 (000) units.
   - Capacity of the new plant production thereafter will be 900 (000) units.
   - Cost of goods sold (COGS) equals variable costs of production plus total transportation costs.
   - Costs to ship from a plant to its own warehouse are zero; however, production costs are applicable.

TABLE D.4 | LOCATION DATA FOR R.U. REDDIE

City	Most Likely Demand First Year[1]	Most Likely Demand After Years 2–10[1]	Current Costs, Building and Equipment[1,2]	Annual Fixed Costs (SGA)[1,3]	Variable Production Costs/Unit	Land[1]
Boston	80	140	$9,500	$600	$3.80	$500
Cleveland	200	260	7,700	300	3.00	400
Chicago	370	430	8,600	400	3.25	600
St. Louis	440	500				
Denver	610	670				

[1] Figures are given in thousands.
[2] Net book value of plant and equipment with remaining depreciable life of 10 years.
[3] Annual fixed costs do not include depreciation on plant and equipment.

7. R.U. Reddie operations and logistics managers determined the shipping plan and cost of goods sold for the option of *not* building a new plant and simply using the existing capacities to their fullest extent (status quo solution).

Year 1	**COGS = $4,692,000**
Boston to Boston	80
Boston to St. Louis	320
Cleveland to Chicago	80
Cleveland to Cleveland	200
Cleveland to St. Louis	120
Chicago to Chicago	290
Chicago to Denver	210
Years 2–10	**COGS = $4,554,000**
Boston to Boston	140
Boston to St. Louis	260
Cleveland to Cleveland	260
Cleveland to St. Louis	140
Chicago to Chicago	430
Chicago to St. Louis	70

QUESTIONS

Your team has been asked to determine whether R.U. Reddie should build a new plant and, if so, where it should be located. Your report should consist of six parts.

1. Write a memo from your team to R.U. Reddie indicating your recommendation and a brief overview of the supporting evidence.
2. Model the location decision as a linear programming problem. The objective function should be to minimize the total variable costs (production plus transportation costs). The variables should be the quantity to ship from each of the plants (including one of the alternative new plants) to each of the warehouses. You should have 20 variables (four plants and five warehouses). You should also have nine constraints (four plant capacity constraints and five warehouse demand constraints). See the addendum for hints. You will need two models—one for Denver and one for St. Louis.
3. Use a software package for linear programming, such as with POM for Windows, to solve for the optimal distribution plan for each alternative (i.e., Denver and St. Louis). A linear programming model is preferable for this case because of the sensitivity analysis information that it provides.
4. Compute the NPV of each alternative. Use the results from the linear models for the COGS for each alternative. (*Hint:* Your analysis will be simplified if you think in terms of incremental cash flows.) Create an easy-to-read spreadsheet for each alternative.
5. Do a sensitivity analysis of the quantitative factors mentioned in the case: Forecast errors (across the board and market shift), errors in COGS estimate, and errors in fixed cost estimates. Do each factor independent of the others and use the "most likely" projections as the base case. Summarize the results in one table.
6. Use the analysis in question 5 to identify the key quantitative variables that determine the superiority of one alternative over another. Rationalize your final recommendation in light of all the considerations R.U. Reddie must make.

Addendum

Here are some hints for your model.

1. The capacity constraint for Boston would look like this:

   $$1B - B + 1B - CL + 1B - CH + 1B - D + 1B - SL \leq 400$$

   The variable B-CH means Boston to Chicago in this example. You will need a total of four capacity constraints, one for each of the existing sites and one for the alternative site you are evaluating. Remember that the new site will have a capacity limit of 500 in the first year, and 900 in the second year.
2. The demand constraint for Boston would look like this:

   $$1B - B + 1CL - B + 1CH - B + 1D - B = 140$$

   The Denver location alternative is depicted in this example (D-B represents the number of units produced in Denver and shipped to Boston). You will need a total of five demand constraints, one for each warehouse location. Notice that the demand constraints have equal signs to indicate that exactly that quantity must be received at each warehouse.
3. Define your variables to be "thousands of units shipped," and omit the three zeros after each demand and capacity value. Then remember to multiply your final decisions and total variable costs by a thousand after you get your solution from the model.
4. Because the capacity and demand changes from year 1 to year 2, you need to run your model twice for each location to get the necessary data. You will also run the model and spread-sheets multiple times for question 5 of the report.

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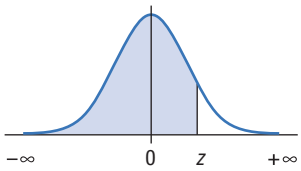
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APPENDIX

1

NORMAL DISTRIBUTION



	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

APPENDIX

2 TABLE OF RANDOM NUMBERS

71509	68310	48213	99928	64650	13229	36921	58732	13459	93487
21949	30920	23287	89514	58502	46185	00368	82613	02668	37444
50639	54968	11409	36148	82090	87298	41396	71111	00076	60029
47837	76716	09653	54466	87987	82362	17933	52793	17641	19502
31735	36901	92295	19293	57582	86043	69502	12601	00535	82697
04174	32342	66532	07875	54445	08795	63563	42295	74646	73120
96980	68728	21154	56181	71843	66134	52396	89723	96435	17871
21823	04027	76402	04655	87276	32593	17097	06913	05136	05115
25922	07122	31485	52166	07645	85122	20945	06369	70254	22806
32530	98882	19105	01769	20276	59401	60426	03316	41438	22012
00159	08461	51810	14650	45119	97920	08063	70819	01832	53295
66574	21384	75357	55888	83429	96916	73977	87883	13249	28870
00995	28829	15048	49573	65277	61493	44031	88719	73057	66010
55114	79226	27929	23392	06432	50200	39054	15528	53483	33972
10614	25190	52647	62580	51183	31338	60008	66595	64357	14985
31359	77469	58126	59192	23371	25190	37841	44386	92420	42965
09736	51873	94595	61367	82091	63835	86858	10677	58209	59820
24709	23224	45788	21426	63353	29874	51058	29958	61220	61199
79957	67598	74102	49824	39305	15069	56327	26905	34453	53964
66616	22137	72805	64420	58711	68435	60301	28620	91919	96080
01413	27281	19397	36231	05010	42003	99865	20924	76151	54089
88238	80731	20777	45725	41480	48277	45704	96457	13918	52375
57457	87883	64273	26236	61095	01309	48632	00431	63730	18917
21614	06412	71007	20255	39890	75336	89451	88091	61011	38072
26466	03735	39891	26361	86816	48193	33492	70484	77322	01016
97314	03944	04509	46143	88908	55261	73433	62538	63187	57352
91207	33555	75942	41668	64650	38741	86189	38197	99112	59694
46791	78974	01999	78891	16177	95746	78076	75001	51309	18791
34161	32258	05345	79267	75607	29916	37005	09213	10991	50451
02376	40372	45077	73705	56076	01853	83512	81567	55951	27156
33994	56809	58377	45976	01581	78389	18268	90057	93382	28494
92588	92024	15048	87841	38008	80689	73098	39201	10907	88092
73767	61534	66197	47147	22994	38197	60844	86962	27595	49907
51517	39870	94094	77092	94595	37904	27553	02229	44993	10468
33910	05156	60844	89012	21154	68937	96477	05867	95809	72827
09444	93069	61764	99301	55826	78849	26131	28201	91417	98172
96896	43769	72890	78682	78243	24061	55449	53587	77574	51580
97523	54633	99656	08503	52563	12099	52479	74374	79581	57143
42568	30794	32613	21802	73809	60237	70087	36650	54487	43718
45453	33136	90246	61953	17724	42421	87611	95369	42108	95369
52814	26445	73516	24897	90622	35018	70087	60112	09025	05324
87318	33345	14546	15445	81588	75461	12246	47858	08983	18205
08063	83575	26294	93027	09988	04487	88364	31087	22200	91019
53400	82078	52103	25650	75315	18916	06809	88217	12245	33053
90789	60614	20862	34475	11744	24437	55198	55219	74730	59820
73684	25859	86858	48946	30941	79017	53776	72534	83638	44680
82007	12183	89326	53713	77782	50368	01748	39033	47042	65758
80208	30920	97774	41417	79038	60531	32990	57770	53441	58732
62434	96122	63019	58439	89702	38657	60049	88761	22785	66093
04718	83199	65863	58857	49886	70275	27511	99426	53985	84077

Glossary

ABC analysis The process of dividing SKUs into three classes, according to their dollar usage, so that managers can focus on items that have the highest dollar value.

acceptable quality level (AQL) A statement of the proportion of defective items (outside of specifications) that the buyer will accept in a shipment.

acceptance sampling The application of statistical techniques to determine whether a quantity of material should be accepted or rejected based on the inspection or test of a sample.

action notice A computer-generated memo alerting planners about releasing new orders and adjusting the due dates of scheduled receipts.

activity The smallest unit of work effort consuming both time and resources that the project manager can schedule and control.

activity-on-node (AON) network An approach used to create a network diagram, in which nodes represent activities and arcs represent the precedence relationships between them.

activity slack The maximum length of time that an activity can be delayed without delaying the entire project, calculated as S = LS – ES or S = LF – EF.

additive seasonal method A method in which seasonal forecasts are generated by adding a constant to the estimate of average demand per season.

advanced planning and scheduling (APS) systems Computer software systems that seek to optimize resources across the supply chain and align daily operations with strategic goals.

aggregate plan See sales and operations plan.

aggregation The act of clustering several similar services or products so that forecasts and plans can be made for whole families.

allowance time The time added to the normal time to adjust for certain factors.

annual plan or financial plan A plan for financial assessment used by a nonprofit service organization.

annuity A series of payments on a fixed amount for a specified number of years.

anticipation inventory Inventory used to absorb uneven rates of demand or supply.

appraisal costs Costs incurred when the firm assess the performance level of its processes.

assemble-to-order strategy A strategy for producing a wide variety of products from relatively few subassemblies and components after the customer orders are received.

assignable causes of variation Any variation-causing factors that can be identified and eliminated.

attributes Service or product characteristics that can be quickly counted for acceptable performance.

auction A marketplace where firms place competitive bids to buy something.

automation A system, process, or piece of equipment that is self-acting and self-regulating.

available-to-promise (ATP) inventory The quantity of end items that marketing can promise to deliver on specified dates.

average aggregate inventory value The total average value of all items held in inventory for a firm.

average outgoing quality (AOQ) The expressed proportion of defects that the plan will allow to pass.

average outgoing quality limit (AOQL) The maximum value of the average outgoing quality over all possible values of the proportion defective.

back office A process with low customer contact and little service customization.

backlog An accumulation of customer orders that a manufacturer has promised for delivery at some future date.

backorder A customer order that cannot be filled when promised or demanded but is filled later.

backorder and stockout Additional costs to expedite past-due orders, the costs of lost sales, and the potential cost of losing a customer to a competitor (sometimes called loss of goodwill).

backward integration A firm's movement upstream toward the sources of raw materials, parts, and services through acquisitions.

balance delay The amount by which efficiency falls short of 100 percent.

Baldrige Performance Excellence Program A program that promotes, recognizes, and publicizes quality strategies and achievements.

bar chart A series of bars representing the frequency of occurrence of data characteristics measured on a yes-or-no basis.

base case The act of doing nothing and losing orders from any demand that exceeds current capacity, or incur costs because capacity is too large.

base-stock system An inventory control system that issues a replenishment order, *Q*, each time a withdrawal is made, for the same amount of the withdrawal.

batch process A process that differs from the job process with respect to volume, variety, and quantity.

benchmarking A systematic procedure that measures a firm's processes, services, and products against those of industry leaders.

bill of materials (BOM) A record of all the components of an item, the parent–component relationships, and the usage quantities derived from engineering and process designs.

bill of resources (BOR) A record of a service firm's parent-component relationships and all of the materials, equipment time, staff, and other resources associated with them, including usage quantities.

binding constraint A constraint that helps form the optimal corner point; it limits the ability to improve the objective function.

Black Belt An employee who reached the highest level of training in a Six Sigma program and spends all of his or her time teaching and leading teams involved in Six Sigma projects.

block plan A plan that allocates space and indicates placement of each operation.

bottleneck A capacity constraint resource (CCR) whose available capacity limits the organization's ability to meet the product volume, product mix, or demand fluctuation required by the marketplace.

brainstorming Letting a group of people, knowledgeable about the process, propose ideas for change by saying whatever comes to mind.

break-even analysis The use of the break-even quantity; it can be used to compare processes by finding the volume at which two different processes have equal total costs.

break-even quantity The volume at which total revenues equal total costs.

bullwhip effect The phenomenon in supply chains whereby ordering patterns experience increasing variance as you proceed upstream in the chain.

business plan A projected statement of income, costs, and profits.

***c*-chart** A chart used for controlling the number of defects when more than one defect can be present in a service or product.

capacity The maximum rate of output of a process or a system.

capacity cushion The amount of reserve capacity a process uses to handle sudden increases in demand or temporary losses of production capacity; it measures the amount by which the average utilization (in terms of total capacity) falls below 100 percent.

capacity gap Positive or negative difference between projected demand and current capacity.

capacity requirement What a process's capacity should be for some future time period to meet the demand of customers (external or internal), given the firm's desired capacity cushion.

capacity requirements planning (CRP) A technique used for projecting time-phased capacity requirements for workstations; its purpose is to match the material requirements plan with the capacity of key processes.

capital intensity The mix of equipment and human skills in a process.

carbon footprint The total amount of greenhouse gasses produced to support operations, usually expressed in equivalent tons of carbon dioxide (CO_2).

cash flow The difference between the flows of funds into and out of an organization over a period of time, including revenues, costs, and changes in assets and liabilities.

catalog hubs A system whereby suppliers post their catalog of items on the Internet and buyers select what they need and purchase them electronically.

causal methods A quantitative forecasting method that uses historical data on independent variables, such as promotional campaigns, economic conditions, and competitors' actions, to predict demand.

cause-and-effect diagram A diagram that relates a key performance problem to its potential causes.

center of gravity A good starting point to evaluate locations in the target area using the load–distance model.

centralized placement Keeping all the inventory of a product at a single location such as a firm's manufacturing plant or a warehouse and shipping directly to each of its customers.

certainty The word that is used to describe that a fact is known without doubt.

channel One or more facilities required to perform a given service.

channel assembly The process of using members of the distribution channel as if they were assembly stations in the factory.

chase strategy A strategy that involves hiring and laying off employees to match the demand forecast.

checklist A form used to record the frequency of occurrence of certain process failures.

close out An activity that includes writing final reports, completing remaining deliverables, and compiling the team's recommendations for improving the project process.

closed-loop supply chain A supply chain that integrates forward logistics with reverse logistics, thereby focusing on the complete chain of operations from the birth to the death of a product.

closeness matrix A table that gives a measure of the relative importance of each pair of operations being located close together.

collaborative planning, forecasting, and replenishment (CPFR) A nine-step process for supply chain integration that allows a supplier and its customers to collaborate on making the forecast by using the Internet.

combination forecasts Forecasts that are produced by averaging independent forecasts based on different methods, different sources, or different data.

common causes of variation The purely random, unidentifiable sources of variation that are unavoidable with the current process.

competitive capabilities The cost, quality, time, and flexibility dimensions that a process or supply chain actually possesses and is able to deliver.

competitive orientation A supplier relation that views negotiations between buyer and seller as a zero-sum game: Whatever one side loses, the other side gains, and short-term advantages are prized over long-term commitments.

competitive priorities The critical dimensions that a process or supply chain must possess to satisfy its internal or external customers, both now and in the future.

complementary products Services or products that have similar resource requirements but different demand cycles.

component An item that goes through one or more operations to be transformed into or become part of one or more parents.

compounding interest The process by which interest on an investment accumulates and then earns interest itself for the remainder of the investment period.

concurrent engineering A concept that brings product engineers, process engineers, marketers, buyers, information specialists, quality specialists, and suppliers together to design a product and the processes that will meet customer expectations.

consistent quality Producing services or products that meet design specifications on a consistent basis.

constraint Any factor that limits the performance of a system and restricts its output.

constraints In linear programming, the limitations that restrict the permissible choices for the decision variables.

consumer's risk (β) The probability of accepting a lot with LTPD quality (a type II error).

continuous flow process The extreme end of high-volume standardized production and rigid line flows, with production not starting and stopping for long time intervals.

continuous improvement The philosophy of continually seeking ways to improve processes based on a Japanese concept called *kaizen*.

continuous review (Q) system A system designed to track the remaining inventory of a SKU each time a withdrawal is made to determine whether it is time to reorder.

control chart A time-ordered diagram that is used to determine whether observed variations are abnormal.

cooperative orientation A supplier relation in which the buyer and seller are partners, each helping the other as much as possible.

core competencies The unique resources and strengths that an organization's management considers when formulating strategy.

core process A set of activities that delivers value to external customers.

corner point A point that lies at the intersection of two (or possibly more) constraint lines on the boundary of the feasible region.

crash cost (CC) The activity cost associated with the crash time.

crash time (CT) The shortest possible time to complete an activity.

critical mass A situation whereby several competing firms clustered in one location attract more customers than the total number who would shop at the same stores at scattered locations.

critical path The sequence of activities between a project's start and finish that takes the longest time to complete.

critical path method (CPM) A network planning method developed in the 1950s as a means of scheduling maintenance shutdowns at chemical-processing plants.

critical ratio (CR) A ratio that is calculated by dividing the time remaining until a job's due date by the total shop time remaining for the job, which is defined as the setup, processing, move, and expected waiting times of all remaining operations, including the operation being scheduled.

cross-docking The packing of products on incoming shipments so that they can be easily sorted at intermediate warehouses for outgoing shipments based on their final destinations; the items are carried from the incoming-vehicle docking point to the outgoing-vehicle docking point without being stored in inventory at the warehouse.

cumulative sum of forecast errors (CFE) A measurement of the total forecast error that assesses the bias in a forecast.

customer contact The extent to which the customer is present, is actively involved, and receives personal attention during the service process.

customer involvement The ways in which customers become part of the process and the extent of their participation.

customer population An input that generates potential customers.

customer relationship process A process that identifies, attracts, and builds relationships with external customers, and facilitates the placement of orders by customers; sometimes referred to as *customer relationship management*.

customization Satisfying the unique needs of each customer by changing service or product designs.

cycle counting An inventory control method, whereby storeroom personnel physically count a small percentage of the total number of items each day, correcting errors that they find.

cycle inventory The portion of total inventory that varies directly with lot size.

cycle-service level See service level.

cycle time The maximum time allowed for work on a unit at each station.

decision theory A general approach to decision making when the outcomes associated with alternatives are often in doubt.
decision tree A schematic model of alternatives available to the decision maker, along with their possible consequences.
decision variables Variables that represent the choices the decision maker can control.
defect Any instance when a process fails to satisfy its customer.
degeneracy A condition that occurs when the number of nonzero variables in the optimal solution is less than the number of constraints.
delivery speed Quickly filling a customer's order.
Delphi method A process of gaining consensus from a group of experts while maintaining their anonymity.
demand management The process of changing demand patterns using one or more demand options.
dependent demand The demand for an item that occurs because the quantity required varies with the production plans for other items held in the firm's inventory.
dependent variable The variable that one wants to forecast.
design team A group of knowledgeable, team-oriented individuals who work at one or more steps in the process, conduct the process analysis, and make the necessary changes.
development speed Quickly introducing a new service or a product.
discount rate The interest rate used in discounting the future value to its present value.
discounting The process of finding the present value of an investment when the future value and the interest rate are known.
diseconomies of scale Occurs when the average cost per unit increases as the facility's size increases.
distribution center A warehouse or stocking point where goods are stored for subsequent distribution to manufacturers, wholesalers, retailers, and customers.
double-sampling plan A plan in which management specifies two sample sizes and two acceptance numbers; if the quality of the lot is very good or very bad, the consumer can make a decision to accept or reject the lot on the basis of the first sample, which is smaller than in the single-sampling plan.
drum-buffer-rope (DBR) A planning and control system that regulates the flow of work-in-process materials at the bottleneck or the capacity constrained resource (CCR) in a productive system.
earliest due date (EDD) A priority sequencing rule that specifies that the job or customer with the earliest due date is the next job to be processed.
earliest finish time (EF) An activity's earliest start time plus its estimated duration, t, or EF = ES + t.
earliest start time (ES) The earliest finish time of the immediately preceding activity.
early supplier involvement A program that includes suppliers in the design phase of a service or product.
economic order quantity (EOQ) The lot size that minimizes total annual inventory holding and ordering costs.
economic production lot size (ELS) The optimal lot size in a situation in which replenishment is not instantaneous.
economies of scale A concept that states that the average unit cost of a service or good can be reduced by increasing its output rate.
economies of scope Economies that reflect the ability to produce multiple products more cheaply in combination than separately.
electronic commerce (e-commerce) The application of information and communication technology anywhere along the supply chain of business processes.
electronic data interchange (EDI) A technology that enables the transmission of routine business documents having a standard format from computer to computer over telephone or direct leased lines.
elemental standard data A database of standards compiled by a firm's analysts for basic elements that they can draw on later to estimate the time required for a particular job, which is most appropriate when products or services are highly customized, job processes prevail, and process divergence is great.
employee empowerment An approach to teamwork that moves responsibility for decisions further down the organizational chart—to the level of the employee actually doing the job.
end item The final product sold to a customer.
enterprise process A companywide process that cuts across functional areas, business units, geographical regions, and product lines.
enterprise resource planning (ERP) systems Large, integrated information systems that support many enterprise processes and data storage needs.
environmental responsibility An element of sustainability that addresses the ecological needs of the planet and the firm's stewardship of the natural resources used in the production of services and products.
Euclidean distance The straight-line distance, or shortest possible path, between two points.
exchange An electronic marketplace where buying firms and selling firms come together to do business.
executive opinion A forecasting method in which the opinions, experience, and technical knowledge of one or more managers are summarized to arrive at a single forecast.
expediting The process of completing a job or finishing with a customer sooner than would otherwise be done.

exponential smoothing method A weighted moving average method that calculates the average of a time series by implicitly giving recent demands more weight than earlier demands.

external customers A customer who is either an end user or an intermediary (e.g., manufacturers, financial institutions, or retailers) buying the firm's finished services or products.

external failure costs Costs that arise when a defect is discovered after the customer receives the service or product.

external suppliers The businesses or individuals who provide the resources, services, products, and materials for the firm's short-term and long-term needs.

facility location The process of determining geographic sites for a firm's operations.

feasible region A region that represents all permissible combinations of the decision variables in a linear programming model.

financial responsibility An element of sustainability that addresses the financial needs of the shareholders, employees, customers, business partners, financial institutions, and any other entity that supplies the capital for the production of services or products or relies on the firm for wages or reimbursements.

finished goods (FG) The items in manufacturing plants, warehouses, and retail outlets that are sold to the firm's customers.

first-come, first served (FCFS) A priority sequencing rule that specifies that the job or customer arriving at the workstation first has the highest priority.

five S (5S) A methodology consisting of five workplace practices—sorting, straightening, shining, standardizing, and sustaining—that are conducive to visual controls and lean production.

fixed automation A manufacturing process that produces one type of part or product in a fixed sequence of simple operations.

fixed cost The portion of the total cost that remains constant regardless of changes in levels of output.

fixed order quantity (FOQ) A rule that maintains the same order quantity each time an order is issued.

fixed schedule A schedule that calls for each employee to work the same days and hours each week.

flexible (or programmable) automation A manufacturing process that can be changed easily to handle various products.

flexible flow The customers, materials, or information move in diverse ways, with the path of one customer or job often crisscrossing the path that the next one takes.

flexible workforce A workforce whose members are capable of doing many tasks, either at their own workstations or as they move from one workstation to another.

flow shop A manufacturer's operation that specializes in medium- to high-volume production and utilizes line or continuous flow processes.

flow time The amount of time a job spends in the service or manufacturing system.

flowchart A diagram that traces the flow of information, customers, equipment, or materials through the various steps of a process.

focus forecasting A method of forecasting that selects the best forecast from a group of forecasts generated by individual techniques.

focused factories The result of a firm's splitting large plants that produced all the company's products into several specialized smaller plants.

forecast A prediction of future events used for planning purposes.

forecast error The difference found by subtracting the forecast from actual demand for a given period.

forward integration Acquiring more channels of distribution, such as distribution centers (warehouses) and retail stores, or even business customers.

forward placement Locating stock closer to customers at a warehouse, DC, wholesaler, or retailer.

front office A process with high customer contact where the service provider interacts directly with the internal or external customer.

future value of an investment The value of an investment at the end of the period over which interest is compounded.

Gantt chart A project schedule, usually created by the project manager using computer software, that superimposes project activities, with their precedence relationships and estimated duration times, on a time line.

geographical information system (GIS) A system of computer software, hardware, and data that the firm's personnel can use to manipulate, analyze, and present information relevant to a location decision.

graphic method of linear programming A type of graphic analysis that involves the following five steps: plotting the constraints, identifying the feasible region, plotting an objective function line, finding a visual solution, and finding the algebraic solution.

graphs Representations of data in a variety of pictorial forms, such as line charts and pie charts.

Green Belt An employee who achieved the first level of training in a Six Sigma program and spends part of his or her time teaching and helping teams with their projects.

green purchasing The process of identifying, assessing, and managing the flow of environmental waste and finding ways to reduce it and minimize its impact on the environment.

gross requirements The total demand derived from *all* parent production plans.

group technology (GT) An option for achieving line-flow layouts with low volume processes; this technique creates cells not limited to just one worker and has a unique way of selecting work to be done by the cell.

heijunka The leveling of production load by both volume and product mix.

hiring and layoff Costs of advertising jobs, interviews, training programs for new employees, scrap caused by the inexperience of new employees, loss of productivity, and initial paperwork. Layoff costs include the costs of exit interviews, severance pay, retaining and retraining remaining workers and managers, and lost productivity.

histogram A summarization of data measured on a continuous scale, showing the frequency distribution of some process failure (in statistical terms, the central tendency and dispersion of the data).

holdout sample Actual demands from the more recent time periods in the time series that are set aside to test different models developed from the earlier time periods.

humanitarian logistics The process of planning, implementing and controlling the efficient, cost-effective flow and storage of goods and materials, as well as related information, from the point of origin to the point of consumption for the purpose of alleviating the suffering of vulnerable people.

hurdle rate The interest rate that is the lowest desired return on an investment; the hurdle over which the investment must pass.

hybrid office A process with moderate levels of customer contact and standard services with some options available.

immediate predecessors Work elements that must be done before the next element can begin.

independent demand items Items for which demand is influenced by market conditions and is not related to the inventory decisions for any other item held in stock or produced.

independent variables Variables that are assumed to affect the dependent variable and thereby "cause" the results observed in the past.

industrial robot Versatile, computer-controlled machine programmed to perform various tasks.

interarrival times The time between customer arrivals.

intermediate item An item that has at least one parent and at least one component.

intermodal shipments Mixing the modes of transportation for a given shipment, such as moving shipping containers or truck trailers on rail cars.

internal customers One or more employees or processes that rely on inputs from other employees or processes in order to perform their work.

internal failure costs Costs resulting from defects that are discovered during the production of a service or product.

internal suppliers The employees or processes that supply important information or materials to a firm's processes.

inventory A stock of materials used to satisfy customer demand or to support the production of services or goods.

inventory holding cost The sum of the cost of capital and the variable costs of keeping items on hand, such as storage and handling, taxes, insurance, and shrinkage.

inventory management The planning and controlling of inventories in order to meet the competitive priorities of the organization.

inventory pooling A reduction in inventory and safety stock because of the merging of variable demands from customers.

inventory position (IP) The measurement of a SKU's ability to satisfy future demand.

inventory record A record that shows an item's lot-size policy, lead time, and various time-phased data.

inventory turnover An inventory measure obtained by dividing annual sales at cost by the average aggregate inventory value maintained during the year.

ISO 9001:2008 A set of standards governing documentation of a quality program.

ISO 14000:2004 Documentation standards that require participating companies to keep track of their raw materials use and their generation, treatment, and disposal of hazardous wastes.

ISO 26000:2010 International guidelines for organizational social responsibility.

jidoka Automatically stopping the process when something is wrong and then fixing the problems on the line itself as they occur.

JIT system A system that organizes the resources, information flows, and decision rules that enable a firm to realize the benefits of JIT principles.

job process A process with the flexibility needed to produce a wide variety of products in significant quantities, with considerable divergence in the steps performed.

job shop A manufacturer's operation that specializes in low- to medium-volume production and utilizes job or batch processes.

Johnson's rule A procedure that minimizes makespan when scheduling a group of jobs on two workstations.

judgment methods A forecasting method that translates the opinions of managers, expert opinions, consumer surveys, and salesforce estimates into quantitative estimates.

judgmental adjustment An adjustment made to forecasts from one or more quantitative models that accounts for recognizing which models are performing particularly well in recent past, or take into account contextual information.

just-in-time (JIT) philosophy The belief that waste can be eliminated by cutting unnecessary capacity or inventory and removing non-value-added activities in operations.

kanban A Japanese word meaning "card" or "visible record" that refers to cards used to control the flow of production through a factory.

labor-limited environment An environment in which the resource constraint is the amount of labor available, not the number of machines or workstations.

latest finish time (LF) The latest start time of the activity that immediately follows.

latest start time (LS) The latest finish time minus its estimated duration, t, or LS = LF – t.

layout The physical arrangement of operations created by the various processes.

lead time The elapsed time between the receipt of a customer order and filling it.

lean systems Operations systems that maximize the value added by each of a company's activities by removing waste and delays from them.

learning curve A line that displays the relationship between processing time and the cumulative quantity of a product or service produced.

learning curve analysis A time estimation technique that takes into account the learning that takes place on an ongoing basis, such as where new products or services are introduced.

level strategy A strategy that keeps the workforce constant, but varies its utilization with overtime, undertime, and vacation planning to match the demand forecast.

line balancing The assignment of work to stations in a line process so as to achieve the desired output rate with the smallest number of workstations.

line flow The customers, materials, or information move linearly from one operation to the next, according to a fixed sequence.

line process A process that lies between the batch and continuous processes on the continuum; volumes are high and products are standardized, which allows resources to be organized around particular products.

linear programming A technique that is useful for allocating scarce resources among competing demands.

linear regression A causal method in which one variable (the dependent variable) is related to one or more independent variables by a linear equation.

linearity A characteristic of linear programming models that implies proportionality and additivity—there can be no products or powers of decision variables.

Little's law A fundamental law that relates the number of customers in a waiting-line system to the arrival rate and waiting time of customers.

load–distance method A mathematical model used to evaluate locations based on proximity factors.

lot A quantity of items that are processed together.

lot-for-lot (L4L) rule A rule under which the lot size ordered covers the gross requirements of a single week.

lot size The quantity of an inventory item management either buys from a supplier or manufactures using internal processes.

lot sizing The determination of how frequently and in what quantity to order inventory.

lot tolerance proportion defective (LTPD) The worst level of quality that the consumer can tolerate.

low-cost operation Delivering a service or a product at the lowest possible cost to the satisfaction of external or internal customers of the process or supply chain.

make-or-buy decision A managerial choice between whether to outsource a process or do it in-house.

make-to-order strategy A strategy used by manufacturers that make products to customer specifications in low volumes.

make-to-stock strategy A strategy that involves holding items in stock for immediate delivery, thereby minimizing customer delivery times.

makespan The total amount of time required to complete a group of jobs.

manufacturing resource planning (MRP II) A system that ties the basic MRP system to the company's financial system and to other core and supporting processes.

market research A systematic approach to determine external consumer interest in a service or product by creating and testing hypotheses through data-gathering surveys.

mass customization The strategy that uses highly divergent processes to generate a wide variety of customized products at reasonably low costs.

mass production A term sometimes used in the popular press for a line process that uses the make-to-stock strategy.

Master Black Belt Full-time teachers and mentors to several Black Belts.

master production schedule (MPS) A part of the material requirements plan that details how many end items will be produced within specified periods of time.

material requirements planning (MRP) A computerized information system developed specifically to help manufacturers manage dependent demand inventory and schedule replenishment orders.

mean absolute deviation (MAD) A measurement of the dispersion of forecast errors.

mean absolute percent error (MAPE) A measurement that relates the forecast error to the level of demand and is useful for putting forecast performance in the proper perspective.

mean squared error (MSE) A measurement of the dispersion of forecast errors.

methods time measurement (MTM) A commonly used predetermined data system.

metrics Performance measures that are established for a process and the steps within it.

minimum-cost schedule A schedule determined by starting with the normal time schedule and crashing activities along the critical path, in such a way that the costs of crashing do not exceed the savings in indirect and penalty costs.

mixed-model assembly A type of assembly that produces a mix of models in smaller lots.

mixed-model line A production line that produces several items belonging to the same family.

mixed strategy A strategy that considers the full range of supply options.

Modified Accelerated Cost Recovery System (MACRS) The only acceptable depreciation method for tax purposes that shortens the lives of investments, giving firms larger early tax deductions.

Monte Carlo simulation A simulation process that uses random numbers to generate simulation events.

most likely time (*m*) The probable time required to perform an activity.

MRP explosion A process that converts the requirements of various final products into a material requirements plan that specifies the replenishment schedules of all the subassemblies, components, and raw materials needed to produce final products.

multiple-dimension rules A set of rules that apply to more than one aspect of a job.

multiplicative seasonal method A method whereby seasonal factors are multiplied by an estimate of average demand to arrive at a seasonal forecast.

nearest neighbor (NN) heuristic A technique that creates a route by deciding the next city to visit on the basis of its proximity.

naïve forecast A time-series method whereby the forecast for the next period equals the demand for the current period, or Forecast = D_t.

nested process The concept of a process within a process.

net present value (NPV) method The method that evaluates an investment by calculating the present values of all after-tax total cash flows and then subtracting the initial investment amount for their total.

network diagram A network planning method, designed to depict the relationships between activities, that consists of nodes (circles) and arcs (arrows).

new service/product development process A process that designs and develops new services or products from inputs received from external customer specifications or from the market in general through the customer relationship process.

nominal value A target for design specifications.

nonnegativity An assumption that the decision variables must be positive or zero.

normal cost (NC) The activity cost associated with the normal time.

normal time (NT) In the context of project management, the time necessary to complete an activity under normal conditions.

normal time (NT) In the context of time study, a measurement found by multiplying the select time (t), the frequency (F) of the work element per cycle, and the rating factor (RF).

normal time for the cycle (NTC) A measurement found by summing the normal time for each element.

objective function An expression in linear programming models that states mathematically what is being maximized or minimized.

offshoring A supply chain strategy that involves moving processes to another country.

one-worker, multiple-machines (OWMM) cell A one-person cell in which a worker operates several different machines simultaneously to achieve a line flow.

on-time delivery Meeting delivery-time promises.

open orders See scheduled receipts (SR).

operating characteristic (OC) curve A graph that describes how well a sampling plan discriminates between good and bad lots.

operations management The systematic design, direction, and control of processes that transform inputs into services and products for internal, as well as external, customers.

operations planning and scheduling The process of balancing supply with demand, from the aggregate level down to the short-term scheduling level.

operations scheduling A type of scheduling in which jobs are assigned to workstations or employees are assigned to jobs for specified time periods.

operations strategy The means by which operations implements the firm's corporate strategy and helps to build a customer-driven firm.

optimistic time (*a*) The shortest time in which an activity can be completed, if all goes exceptionally well.

optional replenishment system A system used to review the inventory position at fixed time intervals and, if the position has dropped to (or below) a predetermined level, to place a variable-sized order to cover expected needs.

order fulfillment process A process that includes the activities required to produce and deliver the service or product to the external customer.

order qualifier Minimal level required from a set of crteria for a firm to do business in a particular market segment.

order winner A criterion customers use to differentiate the services or products of one firm from those of another.

ordering cost The cost of preparing a purchase order for a supplier or a production order for manufacturing.

organizational learning The process of gaining experience with products and processes, achieving greater efficiency through automation and other capital investments, and making other improvements in administrative methods or personnel.

outsourcing Paying suppliers and distributors to perform processes and provide needed services and materials.

overtime The time that employees work that is longer than the regular workday or workweek for which they receive additional pay.

***p*-chart** A chart used for controlling the proportion of defective services or products generated by the process.

pacing The movement of product from one station to the next as soon as the cycle time has elapsed.

parameter A value that the decision maker cannot control and that does not change when the solution is implemented.

parent Any product that is manufactured from one or more components.

Pareto chart A bar chart on which factors are plotted along the horizontal axis in decreasing order of frequency.

part commonality The degree to which a component has more than one immediate parent.

past due The amount of time by which a job missed its due date.

path The sequence of activities between a project's start and finish.

payback method A method for evaluating projects that determines how much time will elapse before the total of after-tax flows will equal, or pay back, the initial investment.

payoff table A table that shows the amount for each alternative if each possible event occurs.

performance rating factor (RF) An assessment that describes *how much* above or below average the worker's performance is on each work element.

periodic order quantity (POQ) A rule that allows a different order quantity for each order issued but issues the order for predetermined time intervals.

periodic review (*P*) sytem A system in which an item's inventory position is reviewed periodically rather than continuously.

perpetual inventory system A system of inventory control in which the inventory records are always current.

pessimistic time (*b*) The longest estimated time required to perform an activity.

phase A single step in providing a service.

pipeline inventory Inventory that is created when an order for an item is issued but not yet received.

plan-do-study-act cycle A cycle, also called the Deming Wheel, used by firms actively engaged in continuous improvement to train their work teams in problem solving.

planned order release An indication of when an order for a specified quantity of an item is to be issued.

planned receipts Orders that are not yet released to the shop or the supplier.

planning horizon The set of consecutive time periods considered for planning purposes.

plants within plants (PWPs) Different operations within a facility with individualized competitive priorities, processes, and workforces under the same roof.

poka-yoke Mistake-proofing methods aimed at designing fail-safe systems that minimize human error.

postponement The strategy of delaying final activities in the provision of a product until the orders are received.

precedence diagram A diagram that allows one to visualize immediate predecessors better; work elements are denoted by circles, with the time required to perform the work shown below each circle.

precedence relationship A relationship that determines a sequence for undertaking activities; it specifies that one activity cannot start until a preceding activity has been completed.

predetermined data approach A database approach that divides each work element into a series of micromotions that make up the element. The analyst then consults a published database that contains the normal times for the full array of possible micromotions.

preemptive discipline A rule that allows a customer of higher priority to interrupt the service of another customer.

preference matrix A table that allows the manager to rate an alternative according to several performance criteria.

present value of an investment The amount that must be invested now to accumulate to a certain amount in the future at a specific interest rate.

presourcing A level of supplier involvement in which suppliers are selected early in a product's concept development stage and are given significant, if not total, responsibility for the design of certain components or systems of the product.

prevention costs Costs associated with preventing defects before they happen.

priority rule A rule that selects the next customer to be served by the service facility.

priority sequencing rule A rule that specifies the job or customer processing sequence when several jobs are waiting in line at a workstation.

process Any activity or group of activities that takes one or more inputs, transforms them, and provides one or more outputs for its customers.

process analysis The documentation and detailed understanding of how work is performed and how it can be redesigned.

process capability The ability of the process to meet the design specifications for a service or product.

process capability index, C_{pk} An index that measures the potential for a process to generate defective outputs relative to either upper or lower specifications.

process capability ratio, C_p The tolerance width divided by six standard deviations.

process chart An organized way of documenting all the activities performed by a person or group of people, at a workstation, with a customer, or on materials.

process choice A way of structuring the process by organizing resources around the process or organizing them around the products.

process divergence The extent to which the process is highly customized with considerable latitude as to how its tasks are performed.

process failure Any performance shortfall, such as error, delay, environmental waste, rework, and the like.

process improvement The systematic study of the activities and flows of each process to improve it.

process simulation The act of reproducing the behavior of a process, using a model that describes each step.

process strategy The pattern of decisions made in managing processes so that they will achieve their competitive priorities.

process structure A process decision that determines the process type relative to the kinds of resources needed, how resources are partitioned between them, and their key characteristics.

producer's risk (α) The risk that the sampling plan will fail to verify an acceptable lot's quality and, thus, reject it (a type I error).

product family A group of services or products that have similar demand requirements and common process, labor, and materials requirements.

production plan a manufacturing firm's sales and operations plan that centers on production rates and inventory holdings.

product-mix problem A one-period type of planning problem, the solution of which yields optimal output quantities (or product mix) of a group of services or products subject to resource capacity and market demand constraints.

productivity The value of outputs (services and products) produced divided by the values of input resources (wages, costs of equipment, and so on).

program An interdependent set of projects that have a common strategic purpose.

program evaluation and review technique (PERT) A network planning method created for the U.S. Navy's Polaris missile project in the 1950s, which involved 3,000 separate contractors and suppliers.

project An interrelated set of activities with a definite starting and ending point, which results in a unique outcome for a specific allocation of resources.

project management A systemized, phased approach to defining, organizing, planning, monitoring, and controlling projects.

projected on-hand inventory An estimate of the amount of inventory available each week after gross requirements have been satisfied.

protection interval The period over which safety stock must protect the user from running out of stock.

pull method A method in which customer demand activates production of the service or item.

purchased item An item that has one or more parents but no components because it comes from a supplier.

purchasing The activity that decides which suppliers to use, negotiates contracts, and determines whether to buy locally.

push method A method in which production of the item begins in advance of customer needs.

quality A term used by customers to describe their general satisfaction with a service or product.

quality at the source A philosophy whereby defects are caught and corrected where they were created.

quality circles Another name for problem-solving teams; small groups of supervisors and employees who meet to identify, analyze, and solve process and quality problems.

quality engineering An approach originated by Genichi Taguchi that involves combining engineering and statistical methods to reduce costs and improve quality by optimizing product design and manufacturing processes.

quality loss function The rationale that a service or product that barely conforms to the specifications is more like a defective service or product than a perfect one.

quality of life A factor that considers the availability of good schools, recreational facilities, cultural events, and an attractive lifestyle.

quantity discount A drop in the price per unit when an order is sufficiently large.

***R*-chart** A chart used to monitor process variability.

radio frequency identification (RFID) A method for identifying items through the use of radio signals from a tag attached to an item.

random number A number that has the same probability of being selected as any other number.

range of feasibility The interval (lower and upper bounds) over which the right-hand-side parameter of a constraint can vary while its shadow price remains valid.

range of optimality The interval (lower and upper bounds) of an objective function coefficient over which the optimal values of the decision variables remain unchanged.

raw materials (RM) The inventories needed for the production of services or goods.
rectified inspection The assumption that all defective items in the lot will be replaced with good items if the lot is rejected and that any defective items in the sample will be replaced if the lot is accepted.
rectilinear distance The distance between two points with a series of 90-degree turns, as along city blocks.
reduced cost How much the objective function coefficient of a decision variable must improve (increase for maximization or decrease for minimization) before the optimal solution changes and the decision variable "enters" the solution with some positive number.
reengineering The fundamental rethinking and radical redesign of processes to improve performance dramatically in terms of cost, quality, service, and speed.
regular time Wages paid to employees plus contributions to benefits.
reorder point (*R*) The predetermined minimum level that an inventory position must reach before a fixed quantity *Q* of the SKU is ordered.
reorder point (ROP) system See continuous review (*Q*) system.
repeatability The degree to which the same work can be done again.
resource flexibility The ease with which employees and equipment can handle a wide variety of products, output levels, duties, and functions.
resource plan A plan that determines the requirements for materials and other resources on a more detailed level than the sales and operations plan.
resource planning A process that takes sales and operations plans; processes information in the way of time standards, routings, and other information on how services or products are produced; and then plans the input requirements.
revenue management Varying price at the right time for different customer segments to maximize revenues yielded by existing supply capacity.
reverse logistics The process of planning, implementing, and controlling the efficient, cost effective flow of products, materials, and information from the point of consumption back to the point of origin for returns, repair, remanufacture, or recycling.
risk-management plan A plan that identifies the key risks to a project's success and prescribes ways to circumvent them.
rotating schedule A schedule that rotates employees through a series of workdays or hours.
route planning An activity that seeks to find the shortest route to deliver a service or product.
SA8000:2008 A list of standards covering nine dimensions of ethical workforce management.
safety stock inventory Surplus inventory that a company holds to protect against uncertainties in demand, lead time, and supply changes.
sales and operations plan (S&OP) A plan of future aggregate resource levels so that supply is in balance with demand.
salesforce estimates The forecasts that are compiled from estimates of future demands made periodically by members of a company's salesforce.
salvage value The cash flow from the sale or disposal of plant and equipment at the end of a project's life.
sample size A quantity of randomly selected observations of process outputs.
sampling plan A plan that specifies a sample size, the time between successive samples, and decision rules that determine when action should be taken.
scatter diagram A plot of two variables showing whether they are related.
schedule A detailed plan that allocates resources over short time horizons to accomplish specific tasks.
scheduled receipts (SR) Orders that have been placed but have not yet been received.
SCOR model A framework that focuses on a basic supply chain of plan, source, make, deliver, and return processes, repeated again and again along the supply chain.
select time ($\bar{t}$) The average observed time based only on representative times.
self-managed team A small group of employees who work together to produce a major portion, or sometimes all, of a service or product.
sensitivity analysis A technique for systematically changing parameters in a model to determine the effects of such changes.
sequencing Determining the order in which jobs or customers are processed in the waiting line at a workstation.
sequential-sampling plan A plan in which the consumer randomly selects items from the lot and inspects them one by one.
service blueprint A special flowchart of a service process that shows which steps have high customer contact.
service facility A person (or crew), a machine (or group of machines), or both necessary to perform the service for the customer.
service level The desired probability of not running out of stock in any one ordering cycle, which begins at the time an order is placed and ends when it arrives in stock.
service system The number of lines and the arrangement of the facilities.
setup cost The cost involved in changing over a machine or workspace to produce a different item.
setup time The time required to change a process or an operation from making one service or product to making another.
shadow price The marginal improvement in *Z* (increase for maximization and decrease for minimization) caused by relaxing the constraint by one unit.

shortest processing time (SPT) A priority sequencing rule that specifies that the job requiring the shortest processing time is the next job to be processed.

shortest route problem A problem whose objective is to find the shortest distance between two cities in a network or map.

simple moving average method A time-series method used to estimate the average of a demand time series by averaging the demand for the n most recent time periods.

simplex method An iterative algebraic procedure for solving linear programming problems.

simulation The act of reproducing the behavior of a system using a model that describes the processes of the system.

single-bin system A system of inventory control in which a maximum level is marked on the storage shelf or bin, and the inventor is brought up to the mark periodically.

single-digit setup The goal of having a setup time of less than 10 minutes.

single-dimension rules A set of rules that bases the priority of a job on a single aspect of the job, such as arrival time at the workstation, the due date, or the processing time.

single-sampling plan A decision to accept or reject a lot based on the results of one random sample from the lot.

Six Sigma A comprehensive and flexible system for achieving, sustaining, and maximizing business success by minimizing defects and variability in processes.

slack The amount by which the left-hand side of a linear programming constraint falls short of the right-hand side.

slack per remaining operations (S/RO) A priority sequencing rule that determines priority by dividing the slack by the number of operations that remain, including the one being scheduled.

social responsibility An element of sustainability that addresses the moral, ethical, and philanthropic expectations that society has of an organization.

sole sourcing The awarding of a contract for a service or item to only one supplier.

special-purpose teams Groups that address issues of paramount concern to management, labor, or both.

staffing plan A sales and operations plan for a service firm, which centers on staffing and other human resource-related factors.

standard deviation(σ) for forecasting A measurement of the dispersion of forecast errors.

standard deviation (σ) for statistical quality control The square root of the variance of a distribution.

standard time (ST) A measurement found by incorporating the normal time and allowances; ST = NTC(1 + A), where A equals the proportion of the normal time added for allowances.

statistical process control (SPC) The application of statistical techniques to determine whether a process is delivering what the customer wants.

steady state The state that occurs when the simulation is repeated over enough time that the average results for performance measures remain constant.

stock-keeping unit (SKU) An individual item or product that has an identifying code and is held in inventory somewhere along the supply chain.

stockout An order that cannot be satisfied, resulting in a loss of the sale.

straight-line depreciation method The simplest method of calculating annual depreciation; found by subtracting the estimated salvage value from the amount of investment required at the beginning of the project, and then dividing by the asset's expected economic life.

subassembly An intermediate item that is *assembled* (as opposed to being transformed by other means) from more than one component.

suggestion system A voluntary system by which employees submit their ideas on process improvements.

supplier relationship process A process that selects the suppliers of services, materials, and information and facilitates the timely and efficient flow of these items into the firm.

supply chain An interrelated series of processes within and across firms that produces a service or product to the satisfaction of customers.

supply chain design Designing a firm's supply chain to meet the competitive priorities of the firm's operations strategy.

supply chain processes Business processes that have external customers or suppliers.

supply chain integration The effective coordination of supply chain processes through the seamless flow of information up and down the supply chain.

supply chain management The synchronization of a firm's processes with those of its suppliers and customers to match the flow of materials, services, and information with customer demand.

support process A process that provides vital resources and inputs to the core processes and therefore is essential to the management of the business.

surplus The amount by which the left-hand side of a linear programming constraint exceeds the right-hand side.

sustainability A characteristic of processes that are meeting humanity's needs without harming future generations.

swim lane flowchart A visual representation that groups functional areas responsible for different sub-processes into lanes.

takt time Cycle time needed to match the rate of production to the rate of sales or consumption.

tardiness See past due.

teams Small groups of people who have a common purpose, set their own performance goals and approaches, and hold themselves accountable for success.

technological forecasting An application of executive opinion to keep abreast of the latest advances in technology.

theoretical minimum (TM) A benchmark or goal for the smallest number of stations possible, where the total time required to assemble each unit (the sum of all work-element standard times) is divided by the cycle time.

theory of constraints (TOC) A systematic management approach that focuses on actively managing those constraints that impede a firm's progress toward its goal.

throughput time Total elapsed time from the start to the finish of a job or a customer being processed at one or more workcenters.

time-based competition A strategy that focuses on the competitive priorities of delivery speed and development speed.

time between orders (TBO) The average elapsed time between receiving (or placing) replenishment orders of Q units for a particular lot size.

time compression The feature of simulation models that allows them to obtain operating characteristic estimates in much less time than is required to gather the same operating data from a real system.

time series The repeated observations of demand for a service or product in their order of occurrence.

time-series analysis A statistical approach that relies heavily on historical demand data to project the future size of demand and recognizes trends and seasonal patterns.

time study A work measurement method using a trained analyst to perform four basic steps in setting a time standard for a job or process: selecting the work elements (or nested processes) within the process to be studied, timing the elements, determining the sample size, and setting the final standard.

time value of money The concept that a dollar in hand can be invested to earn a return so that more than one dollar will be available in the future.

tolerance An allowance above or below the nominal value.

top quality Delivering an outstanding service or product.

total inventory The sum of scheduled receipts and on-hand inventories.

total quality management (TQM) A philosophy that stresses three principles for achieving high levels of process performance and quality: (1) customer satisfaction, (2) employee involvement, and (3) continuous improvement in performance.

tracking signal A measure that indicates whether a method of forecasting is accurately predicting actual changes in demand.

transportation method A more efficient solution technique than the simplex method for solving transportation problems.

transportation method for location problems A quantitative approach that can help solve multiple-facility location problems.

transportation problem A special case of linear programming that has linear constraints for capacity limitations and demand requirements.

traveling salesman problem A problem whose objective is to find the shortest possible route that visits each city exactly once and returns to the starting city.

trend projection with regression A forecasting model that is a hybrid between a time-series technique and the causal method.

two-bin system A visual system version of the Q system in which a SKU's inventory is stored at two different locations.

type I error An error that occurs when the employee concludes that the process is out of control based on a sample result that falls outside the control limits, when in fact it was due to pure randomness.

type II error An error that occurs when the employee concludes that the process is in control and only randomness is present, when actually the process is out of statistical control.

uncontrollable variables Random events that the decision maker cannot control.

undertime The situation that occurs when employees do not have enough work for the regular-time workday or workweek.

usage quantity The number of units of a component that are needed to make one unit of its immediate parent.

utilization The degree to which equipment, space, or the workforce is currently being used, and is measured as the ratio of average output rate to maximum capacity (expressed as a percent).

value analysis A systematic effort to reduce the cost or improve the performance of services or products, either purchased or produced.

value stream mapping (VSM) A qualitative lean tool for eliminating waste or *muda* that involves a current state drawing, a future state drawing, and an implementation plan.

variable cost The portion of the total cost that varies directly with volume of output.

variables Service or product characteristics, such as weight, length, volume, or time that can be measured.

variety Handling a wide assortment of services or products efficiently.

vendor-managed inventories (VMI) A system in which the supplier has access to the customer's inventory data and is responsible for maintaining the inventory on the customer's site.

visual system A system that allows employees to place orders when inventory visibly reaches a certain marker.

volume flexibility Accelerating or decelerating the rate of production of services or products quickly to handle large fluctuations in demand.

waiting line One or more "customers" waiting for service.

warranty A written guarantee that the producer will replace or repair defective parts or perform the service to the customer's satisfaction.

weeks of supply An inventory measure obtained by dividing the average aggregate inventory value by sales per week at cost.

weighted-distance method A mathematical model used to evaluate layouts (of facility locations) based on proximity factors.

weighted moving average method A time-series method in which each historical demand in the average can have its own weight; the sum of the weights equals 1.0.

work breakdown structure (WBS) A statement of all work that has to be completed.

work elements The smallest units of work that can be performed independently.

work-in-process (WIP) Items, such as components or assemblies, needed to produce a final product in manufacturing or service operations.

work measurement The process of creating labor standards based on the judgment of skilled observers.

work sampling method A process that estimates the proportion of time spent by people or machines on different activities, based on observations randomized over time.

work standard The time required for a trained worker to perform a task following a prescribed method with normal effort and skill.

workforce scheduling A type of scheduling that determines when employees work.

$\bar{x}$-chart A chart used to see whether the process is generating output, on average, consistent with a target value set by management for the process or whether its current performance, with respect to the average of the performance measure, is consistent with past performance.

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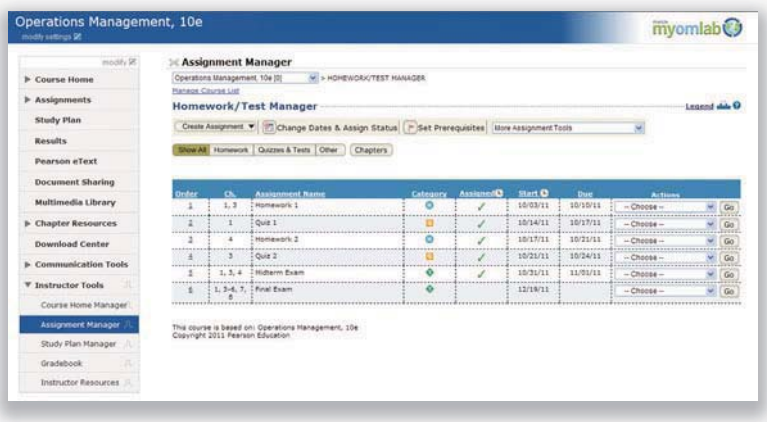
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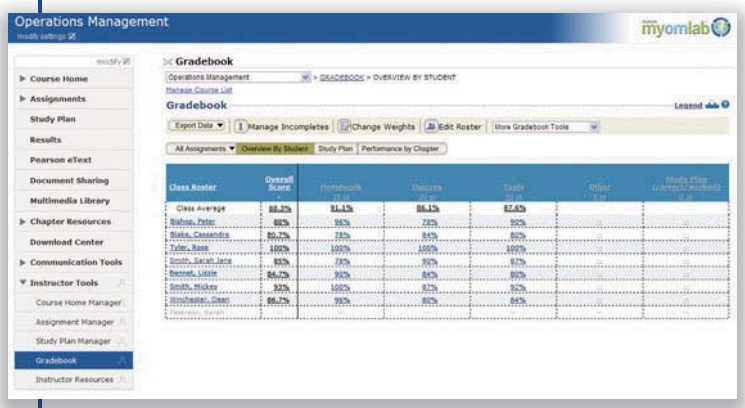
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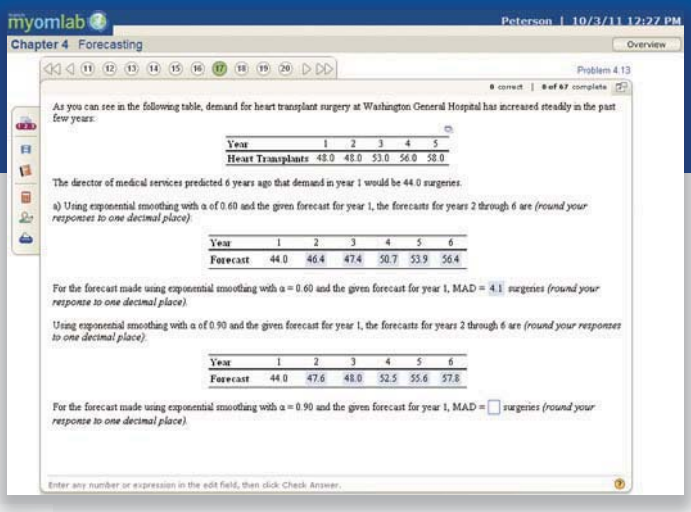
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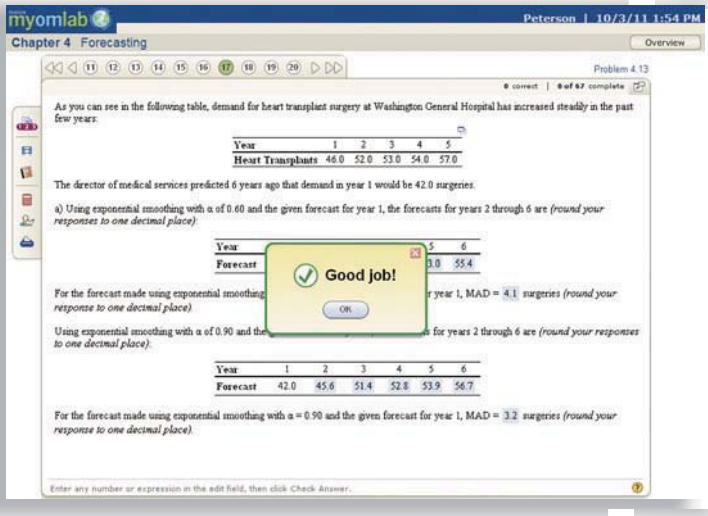
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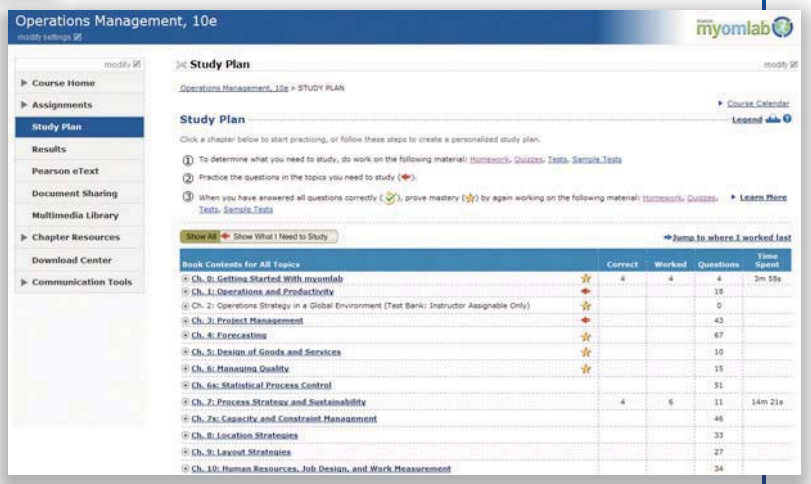
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